

OpenFOAM

The OpenFOAM Foundation

User Guide

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Chapter 1

Introduction

This guide accompanies the release of version 14 of the Open Source Field Operation and Manipulation (OpenFOAM) C++ libraries. It provides a description of the basic operation of OpenFOAM, first through a set of tutorial exercises in chapter 2 and later by more detailed descriptions of different components of OpenFOAM.

OpenFOAM is software for computational fluid dynamics (CFD). It includes a collection of *applications* which perform a range of tasks in CFD. The applications use packaged functionality contained within over 150 *libraries*. As well as performing calculations of the fluid dynamics, there are applications which configure and initialise simulations, manipulate case geometry, generate computational meshes, and process and visualise results.

Applications primarily fall into two categories: *solvers*, which perform the calculations in fluid (or other continuum) mechanics; and *utilities*, that perform the other tasks described above. Prior to version 11 of OpenFOAM, individual solvers were written for numerous specific types of flow. With so many combinations of flow type and additional physics, OpenFOAM included almost 100 solvers at one time. Solvers with names like `simpleFoam` and `pimpleFoam` have been the mainstay of OpenFOAM since the early 1990s.

Version 11 of OpenFOAM, however, introduced *modular solvers* as a major improvement to the original application solvers. The application solvers are replaced by a single `foamRun` solver which describes the steps of a general algorithm for fluid dynamics calculations. `foamRun` loads a solver module, which defines each step to characterise a particular type of flow.

Modular solvers are simpler to use and maintain than application solvers. Their source code is easier to navigate, promoting better understanding. They are more flexible; in particular, there is also a `foamMultiRun` solver which can take two or more domain regions and apply a different solver module to each region. In particular, modules for one or more fluids and solids can be coupled for conjugate heat transfer (CHT) for different flow types, *e.g.* multiphase.

Further details of applications, including modular solvers, are described in chapter 3. General configuration and running of OpenFOAM cases are described in chapter 4. Chapter 5 covers details of the generation of meshes using the mesh generator supplied with OpenFOAM and conversion of mesh data generated by third-party products. Post-processing of results, including visualisation, is in chapter 7. Finally, some aspects of physical modelling, *e.g.* transport and thermophysical modelling, are described in chapter 8.

Chapter 2

Tutorials

This chapter we describes the process of setup, simulation and post-processing for some OpenFOAM test cases, with the principal aim of introducing a user to the basic procedures of running OpenFOAM. The test cases are taken from the *tutorials* directory which contains numerous example cases in OpenFOAM. The directory location is represented by the `$FOAM_TUTORIALS` variable in the OpenFOAM “environment”.

The directory contains numerous cases that demonstrate the use of all the solver modules, other solvers and many utilities supplied with OpenFOAM. Most examples are stored in sub-directories corresponding to each of the modular solvers. For example, the cases that use the `incompressibleFluid` module are stored in `$FOAM_TUTORIALS/incompressibleFluid`. The user can explore these example cases, starting by listing the top-level of the `$FOAM_TUTORIALS` directory, by typing in a terminal

```
ls $FOAM_TUTORIALS
```

The OpenFOAM environment includes a `$FOAM_RUN` variable which represents a directory in the user’s file system at `$HOME/OpenFOAM/<USER>-14/run` where `<USER>` is the account login name and “14” is the OpenFOAM version number. The directory provides a recommended location to store and run simulation cases. The examples presented in this chapter will be copied into the *run* directory. The user should check whether the directory exists by typing

```
ls $FOAM_RUN
```

If a message is returned saying no such directory exists, the user should create the directory by typing

```
mkdir -p $FOAM_RUN
```

Any example case from `$FOAM_TUTORIALS` can then be copied into the *run* directory. For example to try the `motorBike` example for the `incompressibleFluid` solver module, the user can copy it to the *run* directory by typing:

```
cd $FOAM_RUN
cp -r $FOAM_TUTORIALS/incompressibleFluid/motorBike .
```

2.1 Backward-facing step

This tutorial will describe how to pre-process, run and post-process a case involving isothermal, incompressible flow across a backward-facing step. The problem is treated as two dimensional with the geometry shown in Figure 2.1. The domain consists of:

- an inlet opening (left);
- an outlet opening (right);
- an upper wall, which is horizontal before tapering gently toward the outlet;
- a lower wall, which also tapers towards the outlet, but includes an abrupt step within a short distance from the inlet;

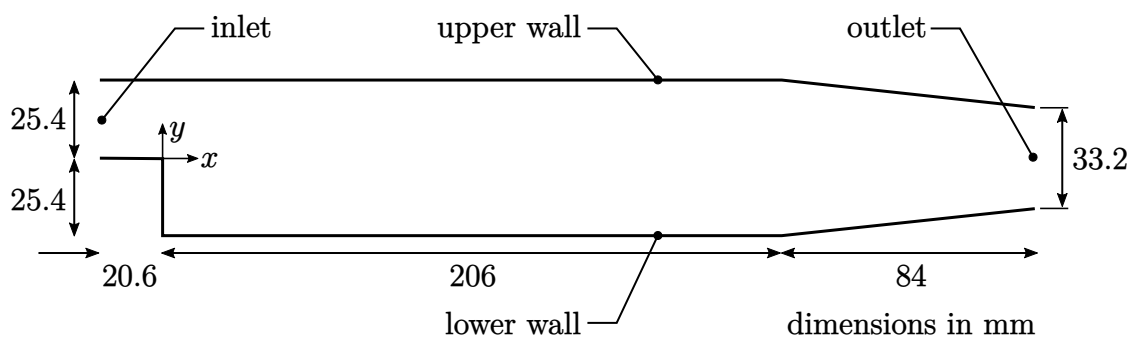


Figure 2.1: Geometry of the backward-facing step.

Flow enters the inlet in the x -direction with a speed of 10 m/s. The flow will be assumed isothermal and incompressible and will be solved using the `incompressibleFluid` modular solver.

2.1.1 Pre-processing

Cases are configured in OpenFOAM by editing input data files. Users should select a suitable file editor to do this, *e.g.* `emacs`, `vi`, `gedit`, `nedit`, *etc.* A case involves multiple data files, corresponding to different parts of the configuration, *e.g.* mesh, fields, properties, control parameters, *etc.* As described in section 4.1, the set of files is stored within a case directory, which is given a suitably descriptive name. This tutorial uses the case `$FOAM_TUTORIALS/incompressibleFluid/pitzDailySteady`, which the user should copy to their `run` directory as follows.

```
cd $FOAM_RUN
cp -r $FOAM_TUTORIALS/incompressibleFluid/pitzDailySteady .
cd pitzDailySteady
```

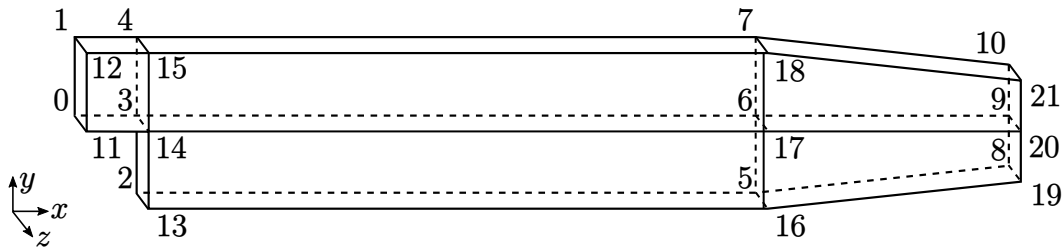



Figure 2.2: Block structure of the mesh for the backward step.

2.1.2 Mesh generation

OpenFOAM always operates in a three dimensional Cartesian coordinate system and all geometries are generated in three dimensions (3D). It solves the case in two dimensions (2D) by specifying a special **empty** boundary condition on boundaries normal to the (3rd) dimension (for which no solution is required).

OpenFOAM includes a simple mesh generator, **blockMesh**, which generates meshes from a *blockMeshDict* file, located in the **system** directory for a given case. The domain is defined using blocks whose vertex locations are specified in the file. The structure of the blocks and respective vertices are shown in Figure 2.2.

The **backwardStep** domain consists of five blocks shown in the figure. The domain depth in the *z* direction (exaggerated in the figure) is 1 mm. The *blockMeshDict* file for this example is reproduced below:

```

1  /*----- C++ -----*\
2  =====
3  \  F ield      | OpenFOAM: The Open Source CFD Toolbox
4  \  O peration  | Website:  https://openfoam.org
5  \  A nd        | Version:  dev
6  \  M anipulation |
7  \*-----*/
8  FoamFile
9  {
10     format      ascii;
11     class        dictionary;
12     location     "system";
13     object       blockMeshDict;
14 }
15 // *****
16
17 // Note: this file is a Copy of $FOAM_TUTORIALS/resources/blockMesh/pitzDaily
18
19 units [mm];
20
21 vertices
22 (
23     (-20.6 0 -0.5)
24     (-20.6 25.4 -0.5)
25     (0 -25.4 -0.5)
26     (0 0 -0.5)
27     (0 25.4 -0.5)
28     (206 -25.4 -0.5)
29     (206 0 -0.5)
30     (206 25.4 -0.5)
31     (290 -16.6 -0.5)
32     (290 0 -0.5)
33     (290 16.6 -0.5)
34
35     (-20.6 0 0.5)
36     (-20.6 25.4 0.5)
37     (0 -25.4 0.5)

```

```

38     (0 0 0.5)
39     (0 25.4 0.5)
40     (206 -25.4 0.5)
41     (206 0 0.5)
42     (206 25.4 0.5)
43     (290 -16.6 0.5)
44     (290 0 0.5)
45     (290 16.6 0.5)
46 );
47
48 negY
49 (
50     (2 4 1)
51     (1 3 0.3)
52 );
53
54 posY
55 (
56     (1 4 2)
57     (2 3 4)
58     (2 4 0.25)
59 );
60
61 posYR
62 (
63     (2 1 1)
64     (1 1 0.25)
65 );
66
67
68 blocks
69 (
70     hex (0 3 4 1 11 14 15 12)
71     (18 30 1)
72     simpleGrading (0.5 $posY 1)
73
74     hex (2 5 6 3 13 16 17 14)
75     (180 27 1)
76     edgeGrading (4 4 4 4 $negY 1 1 $negY 1 1 1 1)
77
78     hex (3 6 7 4 14 17 18 15)
79     (180 30 1)
80     edgeGrading (4 4 4 4 $posY $posYR $posYR $posY 1 1 1 1)
81
82     hex (5 8 9 6 16 19 20 17)
83     (25 27 1)
84     simpleGrading (2.5 1 1)
85
86     hex (6 9 10 7 17 20 21 18)
87     (25 30 1)
88     simpleGrading (2.5 $posYR 1)
89 );
90
91 boundary
92 (
93     inlet
94     {
95         type patch;
96         faces
97         (
98             (0 1 12 11)
99         );
100     }
101     outlet
102     {
103         type patch;
104         faces
105         (
106             (8 9 20 19)
107             (9 10 21 20)
108         );
109     }
110     upperWall
111     {
112         type wall;
113         faces
114         (

```

```

115         (1 4 15 12)
116         (4 7 18 15)
117         (7 10 21 18)
118     );
119 }
120 lowerWall
121 {
122     type wall;
123     faces
124     (
125         (0 3 14 11)
126         (3 2 13 14)
127         (2 5 16 13)
128         (5 8 19 16)
129     );
130 }
131 frontAndBack
132 {
133     type empty;
134     faces
135     (
136         (0 3 4 1)
137         (2 5 6 3)
138         (3 6 7 4)
139         (5 8 9 6)
140         (6 9 10 7)
141         (11 14 15 12)
142         (13 16 17 14)
143         (14 17 18 15)
144         (16 19 20 17)
145         (17 20 21 18)
146     );
147 }
148 );
149
150 // *****

```

The file first contains header information in the form of a banner (lines 1-7), then file information contained in a *FoamFile* sub-dictionary, delimited by curly braces (`{...}`).

For the remainder of the manual:

To save space, file headers, including the banner and *FoamFile* sub-dictionary, will be removed from further verbatim quoting of case files.

The body of the *blockMeshDict* file will be briefly reviewed here, but for further details see section 5.4. The file begins with a **units** entry, which is optional. This entry allows the user to provide an alternative set of units for length. When units are specified, all length values, *e.g.* the **vertices**, in the *blockMeshDict* file must correspond to the specified units. The generated mesh, is however, written in the base units for the underlying unit system, *i.e.* metre (m) for Syst me International (SI).

Units are described generally in section 4.2.6. In this example, the units are specified to be [mm]. The file continues with the coordinates of the block **vertices**, expressed in the specified units.

The file then defines the **blocks** (here, 5 of them). Each block is a hexahedral shape, given by the **hex** entry. The eight vertex labels are listed following the **hex** entry.

The number of cells is specified in each direction for each block by a vector of three integers. For example the first block specifies (18 30 1), which produces 18 cells through the block in the *x*-direction, 30 in the *x*-direction and 1 in the *z*-direction (the unused direction).

The blocks includes mesh grading, which enables the cell lengths to vary across the block. It also includes multi-grading which is described in section 5.4.6 using parameters **negY**, **posY** and **posYR**.

Finally, the mesh splits the boundary into inlet, outlet and wall regions, included in the following patches: `upperWall` and `lowerWall` for the wall boundaries of the domain; `inlet` and `outlet` for the open boundaries. The boundary in the z -normal direction is included in a single patch named `frontAndBack`.

The mesh is generated by running `blockMesh` on this *blockMeshDict* file. From within the case directory, this is done, simply by typing in the terminal:

```
blockMesh
```

The running status of `blockMesh` is reported in the terminal window. Any mistakes in the *blockMeshDict* file are picked up by `blockMesh` and the resulting error message directs the user to the source of the error.

2.1.3 Viewing the mesh

It is sensible to verify the mesh is generated correctly before running the simulation. The mesh can be viewed in `ParaView`, the post-processing tool supplied with OpenFOAM. The `ParaView` post-processing is conveniently launched on OpenFOAM case data by executing the `paraFoam` script from within the case directory.

Any UNIX/Linux executable (application, script, *etc.*) can be run in two ways: as a foreground process, *i.e.* one in which the shell waits until the executable has finished before returning the command prompt; or, as a background process, which allows the shell to accept additional commands while the executable is still running. Since it is convenient to keep `ParaView` open while running other commands from the terminal, we will launch it in the background using the `&` operator by typing

```
paraFoam &
```

This launches the `ParaView` window as shown in Figure 7.1. In the Pipeline Browser, `ParaView` registers `pitzDailySteady.OpenFOAM`, representing the `pitzDailySteady` case.

For the remainder of the manual:

The first time `ParaView` is launched, users are faced with a splash screen which can be permanently deactivated by clicking the relevant checkbox before closing.

Before clicking the Apply button, the user can select some geometry from the **Mesh Parts** panel in the **Properties** window (may require scrolling to find). Because the case is small, it is easiest to select all the data by checking the box adjacent to the **Mesh Parts** panel title, which automatically checks all individual components within the respective panel. The user should then click the **Apply** button to load the geometry into `ParaView`.

The user can control the visual representation of the selected module *either* using the second row of controls at the top of `ParaView` *or* by scrolling down further to the **Display** panel that. The user should make the selections using the second row of controls as shown in Figure 2.3, or as described below from within the **Display** panel.

1. select **Wireframe** from the **Representation** menu;

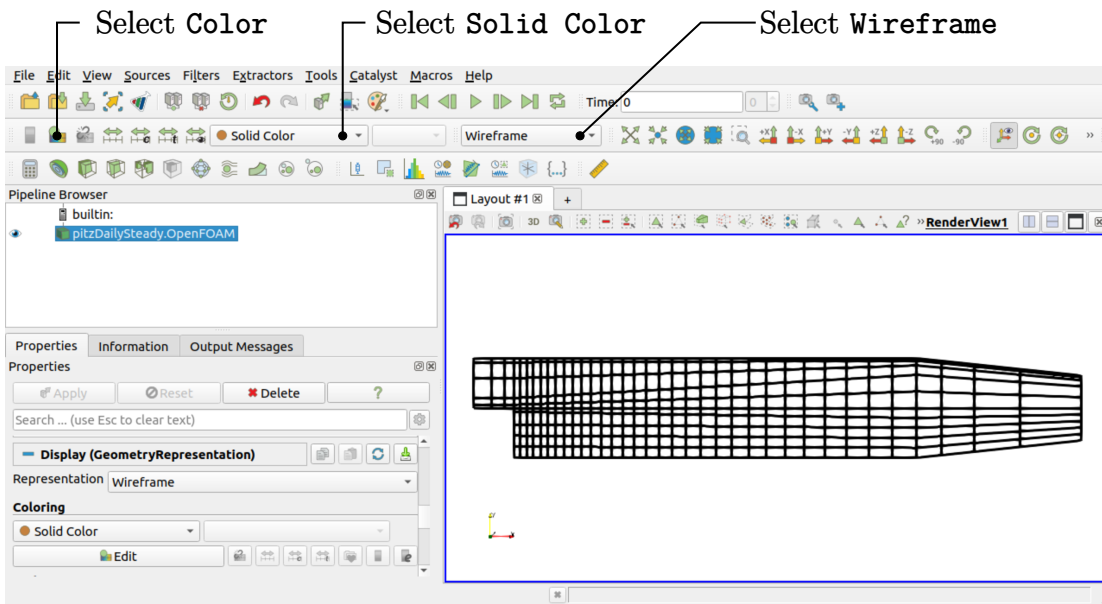


Figure 2.3: Viewing the mesh in ParaView (cell density reduced).

2. in the Coloring section, select Solid Color;
3. click Edit (in Coloring) and select an appropriate colour *e.g.* black (for a white background).

Especially the first time the user starts ParaView, it is **recommended** that they manipulate the view as described in section 7.1.5. In particular, since this is a 2D case, it is recommended that Camera Parallel Projection is selected at the bottom of the View (Render View) panel. The selection can be saved as a user default by clicking the Save current view settings button to the right of the View (Render View) heading (the furthest right of the four buttons). The background colour can be also set in the View (Render View) panel at the bottom of the Properties window.

Note that, **many parameters in the Properties window are only visible by clicking the Advanced Properties gearwheel button** () at the top of the Properties window, next to the search box.

2.1.4 Boundary and initial conditions

Once the mesh generation is complete, the user can look at the configuration of the initial fields for this case. The case starts at time $t = 0$ s, so the initial field data is stored in a 0 sub-directory of the *cavity* directory. The 0 sub-directory contains several files including p and U , which represent the pressure (p) and velocity (U) fields, respectively. Within these files the initial values and boundary conditions must be set. Let us examine the p file below.

```

16
17 dimensions      [kinematicPressure];
18
19 internalField    uniform 0;
20
21 boundaryField

```

```

22 {
23     inlet
24     {
25         type            zeroGradient;
26     }
27     outlet
28     {
29         type            fixedValue;
30         value            uniform 0;
31     }
32
33     upperWall
34     {
35         type            zeroGradient;
36     }
37
38     lowerWall
39     {
40         type            zeroGradient;
41     }
42
43     frontAndBack
44     {
45         type            empty;
46     }
47 }
48
49 // *****
50 // *****

```

This field file contains three principal entries: `dimensions`, `internalField`, and `boundaryField`. The `dimensions` is described in section 4.2.6 and specifies the dimensions of the field, here *kinematic* pressure, *i.e.* corresponding to $\text{m}^2 \text{s}^{-2}$ in SI units.

The `internalField` is described in section 4.2.9. When initialising a field to begin a simulation, the `internalField` is either specified as: `uniform`, described by a single value; or, initialised non-uniform using a field function. When fields are written out in the results of a simulation, they are invariably written with the `nonuniform` identifier followed by a list of values corresponding to all cells (or faces in the case of a surface field).

The `boundaryField` entry is also described in section 4.2.9 and includes boundary conditions and data for all the boundary patches.

For this `pitzDailySteady` case, the initial fields are set to be uniform. Here the pressure is kinematic, and since the solution does not involve energy and thermodynamics, its absolute value is not relevant, so is set to `uniform 0` for convenience.

The boundary includes the `upperWall`, `lowerWall`, `inlet` and `outlet` patches. The walls and inlet are both assigned the `zeroGradient` boundary condition for `p`, meaning “the normal gradient of pressure is zero”. The outlet uses the `fixedValue` boundary condition for `p` with `value` of `uniform 0`. The `frontAndBack` patch, describing the front and back planes of the 2D case, is specified as `empty`.

The user can similarly examine the velocity field in the `0/U` file. The `dimensions` are those expected for velocity, the internal field is initialised as uniform zero, which in the case of velocity must be expressed by 3 vector components, *i.e.* `uniform (0 0 0)` (see section 4.2.5 for more information).

A no-slip condition is assumed on the walls, specified by a `noSlip` condition. The inlet flow speed is 10 m/s in the x -direction so represented by a `fixedValue` condition with `value` of `uniform (10 0 0)`. The outlet reverts to the `zeroGradient` condition. The `frontAndBack` patch must be set to `empty`.

2.1.5 Physical properties

Physical properties and model configurations for the case are stored in dictionary files in the *constant* directory. Properties for this example are specified in the following *physicalProperties* file.

```

16
17 viscosityModel constant;
18
19 nu 1e-05;
20
21 // ***** //
```

First it includes the *viscosityModel* entry which is set to *constant*. With that model, a single kinematic viscosity is then specified by the keyword *nu*, representing the Greek symbol ν phonetically for the kinematic viscosity. The value is set to $\nu = 1 \times 10^{-5} \text{ m}^2 \text{ s}^{-1}$.

2.1.6 Momentum transport

An estimate of the Reynolds number can determine whether the flow is expected to be turbulent. The Reynolds number is defined as:

$$Re = \frac{|\mathbf{U}|L}{\nu} \quad (2.1)$$

where $|\mathbf{U}|$ and L are the characteristic speed and length respectively and ν is the kinematic viscosity. Using the inlet (or step) height $L = 25.4 \text{ mm}$ and $|\mathbf{U}| = 10 \text{ m/s}$, $Re = 25400$. For flow in a pipe, transition typically occurs when $Re \approx > 2000$, so this case can be assumed to be turbulent.

The *momentumTransport* file characterises the viscous stress in the fluid by a variety of models, *e.g.* Newtonian and non-Newtonian fluids, turbulence, visco-elasticity and more. For this example, the file includes the configuration of turbulence modelling as shown below.

```

16
17 simulationType RAS;
18
19 RAS
20 {
21     // Tested with kEpsilon, realizableKE, kOmega, kOmega2006, kOmegaSST, v2f,
22     // ShihQuadraticKE, LienCubicKE.
23     model kEpsilon;
24
25     turbulence on;
26
27     viscosityModel Newtonian;
28 }
29
30
31
32 // ***** //
```

The type of simulation is first specified by the *simulationType* keyword. The *RAS* entry indicates a Reynolds-averaged simulation, the standard form of turbulence modelling. The *RAS* sub-dictionary includes the *model* entry which is set to the well-known $k-\varepsilon$ model by the *kEpsilon* entry. The *turbulence* keyword provides a switch to turn the modelling on and off. The model coefficients have default values which can be overridden with additional entries in the *momentumTransport* file, as described in section 4.2.20. The *viscosityModel* entry confirms the fluid is modelled as Newtonian (which is the default model, so the entry could be omitted).

The k - ε model solves transport equations for: k , the turbulent kinetic energy; and, ε , the turbulent dissipation rate. The initial and boundary conditions for those fields are configured in the *0/k* and *0/epsilon* files, respectively.

In particular, the turbulent fields must be initialised with suitable internal and inlet values. Turbulent kinetic energy k can be calculated by

$$k = \frac{3}{2} (|\mathbf{U}|I)^2 \quad (2.2)$$

from an estimate of *turbulent intensity* $I = \mathbf{U}'_{\text{RMS}}/|\mathbf{U}|$, the ratio of the root-mean-square (RMS) of turbulent fluctuations $\mathbf{U}'_{\text{RMS}}$ to the mean flow speed $|\mathbf{U}|$. This example uses an estimate $I = 5\%$, such that $k = 1.5 \times (10 \times 0.05)^2 = 0.375 \text{ m}^2 \text{ s}^{-2}$. In the *0/k* file, 0.375 is used both for the initial `internalField` and the `inlet value`.

The turbulent dissipation rate ε can be calculated by

$$\varepsilon = C_\mu^{0.75} \frac{k^{1.5}}{l_m}. \quad (2.3)$$

from $C_\mu = 0.09$ and an estimate of Prandtl mixing length l_m . This example uses an estimate $l_m = 10\% \times \text{step height} = 2.54 \text{ mm}$, such that $\varepsilon = 0.09^{0.75} \times 0.375^{1.5} / 0.00254 = 14.855 \text{ m}^2 \text{ s}^{-3}$. In the *0/epsilon* file, 14.855 is used both for the initial `internalField` and the `inlet value`.

The turbulence model is deployed with wall functions to model the behaviour at wall boundaries. Wall functions are applied as boundary conditions on the individual wall patches which enables different wall function models to be applied to different wall regions. The choice of wall function models are specified through the turbulent viscosity field, ν_t in the *0/nut* file.

```

16
17 dimensions      [turbulentViscosity];
18
19 internalField    uniform 0;
20
21 boundaryField
22 {
23     inlet
24     {
25         type      calculated;
26         value      uniform 0;
27     }
28     outlet
29     {
30         type      calculated;
31         value      uniform 0;
32     }
33     upperWall
34     {
35         type      nutkWallFunction;
36         value      uniform 0;
37     }
38     lowerWall
39     {
40         type      nutkWallFunction;
41         value      uniform 0;
42     }
43     frontAndBack
44     {
45         type      empty;
46     }
47 }
48
49
50 // ***** //
```


This case uses standard wall functions, specified by the `nutkWallFunction` type on the `upperWall` and `lowerWall` patches. Alternative wall function models include the rough wall functions, specified through the `nutRoughWallFunction` keyword.

When wall functions are specified through boundary conditions in the `0/nut` file, corresponding conditions must be applied to the wall patches for the turbulence fields. The `0/eps` and `0/k` files show that ε is assigned the `epsilonWallFunction` condition and k is assigned the `kqRWallFunction` condition at the wall patches. The latter is a generic boundary condition that can be applied to any field that are of a turbulent kinetic energy type, *e.g.* k , q or Reynolds Stress R .

2.1.7 Control

Input data relating to the control of time and reading and writing of the solution data are read in from the `controlDict` file. The user should view this file; as a case control file, it is located in the `system` directory.

```

16
17 solver            incompressibleFluid;
18
19 startFrom          startTime;
20
21 startTime           0;
22
23 stopAt             endTime;
24
25 endTime            2000;
26
27 deltaT             1;
28
29 writeControl        timeStep;
30
31 writeInterval       100;
32
33 purgeWrite          0;
34
35 writeFormat         ascii;
36
37 writePrecision      6;
38
39 writeCompression    off;
40
41 timeFormat          general;
42
43 timePrecision       6;
44
45 runTimeModifiable  true;
46
47 // ***** //
```

The file first includes a `solver` entry which describes the solver module used for the simulation. This example uses the `incompressibleFluid` module for steady or transient turbulent flow of incompressible isothermal fluids with optional mesh motion and change.

The start/stop times and the time step for the run must be set. OpenFOAM provides flexible options for time controls which are described in section 4.4. Like most cases, this example starts the simulation at time $t = 0$ which instructs the solver to read its field data from a directory named `0`. Therefore we set the `startFrom` keyword to `startTime` and then specify the `startTime` keyword to be 0.

The aim of the simulation is to reach the steady solution where the recirculation region is fully developed adjacent to the step. The `incompressibleFluid` module can run as a steady-state solver by setting the time derivatives $\partial/\partial t$ to zero in all the equations. This is achieved by setting the `ddtSchemes` to `steadyState` in the `fvSchemes` file, discussed later.

In this mode, the time step, represented by the keyword `deltaT`, is only used as a time increment (since it is no longer used to discretise $\partial/\partial t$). Its value does not affect the solution, so for steady solutions it is set to 1 so that time simply represents the number of solution steps.

The `endTime` keyword sets a time at which the solver application stops running. The value of 2000 provides an adequate number of solution steps to enable the steady solution to converge to a reasonable level of accuracy.

As the simulation progresses we wish to write results at intervals of time of interest for visualisation and other post-processing. The `writeControl` keyword presents several options for setting the time at which the results are written; here the `timeStep` option specifies that results are written every n th time step where the value n is specified under the `writeInterval` keyword. The interval of 100 means results are written at 100, 200, *etc.*

OpenFOAM creates a new directory *named after the current time*, *e.g.* 100, on each occasion that it writes a set of data, as discussed in full in section 4.1. It writes out the results for each solution field, *e.g.* U, p, k, into the time directories.

2.1.8 Discretisation and linear-solver settings

The user specifies the choice of finite volume discretisation schemes in the `fvSchemes` file in the `system` directory. The specification of the linear equation solvers and tolerances and other algorithm controls is made in the `fvSolution` file, also in the `system` directory.

The details of those two files are described in sections 4.5 and 4.6, respectively. For this example, the following points are important:

- the `ddtSchemes` defaults to `steadyState` in `fvSchemes`, to invoke a steady-state calculation;
- steady-state solution uses an algorithm based on SIMPLE whose controls are configured in a SIMPLE sub-dictionary in the `fvSolution` file;
- the SIMPLE sub-dictionary contains the `consistent` switch which is set to `yes`, applying the “consistent” form of the SIMPLE algorithm (SIMPLEC);
- the convergence of SIMPLEC is very sensitive to the `relaxationFactors` in the `fvSolution` file; the values of 0.9 are carefully tuned and are not suitable for the standard SIMPLE algorithm;
- SIMPLEC’s sensitive convergence generally makes it only reliable for cases with simple geometries.

2.1.9 Running an application

To run a simulation with a single domain region, the `foamRun` application is run. It loads the relevant solver module, from the `solver` entry in the `controlDict` file, to perform the calculation. On this occasion, we will run `foamRun` in the terminal foreground by typing the following from within the case directory.

```
foamRun
```

The progress of the simulation is reported in the terminal window. It describes successive solutions steps, giving the initial and final residuals for each equation, and conservation errors.

The SIMPLE algorithm controls in *fvSolution* include a **residualControl** sub-dictionary with a set of tolerances for **p**, **U** and turbulence fields. The **incompressibleFluid** solver terminates When the initial residuals for **all** fields fall below their respective tolerances. In this example, the solver terminates at 285 iterations with the following statement.

```
SIMPLE solution converged in 285 iterations
```

Results from the simulation are written into time directories within the *pitzDailySteady* case directory. The user can list the directory contents with the “ls” command to see the time directories (100, 200 and 285) containing the results.

2.1.10 Time selection in ParaView

Once the results are written to time directories, they can be viewed using ParaView. To visualise these results in ParaView the time selector must first be made aware of the new times so the user can select a relevant time. The time selector is updated by the user clicking the Refresh Times button in the Properties panel *when the top module is selected* in the Pipeline Browser.

In this example, after clicking Refresh Times, the drop down menu in the time selector displays 0, 100, 200 and 285, corresponding to the case time directories. In order to view the solution at 285, the user can use the VCR Controls or Current Time Controls to change the current time to 285.

For the remainder of the manual:

In ParaView, if the window panels, *e.g.* Properties, do not contain the expected entries, ensure that the relevant module is highlighted in blue in the Pipeline Browser. For example, Refresh Times only appears when the top module (here, *pitzDailySteady.OpenFOAM*) is selected.

2.1.11 Colouring surfaces

To view pressure, the user can *either* make selections from the second row of buttons at the top of ParaView as shown in Figure 2.4 *or* scroll down to the Display panel in Properties and make the following selections:

1. select **Surface** from the **Representation** menu;
2. select  **P** in **Coloring**
3. click the **Rescale** button to set the colour scale to the data range, if necessary.

The pressure field should appear as shown above, with the pressure increasing to a maximum at the contraction of the channel towards the outlet. With the point icon ( **P**), the pressure field is interpolated across each cell to give a continuous appearance. Instead if the user

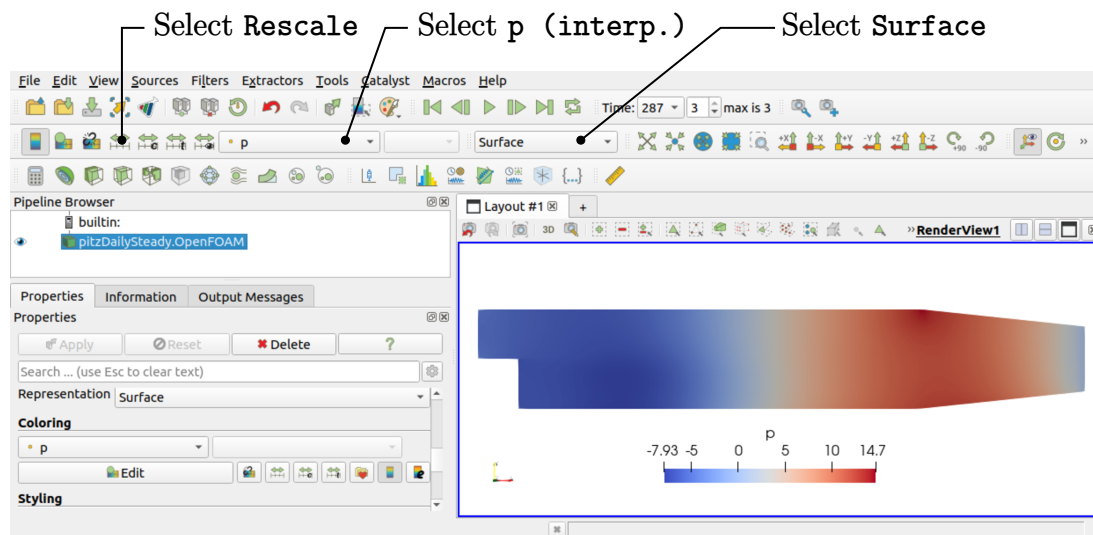


Figure 2.4: Displaying pressure contours for the backward step case.

selects the cell icon, ( **P**), from the **Coloring** menu, a single value for pressure will be attributed to each cell, represented by a single colour with no grading.

A colour legend is included which can be disabled by clicking the **Toggle Color Legend Visibility** button at the left of the second row of buttons at the top of **ParaView**. These buttons are part of the **Active Variable Controls** toolbar, shown in Figure 7.4). The **Edit Color Map** button, second on the left in **Active Variable Controls** toolbar, opens the **Color Map Editor** window, as shown in Figure 2.5, where the user can set a range of attributes of the colour scale and the color bar.

In particular, **ParaView** defaults to using a colour scale of blue to white to red rather than the more common blue to green to red (rainbow). Therefore *the first time* that the user executes **ParaView**, they may wish to change the colour scale. This can be done by selecting the **Choose Preset** button, with the **heart icon**, in the **Color Scale Editor**. The conventional color scale for CFD is **Blue to Red Rainbow** which is only listed if the user types the name in the Search bar or selects the **Rainbow** category from the drop-down menu next to the Search bar.

Images in this manual are created using the **Blue to Red Rainbow** colour scale. **Rainbow Uniform** is a good alternative colour scale which has the advantage of a more uniform distribution of colours. The user can choose either, but if using **Rainbow Uniform**, screen images will appear slightly different to those in this manual.

After selecting a rainbow colour scale, click **Apply** and **Close**, the user can click the **Save as Default** button at the absolute bottom of the panel (file save symbol) so that **ParaView** will always adopt this type of colour bar.

The user can also edit the color legend properties, such as text size, font selection and numbering format for the scale, by clicking the **Edit Color Legend Properties** to the far right of the search bar, as shown in Figure 2.5.

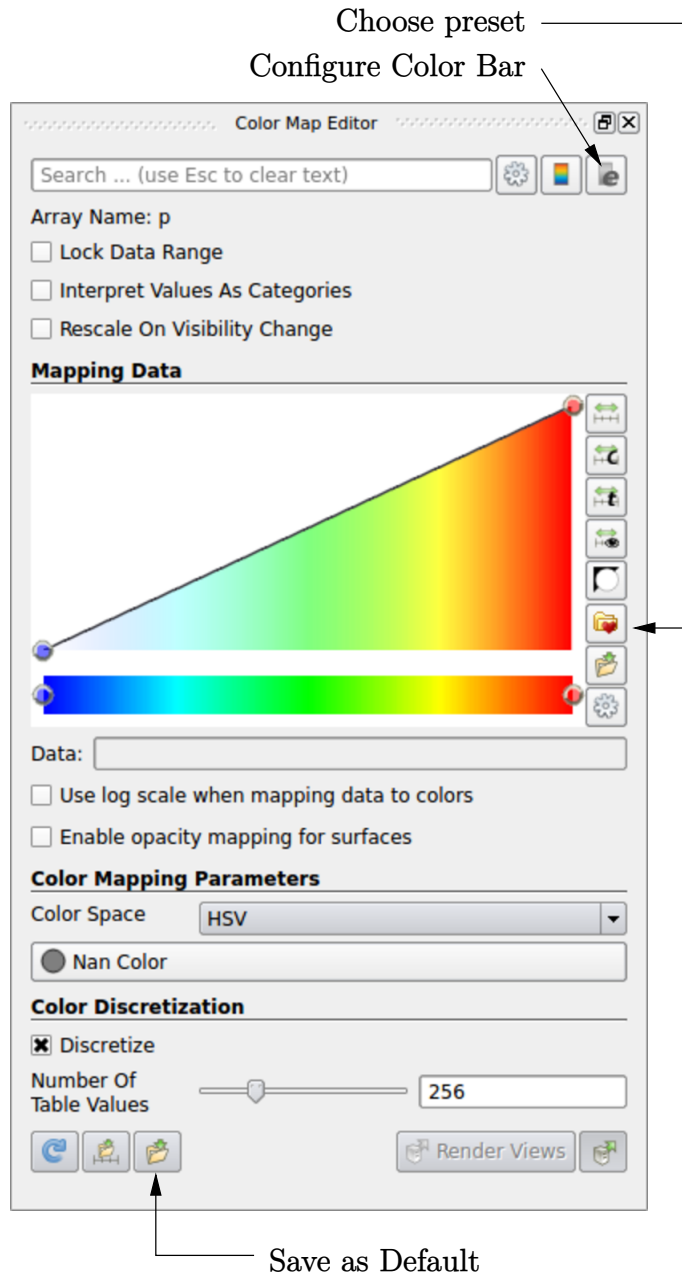


Figure 2.5: Color Map Editor.

2.1.12 Cutting plane (slice)

If the user rotates the image, by holding down the left mouse button in the image window and moving the cursor, they can see that they have now coloured the complete geometry surface by the pressure. In order to produce a 2D contour plot the user should first create a cutting plane, or 'slice'. With the `pitzDailySteady.OpenFOAM` module highlighted in the Pipeline Browser, the user should select the **Slice** filter from the **Filters** menu in the top menu of ParaView (accessible at the top of the screen on some systems). The **Slice** filter can be found in the **Common** sub-menu or among the **Common** and **Data Analysis** buttons, the third row of buttons at the top of ParaView (see Figure 7.4).

Selecting the **Slice** filter creates a new item in the **Pipeline Browser**. In the **Properties** window, the cutting plane should have an origin at a point on the z -axis, *e.g.* $(0, 0, 0)$ and its normal should be set to $(0, 0, 1)$ (click the **Z Normal** button).

When **Apply** is clicked, the slice appears in the **RenderView** window, while the original `pitzDailySteady.OpenFOAM` module disappears. The visibility of each module is enabled and disabled by the **eye** button to the left of each module in the **Pipeline Browser**.

For the remainder of the manual:

In **ParaView**, if items do not appear to be displayed in the **RenderView** window, ensure the relevant module is visible by switching on the **eye** button in the **Pipeline Browser**.

2.1.13 Vector plots

Before drawing vectors of the flow velocity, turn off the display of the **Slice** module by highlighting it in the **Pipeline Browser** and clicking the **eye** button to the left of it. The aim is to generate a vector glyph for velocity at the *centre of each cell*. We therefore first need to filter the cell centres from the mesh geometry as described in section 7.1.7. With the `pitzDailySteady.OpenFOAM` module highlighted in the **Pipeline Browser**, the user should select **Cell Centers** from the **Filters->Alphabetical** menu and then click **Apply**.

With the **Centers** highlighted in the **Pipeline Browser**, the user should then select **Glyph** from the **Filters->Common** menu (or the third row of buttons). The **Properties** window panel should appear as shown in Figure 2.6.

When displaying velocity vectors, there are four principal settings required for the glyphs:

- the glyph type, set to **Arrow**;
- the arrow direction, set by **Orientation Array**;
- the arrow lengths, set by **Scale Array** and **Scale Factor**;
- the number of arrows, set by **Glyph Mode**.

On clicking **Apply**, the glyphs appear as a single colour, *e.g.* white. The user should colour the glyphs by velocity magnitude which, as usual, is controlled by setting **U** in the drop down menu towards the left of the second row of buttons.

The **Legend** is also displayed. The user can configure the legend by clicking **Edit Color Map** (second button from left, second row). This opens the **Color Map Editor** window. The legend can be configured by clicking by the button furthest to the right of the search box. This button opens the **Edit Color Legend Properties** window. The **Advanced Properties** gearwheel button should be checked to the right of the search box.

Titles and labels can be fully configured. For Figure 2.7, the legend title is set to **Velocity U [m/s]**. The labels are specified to 1 fixed decimal place by unchecking **Automatic Label Format** and entering `%-#6.1f` in the **Label Format** text box. **Add Range Labels** is also unchecked.

Sample output is shown in Figure 2.7, zooming in on part of the recirculation region downstream of the step. At the lower wall, glyphs point in a direction opposing that of the flow adjacent to the wall. This glitch is caused by these glyphs being drawn at the face

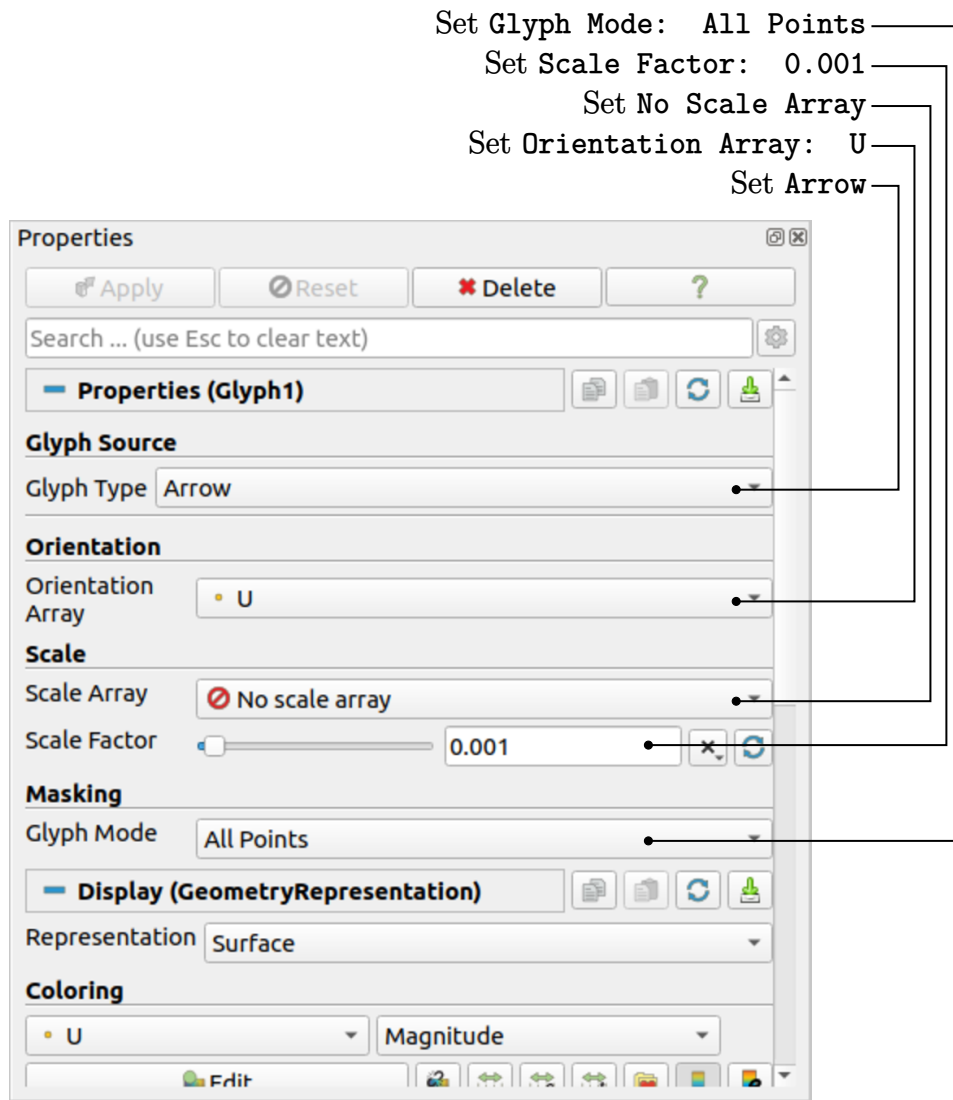


Figure 2.6: Properties panel for the Glyph filter.

centres of the wall boundary patch, where the velocity magnitude is 0 due to the no-slip condition. Without any direction to the vectors, **ParaView** orientates the arrows in a default x -direction. A quick way to remove these vectors is:

- go back to the `pitzDailySteady.OpenFOAM` module at the top of the Pipeline Browser;
- in the Mesh Parts panel, uncheck the `lowerWall` patch;
- click Apply.

2.1.14 Popular filters in ParaView

ParaView includes over 200 filters which can be listed by the **Filters->Alphabetical** menu. Only a small fraction, *e.g.* 10-15, of these filters are relevant for CFD, which we suggest adding to the **Filters->Favourites** menu. To do this, select **Manage Favourites** from the

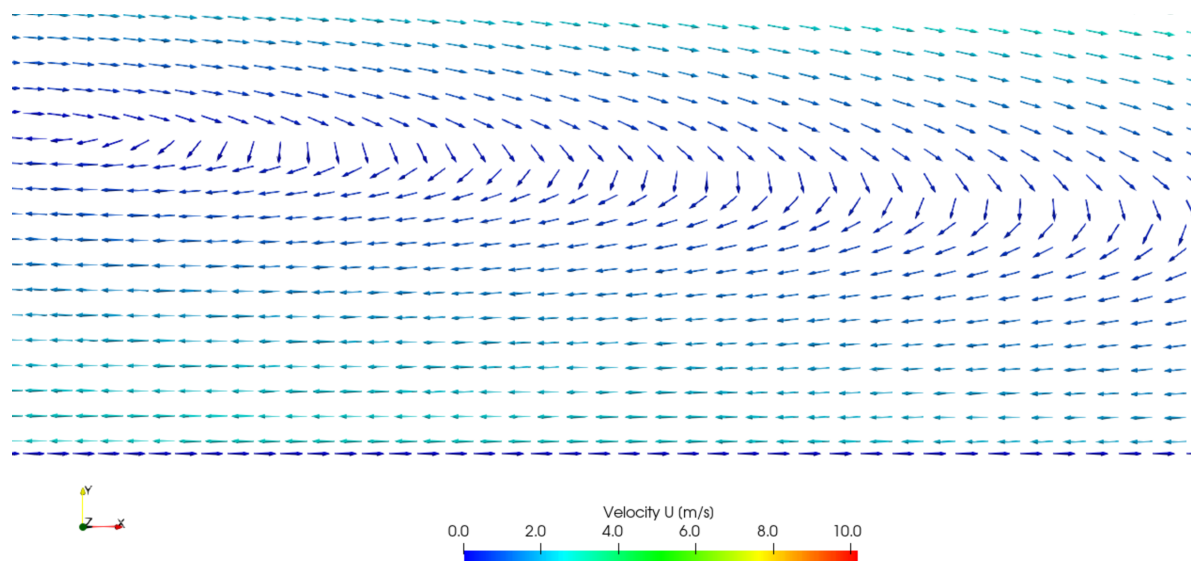


Figure 2.7: Velocities in the backward facing step.

Filters→**Favourites** menu. Search for the following important filters and click **Add>>** to add them to the **Favourites** menu:

- **Extract Block**, to select components of the domain, *e.g.* boundary patches and internal cells;
- **Slice**, to insert a plane through the geometry;
- **Cell Centers** and **Glyph**, principally to draw velocity vectors;
- **Stream Tracer** and **Tube**, to draw streamlines;
- **Contour**, to draw contour lines (on surfaces) and iso-surfaces;
- **Feature Edges**, to capture features on a surface for better image definition.

2.1.15 Contours

Before drawing contour lines of the flow speed, turn off the display of the **Glyph** module by highlighting it in the **Pipeline Browser** and clicking the **eye** button to its left. The user should then highlight the **Slice** module and colour by velocity **U**. We will then aim to draw contour lines on the slice at intervals of **U** of 1, 2, ..., 10.

The contour lines can be drawn by applying the **Contour** filter to the slice. The filter draws lines in 2D, or surfaces in 3D, along constant values of *scalar* quantities. Contours cannot be drawn directly from **U**, since it is a *vector*, so we first need to generate a scalar field of the magnitude of **U**.

The **mag(U)** field can be generated using post-processing with function objects, described in section 7.3. The user should run the **foamPostProcess** utility, calling the **mag** function object using the **-func** option as follows:

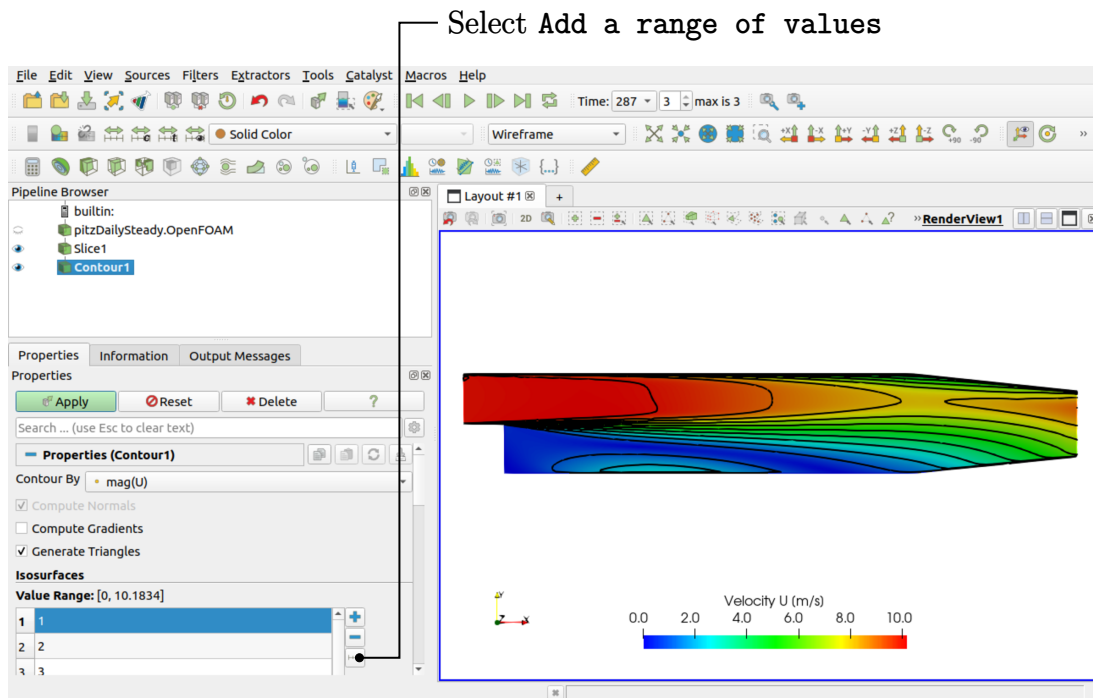


Figure 2.8: Contours in the backward facing step.

```
foamPostProcess -func "mag(U)"
```

The utility loops over all time directories. For each time directory, it reads in U , calculates $\text{mag}(U)$ and writes it out as a field file back into the time directory. The $\text{mag}(U)$ field must then be loaded into ParaView by: first, from the `pitzDailySteady.OpenFOAM` in the Pipeline Browser, clicking Refresh Times; then, scrolling down to the Fields panel, selecting $\text{mag}(U)$ and clicking Apply.

The user should re-select the `Slice` module in the Pipeline Browser, then apply the Contour filter. In the Properties panel, the user should select $\text{mag}(U)$ from the Contour By menu. Under Isosurfaces, the user should first delete the default value by clicking the `-` button, then add a range of 10 values as shown in Figure 2.8. The contours can be displayed with a Wireframe representation with solid black Coloring.

2.1.16 Streamline plots

Before drawing streamlines, the user should turn off the display of the Contour module. To display streamlines for this 2D example, the user should first highlight the `Slice` module in the Pipeline Browser and then apply the Stream Tracer filter.

A new `StreamTracer` module opens which is configured through its Properties window. Tracer is created by tracking lines in the direction of flow, starting from *seed points*. With Integration Direction BOTH, lines are tracked both upstream and downstream of the seed points. The user should scroll down the Properties window to configure the Seeds. The default Seed Type in Line which seeds points along a line drawn between specified

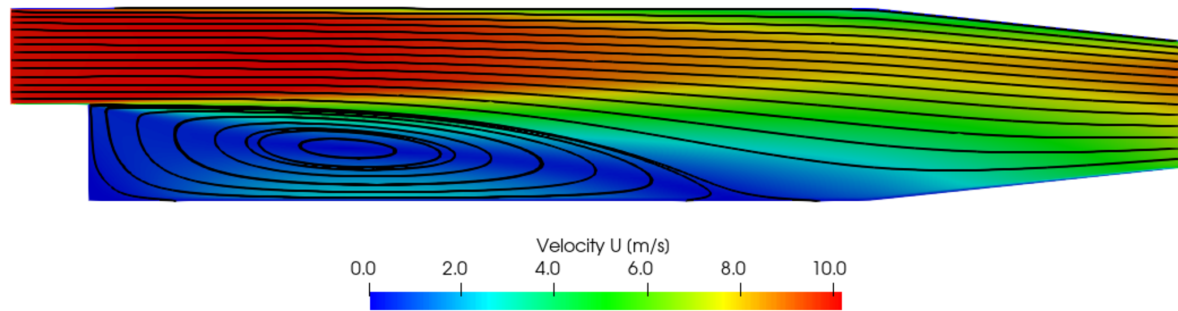


Figure 2.9: Streamlines in the backward facing step.

points. In the **Line Parameters**, the user should can set the two points to $(0, -0.025, 0)$ to $(0.2, 0.025, 0)$. The **Resolution** specifies the number of seed points distributed along the line, which should be reduced to 25. On clicking **Apply** the tracer is generated as shown in Figure 2.9. The user can experiment with the line points and resolution to produce different stream tracer output.

2.1.17 Inlet boundary condition

The user should examine the flow at the inlet boundary of the domain. The velocity condition is specified in $0/U$ as **fixedValue** with a value of $(10\ 0\ 0)$. This value is applied to *all* faces of the inlet boundary patch. The inlet is adjacent to wall boundaries where the **noSlip** condition is applied, giving rise to a sudden change in U , between adjacent boundary faces.

The user should zoom in around the inlet region of the geometry using the right button of the mouse. The **Slice** module should be activated in the **Pipeline Browser** and coloured by *cell values* of p ( **P** selection). In order to highlight the variation in pressure in cells close to the inlet, the user should apply a custom range of $-5 < p < -1$ as shown in Figure 2.10. A custom range is applied by clicking the **Rescale to Custom Data Range** button (), located fifth from the left of the second row of buttons. A panel opens, in which the minimum and maximum values of the range should be entered before clicking the **Rescale** button. The velocity profile can be shown by returning to the **Glyph** module in the **Pipeline Browser** and making it visible. The profile can be illustrated by scaling the arrows by the flow speed. In the **Properties** of the **Glyph**, scroll down to the **Scale** panel and set **Scale Array** to U , with **Scale By Magnitude** and a **Scale Factor** of $8e-05$. The vectors are assigned a **Solid Color** of white.

The figure shows the inlet region adjacent to the lower wall boundary. At the left of the image, the vectors show a uniform profile. Shear at the wall causes the flow to decelerate, starting in the near-wall cell. The deceleration causes an increase in pressure, which produces a driving gradient that redirects the flow slightly away from the wall, to obey mass conservation.

In order to reduce the pressure increase, the boundary condition can be modified so that the inlet velocity is no longer uniform. A **flowRateInletVelocity** boundary condition is a general boundary condition for U , specifying flow at an inlet. Documentation for this boundary condition can be viewed using the **foamInfo** script, a general tool which provides documentation for applications, models and tools in OpenFOAM. It is run for the **flowRateInletVelocity**

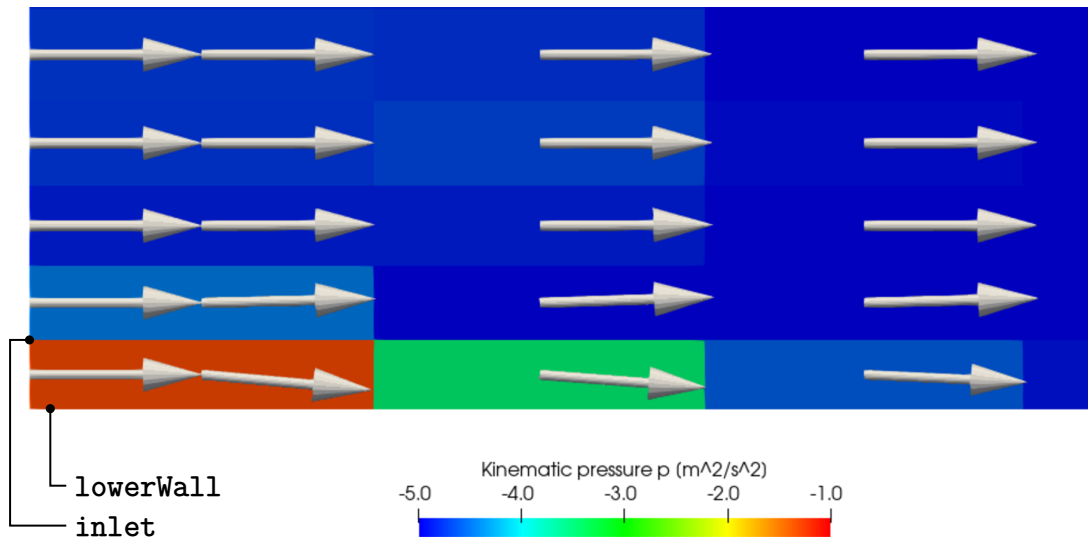


Figure 2.10: Uniform flow at the inlet in the backward facing step.

boundary condition as follows (noting that the name can sometimes be abbreviated for simplicity, here `flowRateInlet`, rather than `flowRateInletVelocity`).

```
foamInfo flowRateInlet
```

It locates and prints the header file of the related code and extracts the `Description` and `Usage` information from the file. It then identifies associated models, *i.e.* other boundary conditions in this case, and lists example cases that use the model. The `foamInfo` script is not perfect, but provides useful information quickly in at least nine times out of ten.

The documentation explains that the `flowRateInletVelocity` condition can specify the flow by a `massFlowRate`, `volumetricFlowRate` or `meanVelocity`. The user should open the `0/U` in their editor and locate the boundary field entry for the `inlet` patch. The condition with a `meanVelocity` can then be applied by changing the `inlet` sub-dictionary as follows:

```
inlet
{
    type          flowRateInletVelocity; // modify
    meanVelocity  10;                    // insert
}
```

The condition evaluates the velocity on the boundary, setting the `value` for all faces on the boundary patch. The `value` entry is therefore redundant for OpenFOAM, but it can be needed by ParaView to display initial values of initialised fields at patches. Therefore, **the value entry is retained** for ParaView's benefit.

After saving the `0/U` file, the user can re-run the simulation to check first that the `flowRateInletVelocity` emulates the original `fixedValue` condition. The `controlDict` file specifies that case starts from time 0 and it will overwrite previous results in time directories. The user can test the new condition simply by re-running the `foamRun` solver.

```
foamRun
```

The solver runs as before, terminating at 285 iterations. The user can return to ParaView and click the Refresh Times button in the `pitzDailySteady.OpenFOAM` module of the Pipeline Browser. There is no change to the results, demonstrating that the `flowRateInletVelocity` is setting a uniform value of (10 0 0) at this stage.

The user should now modify the `flowRateInletVelocity` condition in the `0/U` file by including a profile for the velocity. The documentation highlights two customised profile function, `turbulentBL` and `laminarBL`, which provide *power-law* and *quadratic* profiles for fully-developed turbulent and laminar boundary layers, respectively. In this example, add the `turbulentBL` profile to the boundary condition as follows:

```
inlet
{
    type            flowRateInletVelocity;
    meanVelocity    10;
    profile         turbulentBL; // add
}
```

Before running the simulation again, it is recommended to delete the previous solution time directories. The results should be deleted now because the solver will likely terminate at a different time, so the results from the final time directory from the old case will not be overwritten, potentially causing confusion.

The `foamListTimes` utility provides a quick, simple way to delete solution time directories, *i.e.* retaining the 0 directory. First run `foamListTimes` in the terminal as follows:

```
foamListTimes
```

This returns the list of the solution time directories, 100, 200 and 285, but not 0. The listed directories can be deleted by including the `-rm` remove option to `foamListTimes`, *i.e.* running the following command.

```
foamListTimes -rm
```

With the time directories containing the results now deleted, the `foamRun` solver should be run as before. This time `foamRun` terminates at 279 iterations. The user should return to ParaView and click the Refresh Times button in the `pitzDailySteady.OpenFOAM` module of the Pipeline Browser.

The change in output times may create confusion for ParaView, causing it to query times in the time selector with (?) symbols. The selector entries can be rebuilt by reverting to time 0 or clicking Refresh Times. The user can then select the last time in the sequence (279). The velocity profile and pressure are updated as shown in Figure 2.11. For pressure, the custom range is moved to $-7 < p < -3$. The velocity is no longer uniform at the inlet, but forms a profile according to the `turbulentBL` function. The velocity magnitude at the face adjacent to the lower wall is significantly reduced from previously so the deceleration

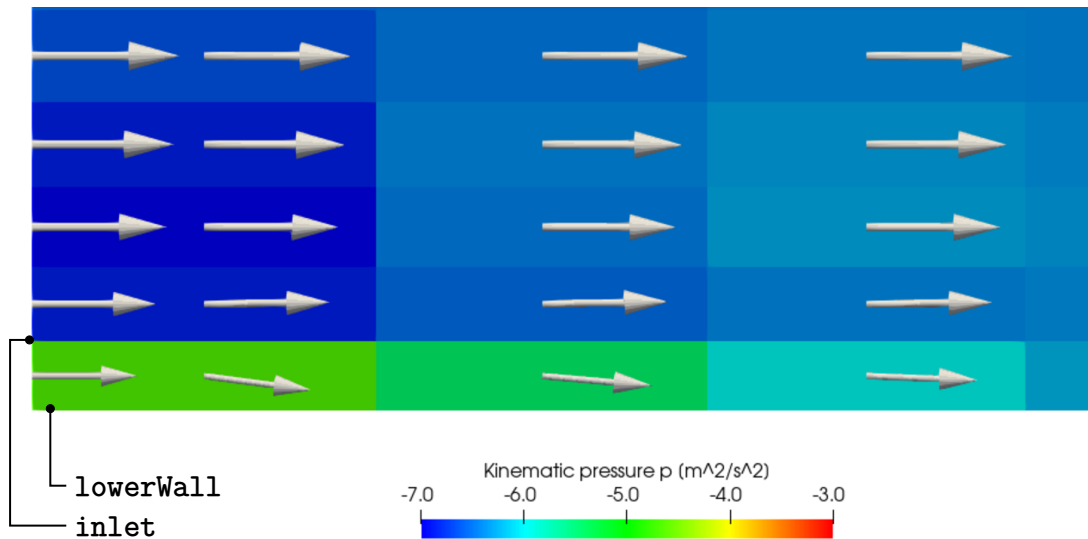


Figure 2.11: Turbulent boundary layer at the inlet in the backward facing step.

due to shear is also reduced in the near-wall cell. The pressure difference between the left corner and surrounding cells is consequently not as high, approximately $+2 \text{ m}^2 \text{ s}^{-2}$ (from -7 to -5) compared to $+4 \text{ m}^2 \text{ s}^{-2}$ (from -5 to -1) previously.

2.1.18 Turbulence model

The backward step case is set up to allow users to try out different turbulence models quickly. There is comment in the *momentumTransport* file in the *constant* directory, listing different turbulence models tested on this case. Some of the models solve equations for fields other than k and ε , *e.g.* ω (omega). To minimise the work when changing models, files for these other fields are already included in the *0* directory. Similarly, entries for schemes and solvers for these fields are included in the *fvSchemes* and *fvSolution* files, respectively (in the *system* directory).

The user should open the *momentumTransport* file in their editor to change the turbulence model. They can change the model to realizable $k-\varepsilon$ by the following setting.

```
model    realizableKE;
```

The simulation can now be re-run using this model by deleting the solution time directories and executing *foamRun* as follows.

```
foamListTimes -rm
foamRun
```

The solver terminates this time at 255 iterations. The user can now return to ParaView and click Refresh Times to view the results at time 255. The results are shown in Figure 2.12 using the slice with the velocity field and the streamlines filters configured earlier.

The realizable $k-\varepsilon$ model is less diffusive than the standard $k-\varepsilon$ model. It captures a secondary vortex at the base of the step which is approximately half the step height. The

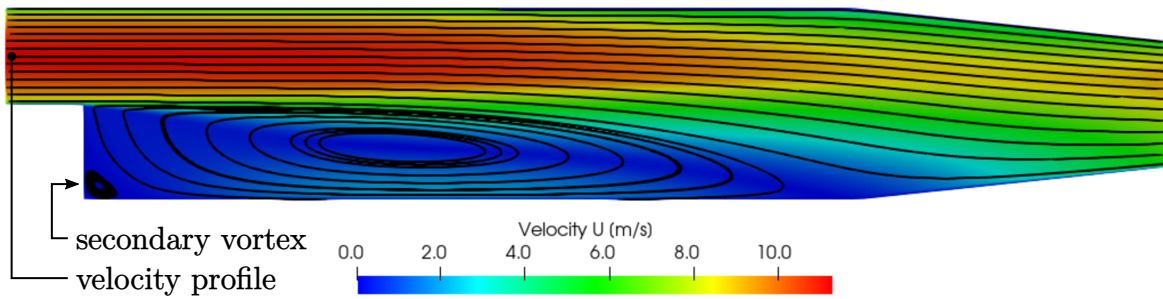


Figure 2.12: Streamlines with the realizable $k-\varepsilon$ model.

recirculation region is also longer, with reattachment occurring at the point the lower wall begins to taper towards the outlet.

The user can also test other turbulence models. The $k-\omega$ SST model is a popular choice in industrial CFD which can be selected by the following setting in the *momentumTransport* file.

```
model      kOmegaSST;
```

It requires initialisation of specific turbulent dissipation rate ω . Item can be calculated similarly to ε in Equation 2.3 using l_m as follows:

$$\omega = C_\mu^{-0.25} \frac{k^{0.5}}{l_m}. \quad (2.4)$$

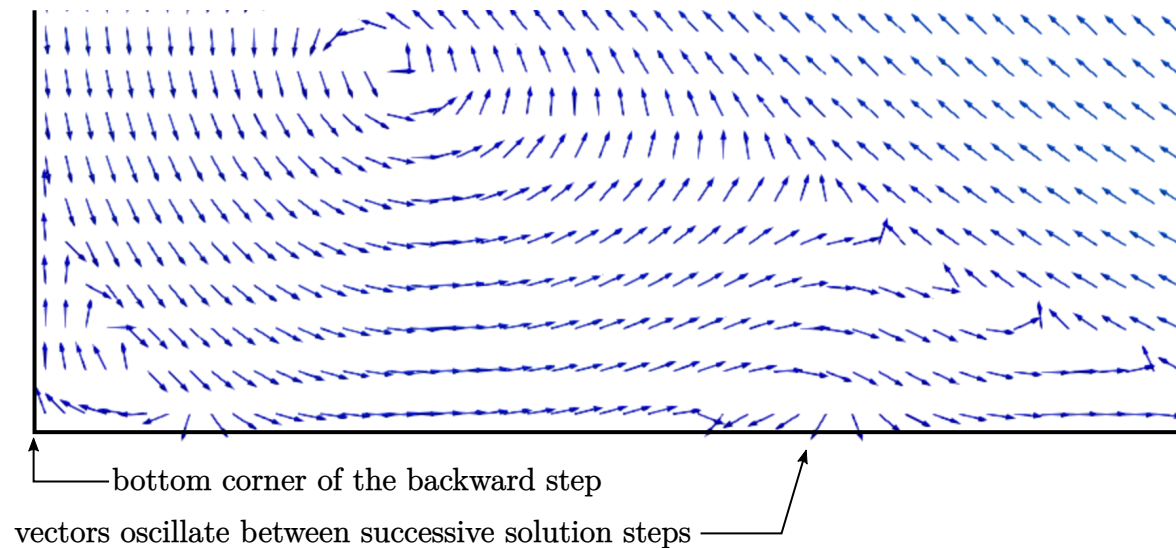
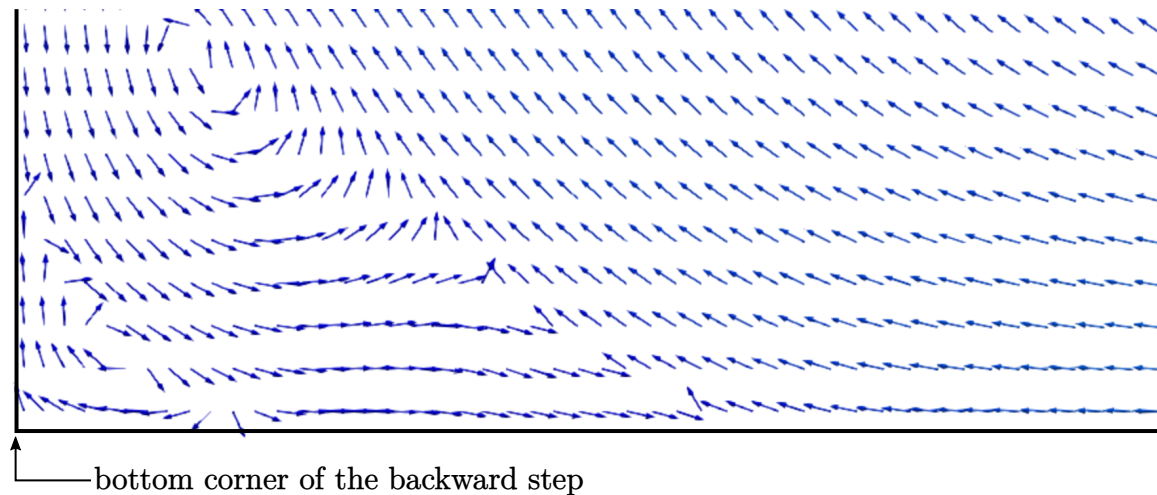
Using the estimate $l_m = 2.54$ mm as before, $\omega = 0.09^{-0.25} \times 0.375^{0.5} / 0.00254 = 440.2 \text{ s}^{-1}$. In the *0/omega* file, 440.2 is used both for the initial *internalField* and the inlet value.

The simulation can now be re-run using this model by deleting the solution time directories and executing *foamRun* as before. The solver does not terminate early by converging to within the tolerances specified in the *residualControl* sub-dictionary of the *fvSolution* file. Instead, it terminates at 2000 iterations, the *endTime* specified in the *controlDict* file.

The results with the $k-\omega$ SST model show a larger secondary vortex at the base of the step. The vortex does not stabilise to a steady-state, but instead oscillates a small amount over successive solution steps. The oscillations can be seen by examining the velocity vectors at different solution steps, *e.g.* 1000, 1100, 1200, *etc.* Figure 2.13 shows the extent of the secondary vortex and indicates where vectors oscillate around the reattachment point of the vortex.

The secondary vortex can be stabilised by a change to the numerical scheme for momentum advection. The user should open the *fvSchemes* file from the *system* directory. The discretisation of the advection terms is specified by the keyword entries of the form “div(phi,...)” in the *divSchemes* sub-dictionary. Momentum advection uses the *linearUpwind* scheme as shown below.

```
div(phi,U)      bounded Gauss linearUpwind grad(U);
```


Figure 2.13: Secondary vortex with $k-\omega$ SST model.Figure 2.14: Secondary vortex with $k-\omega$ SST model with limiting on $\text{grad}(\mathbf{U})$.

The `linearUpwind` scheme interpolates fields from cell centres to faces by extrapolation using the cell gradient. The `grad(U)` entry specifies the form of the gradient calculation, using the scheme specified in the `gradSchemes` sub-dictionary. In the example case, it includes only a `default` scheme for calculating all gradient terms in all equations.

In order to stabilise the secondary vortex, the `cellLimited` scheme can be applied specifically to the discretisation of the velocity gradient `grad(U)`. The syntax is shown below.

```
gradSchemes
{
    default          Gauss linear;
    grad(U)          cellLimited Gauss linear 1;
}
```

After saving the *fvSchemes* file, the simulation can be re-run by deleting the solution time directories and executing *foamRun* as before. With the *cellLimited* scheme applied, the solution converges normally, with the solver terminating at 285 iterations. The secondary vortex stabilises as shown in Figure 2.14.

2.2 Breaking of a dam

In this example we shall solve a problem of a simplified dam break in 2 dimensions using the *incompressibleVoF* modular solver. The feature of the problem is a transient flow of two fluids separated by a sharp interface, or free surface. The two-phase algorithm in *incompressibleVoF* is based on the volume of fluid (VoF) method in which a phase transport equation is used to determine the relative volume fraction of the two phases, or phase fraction α , in each computational cell. Physical properties are calculated as weighted averages based on this fraction. The nature of the VoF method means that an interface between the phases is not explicitly computed, but rather emerges as a property of the phase fraction field. Since the phase fraction can have any value between 0 and 1, the interface is never precisely defined, but occupies a volume around the region where a sharp interface should exist.

The test setup consists of a column of water at rest located behind a membrane on the left side of a tank. At time $t = 0$ s, the membrane is removed and the column of water collapses. During the collapse, the water impacts an obstacle at the bottom of the tank and creates a complicated flow structure, including several captured pockets of air. The geometry and the initial setup is shown in Figure 2.15.

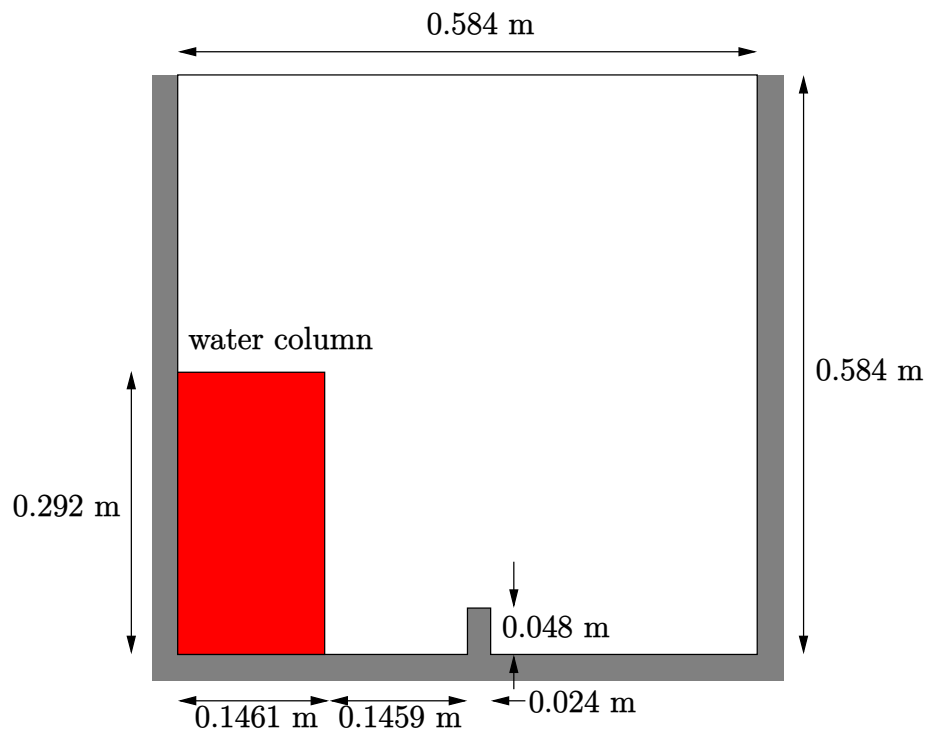


Figure 2.15: Geometry of the dam break.

2.2.1 Mesh generation

The user should go to their *run* directory and copy the *damBreakLaminar* case from the *\$FOAM_TUTORIALS/incompressibleVoF* directory, *i.e.*

```
run
cp -r $FOAM_TUTORIALS/incompressibleVoF/damBreakLaminar .
```

Go into the *damBreakLaminar* case directory and generate the mesh running *blockMesh* as described previously. The *damBreakLaminar* mesh consist of five blocks; the *blockMeshDict* entries are given below. The *blockMeshDict* file includes the optional *scale* keyword, described in section 5.4.2, which provides a scaling factor that is applied to the mesh.

```
16
17   scale 0.146;
18
19   vertices
20   (
21       (0 0 0)
22       (2 0 0)
23       (2.16438 0 0)
24       (4 0 0)
25       (0 0.32876 0)
26       (2 0.32876 0)
27       (2.16438 0.32876 0)
28       (4 0.32876 0)
29       (0 4 0)
30       (2 4 0)
31       (2.16438 4 0)
32       (4 4 0)
33       (0 0 0.1)
34       (2 0 0.1)
35       (2.16438 0 0.1)
36       (4 0 0.1)
37       (0 0.32876 0.1)
38       (2 0.32876 0.1)
39       (2.16438 0.32876 0.1)
40       (4 0.32876 0.1)
41       (0 4 0.1)
42       (2 4 0.1)
43       (2.16438 4 0.1)
44       (4 4 0.1)
45   );
46
47   blocks
48   (
49       hex (0 1 5 4 12 13 17 16) (23 8 1) simpleGrading (1 1 1)
50       hex (2 3 7 6 14 15 19 18) (19 8 1) simpleGrading (1 1 1)
51       hex (4 5 9 8 16 17 21 20) (23 42 1) simpleGrading (1 1 1)
52       hex (5 6 10 9 17 18 22 21) (4 42 1) simpleGrading (1 1 1)
53       hex (6 7 11 10 18 19 23 22) (19 42 1) simpleGrading (1 1 1)
54   );
55
56   defaultPatch
57   {
58       type empty;
59   }
60
61   boundary
62   (
63       leftWall
64       {
65           type wall;
66           faces
67           (
68               (0 12 16 4)
69               (4 16 20 8)
70           );
71       }
72       rightWall
```

```

73     {
74         type wall;
75         faces
76         (
77             (7 19 15 3)
78             (11 23 19 7)
79         );
80     }
81     lowerWall
82     {
83         type wall;
84         faces
85         (
86             (0 1 13 12)
87             (1 5 17 13)
88             (5 6 18 17)
89             (2 14 18 6)
90             (2 3 15 14)
91         );
92     }
93     atmosphere
94     {
95         type patch;
96         faces
97         (
98             (8 20 21 9)
99             (9 21 22 10)
100            (10 22 23 11)
101        );
102    }
103 );
104
105 // *****
106
```

The mesh is written into a set of files in a *polyMesh* directory in the *constant* directory. The user can list the contents of the directory to reveal the set of files.

```
ls constant/polyMesh
```

The files include: *points*, a list of the cell vertices; *faces*, a list of the cell faces; *owner* and *neighbour*, containing the indices of cells connected to a given face; *boundary*, a description of the boundary patches.

2.2.2 Boundary conditions

The *boundary* file can be read and understood by the user. The user should take a look at its contents, either by opening it in a file editor or printing out in the terminal window using the *cat* utility.

```
cat constant/polyMesh/boundary
```

The file contains a list of five boundary patches: *leftWall*, *rightWall*, *lowerWall*, *atmosphere* and *defaultFaces*. The user should notice the *type* of the patches. Firstly, the *atmosphere* is a standard *patch*, *i.e.* has no special attributes, merely an entity on which boundary conditions can be specified. Then, the *defaultFaces* patch is formed of **block faces that are omitted** from the *boundary* sub-dictionary in the *blockMeshDict* file. Those block faces form a patch whose properties are specified in a *defaultPatch* sub-dictionary in the *blockMeshDict* file. In this case, the default *type* is set to *empty* since the patch normal is in the direction we will not solve in this 2D case.

The `leftWall`, `rightWall` and `lowerWall` patches are each a `wall`. Like the generic `patch`, the `wall` type contains no geometric or topological information about the mesh and only differs from the plain `patch` in that it identifies the patch as a wall. This is required by some modelling, *e.g.* turbulent wall functions and some turbulence models which include the distance to the nearest wall in their calculations.

With VoF specifically, surface tension models can include wall adhesion at the contact point between the interface and wall surface. Wall adhesion models can be applied through a special boundary condition on the `alpha` (α) field, *e.g.* the `alphaContactAngle` boundary condition, which requires the user to specify a static contact angle, `theta0`.

This example ignores surface tension effects between the wall and interface. This is done by applying the `zeroGradient` condition to `alpha` on the walls, which is a simpler alternative to setting the static contact angle $\theta_0 = 90^\circ$ with the `alphaContactAngle` boundary condition.

The `top` boundary is free to the atmosphere so needs to permit both outflow and inflow according to the internal flow. We therefore use a combination of boundary conditions for pressure and velocity that does this while maintaining stability. They are:

- `prghTotalPressure`, applied to the pressure field, minus the hydrostatic component, p_{pgh} , given by Equation 6.5;
- `pressureInletOutletVelocity`, applied to velocity $\mathbf{U}$, which sets `zeroGradient` on all components of $\mathbf{U}$, except where there is inflow, in which case a `fixedValue` condition is applied to the *tangential* component;
- `inletOutlet` applied to other fields, which is a `zeroGradient` condition when flow outwards, `fixedValue` when flow is inwards.

At all wall boundaries, the `fixedFluxPressure` boundary condition is applied to the pressure field, which adjusts the pressure gradient so that the boundary flux matches the velocity boundary condition for solvers that include body forces such as gravity and surface tension.

The `defaultFaces` patch representing the front and back planes of the 2D problem, is, as usual, an `empty` type.

2.2.3 Phases

The fluid phases are specified in the `phaseProperties` file in the `constant` directory as follows:

```

16
17 phases          (water air);
18
19 sigma           0.07;
20
21
22 // ***** //
```

It lists two phases, `water` and `air`. Equations for phase fraction are solved for the phases in the list, **except the last phase** listed, *i.e.* `air` in this case. Since there are only two phases, only one phase fraction equation is solved in this case, for the water phase fraction α_{water} , specified in the file `alpha.water` in the `0` directory.

The `phaseProperties` file also contains an entry for the surface tension between the two phases, specified by the keyword `sigma` in units N m^{-1} .

2.2.4 Setting initial fields

Unlike the previous cases, the phase fraction field α_{water} must be initialised to be non-uniform according to:

$$\alpha_{\text{water}} = \begin{cases} 1 & \text{for the water phase} \\ 0 & \text{for the air phase} \end{cases} \quad (2.5)$$

Prior to OpenFOAM v14, non-uniform fields were generally created by the `setFields` utility. Like any typical pre-processing utility, it reads a configuration file — in this case, named *setFieldsDict* (*i.e.* the utility name with *Dict* suffix). Within the file, the user specifies how the non-uniform field is generated.

However, from OpenFOAM v14 onwards, non-uniform fields can be initialised by applying a *field function* to the `internalField` of the field file itself, *e.g.* as shown in *0/alpha.water* below:

```

16 dimensions      [];
17
18
19 internalField
20 {
21     type          zonal;
22     defaultValue   0;
23
24     zones
25     {
26         waterColumn
27         {
28             type          box;
29             box            (0 0 -1) (0.1461 0.292 1);
30             value          1;
31         }
32     }
33 }
34
35
36
37 boundaryField
38 {
39     #includeEtc "caseDicts/setConstraintTypes"
40
41     wall
42     {
43         type          zeroGradient;
44     }
45
46     atmosphere
47     {
48         type          inletOutlet;
49         inletValue     uniform 0;
50     }
51 }
52
53
54
55 // ***** //
```

This example uses a `zonal` field function for the `internalField`. The non-uniform field consists first of a `defaultValue` of 0. That value is then over-ridden by setting a `value` of 1 within a `cellZone` named `waterColumn`. The `cellZone` is defined as described in section 5.6 using the `box` zone generator described in section 5.6.2.

Using `paraFoam`, check that the initial `alpha.water` field corresponds to the desired distribution as in Figure 2.16.

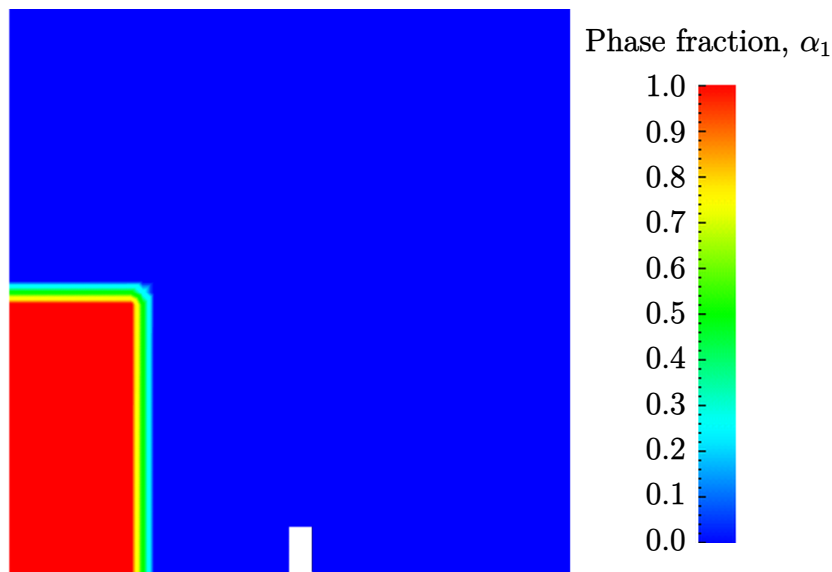


Figure 2.16: Initial conditions for phase fraction `alpha.water`.

2.2.5 Fluid properties

The physical properties for the air and water phases are specified in *physicalProperties.air* and *physicalProperties.water* files, respectively, in the *constant* directory. Physical properties describe characteristics of the fluid in a absence of flow. Each file specifies the viscosity model through the `viscosityModel` keyword, which is set to `constant` to indicate the value is unchanging. The viscosity is then specified by the `nu` keyword in units $\text{m}^2 \text{s}^{-1}$. The density of each fluid is also specified by the keyword `rho` in units kg m^{-3} . The *physicalProperties.air* file is shown below as an example:

```

16
17 viscosityModel constant;
18
19 nu          1.48e-05;
20
21 rho         1;
22
23
24 // ***** //
```

If the viscosity does change according to the flow, *e.g.* as in non-Newtonian or visco-elastic fluids, then those models are specified through the *momentumProperties* file, as described in section 8.3.

2.2.6 Gravity

Gravitational acceleration is uniform across the domain and is specified in a file named *g* in the *constant* directory. Unlike a normal field file, *e.g.* *U* and *p*, *g* is a `uniformDimensionedVectorField` and so simply contains a set of `dimensions` and a value that represents $(0, 9.81, 0) \text{ m s}^{-2}$ for this case:

```

16
17 dimensions    [acceleration];
18 value         (0 -9.81 0);
19
20
21 // ***** //
```

2.2.7 Turbulence modelling

As in the cavity example, the choice of turbulence modelling method is selectable at run-time through the `simulationType` keyword in *momentumTransport* dictionary. In this example, we wish to run without turbulence modelling so we set `laminar`:

```

16
17  simulationType  laminar;
18
19
20  // ***** //
```

2.2.8 Time step control

The simulation of the `damBreakLaminar` case is fully transient so the time step requires attention. The Courant number Co is an important consideration relating to time step. It is dimensionless parameter which can be defined for each cell as:

$$Co = \frac{\delta t |\mathbf{U}|}{\delta x} \quad (2.6)$$

where δt is the time step, $|\mathbf{U}|$ is the magnitude of the velocity through that cell and δx is the cell size in the direction of the velocity. With *explicit* solution, stability requires the maximum $Co < 1$ at least; stricter limits exist depending on the choice of advection scheme. *Implicit* solutions do not have the same stability limit of the maximum Co , but temporal accuracy becomes more relevant as Co increased beyond 1.

Time step control is particularly important with interface-capturing. The `incompressibleVoF` solver module uses the multidimensional universal limiter for explicit solution (MULES), created by Henry Weller, to maintain boundedness of the phase fraction. Co needs to be limited depending on the choice of MULES algorithm. With the original *explicit* MULES algorithm, an upper limit of $Co \approx 0.25$ for stability is typically required. However, there is also the *semi-implicit* version of MULES, specified by the `MULESCorr` switch in the *fvSolution* file. For semi-implicit MULES, there is really no upper limit in Co for stability, but instead the level is determined by requirements of temporal accuracy.

In general it is difficult to specify a fixed time-step to satisfy the Co criterion since $|\mathbf{U}|$ is changing from cell to cell during the simulation. Instead, automatic adjustment of the time step is specified in the *controlDict* by switching `adjustTimeStep` to `on` and specifying the maximum Co for the phase fields, `maxAlphaCo`, and other fields, `maxCo`. In this example, the `maxAlphaCo` and `maxCo` are set to 1.0. The upper limit on time step `maxDeltaT` can be set to a value that will not be exceeded in this simulation, *e.g.* 1.0.

By using automatic time step control, the steps themselves are never rounded to a convenient value. Consequently if we request that OpenFOAM saves results at a fixed number of time step intervals, the times at which results are saved are somewhat arbitrary. However with automatic time step adjustment, results can be written at fixed times using the `adjustableRunTime` option for `writeControl` in the *controlDict* dictionary. With this option, the automatic time stepping procedure further adjusts their time steps so that it ‘hits’ on the exact times specified by the `writeInterval`, set to 0.05 in this example. The *controlDict* dictionary entries are shown below.

```

16
17  solver          incompressibleVoF;
```

```

18
19  startFrom      startTime;
20
21  startTime      0;
22
23  stopAt         endTime;
24
25  endTime        1;
26
27  deltaT         0.001;
28
29  writeControl    adjustableRunTime;
30
31  writeInterval   0.05;
32
33  purgeWrite      0;
34
35  writeFormat     ascii;
36
37  writePrecision  6;
38
39  writeCompression off;
40
41  timeFormat      general;
42
43  timePrecision   6;
44
45  runTimeModifiable yes;
46
47  adjustTimeStep  yes;
48
49  maxCo           1;
50
51  maxAlphaCo      1;
52
53  maxDeltaT       1;
54
55  // *****

```

2.2.9 Discretisation schemes

The MULES method, used by the incompressibleVoF modular solver, maintains boundedness of the phase fraction independently of the underlying numerical scheme, mesh structure, *etc.* The choice of schemes for convection are therefore not restricted to those that are strongly stable or bounded, such as upwind differencing.

The convection schemes settings are made in the *divSchemes* sub-dictionary of the *fvSchemes* dictionary. In this example, the convection term in the momentum equation, $\nabla \cdot (\rho \mathbf{U} \mathbf{U})$, denoted by the `div(rhoPhi,U)` keyword, uses `Gauss linearUpwind grad(U)` to produce good accuracy.

The $\nabla \cdot (\mathbf{U} \alpha)$ term, represented by the `div(phi,alpha)` keyword uses a bespoke `interfaceCompression` scheme where the specified coefficient is a factor that controls the compression of the interface where: 0 corresponds to no compression; 1 corresponds to conservative compression; and, anything larger than 1, relates to enhanced compression of the interface. We generally use a value of 1.0, as in this example.

The other discretised terms use commonly employed schemes so that the *fvSchemes* dictionary entries is as follows.

```

16
17  ddtSchemes
18  {
19      default      Euler;
20  }
21
22  gradSchemes
23  {

```

```

24     default      Gauss linear;
25 }
26
27 divSchemes
28 {
29     div(phi,alpha)  Gauss interfaceCompression vanLeer 1;
30     div(rhoPhi,U)   Gauss linearUpwind grad(U);
31     div(((rho*nuEff)*dev2(T(grad(U))))) Gauss linear;
32 }
33
34 laplacianSchemes
35 {
36     default      Gauss linear uncorrected;
37 }
38
39 interpolationSchemes
40 {
41     default      linear;
42 }
43
44 snGradSchemes
45 {
46     default      uncorrected;
47 }
48
49 // *****
50
51 // *****
52

```

2.2.10 Linear-solver control

In the *fvSolution* file, the sub-dictionary in *solvers* for *alpha.water* contains elements that are specific to the MULES algorithm as shown below.

```

"alpha.water.*"
{
    nAlphaCorr      2;
    nAlphaSubCycles 1;

    MULESCorr       yes;

    MULES
    {
        nIter        10;
        tolerance     1e-2;
    }

    solver          smoothSolver;
    smoother         symGaussSeidel;
    tolerance        1e-8;
    relTol           0;
}

```

MULES calculates two limiters to keep the phase fraction within the lower and upper bounds of 0 and 1. The limiter calculation is iterative, with the maximum number of iterations specified by *nIter* and the tolerance specified by *tolerance*. Here, *nIter* is set to a high value (10) but the *tolerance* setting ensures that number of iterations will not be reached.

The semi-implicit version of MULES is activated by the *MULESCorr* switch. It first calculates an implicit, upwind solution before applying MULES as a higher-order correction. The linear solver must be configured for the implicit, upwind solution, through the *solver*, *smoother*, *tolerance* and *relTol* parameters.

The `nAlphaCorr` keyword which controls the number of iterations of the phase fraction equation within a solution step. The iteration is used to overcome nonlinearities in the advection which are present in this case due to the `interfaceCompression` scheme.

2.2.11 Running the code

Running of the code has been described in the previous tutorial. Try the following, that uses `tee`, a command that enables output to be written to both standard output and files:

```
foamRun | tee log
```

The code will now be run interactively, with a copy of output stored in the `log` file.

2.2.12 Post-processing

Post-processing of the results can now be done in the usual way. The user can monitor the development of the phase fraction `alpha.water` in time, *e.g.* see Figure 2.17.

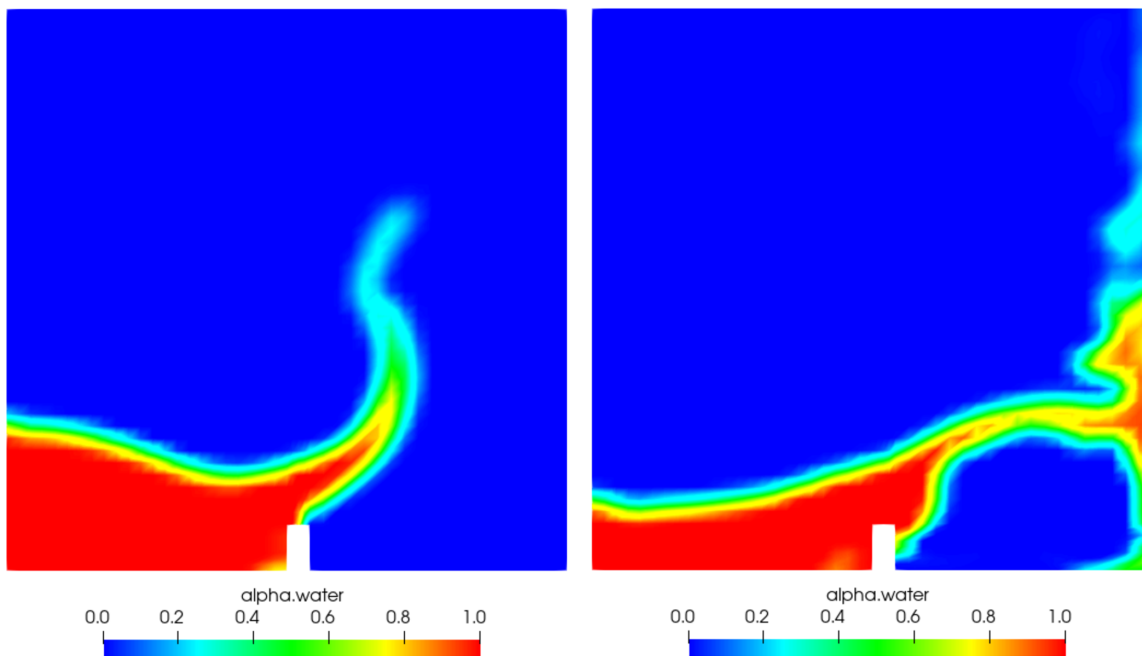


Figure 2.17: Phase fraction α at $t = 0.25$ s (left) and 0.50 s (right).

2.2.13 Running in parallel

The results from the previous example are generated using a fairly coarse mesh. We now wish to increase the mesh resolution and re-run the case. Using a finer mesh, we can then demonstrate the parallel processing capability of OpenFOAM.

The user should first clone the `damBreakLaminar` case, *e.g.* by

```
run
foamCloneCase damBreakLaminar damBreakLaminarFine
```

Change into the new case directory (with `cd`) and change the `blocks` description in the *blockMeshDict* dictionary to

```
blocks
(
    hex (0 1 5 4 12 13 17 16) (46 10 1) simpleGrading (1 1 1)
    hex (2 3 7 6 14 15 19 18) (40 10 1) simpleGrading (1 1 1)
    hex (4 5 9 8 16 17 21 20) (46 76 1) simpleGrading (1 2 1)
    hex (5 6 10 9 17 18 22 21) (4 76 1) simpleGrading (1 2 1)
    hex (6 7 11 10 18 19 23 22) (40 76 1) simpleGrading (1 2 1)
);
```

Here, the entry is presented as printed from the *blockMeshDict* file; in short the user must change the mesh densities, *e.g.* the `46 10 1` entry, and some of the mesh grading entries to `1 2 1`. Once the dictionary is correct, generate the mesh by running `blockMesh`.

Parallel computing uses domain decomposition, in which the geometry and associated fields are broken into pieces and allocated to separate processors for solution. The first step required to run a parallel case is therefore to decompose the domain using the `decomposePar` utility. The `decomposePar` utility needs to be configured using a *decomposeParDict* file located in the *system* directory. Sample configuration files can be found within the *etc* directory in the OpenFOAM installation, which can be copied to the case directory by running the `foamGet` script. A *decomposeParDict* file can therefore be copied by typing:

```
foamGet decomposeParDict
```

It responds by offering one of two files, one lightly annotated and one heavily annotated. The former is suggested by default so hit `Enter` to accept that file. The user should open the *decomposeParDict* file in their editor. The first entry is `numberOfSubdomains` which specifies the number of subdomains into which the case will be decomposed, usually corresponding to the number of processors available for the case. Change the entry as follows.

```
numberOfSubdomains 4;
```

This example uses the `hierarchical` method of decomposition. It requires the `hierarchical` sub-dictionary to be configured according to the following criteria. The domain is split into pieces, or subdomains, in the x , y and z directions, the number of subdomains in each direction being given by the vector **n**. As this geometry is 2 dimensional, the 3rd direction, z , cannot be split, hence n_z must equal 1. The n_x and n_y components of **n** split the domain in the x and y directions and must be specified so that the number of subdomains specified by n_x and n_y equals the specified `numberOfSubdomains`, *i.e.* $n_x n_y = \text{numberOfSubdomains}$.

It is beneficial to keep the number of cell faces adjoining the subdomains to a minimum so, for a square geometry, it is best to keep the split between the x and y directions fairly even. Since we have 4 subdomains, the user should set $n_x = n_y = 2$ as follows.

```
hierarchical
{
    n      (2 2 1);
    order  xyz;
}
```

The user should now save the file and then run **decomposePar** by the following command.

```
decomposePar
```

The terminal output shows that the decomposition is distributed evenly between the subdomains. The decomposition writes the mesh and fields of each sub-domain into separate sub-directories named **processor<N>**, where **N** is the sub-domain ID, *e.g.* 0, 1, 2, *etc.* The user should list the files in the case directory to confirm that four directories **processor0**, **processor1**, **processor2** and **processor3** exist.

This example presents running in parallel with the **openMPI** implementation of the standard message-passing interface (MPI). The following command runs on 4 cores of a local multi-processor CPU.

```
mpirun -np 4 foamRun -parallel
```

The user can consult section 3.4 for more details of how to run a case in parallel. For example, the user may run on more nodes over a network by creating a file that lists the host names of the machines on which the case is to be run as described in section 3.4.3.

2.2.14 Post-processing a case run in parallel

When the case runs in parallel, the results are written into time directories within the **processor<N>** sub-directories. The user can confirm this by listing the time directories for the **processor0** directory.

```
ls processor0
```

It is possible to post-process an individual sub-domain by treating the individual **processor** directory as a case in its own right. For example, to view the **processor1** sub-domain in **ParaView**, the user can launch **paraFoam** by running the following command.

```
paraFoam -case processor1
```

Figure 2.18 shows the mesh from this sub-domain, following the decomposition of the domain using the **simple** method.

The decomposed fields and mesh can also be viewed directly in **ParaView**, by first running the **paraFoam** script as normal.

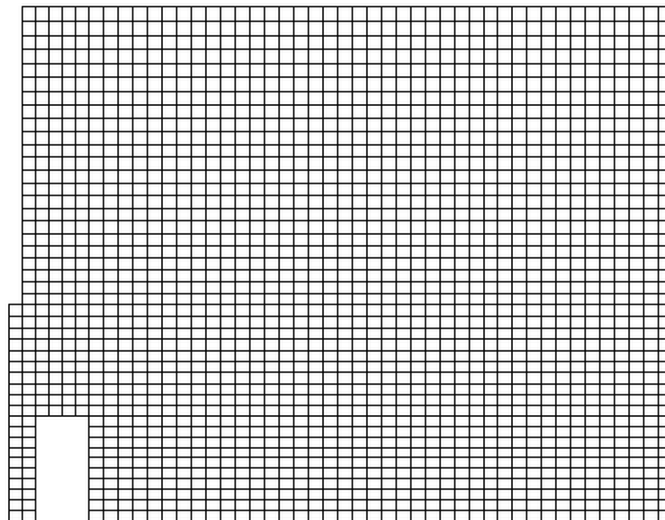


Figure 2.18: Mesh of processor 2 in parallel processed case.

`paraFoam`

When **ParaView** opens, the user can select the **Decomposed Case** before hitting **Apply**. The entire case will then appear in the **RenderView** window.

One further option involves first reassembling the case back to a single domain. The `reconstructPar` utility performs this reassembly, taking the field files from time directories from each `processor` sub-domain and building equivalent field files for the complete domain. The user can test this by running.

`reconstructPar`

The fields are reconstructed and written to solution time directories in the case directory. These fields can be visualised as normal in **ParaView**. The results from the fine mesh are shown in Figure 2.19. The user can see that the resolution of interface has improved significantly compared to the coarse mesh.

2.3 Stress analysis of a plate with a hole

This tutorial describes how to pre-process, run and post-process a case involving linear-elastic, steady-state stress analysis on a square plate with a circular hole at its centre. The plate dimensions are: side length 4 m and radius $R = 0.5$ m. It is loaded with a uniform traction of $\sigma = 10$ kPa over its left and right faces as shown in Figure 2.20. Two symmetry planes can be identified for this geometry and therefore the solution domain need only cover a quarter of the geometry, shown by the shaded area in Figure 2.20.

The problem can be approximated as 2D since the load is applied in the plane of the plate. In a Cartesian coordinate system there are two possible assumptions to take in regard to the behaviour of the structure in the third dimension: (1) the plane stress condition, in which the stress components acting out of the 2D plane are assumed to be negligible; (2)

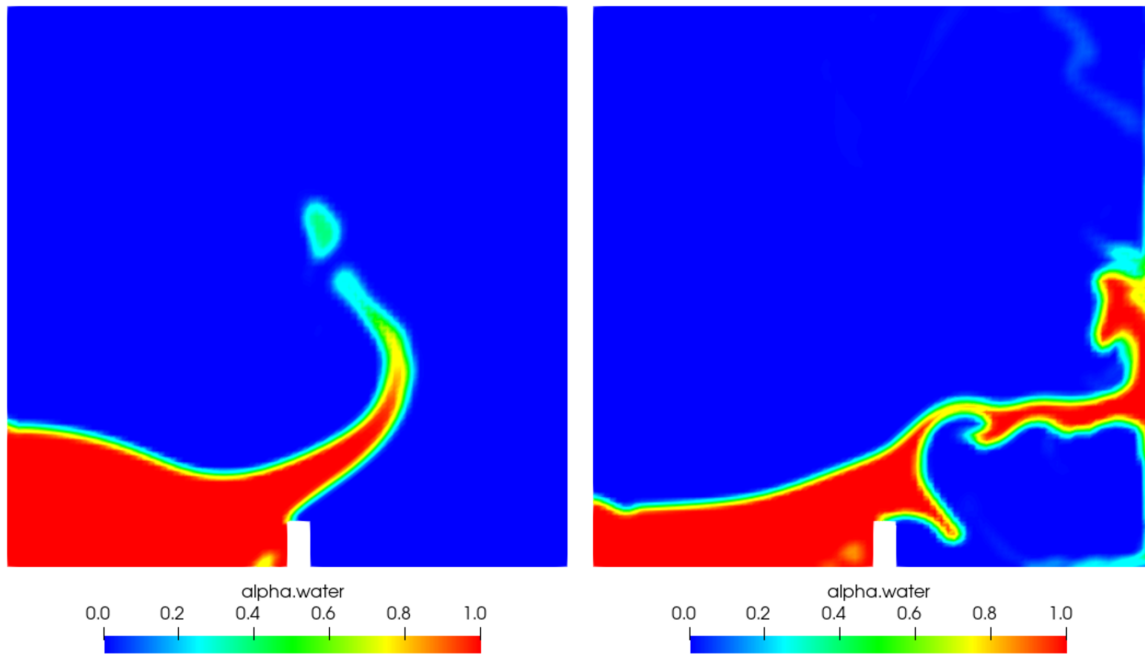


Figure 2.19: Phase fraction α at $t = 0.25$ s (left) and 0.50 s (right).

the plane strain condition, in which the strain components out of the 2D plane are assumed negligible. The plane stress condition is appropriate for solids whose third dimension is thin as in this case; the plane strain condition is applicable for solids where the third dimension is thick.

An analytical solution exists for loading of an infinitely large, thin plate with a circular hole. The solution for the stress normal to the vertical plane of symmetry is

$$(\sigma_{xx})_{x=0} = \begin{cases} \sigma \left(1 + \frac{R^2}{2y^2} + \frac{3R^4}{2y^4} \right) & \text{for } |y| \geq R \\ 0 & \text{for } |y| < R \end{cases} \quad (2.7)$$

Results from the simulation will be compared with this solution. At the end of the tutorial, the user can: investigate the sensitivity of the solution to mesh resolution and mesh grading; and, increase the size of the plate in comparison to the hole to try to estimate the error in comparing the analytical solution for an infinite plate to the solution of this problem of a finite plate.

The example uses the `solidDisplacement` modular solver. The user should go to their `run` directory, copy the `plateHole` case from the `$FOAM_TUTORIALS/solidDisplacement` directory and finally change into the `plateHole` case directory.

```
run
cp -r $FOAM_TUTORIALS/solidDisplacement/plateHole .
cd plateHole
```

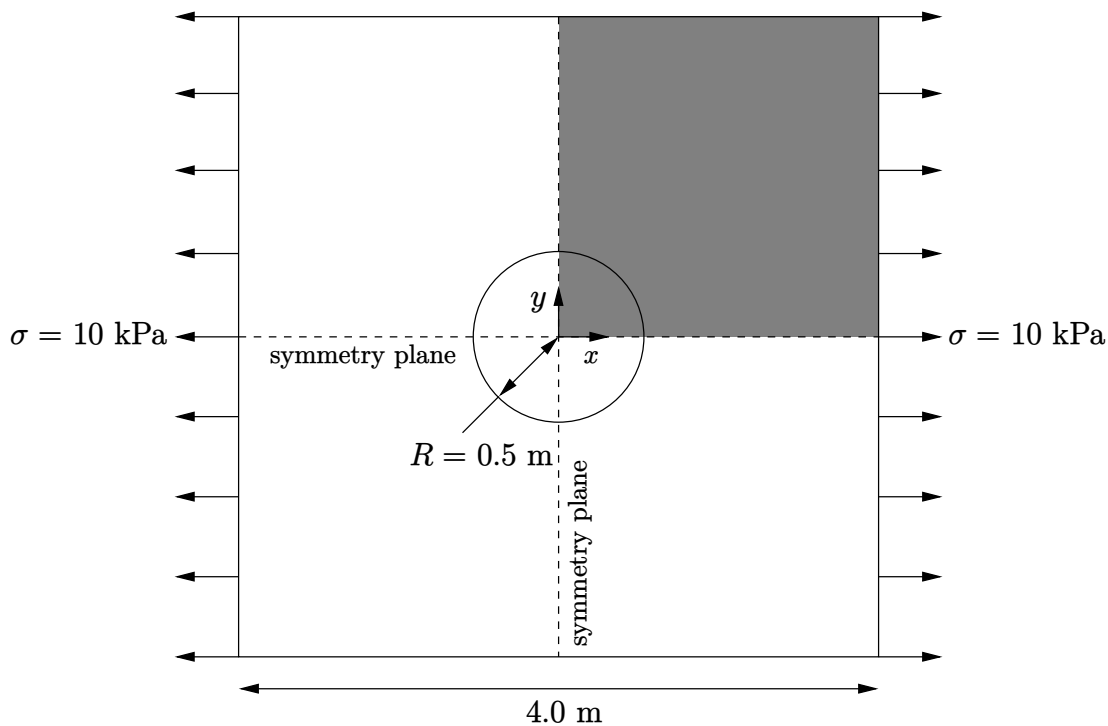


Figure 2.20: Geometry of the plate with a hole.

2.3.1 Mesh generation

The domain consists of four blocks, some of which have arc-shaped edges. The block structure for the part of the mesh in the $x - y$ plane is shown in Figure 2.21. As already mentioned in section 2.1.2, all geometries are generated in 3D in OpenFOAM even if the case is to be as a 2D problem. Therefore a dimension of the block in the z direction has to be chosen; here, 0.5 m is selected. It does not affect the solution since the traction boundary condition is specified as a stress rather than a force, thereby making the solution independent of the cross-sectional area. The user should open the *blockMeshDict* file in an editor, as listed below.

```

16
17 vertices
18 (
19     (0.5 0 0)
20     (1 0 0)
21     (2 0 0)
22     (2 0.707107 0)
23     (0.707107 0.707107 0)
24     (0.353553 0.353553 0)
25     (2 2 0)
26     (0.707107 2 0)
27     (0 2 0)
28     (0 1 0)
29     (0 0.5 0)
30     (0.5 0 0.5)
31     (1 0 0.5)
32     (2 0 0.5)
33     (2 0.707107 0.5)
34     (0.707107 0.707107 0.5)
35     (0.353553 0.353553 0.5)
36     (2 2 0.5)
37     (0.707107 2 0.5)
38     (0 2 0.5)
39     (0 1 0.5)
40     (0 0.5 0.5)

```

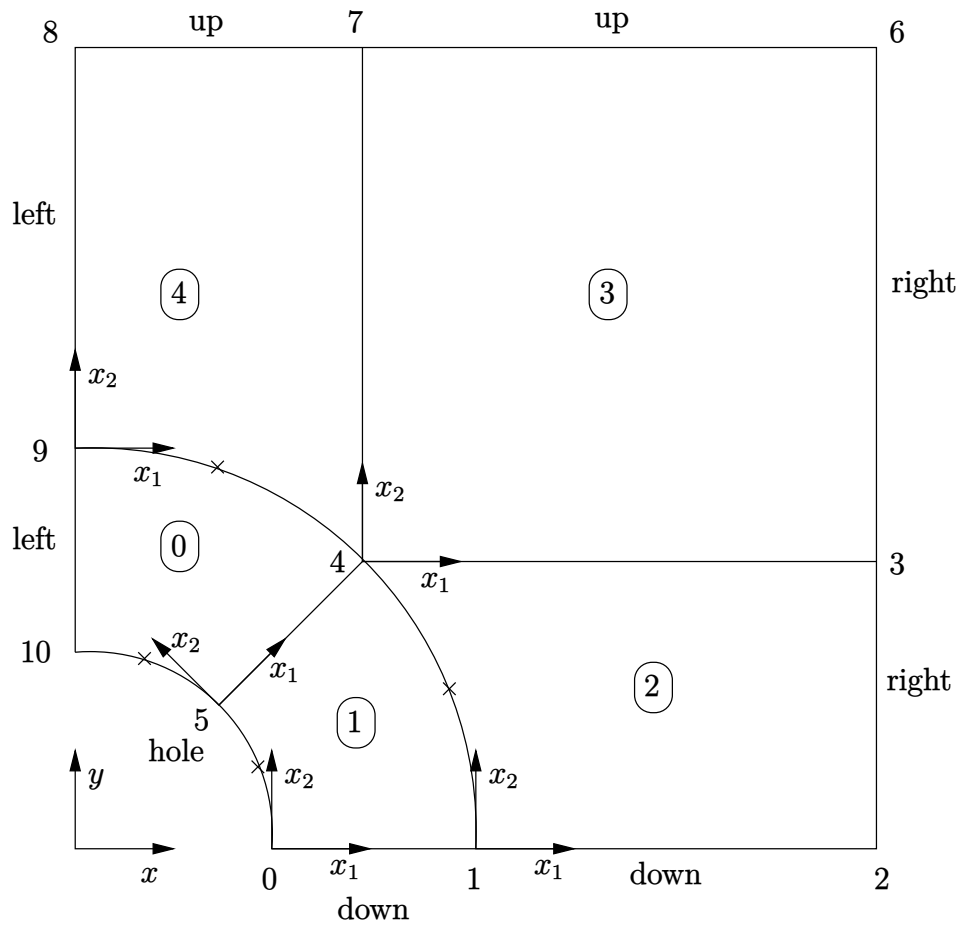


Figure 2.21: Block structure of the mesh for the plate with a hole.

```

41 );
42
43 blocks
44 (
45     hex (5 4 9 10 16 15 20 21) (10 10 1) simpleGrading (1 1 1)
46     hex (0 1 4 5 11 12 15 16) (10 10 1) simpleGrading (1 1 1)
47     hex (1 2 3 4 12 13 14 15) (20 10 1) simpleGrading (1 1 1)
48     hex (4 3 6 7 15 14 17 18) (20 20 1) simpleGrading (1 1 1)
49     hex (9 4 7 8 20 15 18 19) (10 20 1) simpleGrading (1 1 1)
50 );
51
52 edges
53 (
54     arc 0 5 (0.469846 0.17101 0)
55     arc 5 10 (0.17101 0.469846 0)
56     arc 1 4 (0.939693 0.34202 0)
57     arc 4 9 (0.34202 0.939693 0)
58     arc 11 16 (0.469846 0.17101 0.5)
59     arc 16 21 (0.17101 0.469846 0.5)
60     arc 12 15 (0.939693 0.34202 0.5)
61     arc 15 20 (0.34202 0.939693 0.5)
62 );
63
64 boundary
65 (
66     left
67     {
68         type symmetryPlane;
69         faces
70         (
71             (8 9 20 19)

```

```

72         (9 10 21 20)
73     );
74 }
75 right
76 {
77     type patch;
78     faces
79     (
80         (2 3 14 13)
81         (3 6 17 14)
82     );
83 }
84 down
85 {
86     type symmetryPlane;
87     faces
88     (
89         (0 1 12 11)
90         (1 2 13 12)
91     );
92 }
93 up
94 {
95     type patch;
96     faces
97     (
98         (7 8 19 18)
99         (6 7 18 17)
100    );
101 }
102 hole
103 {
104     type patch;
105     faces
106     (
107         (10 5 16 21)
108         (5 0 11 16)
109    );
110 }
111 frontAndBack
112 {
113     type empty;
114     faces
115     (
116         (10 9 4 5)
117         (5 4 1 0)
118         (1 4 3 2)
119         (4 7 6 3)
120         (4 9 8 7)
121         (21 16 15 20)
122         (16 11 12 15)
123         (12 13 14 15)
124         (15 14 17 18)
125         (15 18 19 20)
126    );
127 }
128 );
129
130
131 // *****

```

Until now, we have only specified straight edges in the geometries of previous tutorials but here we need to specify curved edges. These are specified under the `edges` keyword entry which is a list of non-straight edges. The syntax of each list entry begins with the type of curve, including `arc`, `simpleSpline`, `polyLine` *etc.*, described further in section 5.4.4. In this example, all the edges are circular and so can be specified by the `arc` keyword entry. The following entries are the labels of the start and end vertices of the arc and a point vector through which the circular arc passes.

The blocks in this *blockMeshDict* do not all have the same orientation. As can be seen in Figure 2.21 the x_2 direction of block 0 is equivalent to the $-x_1$ direction for block 4. This means care must be taken when defining the number and distribution of cells in each block

so that the cells match up at the block faces.

Six patches are defined: one for each side of the plate, one for the hole and one for the front and back planes. The `left` and `down` patches are both a symmetry plane. Since this is a *geometric* constraint, it is included in the definition of the *mesh*, rather than being purely a specification on the boundary condition of the fields. Therefore they are defined as such using a special `symmetryPlane` type as shown in the *blockMeshDict*.

The `frontAndBack` patch represents the plane which is ignored in a 2D case. Again this is a geometric constraint so is defined within the mesh, using the `empty` type as shown in the *blockMeshDict*. For further details of boundary types and geometric constraints, the user should refer to section 5.3.

The remaining patches are of the regular `patch` type. The mesh should be generated using `blockMesh` and can be viewed in `paraFoam` as described in section 2.1.3. It should appear as in Figure 2.22.

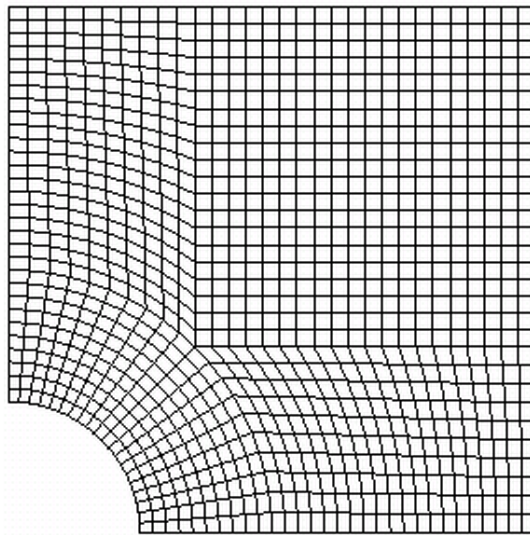


Figure 2.22: Mesh of the hole in a plate problem.

2.3.2 Boundary and initial conditions

Once the mesh generation is complete, the initial field with boundary conditions must be set. For a stress analysis case without thermal stresses, only displacement `D` needs to be set. The *0/D* is as follows:

```

16
17 dimensions      [length];
18
19 internalField    uniform (0 0 0);
20
21 boundaryField
22 {
23     left
24     {
25         type      symmetryPlane;
26     }
27     right
28     {
29         type      tractionDisplacement;
30         traction  uniform (10000 0 0);

```

```

31         pressure      uniform 0;
32         value          uniform (0 0 0);
33     }
34     down
35     {
36         type            symmetryPlane;
37     }
38     up
39     {
40         type            tractionDisplacement;
41         traction        uniform (0 0 0);
42         pressure        uniform 0;
43         value           uniform (0 0 0);
44     }
45     hole
46     {
47         type            tractionDisplacement;
48         traction        uniform (0 0 0);
49         pressure        uniform 0;
50         value           uniform (0 0 0);
51     }
52     frontAndBack
53     {
54         type            empty;
55     }
56 }
57
58 // *****

```

Firstly, it can be seen that the displacement initial conditions are set to $(0,0,0)$ m. The **left** and **down** patches **must** be both of **symmetryPlane** type since they are specified as such in the mesh description in the *constant/polyMesh/boundary* file. Similarly the **frontAndBack** patch is declared **empty**.

The other patches are traction boundary conditions, set by a specialist **tractionDisplacement** boundary type. The traction boundary conditions are specified by a linear combination of: (1) a boundary traction vector under keyword **traction**; (2) a pressure that produces a traction normal to the boundary surface that is defined as negative when pointing out of the surface, under keyword **pressure**. The **up** and **hole** patches are zero traction so the boundary traction and pressure are set to zero. For the **right** patch the traction should be $(1e4, 0, 0)$ Pa and the pressure should be 0 Pa.

2.3.3 Physical properties

The physical properties for the case are set in the *physicalProperties* dictionary in the *constant* directory, shown below:

```

16
17 rho
18 {
19     type            uniform;
20     value           7854;
21 }
22
23 nu
24 {
25     type            uniform;
26     value           0.3;
27 }
28
29 E
30 {
31     type            uniform;
32     value           2e+11;
33 }
34

```

```

35  Cv
36  {
37      type          uniform;
38      value          434;
39  }
40
41  kappa
42  {
43      type          uniform;
44      value          60.5;
45  }
46
47  alphav
48  {
49      type          uniform;
50      value          1.1e-05;
51  }
52
53  planeStress      yes;
54  thermalStress    no;
55
56  // *****
57  // *****

```

The file includes mechanical properties of steel:

- Density $\rho = 7854 \text{ kg m}^{-3}$
- Young's modulus $E = 2 \times 10^{11} \text{ Pa}$
- Poisson's ratio $\nu = 0.3$

The `planeStress` switch is set to `yes` to adopt the plane stress assumption in this 2D case. The `solidDisplacementFoam` solver may optionally solve a thermal equation that is coupled with the momentum equation through the thermal stresses that are generated. The user specifies at run time whether OpenFOAM should solve the thermal equation by the `thermalStress` switch (currently set to `no`). The thermal properties are also specified for steel for this case, *i.e.*:

- Specific heat capacity $C_p = 434 \text{ J kg}^{-1} \text{ K}^{-1}$
- Thermal conductivity $\kappa = 60.5 \text{ W m}^{-1} \text{ K}^{-1}$
- Thermal expansion coefficient $\alpha_v = 1.1 \times 10^{-5} \text{ K}^{-1}$

For thermal calculations, the temperature field variable `T` is present in the `0` directory.

2.3.4 Control

As before, the information relating to the control of the solution procedure are read in from the `controlDict` dictionary. For this case, the `startTime` is 0 s. The time step is not important since this is a steady state case; in this situation it is best to set the time step `deltaT` to 1 so it simply acts as an iteration counter for the steady-state case. The `endTime`, set to 100, then acts as a limit on the number of iterations. The `writeInterval` can be set to 20.

The `controlDict` entries are as follows:

```

16
17  solver          solidDisplacement;
18
19  startFrom        startTime;

```

```

20
21     startTime      0;
22
23     stopAt          endTime;
24
25     endTime         100;
26
27     deltaT          1;
28
29     writeControl     timeStep;
30
31     writeInterval    20;
32
33     purgeWrite       0;
34
35     writeFormat       ascii;
36
37     writePrecision   6;
38
39     writeCompression off;
40
41     timeFormat        general;
42
43     timePrecision     6;
44
45     runTimeModifiable true;
46
47 // *****

```

2.3.5 Discretisation schemes and linear-solver control

Let us turn our attention to the *fvSchemes* dictionary. Firstly, the problem we are analysing is steady-state so the user should select **SteadyState** for the time derivatives in **timeScheme**. This essentially switches off the time derivative terms. Not all solvers, especially in fluid dynamics, work for both steady-state and transient problems but **solidDisplacementFoam** does work, since the base algorithm is the same for both types of simulation.

The momentum equation in linear-elastic stress analysis includes several explicit terms containing the gradient of displacement. The calculations benefit from accurate and smooth evaluation of the gradient. Normally, in the finite volume method the discretisation is based on Gauss's theorem. The Gauss method is sufficiently accurate for most purposes but, in this case, the least squares method will be used. The user should therefore open the **fvSchemes** dictionary in the *system* directory and ensure the **leastSquares** method is selected for the **grad(U)** gradient discretisation scheme in the **gradSchemes** sub-dictionary:

```

16
17     d2dt2Schemes
18     {
19         default          steadyState;
20     }
21
22     ddtSchemes
23     {
24         default          Euler;
25     }
26
27     gradSchemes
28     {
29         default          leastSquares;
30     }
31
32     divSchemes
33     {
34         default          none;
35         div(sigmaD)      Gauss linear;
36     }
37
38     laplacianSchemes

```

```

39 {
40     default      Gauss linear corrected;
41 }
42
43 interpolationSchemes
44 {
45     default      linear;
46 }
47
48 snGradSchemes
49 {
50     default      none;
51 }
52
53 // *****

```

The *fvSolution* dictionary in the *system* directory controls the linear equation solvers and algorithms used in the solution. The user should first look at the *solvers* sub-dictionary and notice that the choice of *solver* for D is GAMG. The solver *tolerance* should be set to 10^{-6} for this problem. The solver relative tolerance, denoted by *relTol*, sets the required reduction in the residuals within each iteration. It is uneconomical to set a tight (low) relative tolerance within each iteration since a lot of terms in each equation are explicit and are updated as part of the segregated iterative procedure. Therefore a reasonable value for the relative tolerance is 0.01, or possibly even higher, say 0.1, or in some cases even 0.9 (as in this case).

```

16
17 solvers
18 {
19     "(D|e)"
20     {
21         solver      GAMG;
22         tolerance    1e-06;
23         relTol       0.9;
24         smoother     GaussSeidel;
25         minCellsPerProcessor 20;
26     }
27 }
28
29 PIMPLE
30 {
31     compactNormalStress yes;
32 }
33
34
35 // *****

```

2.3.6 Running the code

The user can now try running *foamRun* in the background of the terminal. When running an OpenFOAM application in the background, the standard output (log information) should be redirected to a file. In the command below, standard output is written to a file named *log* which can be examined afterwards.

```
foamRun > log &
```

The user should check the convergence information by viewing the generated *log* file. It shows the number of iterations and the initial and final residuals of the displacement in each direction being solved. The final residual should always be less than 0.9 times the initial residual as this iteration tolerance set. By the end of the simulation, the initial residuals have reduced towards the convergence tolerance of 10^{-6} .

2.3.7 Post-processing

The `solidDisplacementFoam` solver outputs the stress field σ as a symmetric tensor field `sigma`. To post-process individual scalar field components, σ_{xx} , σ_{xy} etc., the user can run the `foamPostProcess` utility, calling the `components` function object on the `sigma` field using the `-func` option as follows:

```
foamPostProcess -func "components(sigma)"
```

Components named `sigmaxx`, `sigmaxy` etc. are written to time directories of the case. The σ_{xx} stresses can be viewed in ParaView as shown in Figure 2.23.

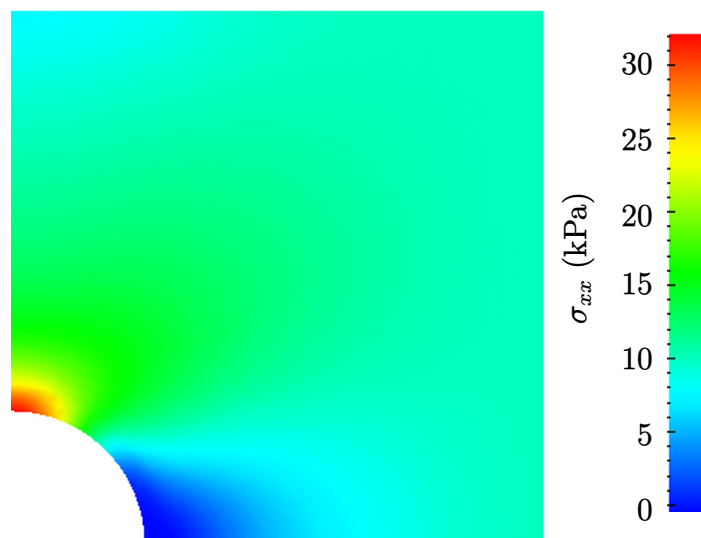


Figure 2.23: σ_{xx} stress field in the plate with hole.

In order to compare the solution to the analytical solution of Equation 2.7, data of σ_{xx} must be extracted along the left edge symmetry plane of our domain. The user may generate the required graph data using the `foamPostProcess` utility with the `graphUniform` function. Unlike earlier examples of `foamPostProcess` where no configuration is required, this example includes a `graphUniform` file pre-configured in the `system` directory. The sample line is set between $(0.0, 0.5, 0.25)$ and $(0.0, 2.0, 0.25)$, and the fields are specified in the `fields` list:

```

9      Writes graph data for specified fields along a line, specified by start and
10     end points. A specified number of graph points are used, distributed
11     uniformly along the line.
12
13     \*-----*/
14
15     start      (0 0.5 0.25);
16     end        (0 2 0.25);
17     nPoints    100;
18
19     fields     (sigmaxx);
20
21     axis       y;
22
23     #includeEtc "caseDicts/functions/graphs/graphUniform.cfg"
24
25     // ***** //
```

The user should execute `postProcessing` with the `graphUniform` function:

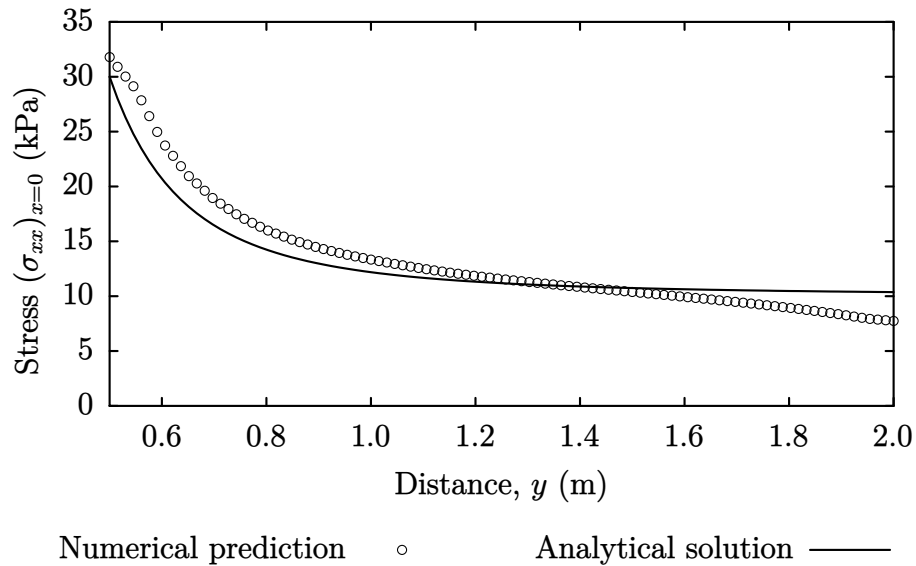


Figure 2.24: Normal stress along the vertical symmetry $(\sigma_{xx})_{x=0}$

```
foamPostProcess -func graphUniform
```

Data is written in raw 2 column format into files within time subdirectories of a *postProcessing/graphUniform* directory, *e.g.* the data at $t = 100$ s is found within the file *graphUniform/100/line.xy*. If the user has **GnuPlot** installed they launch it (by typing **gnuplot**) and then plot both the numerical data and analytical solution as follows:

```
plot [0.5:2] [0:] "postProcessing/graphUniform/100/line.xy",
      1e4*(1+(0.125/(x**2))+(0.09375/(x**4)))
```

An example plot is shown in Figure 2.24.

Chapter 3

Applications and libraries

The examples in Chapter 2 show that OpenFOAM provides a range of software ‘tools’ that are run from a terminal command line. The tools include *applications* which are executable programs written in C++, the base programming language of OpenFOAM. Applications obtain most of the functionality from OpenFOAM’s vast store of pre-compiled *libraries*, also written in C++. Since OpenFOAM is open source software, users have the freedom to create their own applications and libraries. Applications are generally split into two categories:

- *solvers*, e.g. `foamRun`, that perform CFD calculations involving fluid dynamics, energy, etc.;
- *utilities*, e.g. `blockMesh` and `foamPostProcess`, that perform other tasks in CFD like meshing and post-processing.

Prior to version 11 of OpenFOAM, there were many solvers, since separate ones were written for various different types of flow, e.g. `simpleFoam`, `pimpleFoam`, etc. However, most flow solvers are now written as modules, e.g. `incompressibleFluid`, `incompressibleVoF` and e.g. `solid` which are loaded by the general `foamRun` (or `foamMultiRun`) solvers. Rather than existing as an application, each solver module is compiled into a library of its own.

In addition to applications, the tools in OpenFOAM also include *shell scripts*, e.g. `paraFoam`, `foamInfo` and `foamGet`. Many of the scripts help with the configuration of cases.

This chapter gives an overview of applications and libraries, including their creation, modification, compilation and execution.

3.1 The programming language of OpenFOAM

This chapter provides some information to help understand how OpenFOAM applications and libraries are compiled. It provides some background knowledge of C++, the base language of OpenFOAM. Henry Weller chose C++ as the main programming language of OpenFOAM when he created it in the late 1980s.

The idea was to use object-oriented programming to express abstract concepts efficiently, just as verbal language and mathematics can do. For example, in fluid flow, we use the term “velocity field”, which has meaning without any reference to the nature of the flow or any specific velocity data. The term encapsulates the idea of movement with direction and magnitude and relates to other physical properties. In mathematics, “velocity field” can be

replaced by a single symbol, *e.g.* $\mathbf{U}$, and other symbols express operations and functions, *e.g.* “the field of velocity magnitude” by $|\mathbf{U}|$.

CFD deals with partial differential equations in 3 dimensions of space and time. The equations contain: fields of scalars, vectors and tensors; tensor algebra; tensor calculus; and, dimensional units. The solution to these equations involves discretisation procedures, matrices, solvers, and solution algorithms.

Rather than program CFD in terms of intrinsic entities known to a computer, *e.g.* bits, bytes, integers, floating point numbers, OpenFOAM provides *classes* that define the entities encountered in CFD. For example, a velocity field can be defined by a `vectorField` class, allowing a programmer to create an instance, or *object*, of that class. The object can be created with the name `U` to mimic the symbol used in mathematics. Associated functions can be created with names which also try to emulate the simplicity of mathematics, *e.g.* `mag(U)` can represent $|\mathbf{U}|$.

The class structure concentrates code development to contained regions of the code, *i.e.* the classes themselves, thereby making the code easier to manage. New classes can be derived or inherit properties from other classes, *e.g.* the `vectorField` can be derived from a `vector` class and a `Field` class. C++ provides the mechanism of *template classes* such that the template class `Field<Type>` can represent a field of any `<Type>`, *e.g.* `scalar`, `vector`, `tensor`. The general features of the template class are passed on to any class created from the template. Templating and inheritance reduce duplication of code and create class hierarchies that impose an overall structure on the code.

A theme of the OpenFOAM design is that it has a syntax that closely resembles the partial differential equations being solved. For example the equation

$$\frac{\partial \rho \mathbf{U}}{\partial t} + \nabla \cdot \phi \mathbf{U} - \nabla \cdot \mu \nabla \mathbf{U} = -\nabla p$$

is represented by the code

```
solve
(
    fvm::ddt(rho, U)
  + fvm::div(phi, U)
  - fvm::laplacian(mu, U)
  ==
  - fvc::grad(p)
);
```

The equation syntax is most evident in the code for solvers, both the modules and solver applications. Users do not need a deep knowledge of C++ programming to interpret the equations written in a solver. Instead, an understanding of the underlying equations, models and solution method and algorithms is perhaps more helpful, which can be found in *Notes on Computational Fluid Dynamics: General Principles*.

It does help to have a rudimentary understanding of the the principles behind object-orientation and classes, and to have a basic knowledge of some C++ code syntax. To program in OpenFOAM, there is often little need for a user to immerse themselves in the code of any of the OpenFOAM classes. The essence of object-orientation is that the user should

not have to go through the code of each class they use; merely the knowledge of the class' existence and its functionality are sufficient to use the class. A description of each class and its functions is supplied with the OpenFOAM distribution in HTML documentation generated with Doxygen at <https://cpp.openfoam.org>

3.2 Compiling applications and libraries

Compilation is an integral part of code development that requires careful management since every piece of code requires its own set instructions to access dependent components of the OpenFOAM library. In Linux systems there are various tools to help automate the management process, starting with the standard `make` utility. OpenFOAM uses its own `wmake` compilation script that is based on `make`. It is specifically designed for the large number of individual components that are compiled separately in OpenFOAM (approximately 150 applications and 150 libraries).

To understand the compilation process, we first need to explain certain aspects of C++ and its file structure, shown schematically in Figure 3.1. A class is defined through a set of instructions such as object construction, data storage and class member functions. The file that defines these functions — the class *definition* — takes a `.C` extension, *e.g.* a class `nc` would be written in the file `nc.C`. This file can be compiled independently of other code into a binary executable library file known as a shared object library with the `.so` file extension, *i.e.* `nc.so`. When compiling a piece of code, say `newApp.C`, that uses the `nc` class, `nc.C` need not be recompiled, rather `newApp.C` calls the `nc.so` library at runtime. This is known as *dynamic linking*.

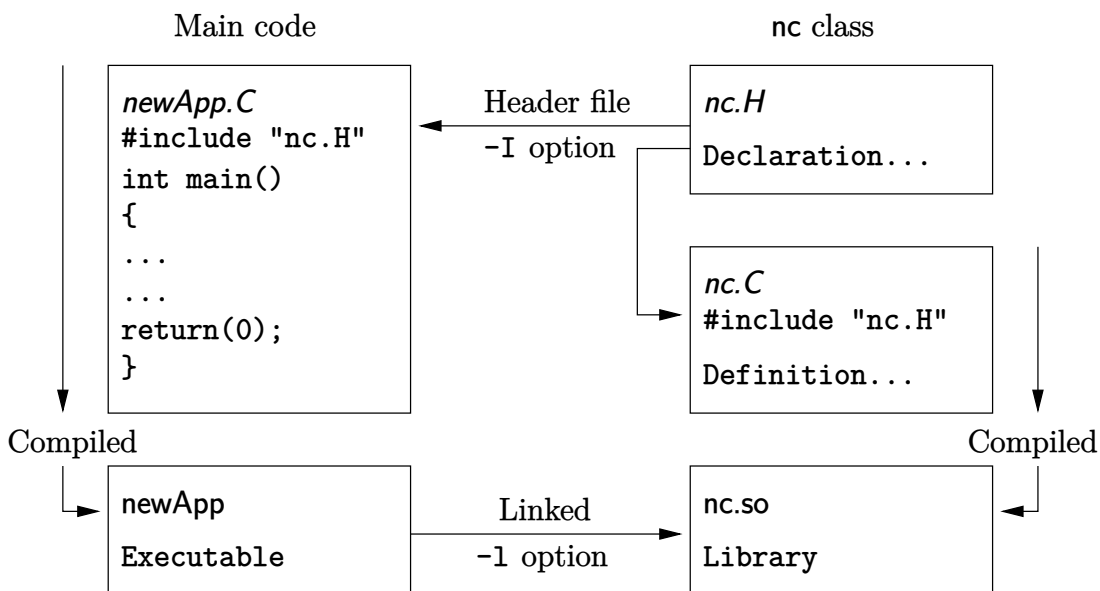


Figure 3.1: Header files, source files, compilation and linking

3.2.1 Header *.H* files

As a means of checking errors, the piece of code being compiled must know that the classes it uses and the operations they perform actually exist. Therefore each class requires a class *declaration*, contained in a header file with a *.H* file extension, *e.g.* *nc.H*, that includes the names of the class and its functions. This file is included at the beginning of any piece of code using the class, using the `#include` directive described below, including the class declaration code itself. Any piece of *.C* code can resource any number of classes and must begin by including all the *.H* files required to declare these classes. Those classes in turn can resource other classes and so also begin by including the relevant *.H* files. By searching recursively down the class hierarchy we can produce a complete list of header files for all the classes on which the top level *.C* code ultimately depends; these *.H* files are known as the *dependencies*. With a dependency list, a compiler can check whether the source files have been updated since their last compilation and selectively compile only those that need to be.

Header files are included in the code using the `#include` directive, *e.g.*

```
#include "otherHeader.H";
```

This causes the compiler to suspend reading from the current file, to read the included file. This mechanism allows any self-contained piece of code to be put into a header file and included at the relevant location in the main code in order to improve code readability. For example, in most OpenFOAM applications the code for creating fields and reading field input data is included in a file *createFields.H* which is called at the beginning of the code. In this way, header files are not solely used as class declarations.

It is `wmake` that performs the task of maintaining file dependency lists amongst other functions listed below.

- Automatic generation and maintenance of file dependency lists, *i.e.* lists of files which are included in the source files and hence on which they depend.
- Multi-platform compilation and linkage, handled through appropriate directory structure.
- Multi-language compilation and linkage, *e.g.* C, C++, Java.
- Multi-option compilation and linkage, *e.g.* debug, optimised, parallel and profiling.
- Support for source code generation programs, *e.g.* lex, yacc, IDL, MOC.
- Simple syntax for source file lists.
- Automatic creation of source file lists for new codes.
- Simple handling of multiple shared or static libraries.

3.2.2 Compiling with wmake

OpenFOAM applications are organised using a standard convention that the source code of each application is placed in a directory whose name is that of the application. The top level source file then takes the application name with the `.C` extension. For example, the source code for an application called `newApp` would reside in a directory `newApp` and the top level file would be `newApp.C` as shown in Figure 3.2. `wmake` then requires the directory

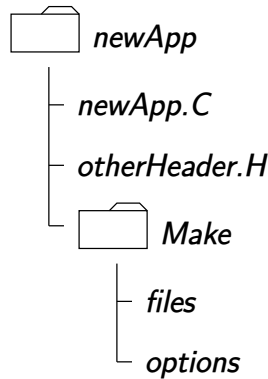


Figure 3.2: Directory structure for an application

must contain a `Make` subdirectory containing 2 files, `options` and `files`, that are described in the following sections.

3.2.3 Including headers

The compiler searches for the included header files in the following order, specified with the `-I` option in `wmake`:

1. the `$WM_PROJECT_DIR/src/OpenFOAM/lnInclude` directory;
2. a local `lnInclude` directory, *i.e.* `newApp/lnInclude`;
3. the local directory, *i.e.* `newApp`;
4. platform dependent paths set in files in the `$WM_PROJECT_DIR/wmake/rules` directory, *e.g.* `/usr/include/X11`;
5. other directories specified explicitly in the `Make/options` file with the `-I` option.

The `Make/options` file contains the full directory paths to locate header files using the syntax:

```

EXE_INC = \
    -I<directoryPath1> \
    -I<directoryPath2> \
    ... \
    -I<directoryPathN>
  
```

Notice first that the directory names are preceded by the `-I` flag and that the syntax uses the `\` to continue the `EXE_INC` across several lines, with no `\` after the final entry.

3.2.4 Linking to libraries

The compiler links to shared object library files in the following directory **paths**, specified with the `-L` option in `wmake`:

1. the `$FOAM_LIBBIN` directory;
2. platform dependent paths set in files in the `$WM_DIR/rules` directory, *e.g.* `$(MPI_ARCH_PATH)/lib`;
3. other directories specified in the `Make/options` file.

The actual library **files** to be linked must be specified using the `-l` option and removing the `lib` prefix and `.so` extension from the library file name, *e.g.* `libnew.so` is included with the flag `-lnew`. By default, `wmake` loads the following libraries:

1. the `libOpenFOAM.so` library from the `$FOAM_LIBBIN` directory;
2. platform dependent libraries specified in set in files in the `$WM_DIR/rules` directory, *e.g.* `libm.so` and `libdl.so`;
3. other libraries specified in the `Make/options` file.

The `Make/options` file contains the full directory paths and library names using the syntax:

```
EXE_LIBS = \
    -L<libraryPath> \
    -l<library1>      \
    -l<library2>      \
    ...               \
    -l<libraryN>
```

To summarise: the directory paths are preceded by the `-L` flag, the library names are preceded by the `-l` flag.

3.2.5 Source files to be compiled

The compiler requires a list of `.C` source files that must be compiled. The list must contain the main `.C` file but also any other source files that are created for the specific application but are not included in a class library. For example, users may create a new class or some new functionality to an existing class for a particular application. The full list of `.C` source files must be included in the `Make/files` file. For many applications the list only includes the name of the main `.C` file, *e.g.* `newApp.C` in the case of our earlier example.

The `Make/files` file also includes a full path and name of the compiled executable, specified by the `EXE =` syntax. Standard convention stipulates the name is that of the application, *i.e.* `newApp` in our example. The OpenFOAM release offers two useful choices for path: standard release applications are stored in `$FOAM_APPBIN`; applications developed by the user are stored in `$FOAM_USER_APPBIN`.

If the user is developing their own applications, we recommend they create an *applications* subdirectory in their `$WM_PROJECT_USER_DIR` directory containing the source code for personal OpenFOAM applications. As with standard applications, the source code for each OpenFOAM application should be stored within its own directory. The only difference between a user application and one from the standard release is that the *Make/files* file should specify that the user's executables are written into their `$FOAM_USER_APPBIN` directory. The *Make/files* file for our example would appear as follows:

```
newApp.C

EXE = $(FOAM_USER_APPBIN)/newApp
```

3.2.6 Running wmake

The `wmake` script is generally executed by typing:

```
wmake <optionalDirectory>
```

The `<optionalDirectory>` is the directory path of the application that is being compiled. Typically, `wmake` is executed from within the directory of the application being compiled, in which case `<optionalDirectory>` can be omitted.

3.2.7 wmake environment variables

For information, the general environment variable settings used by `wmake` are listed below.

- `$WM_PROJECT_INST_DIR`: full path to the installation directory, *e.g.* `$HOME/OpenFOAM`.
- `$WM_PROJECT`: name of the project being compiled, *i.e.* `OpenFOAM`.
- `$WM_PROJECT_VERSION`: version of the project being compiled, *i.e.* `14`.
- `$WM_PROJECT_DIR`: full path to the main directory of the OpenFOAM release, *e.g.* `$HOME/OpenFOAM/OpenFOAM-14`.
- `$WM_PROJECT_USER_DIR`: full path to the equivalent directory for customised developments in the user account, *e.g.* `$HOME/OpenFOAM/${USER}-14`.
- `$WM_THIRD_PARTY`: name of the directory of *ThirdParty* software, *e.g.* `ThirdParty`.
- `$WM_THIRD_PARTY_DIR`: full path to the directory of *ThirdParty* software, *e.g.* `$HOME/OpenFOAM/ThirdParty-14`.

The environment variable settings for the compilation with `wmake` are listed below.

- `$WM_ARCH`: machine architecture, *e.g.* `linux`, `linux64`, `linuxArm64`, `linuxARM7`, `linuxPPC64`, `linuxPPC64le`.

- `$WM_ARCH_OPTION`: 32 or 64 bit architecture.
- `$WM_DIR`: full path of the *wmake* directory.
- `$WM_LABEL_SIZE`: 32 or 64 bit size for labels (integers).
- `$WM_LABEL_OPTION`: `Int32` or `Int64` compilation of labels.
- `$WM_LINK_LANGUAGE`: compiler used to link libraries and executables `c++`.
- `$WM_MPLIB`: parallel communications library, `SYSTEMOPENMPI` = system version of openMPI, alternatives include `OPENMPI`, `SYSTEMMPI`, `MPICH`, `MPICH-GM`, `HPMPI`, `MPI`, `QSMPI`, `INTELMPI` and `SGIMPI`.
- `$WM_OPTIONS`, *e.g.* `linux64GccDPInt32Opt`, formed by combining `$WM_ARCH`, `$WM_COMPILER`, `$WM_PRECISION_OPTION`, `$WM_LABEL_OPTION`, and `$WM_COMPILE_OPTION`.
- `$WM_PRECISION_OPTION`: floating point precision of the compiled binaries, `SP` = single precision, `DP` = double precision.

The environment variable settings relating to the choice of compiler and options with *wmake* are listed below.

- `$WM_CC`: choice of C compiler, `gcc`.
- `$WM_CFLAGS`: extra flags to the C compiler, *e.g.* `-m64 -fPIC`.
- `$WM_CXX`: choice of C++ compiler, `g++`.
- `$WM_CXXFLAGS`: extra flags to the C++ compiler, *e.g.* `-m64 -fPIC -std=c++0x`.
- `$WM_COMPILER`: compiler being used, *e.g.* `Gcc` = `gcc`, `Clang` = LLVM Clang
- `$WM_COMPILE_OPTION`: compilation option, `Debug` = debugging, `Opt` = optimised.
- `$WM_COMPILER_TYPE`: choice of compiler, `system`, or `ThirdParty`, *i.e.* compiled in *ThirdParty* directory.
- `$WM_LDFLAGS`: extra flags for the linker, *e.g.* `-m64`.
- `$WM_LINK_LANGUAGE`: linker language, *e.g.* `c++`.
- `$WM_OSTYPE`: Operating system, `POSIX`.

3.2.8 Removing dependency lists: wclean

When it is run, `wmake` builds a dependency list file with a `.dep` file extension, *e.g.* `newApp.C.dep` in our example, in a `$WM_OPTIONS` sub-directory of the `Make` directory, *e.g.* `Make/linux-GccDPlnt64Opt`. If the user wishes to remove these files, *e.g.* after making code changes, the user can run the `wclean` script by typing:

```
wclean <optionalDirectory>
```

Again, the `<optionalDirectory>` is a path to the directory of the application that is being compiled. Typically, `wclean` is executed from within the directory of the application, in which case the path can be omitted.

3.2.9 Compiling libraries

When compiling a library, there are 2 critical differences in the configuration of the file in the `Make` directory:

- in the `files` file, `EXE =` is replaced by `LIB =` and the target directory for the compiled entity changes from `$FOAM_APPBIN` to `$FOAM_LIBBIN` (and an equivalent `$FOAM_USER_LIBBIN` directory);
- in the `options` file, `EXE_LIBS =` is replaced by `LIB_LIBS =` to indicate libraries linked to library being compiled.

When `wmake` is executed it additionally creates a directory named `InlInclude` that contains soft links to all the files in the library. The `InlInclude` directory is deleted by the `wclean` script when cleaning library source code.

3.2.10 Compilation example: the foamRun application

The source code for application `foamRun` is in the `$FOAM_SOLVERS/foamRun` directory and the top level source file is named `foamRun.C`. The `foamRun.C` source code is:

```

1  /*-----*\
2  \=====|
3  \      | F ield      | OpenFOAM: The Open Source CFD Toolbox
4  \      | O peration  | Website:  https://openfoam.org
5  \      | A nd        | Copyright (C) 2022-2025 OpenFOAM Foundation
6  \      | M anipulation|
7  -----*/
8  License
9      This file is part of OpenFOAM.
10
11      OpenFOAM is free software: you can redistribute it and/or modify it
12      under the terms of the GNU General Public License as published by
13      the Free Software Foundation, either version 3 of the License, or
14      (at your option) any later version.
15
16      OpenFOAM is distributed in the hope that it will be useful, but WITHOUT
17      ANY WARRANTY; without even the implied warranty of MERCHANTABILITY or
18      FITNESS FOR A PARTICULAR PURPOSE. See the GNU General Public License
19      for more details.
20
21      You should have received a copy of the GNU General Public License
22      along with OpenFOAM. If not, see <http://www.gnu.org/licenses/>.
23

```

```

24 Application
25     foamRun
26
27 Description
28     Loads and executes an OpenFOAM solver module either specified by the
29     optional \c solver entry in the \c controlDict or as a command-line
30     argument.
31
32     Uses the flexible PIMPLE (PISO-SIMPLE) solution for time-resolved and
33     pseudo-transient and steady simulations.
34
35 Usage
36     \b foamRun [OPTION]
37
38     - \par -solver <name>
39       Solver name
40
41     - \par -libs '("\lib1.so\" ... "\libN.so\"')
42       Specify the additional libraries loaded
43
44 Example usage:
45     - To run a \c rhoPimpleFoam case by specifying the solver on the
46       command line:
47       \verbatim
48         foamRun -solver fluid
49       \endverbatim
50
51     - To update and run a \c rhoPimpleFoam case add the following entry to
52       the controlDict:
53       \verbatim
54         solver          fluid;
55       \endverbatim
56       then execute \c foamRun
57
58 \*-----*/
59
60 #include "argList.H"
61 #include "solver.H"
62 #include "pimpleSingleRegionControl.H"
63 #include "setDeltaT.H"
64
65 using namespace Foam;
66
67 // * * * * *
68
69 int main(int argc, char *argv[])
70 {
71     argList::addOption
72     (
73         "solver",
74         "name",
75         "Solver name"
76     );
77
78     #include "setRootCase.H"
79     #include "createTime.H"
80
81     // Read the solverName from the optional solver entry in controlDict
82     word solverName
83     (
84         runTime.controlDict().lookupOrDefault("solver", word::null)
85     );
86
87     // Optionally reset the solver name from the -solver command-line argument
88     args.optionReadIfPresent("solver", solverName);
89
90     // Check the solverName has been set
91     if (solverName == word::null)
92     {
93         args.printUsage();
94
95         FatalErrorIn(args.executable())
96             << "solver not specified in the controlDict or on the command-line"
97             << exit(FatalError);
98     }
99     else
100     {

```

```

101         // Load the solver library
102         solver::load(solverName);
103     }
104
105     // Create the default single region mesh
106     #include "createMesh.H"
107
108     // Instantiate the selected solver
109     autoPtr<solver> solverPtr(solver::New(solverName, mesh));
110     solver& solver = solverPtr();
111
112     // Create the outer PIMPLE loop and control structure
113     pimpleSingleRegionControl pimple(solver.pimple);
114
115     // Set the initial time-step
116     setDeltaT(runTime, solver);
117
118     // * * * * *
119
120     Info<< nl << "Starting time loop\n" << endl;
121
122     while (pimple.run(runTime))
123     {
124         solver.preSolve();
125
126         // Adjust the time-step according to the solver maxDeltaT
127         adjustDeltaT(runTime, solver);
128
129         runTime++;
130
131         Info<< "Time = " << runTime.userTimeName() << nl << endl;
132
133         // PIMPLE corrector loop
134         while (pimple.loop())
135         {
136             if (solver.pimple.flow())
137             {
138                 solver.moveMesh();
139                 solver.motionCorrector();
140             }
141
142             if (solver.pimple.models())
143             {
144                 solver.fvModels().correct();
145             }
146
147             solver.prePredictor();
148
149             if (solver.pimple.predictTransport())
150             {
151                 if (solver.pimple.flow())
152                 {
153                     solver.momentumTransportPredictor();
154                 }
155
156                 if (solver.pimple.thermophysics())
157                 {
158                     solver.thermophysicalTransportPredictor();
159                 }
160             }
161
162             if (solver.pimple.flow())
163             {
164                 solver.momentumPredictor();
165             }
166
167             if (solver.pimple.thermophysics())
168             {
169                 solver.thermophysicalPredictor();
170             }
171
172             if (solver.pimple.flow())
173             {
174                 solver.pressureCorrector();

```

```

175         }
176
177         if (solver.pimple.correctTransport())
178         {
179             if (solver.pimple.flow())
180             {
181                 solver.momentumTransportCorrector();
182             }
183
184             if (solver.pimple.thermophysics())
185             {
186                 solver.thermophysicalTransportCorrector();
187             }
188         }
189     }
190
191     solver.postSolve();
192
193     runTime.write();
194
195     Info<< "ExecutionTime = " << runTime.elapsedCpuTime() << " s"
196           << " ClockTime = " << runTime.elapsedClockTime() << " s"
197           << nl << endl;
198 }
199
200 Info<< "End\n" << endl;
201
202 return 0;
203 }
204
205
206 // *****

```

The code begins with a description of the application contained within comments over 1 line (//) and multiple lines (/*...*/). Following that, the code contains several `# include` statements, *e.g.* `# include "argList.H"`, which causes the compiler to suspend reading from the current file, *foamRun.C* to read the *argList.H* file.

foamRun uses the finite volume numerics library and therefore requires the necessary header files, specified by the `EXE_INC = -I...` option, and links to the libraries with the `EXE_LIBS = -l...` option. The *Make/options* therefore contains the following:

```

1  EXE_INC = \
2      -I$(LIB_SRC)/finiteVolume/lnInclude
3
4  EXE_LIBS = \
5      -lfiniteVolume

```

foamRun contains the *foamRun.C* source and the executable is written to the `$FOAM_APPBIN` directory. The application uses functions to initialise and adjust the time step, defined in the *setDeltaT.C* file. The *Make/files* therefore contains:

```

1  setDeltaT.C
2  foamRun.C
3
4  EXE = $(FOAM_APPBIN)/foamRun

```

Following the recommendations of section 3.2.5, the user can compile a separate version of *foamRun* into their local `$FOAM_USER_DIR` directory as follows. First, the user should copy the *foamRun* source code to a local directory, *e.g.* `$FOAM_RUN`.

```

cd $FOAM_RUN
cp -r $FOAM_SOLVERS/foamRun .

```

They should then go into the *foamRun* directory.

```
cd foamRun
```

and edit the *Make/files* file as follows:

```
1  foamRun.C
2
3  EXE = $(FOAM_USER_APPBIN)/foamRun
```

Finally, they should run the `wmake` script.

```
wmake
```

The code should compile and produce a message similar to the following

```
Making dependency list for source file foamRun.C
g++ -std=c++14 -m64...
...
-o ... platforms/linux64GccDPInt32Opt/bin/foamRun
```

If the user tries recompiling without making any changes to the code file, nothing will happen. The user can compile the application from scratch by removing the dependency list with

```
wclean
```

and running `wmake`.

3.2.11 Dynamic linking at run-time

The situation may arise that a user creates a new library, say `new1`, and wishes the features within that library to be available across a range of applications. For example, the user may create a new boundary condition, compiled into `new1`, that would need to be recognised by a range of solver applications, pre- and post-processing utilities, mesh tools, *etc.* Under normal circumstances, the user would need to recompile every application with the `new1` linked to it.

Instead there is a simple mechanism to link one or more shared object libraries dynamically at run-time in OpenFOAM. The user can simply add the optional keyword entry `libs` to the *controlDict* file for a case and enter the full names of the libraries within a list (as quoted string entries). For example, if a user wished to link the libraries `new1` and `new2` at run-time, they would simply need to add the following to the case *controlDict* file:

```
libs
(
    "libnew1.so"
    "libnew2.so"
);
```

3.3 Running applications

Each application is designed to be executed from a terminal command line, typically reading and writing a set of data files associated with a particular case. The data files for a case are stored in a directory named after the case as described in section 4.1; the directory name with full path is here given the generic name *<caseDir>*.

For any application, the form of the command line entry for any can be found by simply entering the application name at the command line with the `-help` option, *e.g.* typing

```
foamRun -help
```

returns the usage

```
Usage: foamRun [OPTIONS]
options:
  -case <dir>           specify alternate case directory, default is cwd
  -fileHandler <handler>
                        override the fileHandler
  -hostRoots <((host1 dir1) .. (hostN dirN))>
                        slave root directories for distributed running
  -libs '("lib1.so" ... "libN.so")'
                        pre-load libraries
  -noFunctionObjects    do not execute functionObjects
  -parallel            run in parallel
  -roots <(dir1 .. dirN)>
                        slave root directories for distributed running
  -solver <name>       Solver name
  -srcDoc              display source code in browser
  -doc                display application documentation in browser
  -help               print the usage
```

If the application is executed from within a case directory, it will operate on that case. Alternatively, the `-case <caseDir>` option allows the case to be specified directly so that the application can be executed from anywhere in the filing system.

Like any UNIX/Linux executable, applications can be run as a background process, *i.e.* one which does not have to be completed before the user can give the shell additional commands. If the user wished to run the `foamRun` example as a background process and output the case progress to a *log* file, they could enter:

```
foamRun > log &
```

3.4 Running applications in parallel

This section describes how to run OpenFOAM in parallel on distributed processors. The method of parallel computing used by OpenFOAM is known as domain decomposition, in

which the geometry and associated fields are broken into pieces and allocated to separate processors for solution. The process of parallel computation involves: decomposition of mesh and fields; running the application in parallel; and, post-processing the decomposed case as described in the following sections. The parallel running uses the public domain **openMPI** implementation of the standard message passing interface (MPI) by default, although other libraries can be used.

3.4.1 Decomposition of mesh and initial field data

The mesh and fields are decomposed using the `decomposePar` utility. The underlying aim is to break up the domain with minimal effort but in such a way to guarantee an economic solution. The geometry and fields are broken up according to a set of parameters specified in a dictionary named `decomposeParDict` that must be located in the `system` directory of the case of interest. An example `decomposeParDict` dictionary can be copied into a case `system` directory using the `foamGet` script.

```
foamGet decomposeParDict
```

The dictionary entries within it are reproduced below.

```

16  numberOfSubdomains 8;
17
18  /*
19  Main methods are:
20  1) Geometric: "simple"; "hierarchical", with ordered sorting, e.g. xyz, yxz
21  2) Scotch: "scotch", when running in serial; "ptscotch", running in parallel
22  */
23
24  method                hierarchical;
25
26  simple
27  {
28      n                  (4 2 1); // total must match numberOfSubdomains
29  }
30
31  hierarchical
32  {
33      n                  (4 2 1); // total must match numberOfSubdomains
34      order              xyz;
35  }
36
37
38
39  // ***** //
```

The user has a choice of four methods of decomposition, specified by the `method` keyword as described below.

simple Simple geometric decomposition in which the domain is split into pieces by direction, *e.g.* 2 pieces in the *x* direction, 1 in *y* *etc.*

hierarchical Hierarchical geometric decomposition which is the same as **simple** except the user specifies the order in which the directional split is done, *e.g.* first in the *y*-direction, then the *x*-direction *etc.*

scotch Scotch decomposition which requires no geometric input from the user and attempts to minimise the number of processor boundaries. The user can specify a weighting for

the decomposition between processors, through an optional **processorWeights** keyword which can be useful on machines with differing performance between processors. There is also an optional keyword entry **strategy** that controls the decomposition strategy through a complex string supplied to Scotch. For more information, see the source code file: *\$FOAM_SRC/parallel/decompose/scotchDecomp/scotchDecomp.C*

For each **method** there are a set of coefficients specified in a sub-dictionary of *decompositionDict*, of the same name as the chosen method, *e.g.* for the **hierarchical** method, the settings are in a sub-dictionary named **hierarchical**. The full set of keyword entries in the *decomposeParDict* dictionary are explained below:

- **numberOfSubdomains**: total number of subdomains N .
- **method**: method of decomposition, **simple**, **hierarchical**, **scotch**.
- **n**: for **simple** and **hierarchical**, number of subdomains in x, y, z ($n_x n_y n_z$)
- **order**: order of **hierarchical** decomposition, **xyz**/**xzy**/**yxz**...
- **processorWeights** option for **scotch**: list of weighting factors (**<wt1>**...**<wtN>**) for allocation of cells to processors; **<wt1>** is the weighting factor for processor 1, *etc.*; weights are normalised so can take any range of values.

The **decomposePar** utility is executed in the normal manner by typing

```
decomposePar
```

3.4.2 File input/output in parallel

Using standard file input/output completion, a set of subdirectories will have been created, one for each processor, in the case directory. The directories are named *processorN* where $N = 0, 1, \dots$ represents a processor number and contains a time directory, containing the decomposed field descriptions, and a *constant/polyMesh* directory containing the decomposed mesh description.

While this file structure is well-organised, for large parallel cases, it generates a large number of files. In very large simulations, users can experience problems including hitting limits on the number of open files imposed by the operating system.

As an alternative, the **collated** file format was introduced in OpenFOAM in which the data for each decomposed field (and mesh) is collated into a single file that is written (and read) on the master processor. The files are stored in a single directory named **processors**.

The file writing can be threaded allowing the simulation to continue running while the data is being written to file — see below for details. NFS (Network File System) is not needed when using the collated format and, additionally, there is a **masterUncollated** option to write data with the original **uncollated** format without NFS.

The controls for the file handling are in the **OptimisationSwitches** of the global *etc/-controlDict* file:


```

OptimisationSwitches
{
    ...

    //- Parallel IO file handler
    // uncollated (default), collated or masterUncollated
    fileHandler uncollated;

    //- collated: thread buffer size for queued file writes.
    // If set to 0 or not sufficient for the file size threading is not used.
    // Default: 2e9
    maxThreadFileBufferSize 2e9;

    //- masterUncollated: non-blocking buffer size.
    // If the file exceeds this buffer size scheduled transfer is used.
    // Default: 2e9
    maxMasterFileBufferSize 2e9;
}

```

The `fileHandler` can be set for a specific simulation by:

- over-riding the global `OptimisationSwitches {fileHandler ...;}` in the case *controlDict* file;
- using the `-fileHandler` command line argument to the solver;
- setting the `$FOAM_FILEHANDLER` environment variable.

A `foamFormatConvert` utility allows users to convert files between the collated and uncollated formats, e.g.

```
mpirun -np 2 foamFormatConvert -parallel -fileHandler uncollated
```

An example case demonstrating the file handling methods is provided in:

`$FOAM_TUTORIALS/fluid/iglooWithFridges`

Collated file handling runs faster with threading, especially on large cases. But it requires threading support to be enabled in the underlying MPI. Without it, the simulation will “hang” or crash. For `openMPI`, threading support is not set by default prior to version 2, but is generally switched on from version 2 onwards. The user can check whether `openMPI` is compiled with threading support by the following command:

```
mpi_info -c | grep -oE "MPI_THREAD_MULTIPLE[^\,]*"
```

When using the collated file handling, memory is allocated for the data in the thread. `maxThreadFileBufferSize` sets the maximum size of memory that is allocated in bytes. If the data exceeds this size, the write does not use threading.

Note: if threading is **not enabled** in the MPI, it must be disabled for collated file handling by setting in the global *etc/controlDict* file:

```
maxThreadFileBufferSize 0;
```

When using the `masterUncollated` file handling, non-blocking MPI communication requires a sufficiently large memory buffer on the master node. `maxMasterFileBufferSize` sets the maximum size of the buffer. If the data exceeds this size, the system uses scheduled communication.

3.4.3 Running a decomposed case

A decomposed OpenFOAM case is run in parallel using the openMPI implementation of MPI. openMPI can be run on a local multiprocessor machine very simply but when running on machines across a network, a file must be created that contains the host names of the machines. The file can be given any name and located at any path. In the following description we shall refer to such a file by the generic name, including full path, *<machines>*.

The *<machines>* file contains the names of the machines listed one machine per line. The names must correspond to a fully resolved hostname in the */etc/hosts* file of the machine on which the openMPI is run. The list must contain the name of the machine running the openMPI. Where a machine node contains more than one processor, the node name may be followed by the entry *cpu=*n** where *n* is the number of processors openMPI should run on that node.

For example, let us imagine a user wishes to run openMPI from machine *aaa* on the following machines: *aaa*; *bbb*, which has 2 processors; and *ccc*. The *<machines>* would contain:

```
aaa
bbb cpu=2
ccc
```

An application is run in parallel using mpirun.

```
mpirun --hostfile <machines> -np <nProcs>
      <foamExec> <otherArgs> -parallel > log &
```

where: *<nProcs>* is the number of processors; *<foamExec>* is the executable, *e.g.* *foamRun*; and, the output is redirected to a file named *log*. For example, if *foamRun* is run on 4 nodes, specified in a file named *machines*, then the following command should be executed:

```
mpirun --hostfile machines -np 4 foamRun -parallel > log &
```

3.4.4 Distributing data across several disks

Data files may need to be distributed if, for example, if only local disks are used in order to improve performance. In this case, the user may find that the root path to the case directory may differ between machines. The paths must then be specified in the *decomposeParDict* dictionary using *distributed* and *roots* keywords. The *distributed* entry should read

```
distributed yes;
```

and the *roots* entry is a list of root paths, *<root0>*, *<root1>*, ..., for each node

```
roots
<nRoots>
(
```

```

    "<root0>"
    "<root1>"
    ...
);

```

where `<nRoots>` is the number of roots.

Each of the *processorN* directories should be placed in the case directory at each of the root paths specified in the *decomposeParDict* dictionary. The *system* directory and *files* within the *constant* directory must also be present in each case directory. Note: the files in the *constant* directory are needed, but the *polyMesh* directory is not.

3.4.5 Post-processing parallel processed cases

When post-processing cases that have been run in parallel the user has two options:

- reconstruction of the mesh and field data to recreate the complete domain and fields, which can be post-processed as normal;
- post-processing each segment of decomposed domain individually.

After a case has been run in parallel, it can be reconstructed for post-processing. The case is reconstructed by merging the sets of time directories from each *processorN* directory into a single set of time directories. The *reconstructPar* utility performs such a reconstruction by executing the command:

```
reconstructPar
```

The user may post-process decomposed cases using the *paraFoam* post-processor, described in section 7.1. The whole simulation can be post-processed by reconstructing the case or alternatively it is possible to post-process a segment of the decomposed domain individually by simply treating the individual processor directory as a case in its own right.

3.5 Solver modules

From OpenFOAM version 11, application solvers, *e.g.* *simpleFoam* have been largely replaced by the generic *foamRun* solver which loads a solver module, *e.g.* *incompressibleFluid* that defines the flow solution. Solver modules are located in the *\$FOAM_MODULES* directory. The current solver modules distributed with OpenFOAM are listed below.

3.5.1 Single-phase modules

fluid Solver module for steady or transient turbulent flow of compressible fluids with heat-transfer for HVAC and similar applications, with optional mesh motion and change.

incompressibleDenseParticleFluid Solver module for transient flow of incompressible isothermal fluids coupled with particle clouds including the effect of the volume fraction of particles on the continuous phase, with optional mesh motion and change.

incompressibleFluid Solver module for steady or transient turbulent flow of incompressible isothermal fluids with optional mesh motion and change.

multicomponentFluid Solver module for steady or transient turbulent flow of compressible multicomponent fluids with optional mesh motion and change.

shockFluid Solver module for density-based solution of compressible flow

XiFluid Solver module for compressible premixed/partially-premixed combustion with turbulence modelling.

3.5.2 Multiphase/VoF flow modules

compressibleMultiphaseVoF Solver module for the solution of multiple compressible, isothermal immiscible fluids using a VOF (volume of fluid) phase-fraction based interface capturing approach, with optional mesh motion and mesh topology changes including adaptive re-meshing.

compressibleVoF Solver module for for 2 compressible, non-isothermal immiscible fluids using a VOF (volume of fluid) phase-fraction based interface capturing approach, with optional mesh motion and mesh topology changes including adaptive re-meshing.

incompressibleDriftFlux Solver module for 2 incompressible fluids using the mixture approach with the drift-flux approximation for relative motion of the phases, with optional mesh motion and mesh topology changes including adaptive re-meshing.

incompressibleMultiphaseVoF Solver module for the solution of multiple incompressible, isothermal immiscible fluids using a VOF (volume of fluid) phase-fraction based interface capturing approach, with optional mesh motion and mesh topology changes including adaptive re-meshing.

incompressibleVoF Solver module for for 2 incompressible, isothermal immiscible fluids using a VOF (volume of fluid) phase-fraction based interface capturing approach, with optional mesh motion and mesh topology changes including adaptive re-meshing.

isothermalFluid Solver module for steady or transient turbulent flow of compressible isothermal fluids with optional mesh motion and change.

multiphaseEuler Solver module for a system of any number of compressible fluid phases with a common pressure, but otherwise separate properties. The type of phase model is run time selectable and can optionally represent multiple species and in-phase reactions. The phase system is also run time selectable and can optionally represent different types of momentum, heat and mass transfer.

3.5.3 Solid modules

solid Solver module for thermal transport in solid domains and regions for conjugate heat transfer, HVAC and similar applications, with optional mesh motion and mesh topology changes.

solidDisplacement Solver module for steady or transient segregated finite-volume solution of linear-elastic, small-strain deformation of a solid body, with optional thermal diffusion and thermal stresses.

3.5.4 Film modules

isothermalFilm Solver module for flow of compressible isothermal liquid films

film Solver module for flow of compressible liquid films

3.5.5 Utility modules

functions Solver module to execute the **functionObjects** for a specified

movingMesh Solver module to move the mesh.

3.5.6 Base classes for solver modules

fluidSolver Base solver module for fluid solvers.

twoPhaseSolver Solver module base-class for for 2 immiscible fluids, with optional mesh motion and mesh topology changes including adaptive re-meshing.

twoPhaseVoFSolver Solver module base-class for for 2 immiscible fluids using a VOF (volume of fluid) phase-fraction based interface capturing approach, with optional mesh motion and mesh topology changes including adaptive re-meshing.

VoFSolver Base solver module base-class for the solution of immiscible fluids using a VOF (volume of fluid) phase-fraction based interface capturing approach, with optional mesh motion and mesh topology changes including adaptive re-meshing.

multiphaseVoFSolver Base solver module for the solution of multiple immiscible fluids using a VOF (volume of fluid) phase-fraction based interface capturing approach, with optional mesh motion and mesh topology changes including adaptive re-meshing.

3.6 Standard solvers

With the introduction of solver modules in OpenFOAM v11, the number of solver applications has much reduced. The applications which are relevant, including **foamRun** and **foamMultiRun**, are located in the **\$FOAM_SOLVERS** directory, reached quickly by typing **sol** at the command line. These solver applications are listed in the following section.

There are also some legacy solver applications, which either have not been replaced yet by new solver modules or are included for teaching purposes. They are provided in the **\$FOAM_APP/legacy** directory are listed in the subsequent section below.

3.6.1 Main solver applications

foamRun Loads and executes an OpenFOAM solver module either specified by the optional **solver** entry in the **controlDict** or as a command-line argument.

foamMultiRun Loads and executes an OpenFOAM solver modules for each region of a multi-region simulation e.g. for conjugate heat transfer.

boundaryFoam Steady-state solver for incompressible, 1D turbulent flow, typically to generate boundary layer conditions at an inlet, for use in a simulation.

chemFoam Solver for chemistry problems, designed for use on single cell cases to provide comparison against other chemistry solvers, that uses a single cell mesh, and fields created from the initial conditions.

potentialFoam Potential flow solver which solves for the velocity potential, to calculate the flux-field, from which the velocity field is obtained by reconstructing the flux.

3.6.2 Legacy solver applications

electrostaticFoam Solver for electrostatics.

magneticFoam Solver for the magnetic field generated by permanent magnets.

mhdFoam Solver for magnetohydrodynamics (MHD): incompressible, laminar flow of a conducting fluid under the influence of a magnetic field.

laplacianFoam Solves a simple Laplace equation, e.g. for thermal diffusion in a solid.

financialFoam Solves the Black-Scholes equation to price commodities.

dsmcFoam Direct simulation Monte Carlo (DSMC) solver for, transient, multi-species flows.

mdEquilibrationFoam Solver to equilibrate and/or precondition molecular dynamics systems.

mdFoam Molecular dynamics solver for fluid dynamics.

adjointShapeOptimizationFoam Steady-state solver for incompressible, turbulent flow of non-Newtonian fluids with optimisation of duct shape by applying "blockage" in regions causing pressure loss as estimated using an adjoint formulation.

icoFoam Transient solver for incompressible, laminar flow of Newtonian fluids.

shallowWaterFoam Transient solver for inviscid shallow-water equations with rotation.

porousSimpleFoam Steady-state solver for incompressible, turbulent flow with implicit or explicit porosity treatment and support for multiple reference frames (MRF).

rhoPorousSimpleFoam Steady-state solver for turbulent flow of compressible fluids, with implicit or explicit porosity treatment and optional sources.

PDRFoam Solver for compressible premixed/partially-premixed combustion with turbulence modelling.

3.7 Standard utilities

The utilities with the OpenFOAM distribution are in the `$FOAM_UTILITIES` directory. The names are reasonably descriptive, *e.g.* `ideasToFoam` converts mesh data from the format written by I-DEAS to the OpenFOAM format. The descriptions of current utilities distributed with OpenFOAM are given in the following sections.

3.7.1 Pre-processing

applyBoundaryLayer Apply a simplified boundary-layer model to the velocity and turbulence fields based on the 1/7th power-law.

boxTurb Makes a box of turbulence which conforms to a given energy spectrum and is divergence free.

createExternalCoupledPatchGeometry Application to generate the patch geometry (points and faces) for use with the `externalCoupled` boundary condition.

dsmcInitialise Initialise a case for `dsmcFoam` by reading the initialisation dictionary system/`dsmcInitialise`.

engineSwirl Generates a swirling flow for engine calculations.

faceAgglomerate Agglomerate boundary faces using the `pairPatchAgglomeration` algorithm. It writes a map from the fine to coarse grid.

foamSetupCHT Sets up a multi-region case using template files for material properties, field and system files.

mapFields Maps volume fields from one mesh to another, reading and interpolating all fields present in the time directory of both cases.

mapFieldsPar Maps volume fields from one mesh to another, reading and interpolating all fields present in the time directory of both cases. Parallel and non-parallel cases are handled without the need to reconstruct them first.

mdInitialise Initialises fields for a molecular dynamics (MD) simulation.

setAtmBoundaryLayer Applies atmospheric boundary layer models to the entire domain for case initialisation.

setFields Set values on a selected set of cells/patchfaces through a dictionary.

setWaves Applies wave models to the entire domain for case initialisation using level sets for second-order accuracy.

snappyHexMeshConfig Preconfigures `blockMeshDict`, `surfaceFeaturesDict` and `snappyHexMeshDict` files based on the case surface geometry files.

viewFactorsGen View factors are calculated based on a face agglomeration array (`finalAgglom` generated by `faceAgglomerate` utility).

3.7.2 Mesh generation

blockMesh A multi-block mesh generator.

extrudeMesh Extrude mesh from existing patch (by default outwards facing normals; optional flips faces) or from patch read from file.

extrude2DMesh Takes 2D mesh (all faces 2 points only, no front and back faces) and creates a 3D mesh by extruding with specified thickness.

extrudeToRegionMesh Extrude **faceZones** (internal or boundary faces) or **faceSets** (boundary faces only) into a separate mesh (as a different region).

snappyHexMesh Automatic split hex mesher. Refines and snaps to surface.

zeroDimensionalMesh Creates a zero-dimensional mesh.

3.7.3 Mesh conversion

ansysToFoam Converts an ANSYS input mesh file, exported from I-DEAS, to OpenFOAM format.

ccm26ToFoam Reads CCM files as written by Prostar/ccm using ccm 2.6

cfx4ToFoam Converts a CFX 4 mesh to OpenFOAM format.

datToFoam Reads in a **datToFoam** mesh file and outputs a points file. Used in conjunction with **blockMesh**.

fluent3DMeshToFoam Converts a Fluent mesh to OpenFOAM format.

fluentMeshToFoam Converts a Fluent mesh to OpenFOAM format including multiple region and region boundary handling.

foamMeshToFluent Writes out the OpenFOAM mesh in Fluent mesh format.

foamToStarMesh Reads an OpenFOAM mesh and writes a pro-STAR (v4) bnd/cel/vrt format.

foamToSurface Reads an OpenFOAM mesh and writes the boundaries in a surface format.

gambitToFoam Converts a GAMBIT mesh to OpenFOAM format.

gmshToFoam Reads .msh file as written by Gmsh.

ideasUnvToFoam I-Deas unv format mesh conversion.

kivaToFoam Converts a KIVA3v grid to OpenFOAM format.

mshToFoam Converts .msh file generated by the Adventure system.

netgenNeutralToFoam Converts neutral file format as written by Netgen v4.4.

plot3dToFoam Plot3d mesh (ascii/formatted format) converter.

sammToFoam Converts a Star-CD (v3) SAMM mesh to OpenFOAM format.

star3ToFoam Converts a Star-CD (v3) pro-STAR mesh into OpenFOAM format.

star4ToFoam Converts a Star-CD (v4) pro-STAR mesh into OpenFOAM format.

tetgenToFoam Converts .ele and .node and .face files, written by tetgen.

vtkUnstructuredToFoam Converts ascii .vtk (legacy format) file generated by vtk/paraview.

writeMeshObj For mesh debugging, writes mesh as three separate OBJ files for visualisation.

3.7.4 Mesh manipulation

autoPatch Divides external faces into patches based on (user supplied) feature angle.

checkMesh Checks validity of a mesh.

createBaffles Makes internal faces into boundary faces. Does not duplicate points.

createNonConformalCouples Utility to create non-conformal couples between non-coupled patches.

createPatch Utility to create patches out of selected boundary faces. Faces come either from existing patches or from a **faceSet**.

createZones Utility to generate zones by executing configured **zoneGenerators**.

deformedGeom Deforms a **polyMesh** using a displacement field **U** and a scaling factor supplied as an argument.

flattenMesh Flattens the front and back planes of a 2D cartesian mesh.

insideCells Picks up cells with cell centre 'inside' of surface. Requires surface to be closed and singly connected.

mergeBaffles Detects faces that share points (baffles) and merges them into internal faces.

mergeMeshes Merges two meshes.

mirrorMesh Mirrors a mesh around a given plane.

objToVTK Read obj line (not surface!) file and convert into vtk.

polyDualMesh Calculates the dual of a **polyMesh**. Adheres to all the feature and patch edges.

refineMesh Utility to refine cells in multiple directions.

renumberMesh Renumbers the cell list in order to reduce the bandwidth, reading and renumbering all fields from all the time directories.

reorderPatches Utility to reorder the patches of a case

singleCellMesh Reads all fields and maps them to a mesh with all internal faces removed (**singleCellFvMesh**) which gets written to region **singleCell**.

splitBaffles Detects faces that share points (baffles) and duplicates the points to separate them.

splitMeshRegions Splits mesh into multiple regions.

stitchMesh Stitches a mesh.

subsetMesh Selects a section of mesh based on a **cellSet**.

transformPoints Transforms the mesh points in the **polyMesh** directory according to the translate, rotate and scale options.

zipUpMesh Reads in a mesh with hanging vertices and zips up the cells to guarantee that all polyhedral cells of valid shape are closed.

3.7.5 Other mesh tools

collapseEdges Collapses short edges and combines edges that are in line.

combinePatchFaces Checks for multiple patch faces on same cell and combines them. Multiple patch faces can result from e.g. removal of refined neighbouring cells, leaving 4 exposed faces with same owner.

refinementLevel Tries to figure out what the refinement level is on refined Cartesian meshes. Run BEFORE snapping.

refineWallLayer Utility to refine cells next to patches.

removeFaces Utility to remove faces (combines cells on both sides).

selectCells Select cells in relation to surface.

splitCells Utility to split cells with flat faces.

3.7.6 Post-processing

engineCompRatio Calculate the geometric compression ratio. Note that if you have valves and/or extra volumes it will not work, since it calculates the volume at BDC and TCD.

foamPostProcess Execute the set of **functionObjects** specified in the selected dictionary (which defaults to *system/controlDict*) or on the command-line for the selected set of times on the selected set of fields.

noise Utility to perform noise analysis of pressure data using the **noiseFFT** library.

pdfPlot Generates a graph of a probability distribution function.

temporalInterpolate Interpolate fields between time-steps e.g. for animation.

3.7.7 Post-processing data converters

foamDataToFluent Translates OpenFOAM data to Fluent format.

foamToEnight Translates OpenFOAM data to EnSight format.

foamToEnightParts Translates OpenFOAM data to Enight format. An Enight part is created for each `cellZone` and patch.

foamToGMV Translates foam output to GMV readable files.

foamToTetDualMesh Converts `polyMesh` results to `tetDualMesh`.

foamToVTK Legacy VTK file format writer.

smapToFoam Translates a STAR-CD SMAP data file into OpenFOAM field format.

3.7.8 Lagrangian post-processing

particleTracks Generates a VTK file of particle tracks for cases that were computed using a tracked-parcel-type cloud.

steadyParticleTracks Generates a VTK file of particle tracks for cases that were computed using a steady-state cloud NOTE: case must be re-constructed (if running in parallel) before use

3.7.9 Surface mesh (e.g. OBJ/STL) tools

surfaceAdd Add two surfaces. Does geometric merge on points. Does not check for overlapping/intersecting triangles.

surfaceAutoPatch Patches surface according to feature angle. Like `autoPatch`.

surfaceBooleanFeatures Generates the *extendedFeatureEdgeMesh* for the interface between a boolean operation on two surfaces. Assumes that the orientation of the surfaces is correct.

surfaceCheck Checks geometric and topological quality of a surface.

surfaceClean Removes baffles - collapses small edges, removing triangles. - converts sliver triangles into split edges by projecting point onto base of triangle.

surfaceCoarsen Surface coarsening using `bunnylod`

surfaceConvert Converts from one surface mesh format to another.

surfaceFeatureConvert Convert between `edgeMesh` formats.

surfaceFeatures Identifies features in a surface geometry and writes them to file, based on control parameters specified by the user.

surfaceFind Finds nearest face and vertex.

- surfaceHookUp** Find close open edges and stitches the surface along them
- surfaceInertia** Calculates the inertia tensor, principal axes and moments of a command line specified **triSurface**. Inertia can either be of the solid body or of a thin shell.
- surfaceLambdaMuSmooth** Smooths a surface using λ/μ smoothing.
- surfaceMeshConvert** Converts between surface formats with optional scaling or transformations (rotate/translate) on a **coordinateSystem**.
- surfaceMeshExport** Export from **surfMesh** to various third-party surface formats with optional scaling or transformations (rotate/translate) on a **coordinateSystem**.
- surfaceMeshImport** Import from various third-party surface formats into **surfMesh** with optional scaling or transformations (rotate/translate) on a **coordinateSystem**.
- surfaceMeshInfo** Miscellaneous information about surface meshes.
- surfaceMeshTriangulate** Extracts surface from a **polyMesh**. Depending on output surface format triangulates faces.
- surfaceOrient** Set normal consistent with respect to a user provided ‘outside’ point. If the **-inside** option is used the point is considered inside.
- surfacePointMerge** Merges points on surface if they are within absolute distance. Since absolute distance use with care!
- surfaceRedistributePar** (Re)distribution of **triSurface**. Either takes an non-decomposed surface or an already decomposed surface and redistributes it so that each processor has all triangles that overlap its mesh.
- surfaceRefineRedGreen** Refine by splitting all three edges of triangle (‘red’ refinement). Neighbouring triangles which are not marked for refinement get split in half (‘green’ refinement).
- surfaceRenamePatch** Rename a patch in a surface geometry file, where the patch can be identified by name or a point close to the patch.
- surfaceSplitByPatch** Writes regions of **triSurface** to separate files.
- surfaceSplitByTopology** Strips any baffle parts of a surface. A baffle region is one which is reached by walking from an open edge, and stopping when a multiply connected edge is reached.
- surfaceSplitNonManifolds** Takes multiply connected surface and tries to split surface at multiply connected edges by duplicating points. Introduces concept of - **borderEdge**. Edge with 4 faces connected to it. - **borderPoint**. Point connected to exactly 2 **borderEdges**. - **borderLine**. Connected list of **borderEdges**.
- surfaceSubset** A surface analysis tool which sub-sets the **triSurface** to choose only a part of interest. Based on **subsetMesh**.

surfaceToPatch Reads surface and applies surface regioning to a mesh. Uses **boundaryMesh** to do the hard work.

surfaceTransformPoints Transform (scale/rotate) a surface. Like **transformPoints** but for surfaces.

3.7.10 Parallel processing

decomposePar Automatically decomposes a mesh and fields of a case for parallel execution of OpenFOAM.

reconstructPar Reconstructs fields of a case that is decomposed for parallel execution of OpenFOAM.

redistributePar Redistributes existing decomposed mesh and fields according to the current settings in the **decomposeParDict** file.

3.7.11 Thermophysical-related utilities

adiabaticFlameT Calculates the adiabatic flame temperature for a given fuel over a range of unburnt temperatures and equivalence ratios.

chemkinToFoam Converts CHEMKINIII thermodynamics and reaction data files into OpenFOAM format.

equilibriumCO Calculates the equilibrium level of carbon monoxide.

equilibriumFlameT Calculates the equilibrium flame temperature for a given fuel and pressure for a range of unburnt gas temperatures and equivalence ratios; the effects of dissociation on O₂, H₂O and CO₂ are included.

mixtureAdiabaticFlameT Calculates the adiabatic flame temperature for a given mixture at a given temperature.

3.7.12 Miscellaneous utilities

foamDictionary Interrogates and manipulates dictionaries.

foamFormatConvert Converts all IOobjects associated with a case into the format specified in the **controlDict**.

foamListTimes List times using **timeSelector**.

foamToC Run-time selection table of contents printing and interrogation.

foamUnits Print information about dimensions and unit conversions

patchSummary Writes fields and boundary condition info for each patch at each requested time instance.

3.7.13 Deprecated utilities

changeDictionary Utility to change dictionary entries, e.g. can be used to change the patch type in the field and **polyMesh**/boundary files.

topoSet Operates on **cellSets**/**faceSets**/**pointSets** through a dictionary.

Chapter 4

OpenFOAM cases

This chapter deals with the file structure and organisation of OpenFOAM cases. Normally, a user would assign a name to a case, *e.g.* the tutorial case of aerodynamics of a motorbike is simply named `motorBike`. This name becomes the name of a directory in which all the case files and sub-directories are stored.

When running a simulation, a case directory can be located anywhere on a user's filing system. However, we recommend putting cases within a *run* subdirectory of the user's filing system, *i.e.* `$HOME/OpenFOAM/${USER}-14` as described at the beginning of chapter 2. The `$FOAM_RUN` environment variable is set to `$HOME/OpenFOAM/${USER}-14/run` by default and the user can quickly move to that directory by executing a preset alias, `run`, at the command line.

The tutorial cases that accompany the OpenFOAM distribution provide useful examples of the case directory structures. The tutorials are located in the `$FOAM_TUTORIALS` directory, reached quickly by executing the `tut` alias at the command line. Users can view tutorial examples at their leisure while reading this chapter.

4.1 File structure of OpenFOAM cases

The basic directory structure of an OpenFOAM case, containing the minimum set of files required to run an application, is shown in Figure 4.1 and described as follows:

constant directory that contains a full description of the case mesh in a subdirectory *polyMesh* and files specifying properties and models for the application concerned, *e.g.* *physicalProperties* and *momentumTransport*.

system directory for setting parameters associated with the solution procedure itself. It contains *at least* the following three files: *controlDict* where run control parameters are set including start/end time, time step and parameters for data output; *fvSchemes* where discretisation schemes used in the solution are selected; and, *fvSolution* where the equation solvers, tolerances and other algorithm controls are set for the run.

'time' directories containing individual files of data for particular fields, *e.g.* velocity and pressure. The data can be: either, initial values and boundary conditions that the user must specify to define the problem; or, results written to file by OpenFOAM. Fields must always be initialised, even when the solution does not strictly require it, as in

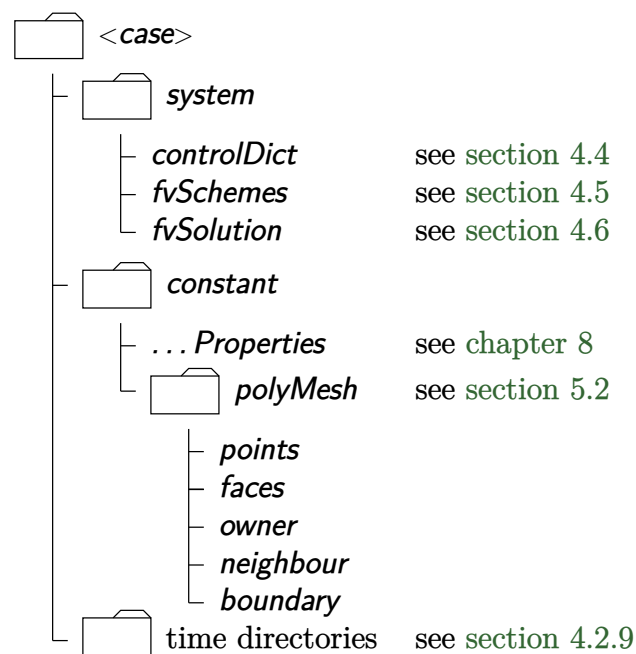


Figure 4.1: Case directory structure

steady-state problems. The name of each time directory is based on the simulated time at which the data is written and is described fully in section 4.4. Since we usually start our simulations at time $t = 0$, the initial conditions are usually stored in a directory named *0*. For example, in the *motorBike* tutorial, the velocity field $\mathbf{U}$ and pressure field p are initialised from files *0/U* and *0/p* respectively.

4.2 Input/output file format

OpenFOAM needs to read a range of data structures such as strings, words, scalars, vectors, tensors, lists and fields. The input/output (I/O) format of files is extremely flexible, following a consistent set of rules that make the files easy to interpret. The OpenFOAM file format is described in the following sections.

4.2.1 General syntax rules

The format resembles C++ code, following the general principles below.

- Files have free form, with no particular meaning assigned to any column and no need to indicate continuation across lines.
- Lines have no particular meaning except to a `//` comment delimiter which makes OpenFOAM ignore any text that follows it until the end of line.
- A comment over multiple lines is done by enclosing the text between `/*` and `*/` delimiters.

4.2.2 Dictionaries

OpenFOAM mainly uses *dictionaries* to specify data, in which data entries are retrieved by means of *keywords*. Each keyword entry follows the general format, beginning with the keyword and ending in semi-colon (;).

```
<keyword> <dataEntry1> ... <dataEntryN>;
```

Many entries include only a single data entry as shown below.

```
<keyword> <dataEntry>;
```

Most data files, *e.g.* *controlDict*, are themselves dictionaries since they contain a series of keyword entries. Any dictionary can contain one or more sub-dictionaries, usually denoted by a dictionary *name* and its keyword entries contained within curly braces {} as follows.

```
<dictionaryName>
{
    ... keyword entries ...
}
```

(Sub-)dictionaries can be nested within others, as shown in the following example. The extract, from an *fvSolution* dictionary file, containing two dictionaries, *solvers* and *PIMPLE*. The *solvers* dictionary contains nested sub-dictionary for different matrix equations based on different solution variables, *e.g.* *p*, *U* and *k* (with some entries using regular expressions described in section 4.2.13).

```

16
17 solvers
18 {
19     p
20     {
21         solver      GAMG;
22         tolerance    1e-7;
23         relTol       0.01;
24
25         smoother     DICGaussSeidel;
26     }
27
28     pFinal
29     {
30         $p;
31         relTol       0;
32     }
33
34     "(U|k|epsilon)"
35     {
36         solver      smoothSolver;
37         smoother     symGaussSeidel;
38         tolerance    1e-05;
39         relTol       0.1;
40     }
41
42     "(U|k|epsilon)Final"
43     {
44         $U;
45         relTol       0;
46     }
47 }
48
```

```

49 PIMPLE
50 {
51     nNonOrthogonalCorrectors 0;
52     nCorrectors             2;
53 }
54
55
56
57 // *****

```

4.2.3 The data file header

All data files that are read and written by OpenFOAM begin with a dictionary named `FoamFile` containing a standard set of keyword entries, listed below:

- **version:** I/O format version, optional, defaults to 2.0
- **format:** data format, `ascii` or `binary`
- **class:** class relating to the data, either `dictionary` or a field, *e.g.* `volVectorField`
- **object:** filename, *e.g.* `controlDict` (mandatory, but not used)
- **location:** path to the file (optional)

An example header for a *controlDict* file is shown below.

```

FoamFile
{
    format      ascii;
    class       dictionary;
    location     "system";
    object       controlDict;
}

```

4.2.4 Lists

OpenFOAM applications contain lists, *e.g.* a list of vertex coordinates for a mesh description. Lists are commonly found in I/O and have a format of their own in which the entries are contained within round braces (). When a user specifies a list in an input file, *e.g.* the `vertices` list in a *blockMeshDict* file, it just includes the `vertices` keyword and the data in (), *e.g.*

```

vertices
(
    ... entries ...
);

```

When OpenFOAM writes out a list, it invariably prefixes it with the number of elements in the list. For example the *points* file for the mesh in the `pizDailySteady` case contains the following (abbreviated) list, where 25012 denotes the number of vertex points in the mesh.

```

25012
(
    (-0.0206 0 -0.0005)
    (-0.01901716308 0 -0.0005)
    ... entries ...
);

```

In some cases, when OpenFOAM writes out a list, it further prefixes it with the class name of the list. For example, the `inGroups` entry in a *boundary* file of a mesh contains a list where each group name is a *word*. The entry for the `lowerWall` patch from the `pizDailySteady` case is shown below, indicating the `List<word>` class with a single (1) element.

```

lowerWall
{
    type                wall;
    inGroups             List<word> 1(wall); // Note!
    nFaces               250;
    startFace            24480;
}

```

4.2.5 Scalars, vectors and tensors

A scalar is a single number represented as such in a data file. A **vector** contains three values, expressed using the simple List format so that the vector (1.0, 1.1, 1.2) is written:

```
(1.0 1.1 1.2)
```

In OpenFOAM, a tensor contains nine elements, such that the identity tensor can be written:

```
(1 0 0 0 1 0 0 0 1)
```

The values are ordered according to the following components, for an Cartesian co-ordinate system.

```
(xx xy xz yx yy yz zx zy zz)
```

OpenFOAM also supports specifically a symmetric tensor for which the off-diagonal components are equal ($xy = yx$, $xz = zx$, $yz = zy$). The duplicate components are not stored, so the symmetric tensor is specified with six components only, as follows.

```
(xx xy xz yy yz zz)
```

4.2.6 Dimensional units

In science, properties are represented in some chosen units, *e.g.* mass in kilograms (kg), volume in cubic metres (m³), pressure in Pascals (kg m⁻¹ s⁻²). Algebraic operations must be performed on these properties using consistent units of measurement. In particular, addition, subtraction and equality are only physically meaningful for properties of the same dimensional units. As a safeguard against implementing a meaningless operation, OpenFOAM attaches dimensions to field data and physical properties and performs dimension checking on any operation.

Dimensions are described by the `dimensionSet` class which are represented with a unique I/O syntax using square brackets, *e.g.*

```
[0 2 -1 0 0 0 0]
```

The syntax above uses the base syntax where each integer corresponds to the power of a base unit of measurement listed in sequence below, accompanied by the corresponding units for the *Système International* (SI) and the *United States Customary System* (USCS), respectively.

1. mass, *e.g.* kilogram (kg), pound-mass (lbm);
2. length, *e.g.* metre (m), foot (ft);
3. time, *e.g.* second (s);
4. temperature, *e.g.* Kelvin (K), degree Rankine (°R);
5. quantity, *e.g.* mole (mol);
6. current, *e.g.* ampere (A);
7. luminous intensity, *e.g.* candela (cd).

Dimensional units can alternatively be specified by name, starting with the base units, named **mass**, **length**, **time**, **temperature**, **moles**, **current**, and **luminousIntensity**. Dimensional units can be expressed using these names, rather than the array of indices, *e.g.* dimensions of length can be written

```
[length]
```

instead of `[0 1 0 0 0 0 0]`. The example, `[0 2 -1 0 0 0 0]`, can be written as

```
[sqr(length)/time]
```

where `sqr(length)` denotes units of `length*length`. There are also names for “composite” dimensional units that are commonly used. For example, **area** represents `sqr(length)`, so the previous example could be written

```
[area/time]
```

In fact, these dimensions are those of kinematic viscosity, for which a named dimension is predefined by

```
[kinematicViscosity]
```

We recommend named dimensions since they are easier to comprehend. Since v14, all example cases in OpenFOAM use named dimensional units only in input files, rather than the array of index notation. The user can list available named dimensions can be listed by running the `foamUnits` utility as follows (see section 4.7.7 for more details).

```
foamUnits
```

They can then obtain further information for any given dimension by specifying the dimension as an argument to `foamUnits`, *e.g.*

```
foamUnits pressure
```

returns, the fundamental dimensions in named and index form, and a list of corresponding units in the default unit set.

```
Dimension [pressure]
+ Dimensions = [mass length^-1 time^-2]
+ Exponents = [1 -1 -2 0 0 0 0]
+ Standard Units = [kg m^-1 s^-2] [Pa]
+ Other Units = [psi] [atm] [kPa] [bar] [MPa]
```

Dimensions do not themselves suggest any particular set of units, *e.g.* SI or USCS. Instead, the set of units is specified in the `$FOAM_ETC/configDict` by the `set` keyword in the `units` sub-dictionary. The default `set` is SI, with two pre-configured alternatives of CGS (centimetre-gram-second) and USCS. The `set` choice can be overridden as described in section 4.3. Indeed, an alternative custom unit set can even be specified, with any defined units, mimicking the syntax in the `configDict` file.

4.2.7 Units and unit conversion

Numerical parameters can be specified with accompanying units, written using their name contained within square brackets, *e.g.* “[mm]” for millimetre units. Standard available units are listed by running the `foamUnits` utility as follows.

```
foamUnits
```

The listed units are those defined in the `units` sub-dictionary of the `$FOAM_ETC/configDict` file. They begin with the fundamental units of the *chosen unit set*, *i.e.* kg, m, s, K, kmol, A and Cd, for the default SI units. A unit can be specified that combines different units with exponents, divisions and/or multipliers, *e.g.* “[cm s^-1]” or “[cm/s]” for a speed in centimetres per second. These units could be used to specify a velocity for a `fixedValue` condition on an `input` patch as follows.

```
inlet
{
    type            fixedValue;
    value           uniform (1000 0 0) [cm/s];
}
```

The way units work is that when data is read that includes units, the numerical value is converted into the base unit by multiplying by a factor. So in the example above, the value (1000 0 0) in [cm/s] is converted into (10 0 0) in the base [m/s] units by multiplying by a conversion factor of 0.01. The conversion factor applied to a given unit can be printed using `foamUnits` as described in section 4.7.7.

Units can be applied from a different base set to the only being used. For example, a parameter could be specified in [ft/min], i.e. the USCS feet per second, when the set being used in SI. The example below shows the `meanVelocity` being specified in this way for the `flowRateInletVelocity` boundary condition.

```
inlet
{
    type            flowRateInletVelocity;
    meanVelocity    30 [ft/s];
}
```

There are many units derived from more complex combinations of base units, e.g. [N] for newton, corresponding to [kg m s⁻²], and [J] for joule, corresponding to [N m]. The [cal] unit for calorie applies a further conversion to [J]. With the `flowRateInletVelocity` boundary condition example above, a `volumetricFlowRate` could be specified in litres per second by [l/s] as follows.

```
inlet
{
    type            flowRateInletVelocity;
    volumetricFlowRate 0.2 [l/s];
}
```

While the `volumetricFlowRate` parameter above is specified by a single value, it is in fact an example of a `Function1`, which is described in section 6.4.4 (about time-varying boundary conditions). A `Function1` is function of one variable; in this case the variable is time, allowing the user to prescribe a flow rate that varies in time. The single value syntax is a shorthand to represent a constant `Function1`.

There is a wide choice of available functions listed in section 6.4.4. The `volumetricFlowRate` could, for example, follow a sine wave with a mean value of 0.15 l/s, amplitude 0.05 l/s and frequency 20 mHz with the following configuration

```
inlet
{
```

```

    type          flowRateInletVelocity;
    volumetricFlowRate
    {
        type      sine;
        level     0.15 [l/s];
        amplitude 0.05 [l/s];
        frequency 0.002 [Hz];
    }
}

```

This shows that the parameters used by the `sine Function1` can include units. Two `Function1`s whose unit handling is slightly more subtle are `table` and `polynomial`. In both cases they provide a `units` entry in which the user can provide the units for the values and the function variable, *i.e.* time t , for example.

For the `table` function, the user specifies values at different values of the function variable. For example, a table entry that specifies a value of 0.2 at $t = 0.1$ s would be represented as (0.1 0.2) in OpenFOAM format.

Imagine we want the `volumetricFlowRate` to increase from 0 l/s at $t = 0$ to 0.2 l/s at $t = 0.1$ s. This could be achieved using a `table` with two entries, since the `table` function interpolates values linearly between successive data points, and clamps to the final value at times beyond the last point. The configuration for this example, below, shows how the optional `units` works.

```

inlet
{
    type          flowRateInletVelocity;
    volumetricFlowRate
    {
        type      table;
        units     ([ms] [l/s]);
        values
        (
            (0      0)
            (100    0.2)
        );
    }
}

```

Notice that the `units` entry sets milliseconds for the time variable, so ramps the flow rate between 0 and 0.2 l/s over 100 ms, or 0.1 s. The values themselves are specified in l/s as in the earlier example.

While units are generally associated with dimensioned quantities, the following units that are available to perform conversions between dimensionless quantities.

- angle: fundamental unit is radian “[rad]” with scaled units of degree “[deg]” and rotations “[rot]”.

- fractions: fundamentally without units, but can be represented by the scaled unit of percentage “[%]”.

Finally, users can define custom units of their own. To do so, they need to add entries to a *configDict*, discussed in section 4.3. Essentially, they should follow the instructions to override global controls in section 4.3.1 in which the sample *configDict* is copied into their case *system* directory. They can then add custom units to the **units** section of that file. Below is an example which creates a niche USCS unit, acre-foot per fortnight “[aff]”, sometimes used in irrigation.

```
units
{
    day          24  [hr];
    fortnight    14  [day];
    chain         22  [yd];
    acre          10  [chain^2];
    aff           1   [acre*ft/fortnight];
}
```

4.2.8 Dimensioned types

Most physical properties are defined with their respective dimensions. They are represented by the **dimensioned** class which strictly includes three components: a **word** name; a **dimensionSet** and a value (scalar, vector, *etc.*). In the distant past, input files would include entries with all three data entries, but over time the **word** and **dimensionSet** components have become hard-coded so that the user only needs to supply the value to be read at input.

Since OpenFOAM v14, there are no example cases with the older input syntax, so the basic input syntax is shown below for parameters for density ρ and kinematic viscosity ν .

```
rho    1000;
nu     1e-5;
```

The value of a dimensioned parameter can also include units. When they do so, the units are checked for consistency against the hard-coded dimensions of the **dimensioned** parameter. If the user wishes to be reminded of the dimensions, they can do so indirectly by including the base units (for which the resulting conversion factor is unity) as follows.

```
rho    1000 [kg m^-3];
nu     1e-5 [m^2/s];
```

Ultimately, the checking of supplied units differentiates a **dimensioned** parameter, *e.g.* **dimensionedScalar**, from its non-dimensioned counterpart, *i.e.* **scalar**. The user has the convenience of specifying a value in a non-base unit, *e.g.* in centistokes for ν , with the reassurance that the chosen units are appropriate for the given parameter.

```
nu     10 [cSt];
```


4.2.9 Fields

Field files, *e.g.* U and p , that are read from and written into the time directories, possess their own customised I/O with the following key entries.

- **dimensions**: the dimensions of the field, *e.g.* `[pressure]` (or `[1 -1 -2 0 0 0 0]`).
- **internalField**: values within the internal field, *e.g.* within each cell of a mesh.
- **boundaryField**: condition (**type**) and data for each patch of the mesh boundary.
- **sources**, **optional** (usually not used): field values required when a fluid mass (or volume) source is introduced internally, rather than at a boundary, see section 6.2.

This section provides an overview of the **basic syntax** of the **internalField** and **boundaryField** entries in field files. The **details** of field initialisation, sources and boundary conditions are then presented in Chapter 6.

For the **internalField**, there are three way it can be specified, listed below.

- **uniform**: typically for initialisation, the field is uniform so can be described by a single value.
- **nonuniform**: typically for results that are written out, a list of values is specified, one value for each element of the field.
- **field function**: used to initialise a nonuniform field, as described in section 6.1.

An example of a **uniform** field, *e.g.* to initialise kinematic pressure (p), is shown below. The uniform value is 100000 and follows the **uniform** keyword.

```
internalField uniform 100000;
```

A uniform field can also be initialised using a set of units, *e.g.* for a pressure of 1 bar:

```
internalField uniform 1 [bar];
```

When results are written out, the field is non-uniform with different values for each element (*i.e.* cells or faces). The output then uses the **nonuniform** keyword, followed by a suitable list of values. The abbreviated example below is from an output p file for a mesh of 12225 cells.

```
internalField nonuniform List<scalar>
12225
(
-4.92806
-5.42676
...
);
```

The `boundaryField` is a dictionary containing a set of entries corresponding to each patch listed in the mesh boundary, described in section 5.3. Each entry in `boundaryField` is itself a sub-dictionary containing a list of keyword entries, beginning with the mandatory `type` entry which specifies the patch field condition for the field. Additional keyword entries are then generally required, corresponding to the specified `type`. They can include *field* data to specify a property on each patch face, that require the `uniform/nonuniform` syntax described above.

A selection of patch field conditions in OpenFOAM are described in sections 6.3, 6.4, 6.5 and 6.6. The selection is by no means exhaustive; rather a complete list of boundary conditions can be printed using the `foamToC` utility (see section 4.7.6) with the `-scalarBCs` and `-vectorBCs` options, *i.e.*

```
foamToC -scalarBCs
foamToC -vectorBCs
```

An example field file for velocity `U`, containing simple entries for `dimensions`, `internalField` and `boundaryField`, is shown below:

```

16
17 dimensions      [velocity];
18
19 internalField    uniform (0 0 0);
20
21 boundaryField
22 {
23     inlet
24     {
25         type      fixedValue;
26         value      uniform (10 0 0);
27     }
28
29     outlet
30     {
31         type      zeroGradient;
32     }
33
34     upperWall
35     {
36         type      noSlip;
37     }
38
39     lowerWall
40     {
41         type      noSlip;
42     }
43
44     frontAndBack
45     {
46         type      empty;
47     }
48 }
49
50 // ***** //
```

4.2.10 Macro expansion

The configuration of case files can benefit from a macro syntax which uses the dollar (\$) symbol in front of a keyword to expand the data associated with the keyword. The keyword can relate to any type of entry, *e.g.* in the following example, `grad(U)` expands to “`cellLimited Gauss linear 1`”.

```

    limited    cellLimited Gauss linear 1;
    grad(U)    $limited;

```

Variables can be accessed within different levels of sub-dictionaries, or scope. Scoping is performed using a '/' (slash) syntax, illustrated by the following example, where the `timeScheme` entry expands to "Euler", the entry for `default` in a sub-dictionary named `ddtSchemes`.

```

ddtSchemes
{
    default    Euler;
}
timeScheme $ddtSchemes/default;

```

Entries can relate to scalar values. In the following example, the `yMin` keyword expands to the value `-10`.

```

xMin -10;
yMin $xMin;
xMax #neg $xMin;

```

The `xMax` entry demonstrates the `#neg` directive which negates the value that follows it, such that it expands to `10` in the example above.

There are further syntax rules for macro expansions:

- to traverse up one level of sub-dictionary, use the `'..'` (double-dot) prefix, see below;
- to traverse up two levels use `'../..'` prefix, *etc.*;
- to traverse to the top level dictionary use the `'!'` (exclamation mark) prefix (most useful), see below;
- to traverse into a separate file named `otherFile`, use `'otherFile!'`, see below;
- for multiple levels of macro substitution, each specified with the `'$'` dollar syntax, `'{'` brackets are required to protect the expansion, see below.

When accessing parameters from another file, the `$FOAM_CASE` environment variable is useful to specify the path to the file as described in section 4.2.12 and illustrated below.

```

a 10;
b a;
c ${$b}; // returns 10, since $b returns "a", and $a returns 10

subdictA
{
    a 20;
}

```

```

subdictB
{
    // double-dot takes scope up 1 level, then into "subdictA" => 20
    b ../../subdictA/a;

    subsubdict
    {
        // exclamation mark takes scope to top level => 10
        b !$a;

        // "a" from another file named "otherFile"
        c $otherFile!a;

        // "a" from another file "otherFile" in the case directory
        d ${${FOAM_CASE}/otherFile!a};
    }
}

```

4.2.11 Including files

Directives are commands that begin with the hash (#) symbol which provide further flexibility when configuring case files. There is a set of directive commands for reading a data file from within another data file. If a case requires a single value of pressure of 100 kPa, used in different input files, we could create a file, *e.g.* named *initialConditions*, which contains the following entry:

```
pressure 1e+05;
```

In order to use this pressure for internal and initial boundary fields, the user could simply include the *initialConditions* file using the `#include` directive, then use a macro expansion on the `pressure` keyword, as follows.

```

#include "initialConditions"
internalField uniform $pressure;
boundaryField
{
    patch1
    {
        type fixedValue;
        value $internalField;
    }
}

```

This example works if the included file is in the same directory as the file that includes it. Otherwise, more generally the path to the file is required, *e.g.* if *initialConditions* is in the *constant* directory:

```
#include "$FOAM_CASE/constant/initialConditions"
```

Here `$FOAM_CASE` represents is the path of the case directory as described in the following section. The following special forms of the `#include` directive also exist.

- `#includeIfPresent`: reads a file if it exists.
- `#includeEtc`: reads a file with the `$FOAM_ETC` directory as the starting path.

The `#include` directives can be called with arguments contained in brackets `"(...)"`. They can include parameters using the syntax `<parameter>=<value>` that are *specifically* used in *macro expansions only*. For example if a *main* contains the following entries,

```
rhoInf    $rho;  
UInf      $U;
```

then the `rhoInf` and `UInf` parameters can be created by including the file using arguments for `rho` and `U`, as follows.

```
#include "main"(rho=1.2,U=25)
```

This works *if* the *main* file does *not* itself contain entries for `rho` and `U`.

There are then three specialised include directives for `functionObjects`, `fvModels` and `fvConstraints`.

- `#includeFunc`: reads a file containing a single `functionObject` configuration, first searching the case *system* directory, followed by the `$FOAM_ETC` directory.
- `#includeModel`: reads a file containing a single `fvModel` configuration, first searching the case *constant* directory, followed by the `$FOAM_ETC` directory.
- `#includeConstraint`: reads a file containing a single `fvConstraint` configuration, first searching the case *system* directory, followed by the `$FOAM_ETC` directory.

These directives can also be called with arguments contained in brackets `"(...)"`. However, the arguments are treated differently from the previous directives. For the specialised directives, `<parameter>=<value>` arguments simply add or modify a keyword entry `<parameter>`, assigning the `<value>`.

Function arguments are used extensively with the specialised include directives. The following `grep` command lists several files in the example cases that demonstrate this.

```
grep -rl '#includeFunc.*(' $FOAM_TUTORIALS
```

The use of parameters for `#includeFunc` is also mentioned in section 7.2.1.

4.2.12 Environment variables

Environment variables can be used in input files. For example, the `$FOAM_RUN` environment variable can be used to identify the *run* directory, as described in the introduction to Chapter 2. This could be used to include a file, *e.g.* by

```
#include "$FOAM_RUN/pitzDailySteady/0/U"
```

In addition to environment variables like `$FOAM_RUN`, set within the operating system, a number of “internal” environment variables are recognised, including the following.

- `$FOAM_CASE`: the path and directory of the running case.
- `$FOAM_CASENAME`: the directory *name* of the running case.
- `$FOAM_APPLICATION`: the name of the running application.

4.2.13 Regular expressions

As discussed, data is looked up from files using keywords. If a particular keyword does not exist, the I/O system will try to match the keyword with any [POSIX regular expression](#), specified inside double-quotations ("`...`") in the input file.

In some cases, when the I/O system searches for a keyword in a case file, a `*` can be used to match the keyword

When running an application, data is initialised by looking up keywords from dictionaries. The user can either provide an entry with a keyword that directly matches the one being looked up, or can provide a that matches the keyword, specified inside double-quotations ("`...`").

Regular expressions have an extensive syntax for various matches of text patterns but in OpenFOAM input files there are only two expressions that are generally used. Firstly, `'.'` denoting “any character”, and `'*'` denoting “repeated any number of times, including 0 times” is often used in combination to match “any characters”. For example, to specify a `noSlip` boundary condition for any patch whose name ends `Wall...`, the user could specify in the `boundaryField` for *U*:

```
".*Wall"
{
    type    noSlip;
}
```

The other common regular expression uses `()` to group expressions. For example, to a `noSlip` boundary condition on two wall patches named `upper` and `lower`, the user could specify:

```
"(upper|lower)"
{
    type    noSlip;
}
```

4.2.14 Keyword ordering

The order in which keywords are listed does not matter, except when *the same keyword is specified multiple times*. Where the same keyword is duplicated, the last instance is used. The most common example of a duplicate keyword occurs when a keyword is included from the file or expanded from a macro, and then overridden. The example below demonstrates this, where `pFinal` adopts all the keyword entries, including `relTol 0.05` in the `p` sub-dictionary by the macro expansion `$p`, then overrides the `relTol` entry.

```
p
{
    solver          PCG;
    preconditioner   DIC;
    tolerance        1e-6;
    relTol           0.05;
}
pFinal
{
    $p;
    relTol           0;
}
```

Where a data lookup matches both a keyword and a regular expression, the keyword match takes precedence irrespective of the order of the entries.

4.2.15 Inline calculations

There are further directives that enable calculations from within input files:

- `#calc`, for one statement calculations, described below;
- `#stream`, for multi-statement calculations, also described below;
- `#codeStream`, for complex multi-statement calculations, described in section 4.2.17;
- `#codeDict`, for creating multiple dictionary entries, described in section 4.2.18.

In the descriptions above, the term “statement” means a code statement, *i.e.* one piece of code ending in “;”. It can be thought of as “line”, but a single statement can be split over multiple lines, so “statement” is more precise. Examples of the use of these directives can be found in files in the *test/dictionary* directory in the OpenFOAM installation.

The `#calc` directive is used for simple one-line calculations, typically to set a parameter in an input file. An example is found at the beginning of the *0/U* in the *\$FOAM_TUTORIALS/incompressibleFluid/drivaerFastback* case.

```
Uinlet          16;

wheelRadius      0.318;
```

```
wheelBase      2.786;
wheelSpeed     #calc "$Uinlet / $wheelRadius";
```

The `#calc` directive is used here to calculate the rotational speed of the wheels `wheelSpeed` from on the rim speed `Uinlet` and the radius `wheelRadius`. It works by including the code: a) within quotations "..."; or, b) within hash-bracket delimiters `#{...#}`;. The code usually includes other keyword entries expanded with macros (`$`), as in the case above.

Care is required with calculations involving a division because the `/` character is otherwise used to identify keywords in sub-dictionaries, `"$a/b"` looks for a keyword `b` within a sub-dictionary named `a`. Where a division is required, the user can put spaces around the `/`, *e.g.*

```
a      3.0;
b      2.0;
c      #calc "$a / $b";
```

or they can apply brackets around the first variable, *e.g.*

```
c      #calc "${a}/$b";
```

The example above involves scalar values only. However, the `#calc` and other inline code frameworks can use variables that represent other OpenFOAM classes, or *types*, such as `vector`, `tensor`, `List`, `Field`, `string` *etc.*. To create a typed variable, the type is specified inside angled brackets `<>`, immediately after the `$` symbol, *e.g.* `$<vector>var` or `$<vector>{var}` substitutes a variable named `var` as a `vector`.

For example, the following code calculates $c = \mathbf{a} \cdot \mathbf{b}$ using `#calc`.

```
a      (1 2 3);
b      (1 1 0);
c      #calc "$<vector>a & $<vector>b";
```

Even in the previous example, care is required over the type of the entries. For example, if a file includes,

```
a      3;
b      2;
c      #calc "$a / $b"; // c = 1, since a and b are integers
```

the resulting value of `c` is 1, since `a` and `b` are interpreted as integers. As well as adding expressing the values as decimals in the original example, the correct value of `c` can be obtained by using a scalar typed variable. All the following examples would work as required.

```
a      3;
b      2;
c      #calc "$a / $<scalar>b";
c      #calc "$<scalar>a / $b";
c      #calc "$<scalar>a / $<scalar>b";
```


A critical feature of using typed macros in the calculation environments is that they support unit conversions, whereas standard macros do not.

```

nu    10 [cSt];
rho   1000 [kg/m^3];

mu    #calc "$<scalar>nu*$<scalar>rho";
// mu    #calc "$nu*$rho"; // will not compile

```

In the example shown, the `nu` and `rho` parameters include units. The standard macro will expand as a string with both a `scalar` and `string`, when the multiplication operation will be meaningless. With the typed macro, however, the expanded variable is specifically a `scalar` including the conversion from the units. The multiplication will be performed with converted scalars yielding the correct result in the base units.

The hash-bracket delimiters, `#{...#};`, (instead of `"..."`) have the advantages that: a) they support code written across multiple lines; and, b) they avoid problems with `string` typed variables that may contain quotation marks as in the example below.

```

s "field";
fieldName #calc
#{
    $<string>s + "Name"
#};

```

If the inline calculation involves more than one code statement, the `#stream` directive can be used instead of `#calc`. The `#stream` environment provides a reference to an `Ostream` named `os`. Therefore a calculated value is “set” to the relevant keyword by streaming them to `os`, by the following code (where `value` is the variable for the calculated value).

```
os << value;
```

The `#stream` environment also provides `dict`, a `const` reference to the current dictionary itself. Taking the earlier example with the inner product of two vectors, we can demonstrate `#stream` by adding a second code statement to print an `Info` message to the terminal.

```

a      (1 2 3);
b      (1 1 0);
c #stream
#{
    Info<< "Calculating c = " << dict.lookup<vector>("a")
        << " & " << dict.lookup<vector>("b") << endl;
    os << ($<vector>a & $<vector>b);
#};

```

where the `Info` to the terminal:

Calculating $c = (1 \ 2 \ 3) \ \& \ (1 \ 1 \ 0)$

It shows within the code block multiple (two) code statements (ending “;”). In the final statement the calculated result is streamed to `os` to set the `c` entry.

4.2.16 Multiple inline code entries with `#codeBlock`

When code is included in an input file, using directives such as `#calc` and `#stream`, the code is compiled dynamically into a shared-object library at run-time, which is linked to the running application. The code is written and compiled in a local directory named *dynamicCode* containing sub-directories for each compiled library. The sub-directory / library names for `#calc` and `#stream` code are created using a SHA (secure hash algorithm) string of hexadecimal characters prefixed with an underscore, *e.g.* `_11de73e6b530658843ed2ff917b9c23fc98ab0e7`.

The compiled code can be found within those directories. The code begins as template files, copied from `$FOAM_ETC/codeTemplates`. The code written in code blocks, *e.g.* inside `{...#};`, in the input file is then injected into relevant locations within the copied template files. Once copied, the files are compiled.

Normally, an individual library is created by each directive. In some cases, there may be a large number of directives, so generating an individual library for each one can be cumbersome. Instead, to generate a single library, the entries can be placed between a `#codeBlock` and `#endcodeBlock` directive. The `$FOAM_TUTORIALS/incompressibleFluid/venturiTube` example case uses `#codeBlock` in the *system/blockMeshDict* file to wrap 8 `#calc` directives into a single library, as shown below. When generating the mesh for this case, it is clear that only one library is generated.

```
#codeBlock

radIn      #calc "$diameter / 2.0";
boxIn      #calc "3.0*${<scalar>}radIn / 10.0";

radVen     #calc "${<scalar>}radIn / 2.0";
boxVen     #calc "${<scalar>}boxIn / 2.0";

xInSt      0;
xInEnd     $diameter;
xVenSt     #calc "${<scalar>}xInEnd + 1.25*$diameter";
xVenEnd    #calc "${<scalar>}xVenSt + 0.5*$diameter";
xOutSt     #calc "${<scalar>}xVenEnd + 2.5*$diameter";
xOutEnd    #calc "${<scalar>}xOutSt + $diameter";

#endcodeBlock
```

Calculations in written code can include standard C++ functions such as trigonometric functions, *e.g.* `sin`. They can also include functions and classed from OpenFOAM. By

default, the dynamic code framework includes, as standard, some header files relating to dictionaries, fields, IO and units.

If the code requires a header file that is not included as standard, it can be included by the `#codeInclude` directive, which is available **withing** `#codeBlock...#endCodeBlock`. Multiple `#codeInclude` entries are permitted when more than one addition header file is required.

The `$FOAM_TUTORIALS/fluid/aerofoilNACA0012Steady` example demonstrates this, where the inlet velocity is calculated using an angle of attack using the code below. It uses the `transform` function from the `transform.H` header file, to rotate unit vectors by the angle of attack to set the lift and drag directions.

```

speed          250;
angleOfAttack  5 [deg];

#codeBlock
#codeInclude   "transform.H"
angle          #calc "$<scalar>angleOfAttack" // unit converted
liftDir       #calc "transform(Ry(-$angle), vector(0, 0, 1))";
dragDir       #calc "transform(Ry( $angle), vector(1, 0, 0))";

Uinlet        #calc "$speed*$<vector>dragDir";
#endCodeBlock

```

4.2.17 Inline code with `#codeStream`

The `#codeStream` directive is like `#stream` in that: a) it is designed to perform calculations involving more than one code statement; and, b) the resulting calculation is streamed to `os`, the reference to the `Ostream`.

But in addition to this, the code in `#codeStream` is written into a special code block that uses the hash-bracket delimiters, as shown below.

```

#codeStream
{
    code
    #{
        // ... code here ...
        os << value;
    };
}

```

In addition to this, `#codeStream` supports three other *optional* code blocks for the inclusion of multiple header files and compilations flags for locating header files and linking additional libraries associated with those header files.

- `codeInclude`: specifies additional C++ `#include` statements to include code files.
- `codeOptions`: specifies any extra compilation flags to be added to `EXE_INC` in *Make/options*.

- `codeLibs`: specifies any extra compilation flags to be added to `LIB_LIBS` in *Make/options*.

All the code blocks are written with hash-bracket delimiters. It is possible to create an empty set of optional code blocks, as shown below, when writing to write a `#codeStream` entry.

```
codeInclude
#{
#};

codeOptions
#{
#};

codeLibs
#{
#};
```

If a particular header file then needs including, *e.g.* `IObject.H`, the relevant entries can be added to these blocks using the `-coded` option to the `foamFind` script, described in section 4.7.8. For example, if the `#codeStream` entry was in a *system/blockMeshDict* file, the user could run the following command.

```
foamFind IObject.H -coded system/blockMeshDict
```

The resulting entries in `#codeStream` would be modified to include the following entries.

```
codeInclude
#{
    #include "IObject.H"
#};

codeOptions
#{
    -I$(LIB_SRC)/finiteVolume/lnInclude
#};

codeLibs
#{
    -lfiniteVolume
#};
```

Code, like any string, can be written across multiple lines by enclosing it within hash-bracket delimiters, *i.e.* `#{...#}`. Anything in between these two delimiters becomes a string with all newlines, quotes, *etc.* preserved.

An example of `#codeStream` is given below, where the code calculates moment of inertia of a box shaped geometry.

```
momentOfInertia #codeStream
{
    codeInclude
    #{
        #include "diagTensor.H"
    #};

    code
    #{
        scalar sqrLx = sqr($Lx);
        scalar sqrLy = sqr($Ly);
        scalar sqrLz = sqr($Lz);
        os <<
            $mass
            *diagTensor(sqrLy + sqrLz, sqrLx + sqrLz, sqrLx + sqrLy)/12.0;
    #};
};
```

4.2.18 Inline code with #codeDict

The last of the code-related conditionals is `#codeDict`. It is broadly similar to `#codeStream`, with a main code block and optional `codeInclude`, `codeOptions` and `codeLibs` blocks. However, there is one main difference. Instead of streaming values to `os` (the `Ostream`), `#codeDict` is able to add and modify dictionary entries through a **non-const** reference `dict` to the dictionary.

Below is an example with `#codeDict`.

```
nu    10 [cSt];
rho   1000 [kg/m^3];
#codeDict
{
    code
    #{
        dict.add
        (
            "mu",
            dict.lookup<scalar>("nu")*dict.lookup<scalar>("rho")
        );
    #};
}
#endcodeBlock
```

4.2.19 Conditionals

Input files support two conditional directives:

- `#if (...) ...#elif (...) ...#else ...#endif`
- `#ifeq (...) ...#else ...#endif`

Note: the syntax for the `#if` conditional changes in OpenFOAM v14, using `(...)` to delimit the condition. The condition is something which returns a Boolean (`true/false`). This is generally provided by some code executed by a `#calc` directive.

Conditionals are particularly useful in the control files for schemes and solvers/algorithms, *fvSchemes* and *fvSolution* respectively, since they can provide different settings based on some condition. A purely hypothetical example for the `#if` directive is below, where the `default` Laplacian scheme is different based on the non-orthogonality angle of a face.

```
laplacianSchemes
{
    #if (#calc "$angle < 75")
        default Gauss linear corrected;
    #else
        default Gauss linear limited corrected 0.5;
    #endif
}
```

The `#ifEq` compares a word or string, and executes based on a match. This is commonly used in *fvSchemes* and *fvSolution* files to set different schemes and other controls based on whether the case is steady-state or transient. To do so, it tests whether the `default` time scheme is set to `steadyState` and provides settings accordingly.

```
divSchemes
{
    #ifeq ($!ddtSchemes/default steadyState)
        div(phi,U)    bounded Gauss linearUpwind limited;
        turbulence    bounded Gauss limitedLinear 1;
    #else
        div(phi,U)    Gauss linearUpwind limited;
        turbulence    bounded Gauss limitedLinear 1;
    #endif
    ...
}
```

4.2.20 Checking input parameters

The following three directives help a user check their input parameters.

- `#dump`: prints the dictionary (file) contents preceding the directive.
- `#print`: prints the dictionary (entire file or sub-dictionary) including *all default entries*.
- `#exit`: causes the application to stop at the point it is read.

An example of the use of `#dump` and `#exit` is as follows. Imagine a user is creating a mesh with `blockMesh` with a parameterised *blockMeshDict* file, like the one in the `$FOAM_TUTORIALS/incompressibleFluid/venturiTube` example case. The file (in the *system* directory) contains a lot of calculated parameters like those beginning

```

diameter    0.1;

#codeBlock
radIn       #calc "$diameter / 2.0";
boxIn       #calc "3.0*${<scalar>radIn / 10.0";
...

```

In order to check those entries are calculating as expected, the user could add `#dump` and `#exit` at the end of these parameters (just before `vertices`).

```

...
boxInN      #neg $boxIn;
radVenN     #neg $radVen;
boxVenN     #neg $boxVen;

#dump
#exit

vertices
(
    ...

```

Then when running the `blockMesh`, the application will dump the parameters preceding `#dump` to the terminal and then terminate immediately. The terminal output will include the contents of the `blockMeshDict` dictionary with all the calculated entries visible.

```

blockMeshDict
{
    diameter      0.1;
    radIn         0.05;
    boxIn         0.015;
    ...
    ...
    boxInN        -0.015;
    radVenN       -0.025;
    boxVenN       -0.0075;
}

```

The `#print` directive is extremely useful for revealing default parameters used in a dictionary. A common use of this is revealing the coefficients of a turbulence model. For example, in a case, *e.g.* `$FOAM_TUTORIALS/incompressibleFluid/pitzDaily`, the user could add `#print` to the entire file or within the `RAS` sub-dictionary. In the latter case, *e.g.*

```

simulationType RAS;

```

```

RAS
{
    model            kEpsilon;
    turbulence       on;
    viscosityModel   Newtonian;

    #print
}

```

the code prints the following when `foamRun` is executed.

```

Selecting RAS turbulence model kEpsilon
Selecting generalised Newtonian model Newtonian
constant/momentumTransport!RAS
{
    model            kEpsilon;
    turbulence       on;
    viscosityModel   Newtonian;
    /* defaults */
    Cmu              0.09;
    C1               1.44;
    C2               1.92;
    C3               0;
    sigmaK           1;
    sigmaEps         1.3;
}

```

Another useful example is printing default values in the *fvSolution* file. Specifically, there are a number of controls in the PIMPLE sub-dictionary that have default values that are worth revealing. For example in the `$FOAM_TUTORIALS/fluid/aerofoilNACA0012Steady`, `#print` could be added to the PIMPLE sub-dictionary which otherwise includes only a small number of controls, as follows.

```

PIMPLE
{
    residualControl
    {
        p            1e-6;
        U            1e-5;
        "(k|omega|h)" 1e-5;
    }

    nNonOrthogonalCorrectors 0;

    #print
}

```


When the simulation is run with `foamRun`, the full extent of the default entries in the `PIMPLE` sub-dictionary is revealed.

```
system/fvSolution!PIMPLE
{
    residualControl
    {
        p            1e-06;
        U            1e-05;
        "(k|omega|h)" 1e-05;
    }
    nNonOrthogonalCorrectors 0;
    /* defaults */
    models                true;
    thermophysics         true;
    flow                  true;
    momentumPredictor     true;
    transonic             false;
    consistent            false;
    nCorrectors           1;
    moveMeshOuterCorrectors false;
    simpleRho             true;
    transportPredictionFirst true;
    transportCorrectionFinal true;
}
```

4.3 Global controls

OpenFOAM includes global control parameters that are configured in a file named *configDict*, located in the OpenFOAM installation within the *etc* directory. That directory is represented by the environment variable `$FOAM_ETC`, so the file can be printed to a terminal by typing the following command. The file contents are also reproduced below for reference.

```
cat $FOAM_ETC/configDict

9  {
10     format      ascii;
11     class       dictionary;
12     object      configDict;
13 }
14 // * * * * *
15
16 documentation
17 {
18     docBrowser      "firefox";
19     doxyDocDirs
20     (
21         "$WM_PROJECT_USER_DIR/html"
22         "~OpenFOAM/html"
23         "$WM_PROJECT_DIR/doc/Doxygen/html"
24         "http://cpp.openfoam.org/dev"
```

```

25     );
26     doxySourceFileExt "_8C.html";
27 }
28
29 InfoSwitches
30 {
31     writePrecision          6;
32     writeJobControl         0;
33     writeJobInfo            0;
34     writeDictionaries       0;
35
36     // Allow case-supplied C++ code (#codeStream, codedFixedValue)
37     allowSystemOperations    1;
38 }
39
40 OptimisationSwitches
41 {
42     // On NFS mounted file system: maximum wait for files to appear/get
43     // updated. Set to 0 on distributed case.
44     fileModificationSkew 10;
45
46     //- Modification checking:
47     // - timeStamp          : use modification time on file
48     // - inotify             : use inotify framework
49     // - timeStampMaster    : do time stamp (and file reading) only on master.
50     // - inotifyMaster       : do inotify (and file reading) only on master.
51     fileModificationChecking timeStampMaster;
52
53     //- Parallel IO file handler
54     // uncollated (default), collated or masterUncollated
55     fileHandler uncollated;
56
57     //- collated: thread buffer size for queued file writes.
58     // If set to 0 or not sufficient for the file size threading is not used.
59     // Default: 2e9
60     maxThreadFileBufferSize 2e9;
61
62     //- masterUncollated: non-blocking buffer size.
63     // If the file exceeds this buffer size scheduled transfer is used.
64     // Default: 2e9
65     maxMasterFileBufferSize 2e9;
66
67     commsType      nonBlocking; // scheduled; // blocking;
68     floatTransfer  0;
69     nProcsSimpleSum 0;
70
71     // Force dumping (at next timestep) upon signal (-1 to disable)
72     writeNowSignal -1; // 10; // SIGUSR1
73
74     // Force dumping and exit (at next timestep) upon signal (-1 to disable)
75     stopAtWriteNowSignal -1; // 12; // SIGUSR2
76 }
77
78 DebugSwitches
79 {
80     dimensionSet      1;
81     fileName          2;
82     GAMGAgglomeration 1;
83     level             2;
84     lduMatrix          1;
85     SolverPerformance 1;
86     vtkUnstructuredReader 1;
87 }
88
89 units
90 {
91     set      SI;
92
93     SI
94     {
95         kg      [1 0 0 0 0 0 0];
96         m        [0 1 0 0 0 0 0];
97         s        [0 0 1 0 0 0 0];
98         K        [0 0 0 1 0 0 0];
99         kmol     [0 0 0 0 1 0 0];

```

```

100     A      [0 0 0 0 0 1 0];
101     Cd      [0 0 0 0 0 0 1];
102 }
103
104 CGS
105 {
106     g      [1 0 0 0 0 0 0];
107     cm      [0 1 0 0 0 0 0];
108     s      [0 0 1 0 0 0 0];
109     K      [0 0 0 1 0 0 0];
110     mol     [0 0 0 0 1 0 0];
111     A      [0 0 0 0 0 1 0];
112     Cd      [0 0 0 0 0 0 1];
113 }
114
115 USCS
116 {
117     lb      [1 0 0 0 0 0 0];
118     ft      [0 1 0 0 0 0 0];
119     s      [0 0 1 0 0 0 0];
120     R      [0 0 0 1 0 0 0];
121     lbmol   [0 0 0 0 1 0 0];
122     A      [0 0 0 0 0 1 0];
123     Cd      [0 0 0 0 0 0 1];
124 }
125
126 // Conversions between fundamental units
127 g      1e-3      [kg];
128 lb      0.45359237 [kg];
129 cm      1e-2      [m];
130 ft      0.3048    [m];
131 K      1.8        [R];
132 kmol    1000      [mol];
133 lbmol    453.59237 [mol];
134
135 // Derived units
136 Hz      1          [s^-1];
137 N      1          [kg m s^-2];
138 pdl     1          [lb ft s^-2];
139 Pa      1          [N m^-2];
140 J      1          [N m];
141 W      1          [J s^-1];
142 V      1          [W/A];
143
144 // Scaled units
145 um      1e-6      [m];
146 mm      1e-3      [m];
147 km      1e3       [m];
148 in      25.4      [mm];
149 yd      3         [ft];
150 mi      1760      [yd];
151 us      1e-6      [s];
152 ms      1e-3      [s];
153 min     60        [s];
154 hr      3600      [s];
155 lbf     32.174049 [pdl];
156
157 // Derived scaled units
158 l      1e-3       [m^3];
159 ml     1e-6       [m^3];
160 rpm    1          [rot/min];
161 bar    1e5        [Pa];
162 atm    101325     [Pa];
163 kPa    1e3        [Pa];
164 MPa    1e6        [Pa];
165 psi    1          [lbf/in^2];
166 cal    4.184      [J];
167 kcal   4184       [J];
168 cSt    1e-6       [m^2/s];
169 cP     1e-3       [kg/m/s];
170 }
171
172 constants
173 {
174     pStd    1e5      [kg m^-1 s^-2];
175     Tstd    298.15   [K];

```

```

176 }
177
178 // *****

```

The file contains sub-dictionaries for the following categories of control parameter.

- **documentation**: for opening OpenFOAM documentation in a web browser.
- **InfoSwitches**: controls information printed to standard output, *i.e.* the terminal.
- **OptimisationSwitches**: for parallel communication and I/O, see section 3.4.2.
- **DebugSwitches**: messaging switches designed to help debug code failures.
- **units**: defines units and conversion factors, *e.g.* cal for calorie, described in section 4.2.7.
- **constants**: defines engineering constants.

While some of these categories are described elsewhere, the remaining categories will be described below, beginning with **DebugSwitches**. It relates to the system of messaging, in OpenFOAM, that is written to the terminal when an application runs. In the **DebugSwitches** category is a vast number of switches that control the printing of messages to help debug faults. The complete set of switches, including the **InfoSwitches** and **OptimisationSwitches**, can be printed by running **foamToC** (see section 4.7.6) with the **-switches** option, *i.e.*

```
foamToC -switches
```

This shows the extent of the number of switches and that, with a value of 0, almost all of them are deactivated by default. Any switch can be activated by setting its value to 1. A small number of **DebugSwitches** control messaging at three levels, 0, 1 and 2, most notably the overall **level** switch and **lduMatrix** which provides messaging for solver convergence during a run.

In the **OptimisationSwitches** category, there are some switches that control certain operational and optimisation issues. The **fileModificationSkew** switch is particularly important when running over a network file system (NFS). OpenFOAM scans the write time of data files to check for modification. On an NFS there can be some disparity in the clock settings on different machines, so field data files may appear to be modified ahead of time. This can cause a problem if OpenFOAM views the files as newly modified and attempts to re-read the data. The **fileModificationSkew** keyword is the time in seconds that OpenFOAM will subtract from the file write time when assessing whether the file has been newly modified. This and other switches are listed below:

- **fileModificationSkew**: a time in seconds that should be set higher than the maximum delay in NFS updates and clock difference for running OpenFOAM over a NFS.
- **fileModificationChecking**: method of checking whether files have been modified during a simulation, either reading the **timeStamp** or using **inotify**; versions that read only master-node data also exist, termed **timeStampMaster** and **inotifyMaster**.

- `commsType`: parallel communications type, `nonBlocking`, `scheduled` or `blocking`.
- `floatTransfer`: if 1, will compact numbers to `float` precision before transfer; default is 0.
- `nProcsSimpleSum`: optimises the global sum for parallel processing, by setting the number of processors above which a hierarchical sum is performed rather than a linear sum.

There are also some remaining switches in `OptimisationSwitches` that relate to file input/output when running in parallel, discussed in section 3.4.2.

From the `InfoSwitches`, the most important switch is `allowSystemOperations` which is enabled in the `configDict` file. Turning this switch off disables run-time compilation of code used for inline calculations and other inline code, described in section 4.2.15 and subsequent sections.

4.3.1 Overriding global controls

A user may wish to override one or more settings in the `configDict` file, either for all simulations or for a specific simulation. For example, they may wish to change the base set of units from the default (SI) to the US Customary system (USCS). This requires the `set` entry to be changed to USCS in the `units` sub-dictionary, as shown below.

```
units
{
    set    USCS;
}
```

To make this change for all simulations, the user could modify this setting in the `etc/configDict` file in the OpenFOAM installation. However, it is more flexible to configure changes in a file in their user directory. OpenFOAM provides a set of directory locations, where a local `configDict` file can be added, which it looks up in an order of precedence. To list the locations, run the following command.

```
foamEtcFile -list
```

The listed locations include a local `$HOME/.OpenFOAM` directory and follow a descending order of precedence, *i.e.* the last location listed (`etc`) is lowest precedence.

Therefore, if a user wished to work permanently in USCS units, they can maintain a `configDict` file in their `$HOME/.OpenFOAM` directory that includes the entry shown above. OpenFOAM would read the keyword entries from the global `configDict` file but override the `set` entry from the local file.

Alternatively, if a user wished to work on a *single case* in USCS units, they can add the same entry into a `configDict` file in the `system` directory for their *case*. There is a sample `configDict` file that can be copied into their `system` directory by running the `foamGet` script (see section 4.7.4) as follows.

```
foamGet configDict
```

Once the file is copied into their case, they can edit it accordingly. The file is annotated with example settings that are most commonly overridden.

4.4 Time and data input/output control

The OpenFOAM solvers begin all runs by setting up a database. The database controls I/O and, since output of data is usually requested at intervals of time during the run, time is an inextricable part of the database. The *controlDict* dictionary sets input parameters *essential* for the creation of the database. The keyword entries in *controlDict* are listed in the following sections. Only the time control and `writeInterval` entries are mandatory, with the database using default values for any of the optional entries that are omitted. Example entries from a *controlDict* dictionary are given below:

```

16
17 solver      incompressibleFluid;
18
19 startFrom    latestTime;
20
21 startTime    0;
22
23 stopAt       endTime;
24
25 endTime      0.3;
26
27 deltaT       0.0001;
28
29 writeControl  adjustableRunTime;
30
31 writeInterval 0.01;
32
33 purgeWrite    0;
34
35 writeFormat   ascii;
36
37 writePrecision 6;
38
39 writeCompression off;
40
41 timeFormat    general;
42
43 timePrecision 6;
44
45 runTimeModifiable yes;
46
47 adjustTimeStep yes;
48
49 maxCo         5;
50
51 // ***** //
```

4.4.1 Modules

solver Choice of solver module for the simulation, *e.g.* `incompressibleFluid`

regionSolvers Dictionary of solvers for different domain regions, *e.g.* heat transfer of water flowing over a plate might use the `fluid` and `solid` modules as follows:

```
regionSolvers
```

```
{
    water fluid;
    plate solid;
}
```

libs List of additional libraries (existing on `$LD_LIBRARY_PATH`) to be loaded at run-time, *e.g.* ("`libNew1.so`" "`libNew2.so`")

4.4.2 Time control

startFrom Controls the start time of the simulation.

- **firstTime**: Earliest time step from the set of time directories.
- **startTime**: Time specified by the **startTime** keyword entry.
- **latestTime**: Most recent time step from the set of time directories.

startTime Start time for the simulation with **startFrom** **startTime**;

stopAt Controls the end time of the simulation.

- **endTime**: Time specified by the **endTime** keyword entry.
- **writeNow**: Stops simulation on completion of current time step and writes data.
- **noWriteNow**: Stops simulation on completion of current time step and does not write out data.
- **nextWrite**: Stops simulation on completion of next scheduled write time, specified by **writeControl**.

endTime End time for the simulation when **stopAt** **endTime**; is specified.

deltaT Time step of the simulation.

4.4.3 Data writing

writeControl Controls the timing of write output to file.

- **timeStep**: Writes data every **writeInterval** time steps.
- **runTime**: Writes data every **writeInterval** seconds of simulated time.
- **adjustableRunTime**: Writes data every **writeInterval** seconds of simulated time, adjusting the time steps to coincide with the **writeInterval** if necessary — used in cases with automatic time step adjustment.
- **cpuTime**: Writes data every **writeInterval** seconds of CPU time.
- **clockTime**: Writes data out every **writeInterval** seconds of real time.

writeInterval Scalar used in conjunction with **writeControl** described above.

purgeWrite Integer representing a limit on the number of time directories that are stored by overwriting time directories on a cyclic basis. For example, if the simulations starts at $t = 5s$ and $\Delta t = 1s$, then with **purgeWrite 2**;, data is first written into 2 directories, 6 and 7, then when 8 is written, 6 is deleted, and so on so that only 2 new results directories exists at any time. *To disable the purging, specify **purgeWrite 0**; (default).*

writeFormat Specifies the format of the data files.

- **ascii** (default): ASCII format, written to **writePrecision** significant figures.
- **binary**: binary format.

writePrecision Integer used in conjunction with **writeFormat** described above, 6 by default.

writeCompression Switch to specify whether files are compressed with **gzip** when written: on/off (yes/no, true/false)

timeFormat Choice of format of the naming of the time directories.

- **fixed**: $\pm m.d\text{d}\text{d}\text{d}\text{d}\text{d}$ where the number of *ds* is set by **timePrecision**.
- **scientific**: $\pm m.d\text{d}\text{d}\text{d}\text{d}\text{d}e\pm xx$ where the number of *ds* is set by **timePrecision**.
- **general** (default): Specifies **scientific** format if the exponent is less than -4 or greater than or equal to that specified by **timePrecision**.

timePrecision Integer used in conjunction with **timeFormat** described above, 6 by default.

graphFormat Format for graph data written by an application.

- **raw** (default): Raw ASCII format in columns.
- **gnuplot**: Data in **gnuplot** format.
- **csv**: Comma-separated values.
- **vtk**: Visualisation Toolkit (VTK) format.
- **ensight**: Ensign format.

4.4.4 Other settings

beginTime Optional entry to for cases with an unusual start time that causes inconvenient write times. With **beginTime**, the write times are multiples of **writeInterval**, starting at the **beginTime**. For example, if the start time of 1.52 and a **writeInterval** of 1, results would be written at 2.52, 3.52, ... If **beginTime** is set to 0 (or 1), the write times would be 2, 3, *etc.*

Switch used by some solvers to adjust the time step during the simulation, usually according to **maxCo**.

adjustTimeStep Switch used by some solvers to adjust the time step during the simulation, usually according to **maxCo**.

maxCo Maximum Courant number, *e.g.* 0.5

runTimeModifiable Switch for whether dictionaries, *e.g.* *controlDict*, are re-read during a simulation at the beginning of each time step, allowing the user to modify parameters during a simulation.

functions Dictionary of functions, *e.g.* *probes* to be loaded at run-time; see examples in *\$FOAM_TUTORIALS*

4.5 Numerical schemes

The *fvSchemes* dictionary in the *system* directory sets the numerical schemes for terms, such as derivatives in equations, that are calculated during a simulation. This section describes how to specify the schemes in the *fvSchemes* dictionary. Details of the schemes are described in Chapter 3 of *Notes on Computational Fluid Dynamics: General Principles*.

The aim for *fvSchemes* is to provide an unrestricted choice of schemes to the user for everything from derivatives, *e.g.* gradient ∇ , to interpolations of values from one set of points to another. OpenFOAM uses the finite volume method so spatial derivatives are based on Gaussian integration which sums values on cell faces, which must be interpolated from cell centres. The user has a wide range of options for interpolation schemes, with certain schemes being specifically designed for particular derivative terms, especially the advection divergence $\nabla \cdot$ terms.

The set of terms, for which numerical schemes must be specified, are subdivided within the *fvSchemes* dictionary into the categories below, using Ψ as an example field variable.

- **timeScheme**: first and second time derivatives, *e.g.* $\partial\Psi/\partial t, \partial^2\Psi/\partial^2 t$
- **gradSchemes**: gradient $\nabla\Psi$
- **divSchemes**: divergence $\nabla \cdot \Psi$
- **laplacianSchemes**: Laplacian $\nabla \cdot \Gamma \nabla \Psi$, with diffusivity Γ
- **interpolationSchemes**: cell to face interpolations of values.
- **snGradSchemes**: component of gradient normal to a cell face.
- **wallDist**: distance to wall calculation, where required.

Each keyword is the name of a sub-dictionary which contains terms of a particular type, *e.g.* **gradSchemes** contains all the gradient derivative terms such as **grad(p)** (which represents ∇p). Further examples can be seen in the extract from an *fvSchemes* dictionary below:

```

16
17 ddtSchemes
18 {
19     default      Euler;
20 }
21
22 gradSchemes
23 {
24     default      Gauss linear;
25 }
```

```

26
27 divSchemes
28 {
29     default                none;
30
31     div(phi,U)              Gauss linearUpwind grad(U);
32     div(phi,k)              Gauss upwind;
33     div(phi,epsilon)        Gauss upwind;
34     div(phi,R)              Gauss upwind;
35     div(R)                  Gauss linear;
36     div(phi,nuTilda)        Gauss upwind;
37
38     div((nuEff*dev2(T(grad(U))))) Gauss linear;
39 }
40
41 laplacianSchemes
42 {
43     default                Gauss linear corrected;
44 }
45
46 interpolationSchemes
47 {
48     default                linear;
49 }
50
51 snGradSchemes
52 {
53     default                corrected;
54 }
55
56
57 // ***** //

```

The example shows *fvSchemes* with 6 ... *Schemes* subdictionaries, each containing keyword entries including: a **default** entry; other entries for the particular term specified, *e.g.* `div(phi, k)` for $\nabla \cdot (Uk)$. If a default scheme is specified in a particular ... *Schemes* sub-dictionary, it is assigned to all of the terms to which the sub-dictionary refers, *e.g.* specifying a default in *gradSchemes* sets the scheme for all gradient terms in the application, *e.g.* ∇p , ∇U . With a default specified, the specific terms are not required in that sub-dictionary, *i.e.* the entries for `grad(p)`, `grad(U)` are omitted in this example. Specifying a particular will however override the **default** scheme.

The user can specify no default scheme by the **none** entry, as in the *divSchemes* in the example above. The user is then obliged to specify all terms in that sub-dictionary individually. Setting **default** to **none** ensures the user specifies all terms individually which is common for *divSchemes* which requires precise configuration.

OpenFOAM includes a vast number of discretisation schemes, from which only a few are typically recommended for real-world, engineering applications. The user can get help with scheme selection by interrogating the tutorial cases for example scheme settings. They should look at the schemes used in relevant cases, *e.g.* for running a large-eddy simulation (LES), look at schemes used in tutorials running LES. Additionally, `foamSearch` is a useful tool to list the schemes used in all the tutorials. For example, to print all the **default** entries for *ddtSchemes* for cases in the `$FOAM_TUTORIALS` directory, the user can type:

```
foamSearch $FOAM_TUTORIALS fvSchemes ddtSchemes/default
```

which returns:

```

default        backward;
default        CrankNicolson 0.9;

```

```

default      Euler;
default      localEuler;
default      none;
default      steadyState;

```

The schemes listed using `foamSearch` are described in the following sections.

4.5.1 Time schemes

The first time derivative ($\partial/\partial t$) terms are specified in the *ddtSchemes* sub-dictionary. The discretisation schemes for each term can be selected from those listed below.

- **steadyState**: sets time derivatives to zero.
- **Euler**: transient, first order implicit, bounded.
- **backward**: transient, second order implicit, potentially unbounded.
- **CrankNicolson**: transient, second order implicit, bounded; requires an off-centering coefficient ψ where:

$$\psi = \begin{cases} 1 & \text{corresponds to pure CrankNicolson,} \\ 0 & \text{corresponds to Euler;} \end{cases}$$

generally $\psi = 0.9$ is used to stabilise the scheme for practical engineering problems.

- **localEuler**: pseudo transient for accelerating a solution to steady-state using local-time stepping; first order implicit.

Any second time derivative ($\partial^2/\partial t^2$) terms are specified in the *d2dt2Schemes* sub-dictionary. Only the **Euler** scheme is available for *d2dt2Schemes*.

4.5.2 Gradient schemes

The *gradSchemes* sub-dictionary contains gradient terms. The **default** discretisation scheme that is primarily used for gradient terms is:

```

default      Gauss linear;

```

The **Gauss** entry specifies the standard finite volume discretisation with Gaussian integration which requires the interpolation of values from cell centres to face centres. The interpolation scheme is then given by the **linear** entry, meaning linear interpolation or central differencing.

In some tutorial cases, particular involving poorer quality meshes, the discretisation of specific gradient terms is then overridden to improve boundedness and stability. The terms that are overridden in those cases are the velocity gradient

```

grad(U)      cellLimited Gauss linear 1;

```

and, less frequently, the gradient of turbulence fields, *e.g.*

```
grad(k)          cellLimited Gauss linear 1;
grad(epsilon)    cellLimited Gauss linear 1;
```

They use the `cellLimited` scheme which limits the gradient such that when cell values are extrapolated to faces using the calculated gradient, the face values do not fall outside the bounds of values in surrounding cells. A limiting coefficient is specified after the underlying scheme for which 1 guarantees boundedness and 0 applies no limiting; 1 is invariably used.

Other schemes that are rarely used are as follows.

- **leastSquares**: a second-order, least squares distance calculation using all neighbour cells.
- **Gauss cubic**: third-order scheme that appears in `solidDisplacement` and `dnsFoam` examples.

4.5.3 Divergence schemes

The *divSchemes* sub-dictionary contains divergence terms, *i.e.* terms of the form $\nabla \cdot \dots$, excluding Laplacian terms (of the form $\nabla \cdot \Gamma \nabla \Psi$). This includes both advection terms, *e.g.* $\nabla \cdot (\mathbf{U}k)$, where velocity $\mathbf{U}$ provides the advective flux, and other terms, that are often diffusive in nature, *e.g.* $\nabla \cdot \nu (\nabla \mathbf{U})^T$.

The fact that terms that are fundamentally different reside in one sub-dictionary means that the `default` scheme is generally set to `none` in *divSchemes*. The non-advective terms then generally use the `Gauss` integration with `linear` interpolation, *e.g.*

```
div(U)          Gauss linear;
```

The treatment of advective terms is one of the major challenges in CFD numerics and so the options are more extensive. The keyword identifier for the advective terms are usually of the form `div(phi, ...)`, where `phi` denotes the (volumetric) flux of velocity on the cell faces for constant-density flows and the mass flux for compressible flows, *e.g.* `div(phi,U)` for the advection of velocity, `div(phi,e)` for the advection of internal energy, *etc.* For advection of velocity, the user can run the `foamSearch` script to extract the `div(phi,U)` keyword from all tutorials.

```
foamSearch $FOAM_TUTORIALS fvSchemes "divSchemes/div(phi,U)"
```

The schemes are all based on `Gauss` integration, using the flux `phi` and the advected field being interpolated to the cell faces by one of a selection of schemes, *e.g.* `linear`, `linearUpwind`, *etc.* There is a `bounded` variant of the discretisation, discussed later.

Ignoring ‘V’-schemes (with keywords ending “V”), and rarely-used schemes such as `Gauss cubic` and `vanLeerV`, the interpolation schemes used in the tutorials are as follows.

- **linear**: second order, unbounded.

- **linearUpwind**: second order, upwind-biased, unbounded (but much less so than **linear**), that requires discretisation of the velocity gradient to be specified.
- **LUST**: blended 75% **linear**/ 25%**linearUpwind** scheme, that requires discretisation of the velocity gradient to be specified.
- **limitedLinear**: **linear** scheme that limits towards **upwind** in regions of rapidly changing gradient; requires a coefficient, where 1 is strongest limiting, tending towards **linear** as the coefficient tends to 0.
- **upwind**: first-order bounded, generally too inaccurate for velocity but more often used for transport of scalar fields.

Example syntax for these schemes is as follows.

```
div(phi,U)      Gauss linear;
div(phi,U)      Gauss linearUpwind grad(U);
div(phi,U)      Gauss LUST grad(U);
div(phi,U)      Gauss LUST unlimitedGrad(U);
div(phi,U)      Gauss limitedLinear 1;
div(phi,U)      Gauss upwind;
```

‘**V**’-schemes are specialised versions of schemes designed for vector fields. They differ from conventional schemes by calculating a single limiter which is applied to all components of the vectors, rather than calculating separate limiters for each component of the vector. The ‘**V**’-schemes’ single limiter is calculated based on the direction of most rapidly changing gradient, resulting in the strongest limiter being calculated which is most stable but arguably less accurate. Example syntax is as follows.

```
div(phi,U)      Gauss limitedLinearV 1;
div(phi,U)      Gauss linearUpwindV grad(U);
```

The **bounded** variants of schemes relate to the treatment of the material time derivative which can be expressed in terms of a spatial time derivative and convection, *e.g.* for field e in incompressible flow

$$\frac{De}{Dt} = \frac{\partial e}{\partial t} + \mathbf{U} \cdot \nabla e = \frac{\partial e}{\partial t} + \nabla \cdot (\mathbf{U}e) - (\nabla \cdot \mathbf{U})e \quad (4.1)$$

For numerical solution of incompressible flows, $\nabla \cdot \mathbf{U} = 0$ at convergence, at which point the third term on the right hand side is zero. Before convergence is reached, however, $\nabla \cdot \mathbf{U} \neq 0$ and in some circumstances, particularly steady-state simulations, it is better to include the third term within a numerical solution to help maintain boundedness of the solution variable and promote better convergence. The **bounded** variant of the **Gauss** scheme provides this, automatically including the discretisation of the third-term with the advection term. Example syntax is as follows, as seen in *fvSchemes* files for steady-state cases.

```
div(phi,U)      bounded Gauss limitedLinearV 1;
div(phi,U)      bounded Gauss linearUpwindV grad(U);
```

The schemes used for advection of scalar fields are similar to those for advection of velocity, although in general there is greater emphasis placed on boundedness than accuracy when selecting the schemes. For example, a search for schemes for advection of internal energy (*e*) reveals the following.

```
foamSearch $FOAM_TUTORIALS fvSchemes "divSchemes/div(phi,e)"

div(phi,e)      bounded Gauss upwind;
div(phi,e)      Gauss limitedLinear 1;
div(phi,e)      Gauss linearUpwind limited;
div(phi,e)      Gauss LUST grad(e);
div(phi,e)      Gauss upwind;
div(phi,e)      Gauss vanAlbada;
```

In comparison with advection of velocity, there are no cases set up to use **linear**. The **limitedLinear** and **upwind** schemes are commonly used, with the additional appearance of **vanLeer**, another limited scheme, with less strong limiting than **limitedLinear**.

There are specialised versions of the limited schemes for scalar fields that are commonly bounded between 0 and 1, *e.g.* the laminar flame speed regress variable *b*. A search for the discretisation used for advection in the laminar flame transport equation yields:

```
div(phiSt,b)    Gauss limitedLinear01 1;
```

The underlying scheme is **limitedLinear**, specialised for stronger bounding between 0 and 1 by adding 01 to the name of the scheme.

The **multivariateSelection** mechanism also exists for grouping multiple equation terms together, and applying the same limiters on all terms, using the strongest limiter calculated for all terms. A good example of this is in a set of mass transport equations for fluid species, where it is good practice to apply the same discretisation to all equations for consistency. The example below comes from the **smallPoolFire3D** tutorial in *\$FOAM_TUTORIALS/multi-compon/entFluid*, in which the equation for enthalpy *h* is included with the specie mass transport equations in the calculation of a single limiter.

```
div(phi,Yi_h)    Gauss multivariateSelection
{
    O2 limitedLinear01 1;
    CH4 limitedLinear01 1;
    N2 limitedLinear01 1;
    H2O limitedLinear01 1;
    CO2 limitedLinear01 1;
    h limitedLinear 1 ;
}
```

4.5.4 Surface normal gradient schemes

It is worth explaining the *snGradSchemes* sub-dictionary that contains surface normal gradient terms, before discussion of *laplacianSchemes*, because they are required to evaluate a

Laplacian term using Gaussian integration. A surface normal gradient is evaluated at a cell face; it is the component, normal to the face, of the gradient of values at the centres of the 2 cells that the face connects.

A search for the `default` scheme for *snGradSchemes* reveals the following entries.

```
default      corrected;
default      limited corrected 0.33;
default      limited corrected 0.5;
default      orthogonal;
default      uncorrected;
```

The basis of the gradient calculation at a face is to subtract the value at the cell centre on one side of the face from the value in the centre on the other side and divide by the distance. The calculation is second-order accurate for the gradient *normal to the face* if the vector connecting the cell centres is orthogonal to the face, *i.e.* they are at right-angles. This is the `orthogonal` scheme.

Orthogonality requires a regular mesh, typically aligned with the Cartesian co-ordinate system, which does not generally occur with real world, engineering geometries. Therefore, to maintain second-order accuracy, an explicit non-orthogonal correction can be added to the orthogonal component, known as the `corrected` scheme. The correction increases in size as the non-orthogonality, *i.e.* the angle α between the cell-cell vector and face normal vector, increases.

As α tends towards 90° , typically beyond 75° , the explicit correction can be so large to cause a solution to go unstable. The solution can be stabilised by applying the `limited` scheme to the correction which requires a coefficient ψ , $0 \leq \psi \leq 1$ where

$$\psi = \begin{cases} 0 & \text{corresponds to } \texttt{uncorrected}, \\ 0.333 & \text{non-orthogonal correction} \leq 0.5 \times \text{orthogonal part}, \\ 0.5 & \text{non-orthogonal correction} \leq \text{orthogonal part}, \\ 1 & \text{corresponds to } \texttt{corrected}. \end{cases} \quad (4.2)$$

Typically, *psi* is chosen to be 0.33 or 0.5, where 0.33 offers greater stability and 0.5 greater accuracy.

The `corrected` scheme applies under-relaxation in which the implicit orthogonal calculation is increased by $\cos^{-1}\alpha$, with an equivalent boost within the non-orthogonal correction. The `uncorrected` scheme is equivalent to the `corrected` scheme, without the non-orthogonal correction, so is like `orthogonal` but with the additional $\cos^{-1}\alpha$ under-relaxation.

Generally the `uncorrected` and `orthogonal` schemes are only recommended for meshes with very low non-orthogonality (*e.g.* maximum 5°). The `corrected` scheme is generally recommended, but for maximum non-orthogonality above 75° , `limited` may be required. At non-orthogonality above 85° , convergence is generally hard to achieve.

4.5.5 Laplacian schemes

The *laplacianSchemes* sub-dictionary contains Laplacian terms. A typical Laplacian term is $\nabla \cdot (\nu \nabla \mathbf{U})$, the diffusion term in the momentum equations, which corresponds to the keyword

`laplacian(nu,U)` in *laplacianSchemes*. The **Gauss** scheme is the only choice of discretisation and requires a selection of both an interpolation scheme for the diffusion coefficient, *i.e.* ν in our example, and a surface normal gradient scheme, *i.e.* $\nabla\mathbf{U}$. To summarise, the entries required are:

```
Gauss <interpolationScheme> <snGradScheme>
```

The user can search for the **default** scheme for *laplacianSchemes* in all the cases in the *\$FOAM_TUTORIALS* directory.

```
foamSearch $FOAM_TUTORIALS fvSchemes laplacianSchemes/default
```

It reveals the following entries.

```
default      Gauss linear corrected;
default      Gauss linear limited corrected 0.33;
default      Gauss linear limited corrected 0.5;
default      Gauss linear orthogonal;
default      Gauss linear uncorrected;
```

In all cases, the **linear** interpolation scheme is used for interpolation of the diffusivity. The cases uses the same array of **snGradSchemes** based on the maximum non-orthogonality in the mesh, as described in section 4.5.4.

4.5.6 Interpolation schemes

The *interpolationSchemes* sub-dictionary contains terms that are interpolations of values typically from cell centres to face centres, primarily used in the interpolation of velocity to face centres for the calculation of flux **phi**. There are numerous interpolation schemes in OpenFOAM, but a search for the **default** scheme in all the tutorial cases reveals that **linear** interpolation is used in almost every case, except for one stress analysis example which uses **cubic** interpolation.

4.6 Solution and algorithm control

Once a matrix equation is constructed according to the schemes in the previous section, a linear solver is applied. A set of equations is also framed within an algorithm containing loops and controls. For information about the main algorithms and solvers in OpenFOAM, refer to Chapter 5 of *Notes on Computational Fluid Dynamics: General Principles*.

The controls for algorithms and solvers are found in the *fvSolution* dictionary in the *system* directory. Below is an example set of entries from the *fvSolution* dictionary for a case using the *incompressibleFluid* modular solver.

```
16
17 solvers
18 {
19     p
20     {
```



```

21         solver          GAMG;
22         tolerance       1e-7;
23         relTol          0.01;
24
25         smoother        DICGaussSeidel;
26     }
27
28     pFinal
29     {
30         $p;
31         relTol          0;
32     }
33
34     "(U|k|epsilon)"
35     {
36         solver          smoothSolver;
37         smoother        symGaussSeidel;
38         tolerance       1e-05;
39         relTol          0.1;
40     }
41
42     "(U|k|epsilon)Final"
43     {
44         $U;
45         relTol          0;
46     }
47 }
48
49 PIMPLE
50 {
51     nNonOrthogonalCorrectors 0;
52     nCorrectors              2;
53 }
54
55
56
57 // *****

```

fvSolution contains a set of sub-dictionaries, described in the remainder of this section that includes: *solvers*; *relaxationFactors*; and, *SIMPLE* for steady-state cases or *PIMPLE* for transient or pseudo-transient cases.

4.6.1 Linear solver control

The first sub-dictionary in our example is *solvers*. It specifies each linear solver that is used for each discretised equation; here, the term *linear* solver refers to the method of number-crunching to solve a matrix equation, as opposed to an *modular* solver, such as *incompressibleFluid* which describes the entire set of equations and algorithms to solve a particular problem. The term ‘linear solver’ is abbreviated to ‘solver’ in much of what follows; hopefully the context of the term avoids any ambiguity.

The syntax for each entry within *solvers* starts with a keyword that for the variable being solved in the particular equation. For example, *incompressibleFluid* solves equations for pressure *p*, velocity *U* and often turbulence fields, hence the entries for *p*, *U*, *k* and *epsilon*. The keyword introduces a sub-dictionary containing the type of solver and the parameters that the solver uses. The solver is selected through the *solver* keyword from the options listed below. The parameters, including *tolerance*, *relTol*, *preconditioner*, *etc.* are described in following sections.

- **PCG/PBiCGStab**: Stabilised preconditioned (bi-)conjugate gradient, for both symmetric and asymmetric matrices.

- **PCG/PBiCG**: preconditioned (bi-)conjugate gradient, with **PCG** for symmetric matrices, **PBiCG** for asymmetric matrices.
- **smoothSolver**: solver that uses a smoother.
- **GAMG**: generalised geometric-algebraic multi-grid.
- **diagonal**: diagonal solver for explicit systems.

The solvers distinguish between symmetric matrices and asymmetric matrices. The symmetry of the matrix depends on the terms of the equation being solved, *e.g.* time derivatives and Laplacian terms form coefficients of a symmetric matrix, whereas an advective derivative introduces asymmetry. If the user specifies a symmetric solver for an asymmetric matrix, or vice versa, an error message will be written to advise the user accordingly, *e.g.*

```
--> FOAM FATAL IO ERROR : Unknown asymmetric matrix solver PCG
Valid asymmetric matrix solvers are :
4
(
  GAMG
  PBiCG
  PBiCGStab
  smoothSolver
)
```

4.6.2 Solution tolerances

The finite volume method generally solves equations in a segregated, decoupled manner, meaning that each matrix equation solves for only one variable. The matrices are consequently sparse, meaning they predominately include coefficients of 0. The solvers are generally iterative, *i.e.* they are based on reducing the equation residual over successive solutions. The residual is ostensibly a measure of the error in the solution so that the smaller it is, the more accurate the solution. More precisely, the residual is evaluated by substituting the current solution into the equation and taking the magnitude of the difference between the left and right hand sides; it is also normalised to make it independent of the scale of the problem being analysed.

Before solving an equation for a particular field, the initial residual is evaluated based on the current values of the field. After each solver iteration the residual is re-evaluated. The solver stops if *any one* of the following conditions are reached:

- the residual falls below the *solver tolerance*, **tolerance**;
- the ratio of current to initial residuals falls below the *solver relative tolerance*, **relTol**;
- the number of iterations exceeds a *maximum number of iterations*, **maxIter**;

The solver tolerance should represent the level at which the residual is small enough that the solution can be deemed sufficiently accurate. The solver relative tolerance limits the

relative improvement from initial to final solution. In transient simulations, it is usual to set the solver relative tolerance to 0 to force the solution to converge to the solver tolerance in each time step. The tolerances, `tolerance` and `relTol` must be specified in the dictionaries for all solvers; `maxIter` is optional and defaults to a value of 1000.

Equations are very often solved multiple times within one solution step, or time step. For example, when using the PIMPLE algorithm, a pressure equation is solved according to the number specified by `nCorrectors`, as described in section 4.6.7. Where this occurs, the solver is very often set up to use different settings when solving the particular equation for the final time, specified by a keyword that adds `Final` to the field name. For example, in the transient `pitzDaily` example for the `incompressibleFluid` solver, the solver settings for pressure are as follows.

```
p
{
    solver          GAMG;
    tolerance       1e-07;
    relTol          0.01;
    smoother        DICGaussSeidel;
}

pFinal
{
    $p;
    relTol          0;
}
```

If the case is specified to solve pressure 4 times within one time step, then the first 3 solutions would use the settings for `p` with `relTol` of 0.01, so that the cost of solving each equation is relatively low. Only when the equation is solved the final (4th) time, it solves to a residual level specified by `tolerance` (since `relTol` is 0, effectively deactivating it) for greater accuracy, but at greater cost.

4.6.3 Preconditioned conjugate gradient solvers

There are a range of options for preconditioning of matrices in the conjugate gradient solvers, represented by the `preconditioner` keyword in the solver dictionary, listed below. Note that the DIC/DILU preconditioners are exclusively specified in the tutorials in OpenFOAM.

- DIC/DILU: diagonal incomplete-Cholesky (symmetric) and incomplete-LU (asymmetric)
- FDIC: slightly faster diagonal incomplete-Cholesky (DIC with caching, symmetric)
- GAMG: geometric-algebraic multi-grid.
- `diagonal`: diagonal preconditioning, not generally used.
- `none`: no preconditioning.

4.6.4 Smooth solvers

The solvers that use a smoother require the choice of smoother to be specified. The smoother options are listed below. The `symGaussSeidel` and `GaussSeidel` smoothers are preferred in the tutorials.

- `GaussSeidel`: Gauss-Seidel.
- `symGaussSeidel`: symmetric Gauss-Seidel.
- `DIC/DILU`: diagonal incomplete-Cholesky (symmetric), incomplete-LU (asymmetric).
- `DICGaussSeidel`: diagonal incomplete-Cholesky/LU with Gauss-Seidel (symmetric/-asymmetric).

When using the smooth solvers, the user can optionally specify the number of sweeps, by the `nSweeps` keyword, before the residual is recalculated. Without setting it, it reverts to a default value of 1.

4.6.5 Geometric-algebraic multi-grid solvers

The geometric-algebraic multi-grid (GAMG) method uses the principle of: generating a quick solution on a mesh with a small number of cells; mapping this solution onto a finer mesh; using it as an initial guess to obtain an accurate solution on the fine mesh. GAMG is faster than standard methods when the increase in speed by solving first on coarser meshes outweighs the additional costs of mesh refinement and mapping of field data. In practice, GAMG starts with the original mesh and coarsens/refines the mesh in stages. The coarsening is limited by a specified minimum number of cells at the most coarse level.

The agglomeration of cells is performed by the method specified by the `agglomerator` keyword. The tutorials all use the default `faceAreaPair` method. The agglomeration can be controlled using the following optional entries, most of which default in the tutorials.

- `cacheAgglomeration`: switch specifying caching of the agglomeration strategy (default `true`).
- `nCellsInCoarsestLevel`: approximate mesh size at the most coarse level in terms of the number of cells (default 10).
- `directSolveCoarset`: use a direct solver at the coarsest level (default `false`).
- `mergeLevels`: keyword controls the speed at which coarsening or refinement is performed; the default is 1, which is safest, but for simple meshes, the solution speed can be increased by coarsening/refining 2 levels at a time, *i.e.* setting `mergeLevels` 2.

Smoothing is specified by the `smoother` as described in section 4.6.4. The number of sweeps used by the smoother at different levels of mesh density are specified by the following optional entries.

- `nPreSweeps`: number of sweeps as the algorithm is coarsening (default 0).

- **preSweepsLevelMultiplier**: multiplier for the number of sweeps between each coarsening level (default 1).
- **maxPreSweeps**: maximum number of sweeps as the algorithm is coarsening (default 4).
- **nPostSweeps**: number of sweeps as the algorithm is refining (default 2).
- **postSweepsLevelMultiplier**: multiplier for the number of sweeps between each refinement level (default 1).
- **maxPostSweeps**: maximum number of sweeps as the algorithm is refining (default 4).
- **nFinestSweeps**: number of sweeps at finest level (default 2).

4.6.6 Solution under-relaxation

The *fvSolution* file usually includes a *relaxationFactors* sub-dictionary which controls under-relaxation. Under-relaxation is used to improve stability of a computation, particularly for steady-state problems. It works by limiting the amount which a variable changes from one iteration to the next, either by manipulating the matrix equation prior to solving for a field, or by modifying the field afterwards. An under-relaxation factor α , $0 < \alpha \leq 1$ specifies the amount of under-relaxation, as described below.

- No specified α : no under-relaxation.
- $\alpha = 1$: guaranteed matrix diagonal equality/dominance.
- α decreases, under-relaxation increases.
- $\alpha = 0$: solution does not change with successive iterations.

An optimum choice of α is one that is small enough to ensure stable computation but large enough to move the iterative process forward quickly; values of α as high as 0.9 can ensure stability in some cases and anything much below, say, 0.2 can be prohibitively restrictive in slowing the iterative process.

Relaxation factors for under-relaxation of fields are specified within a *field* sub-dictionary; relaxation factors for equation under-relaxation are within a *equations* sub-dictionary. An example is shown below from a case with the *incompressibleFluid* solver module running in steady-state mode. The factors are specified for pressure *p*, pressure *U*, and turbulent fields grouped using a regular expression.

```

54
55 relaxationFactors
56 {
57     fields
58     {
59         p                0.3;
60     }
61     equations
62     {
63         U                0.7;
64         "(k|omega|epsilon).*" 0.7;
65     }
66 }
67
68 // *****
```

For a transient case with the `incompressibleFluid` module, under-relaxation would simply slow convergence of the solution within each time step. Instead, the following setting is generally adopted to ensure diagonal equality which is generally required for convergence of a matrix equation.

```

60 relaxationFactors
61 {
62     equations
63     {
64         "." "*"          1;
65     }
66 }
67
68
69 // ***** //
```

4.6.7 SIMPLE and PIMPLE algorithms

Most fluid solver modules use algorithms to couple equations for mass and momentum conservation known as:

- SIMPLE (semi-implicit method for pressure-linked equations);
- PIMPLE, a combination of PISO (pressure-implicit split-operator) and SIMPLE.

Within in time, or solution, step, the algorithms solve a pressure equation, to enforce mass conservation, with an explicit correction to velocity to satisfy momentum conservation. They optionally begin each step by solving the momentum equation — the so-called momentum predictor.

While all the algorithms solve the same governing equations (albeit in different forms), the algorithms principally differ in how they loop over the equations. The looping is controlled by input parameters that are listed below. They are set in a dictionary named after the algorithm, *i.e.* *SIMPLE*, or *PIMPLE*.

- **nCorrectors**: used by PIMPLE, sets the number of times the algorithm solves the pressure equation and momentum corrector in each step; typically set to 2 or 3.
- **nNonOrthogonalCorrectors**: used by all algorithms, specifies repeated solutions of the pressure equation, used to update the explicit non-orthogonal correction, described in section 4.5.4, of the Laplacian term $\nabla \cdot ((1/A)\nabla p)$; typically set to 0 for steady-state and 1 for transient cases.
- **nOuterCorrectors**: used by PIMPLE, it enables looping over the entire system of equations within on time step, representing the total number of times the system is solved; must be ≥ 1 and is typically set to 1.
- **momentumPredictor**: switch that controls solving of the momentum predictor; typically set to `off` for some flows, including low Reynolds number and multiphase.

4.6.8 Pressure referencing

In a closed incompressible system, pressure is relative: it is the pressure range that matters not the absolute values. In these cases, the solver sets a reference level of `pRefValue` in cell `pRefCell`. These entries are generally stored in the *SIMPLE* or *PIMPLE* sub-dictionary.

4.7 Case management tools

There are a set of applications and scripts that help with managing case files and help the user find and set keyword data entries in case files. The tools are described in the following sections.

4.7.1 General file management

There are some tools for general management of case files, including `foamListTimes`, `foamCleanCase` and `foamCloneCase`. A case includes configuration files for various processes such as meshing, case initialisation, simulation and post-processing. Each process generates new data files in various directories, *e.g.* mesh data is stored in `constant/polyMesh`, CFD results in time directories, and further post-processing in a `postProcessing` directory.

The `foamListTimes` utility lists the time directories for a case, omitting the `0` directory by default. Prior to re-running a CFD simulation, it can be useful to delete the results from the previous simulation. The `foamListTimes` utility provides this function through the `-rm` option which deletes the listed time directories, executed by the following command.

```
foamListTimes -rm
```

The `foamCleanCase` script aims to reset the case files to their original state, removing all files generated during a workflow including the meshing, post-processing. It deletes directories including: `postProcessing` and `VTK`; the `constant/polyMesh` directory; `processor` directories from parallel decomposition; and, `dynamicCode` for run-time compiled code. The script is simply run as follows.

```
foamCleanCase
```

The `foamCloneCase` script creates a new case, by copying the `0`, `system` and `constant` directories from an existing case. The copied case is ready to run since the mesh is copied through the `constant` directory. If the original case is set up to run in parallel, the `processor` directories can also be copied using the `-processor` option. The basic command is executed simply as follows, where `oldCase` refers to an existing case directory.

```
foamCloneCase oldCase newCase
```

4.7.2 The foamDictionary utility

The `foamDictionary` utility offers several options for printing, editing and adding keyword entries in case files. The utility is executed with a case dictionary file as an argument, *e.g.* from within a case directory on the `fvSchemes` file.

```
foamDictionary system/fvSchemes
```

Without options, the utility prints the entries from the file, removing comments, *e.g.* as follows for the `fvSchemes` file in the `pitzDailySteady` tutorial case.

```

FoamFile
{
    format            ascii;
    class              dictionary;
    location           "system";
    object             fvSchemes;
}

ddtSchemes
{
    default           steadyState;
}

gradSchemes
{
    default           Gauss linear;
    grad(U)           cellLimited Gauss linear 1;
}

divSchemes
{
    default           none;
    div(phi,U)        bounded Gauss linearUpwind grad(U);
    turbulence         bounded Gauss limitedLinear 1;
    div(phi,k)         $turbulence;
    div(phi,epsilon)   $turbulence;
    div(phi,omega)     $turbulence;
    div(phi,v2)        $turbulence;
    div((nuEff*dev2(T(grad(U)))) Gauss linear;
    div(nonlinearStress) Gauss linear;
}

laplacianSchemes
{
    default           Gauss linear corrected;
}

interpolationSchemes
{
    default           linear;
}

snGradSchemes
{
    default           corrected;
}

```

The output includes the macros before expansion, indicated by the \$ symbol, *e.g.* \$turbulence. The macros can be expanded by the `-expand` option as shown below

```
foamDictionary -expand system/fvSchemes
```

The `-entry` option prints the entry for a particular keyword, expanding the macros by default, *e.g.* `divSchemes` in the example below

```
foamDictionary -entry divSchemes system/fvSchemes
```

The example clearly extracts the `divSchemes` dictionary.

```

divSchemes
{
    default           none;
    div(phi,U)        bounded Gauss linearUpwind grad(U);
}

```



```

turbulence      bounded Gauss limitedLinear 1;
div(phi,k)      $turbulence;
div(phi,epsilon) $turbulence;
div(phi,omega)  $turbulence;
div(phi,v2)     $turbulence;
div((nuEff*dev2(T(grad(U)))) Gauss linear;
div(nonlinearStress) Gauss linear;
}

```

The “/” syntax allows access to keywords with levels of sub-dictionary. For example, the `div(phi,U)` keyword can be accessed within the `divSchemes` sub-dictionary by the following command.

```
foamDictionary -entry "divSchemes/div(phi,U)" system/fvSchemes
```

The example returns the single `divSchemes/div(phi,U)` entry.

```
div(phi,U)      bounded Gauss linearUpwind grad(U);
```

The `-value` option prints only the entry.

```
foamDictionary -entry "divSchemes/div(phi,U)" -value system/fvSchemes
```

The example removes the keyword and terminating semicolon, leaving just the data.

```
bounded Gauss linearUpwind grad(U)
```

The `-keywords` option prints only the keywords.

```
foamDictionary -entry divSchemes -keywords system/fvSchemes
```

The example produces a list of keywords inside the `divSchemes` dictionary.

```

default
div(phi,U)
div(phi,k)
div(phi,epsilon)
div(phi,omega)
div(phi,v2)
div((nuEff*dev2(T(grad(U))))
div(nonlinearStress)

```

`foamDictionary` can set entries with the `-set` option. If the user wishes to change the `div(phi,U)` to the upwind scheme, they can enter the following.

```
foamDictionary -entry "divSchemes/div(phi,U)" \
-set "bounded Gauss upwind" system/fvSchemes
```

An alternative “=” syntax can be used with the `-set` option which is particularly useful when modifying multiple entries:

```
foamDictionary -set "startFrom=startTime, startTime=0" \
system/controlDict
```

`foamDictionary` can add entries with the `-add` option. If the user wishes to add an entry named `turbulence` to `divSchemes` with the upwind scheme, they can enter the following.

```
foamDictionary -entry "divSchemes/turbulence" \
-add "bounded Gauss upwind" system/fvSchemes
```

4.7.3 The foamSearch script

The `foamSearch` script, demonstrated extensively in section 4.5, uses `foamDictionary` to extract and sort keyword entries from all files of a specified name in a specified dictionary. The `-c` option counts the number of entries of each type, *e.g.* the user could search for the choice of `solver` for the `p` equation in all the *fvSolution* files in the tutorials.

```
foamSearch -c $FOAM_TUTORIALS fvSolution solvers/p/solver
```

The search shows **GAMG** to be the most common choice in all the tutorials.

```
64 solver      GAMG;
 2 solver      PBiCGStab;
26 solver      PCG;
 5 solver      smoothSolver;
```

4.7.4 The foamGet script

The `foamGet` script copies configuration files into a case quickly and conveniently. The user must be inside a case directory to run the script or identify the case directory with the `-case` option. Its operation can be described using the `pitzDailySteady` case described in section 2.1. The example begins by copying the case directory as follows:

```
run
cp -r $FOAM_TUTORIALS/modules/incompressibleFluid/pitzDailySteady .
```

The mesh is generated for the case by going into the case directory and running `blockMesh`:

```
cd pitzDailySteady
blockMesh
```

The user might decide before running the simulation to configure some automatic post-processing as described in section 7.2. The user can list the pre-configured function objects by the following command.

```
foamPostProcess -list
```

From the output, the user could select the `patchFlowRate` function to monitor the flow rate at the outlet patch. The *patchFlowRate* configuration file can be copied into the *system* directory using `foamGet`:

```
foamGet patchFlowRate
```

In order to monitor the flow through the `outlet` patch, the `patch` entry in *patchFlowRate* file should be set as follows:

```
patch    outlet;
```

The *patchFlowRate* configuration is then included in the case by adding to a *functions* file *system* directory, as discussed in section 7.2.

```
functions
{
    ...
    //#includeFunc writeObjects(kEpsilon:G) // existing entry
    #includeFunc patchFlowRate
}
```

4.7.5 The foamInfo script

The *foamInfo* script provides quick information and examples relating to thing in OpenFOAM that the user can “select”. The selections include models (including boundary conditions and packaged function objects), solver modules, applications and scripts. For example, *foamInfo* prints information about the *incompressibleFluid* solver module by typing the following:

```
foamInfo incompressibleFluid
```

Information for the *flowRateInletVelocity* boundary condition can similarly be obtained by typing the following command.

```
foamInfo flowRateInletVelocity
```

The output includes: the location of the source code header file for this boundary condition; the description and usage details from the header file; a list of other models of the same type, *i.e.* other boundary conditions; and, a list of example cases that use the boundary condition. This example is demonstrated in section 2.1.17.

When the user requests information about a model with *foamInfo*, it attempts to provide a list of other models in the “family”. For example, if the user requests information about the *kEpsilon* turbulence model by

```
foamInfo kEpsilon
```

the output includes the following

```
Model
```

```
This appears to be the 'kEpsilon' model of the 'RAS' family.
```

```
The models in the 'RAS' family are:
```

```
+ kEpsilon
```

```
+ kOmega
```

```
+ kOmega2006
```

```

+ kOmegaSST
+ kOmegaSSTLM
+ kOmegaSSTSAS
+ LaunderSharmaKE
+ LRR
+ RAS
+ realizableKE
+ RNGkEpsilon
+ SpalartAllmaras
+ SSG
+ v2f

```

It provides fairly complete lists for `fvModels` and `fvConstraints`, which can be demonstrated by typing the following commands.

```

foamInfo clouds
foamInfo limitTemperature

```

It also lists options used in input files, *e.g.* for `Function1` entries in boundary conditions, `searchableSurface` entries in the configuration of `snappyHexMesh`. Users could try searching for specific models within those families or the families themselves, *e.g.*

```

foamInfo linearRamp
foamInfo triSurfaceMesh
foamInfo Function1
foamInfo searchableSurfaces

```

4.7.6 The foamToC utility

The `foamToC` utility lists all the options in OpenFOAM which the user can select through input files. The functionality overlaps with `foamInfo` to some extent but `foamToC` produces more definitive reporting since it is an OpenFOAM application which directly interrogates the run-time selection tables in the compiled libraries. The “ToC” in the name is an abbreviation for “Table of Contents.”

As well as providing general options to interrogate anything in OpenFOAM, `foamToC` includes specific options that replicate most of the “`-list...`” options included in application solvers prior to v11. These options included: `-listScalarBCs` and `-listVectorBCs` to list boundary conditions; `-listFunctionObjects` to list `functionObjects`; `-listFvModels` to list `fvModels`; and, `-listFvConstraints` to list `fvConstraints`. The equivalent options in `foamToC` are listed below, with an additional `-solvers` option:

- `-scalarBCs` and `-vectorBCs` to list boundary conditions;
- `-functionObjects` to list `functionObjects`;
- `-fvModels` to list `fvModels`;

- `-fvConstraints` to list `fvConstraints`; and,
- `-solvers` to list the solver modules.

For example, with the last option, `foamToC` prints the following output

```
>> foamToC -fvConstraints
```

```
fvConstraints:
```

```
Contents of table fvConstraint:
```

bound	libfvConstraints.so
fixedTemperatureConstraint	libfvConstraints.so
fixedValueConstraint	libfvConstraints.so
limitMag	libfvConstraints.so
limitPressure	libfvConstraints.so
limitTemperature	libfvConstraints.so
meanVelocityForce	libfvConstraints.so
patchMeanVelocityForce	libfvConstraints.so
zeroDimensionalFixedPressure	libfvConstraints.so

Each `fvConstraint` is listed in the left column with the library to which it belongs in the right column. The options listed above essentially invoke the more general `-table` option that lists the contents of a run-time selection table. The `-fvConstraints` option is equivalent to the following command which lists the items in the `fvConstraint` table.

```
foamToC -table fvConstraint
```

All the selection tables in OpenFOAM are listed by running `foamToC` with the `-tables` option as shown below.

```
foamToC -tables
```

This is also the default response of `foamToC` without options, *i.e.*

```
foamToC
```

By default `foamToC` loads **all** the libraries in OpenFOAM to produce a complete list of tables. The user can control the libraries that are loaded with the following options.

- `-noLibs`: only loads the core `libOpenFOAM.so` library by default.
- `-solver <solver>`: only loads libraries associated with the specified solver module `<solver>`.
- `-libs '("lib1.so" ... "libN.so")'`: *additionally* pre-loads specified libraries, *e.g.* customised libraries of the user.

An important use of `foamToC` is to enable users to find alternative models to the one currently configured for their case. The challenge is to find the table that contains the models they wish to list. The `-search` option helps with this, since it takes an entry, *e.g.* a model, and reports the table in which it belongs. For example, if the user was familiar with the `BirdCarreau` model for viscosity and wished to list alternative non-Newtonian models, they could first issue the following command.

```
>> foamToC -search BirdCarreau
```

BirdCarreau is in table

```
generalisedNewtonianViscosityModel  libmomentumTransportModels.so
```

Having identified the table, the user can then list its contents using the `-table` option.

```
>> foamToC -table generalisedNewtonianViscosityModel
```

Contents of table `generalisedNewtonianViscosityModel`:

BirdCarreau	libmomentumTransportModels.so
Casson	libmomentumTransportModels.so
CrossPowerLaw	libmomentumTransportModels.so
HerschelBulkley	libmomentumTransportModels.so
Newtonian	libmomentumTransportModels.so
powerLaw	libmomentumTransportModels.so
strainRateFunction	libmomentumTransportModels.so

If the user wished to list the solver modules, they can run

```
foamToC -solvers
```

which is equivalent to running

```
foamToC -table solver
```

With turbulence models, a search for `kEpsilon` lists the following set of tables (the corresponding libraries are not shown here).

`kEpsilon` is in table

```
RAS
```

```
RAScompressibleMomentumTransportModel
RASincompressibleMomentumTransportModel
RASphaseCompressibleMomentumTransportModel
RASphaseIncompressibleMomentumTransportModel
```

The output indicates the model exists in different tables corresponding to both incompressible and compressible flows, and both single phase and multi-phase flows (`phase` indicates multi-phase in the table name). For single phase, incompressible flows, the available RAS turbulence models can be listed as follows.

```
foamToC -table RASincompressibleMomentumTransportModel
```

4.7.7 The foamUnits utility

In OpenFOAM v12, a `foamUnits` *script* was introduced which provides details of named units and dimensional units. In OpenFOAM v14, `foamUnits` became a utility application which is more functional especially for reporting factors used in unit conversion.

`foamUnits` helps to use named dimensions and units in input files, described in section 4.2.6) and 4.2.7, respectively. The simplest use of `foamUnits` involves running it without options to list both the named dimensions and units, as follows.

```
foamUnits
```

The user can then select the name of a dimension or unit from the list and obtain further information by adding the name as an argument to `foamUnits`. For example, for the *dimensions* of [power],

```
foamUnits power
```

returns the fundamental dimensions in descriptive and exponent form and standard units that match those dimensions.

```
Dimension [power]
+ Dimensions = [mass length^2 time^-3]
+ Exponents = [1 2 -3 0 0 0 0]
+ Standard Units = [kg m^2 s^-3] [W]
```

And for the *unit* of calorie [cal],

```
foamUnits cal
```

returns the dimensions of the unit, the standard units and the conversion factor to those units:

```
Unit [cal]
+ Dimensions = [mass length^2 time^-2]
+ Standard Unit = [kg m^2 s^-2]
+ Conversion Factor = 4.184
```

Should the user wish to do so, details for *all* the named dimensions and units can be printed by applying the `-all` option to `foamUnits`, *i.e.*

```
foamUnits -all
```

`foamUnits` can also provide conversion factors both ways between two units with the same dimensions. To do so, the user specifies the two sets of units as arguments to `foamUnits`. For example, the following command

```
foamUnits cal "lbf*m"
```

reports the conversion factors between two energies, [cal] and [lbf m] (pound force metres), the latter being an unconventional mixture of units for from different systems (USCS and SI).

```
Units [cal] [lbf*m]
+ Dimensions = [mass length^2 time^-2]
+ Standard Unit = [kg m^2 s^-2]
+ Conversion: 1 [cal] = 0.940601 [lbf*m]
+ Conversion: 1 [lbf*m] = 1.06315 [cal]
```

Finally, the `-value` option prints only the conversion factor with units. For a unit comparison it prints only the factor from the first to second unit, *e.g.*

```
foamUnits -value cal "lbf*m"
```

returns 0.940601. With a single unit argument, it prints the conversion factor to the base units, *e.g.*

```
foamUnits -value cal
```

returns 4.184.

4.7.8 The foamFind script

The `foamFind` script searches for files in OpenFOAM. It can simply provide the location of files by printing their filename including the path. Alternatively it can print the entire file into the terminal, print lines matching a search string or open the file in a text editor. The script is principally used to locate source code files (so perhaps does not fall much under “Case Management”, the title of this section of the guide).

For example, if a user wished to locate the file *wallHeatFlux.C*, they could simply type

```
foamFind wallHeatFlux.C
```

This would simply return the location of the file, *e.g.*

```
File: $FOAM_SRC//functionObjects/field/wallHeatFlux/wallHeatFlux.C
```

The `-print` option would additionally print the contents of the file to the terminal; the argument positioning is flexible:

```
foamFind -print wallHeatFlux.C
foamFind wallHeatFlux.C -print # Alternatively
```

To view the printed file slowly, the user could pipe the output to the UNIX `less` command

```
foamFind wallHeatFlux.C -print | less
```


`less` provides some keystroke to scroll, search and quit:

- click the space bar, page up, page down keys to scroll;
- enter line number after “:” to jump to a line;
- enter “/” followed by a string to search for the string, *e.g.* `/calc` to search for “calc”;
- enter “q” to quit.

A file can also be opened in a file editor using the `-editor` option. For this option to work, the user’s environment must include an `$EDITOR` environment variable set to the user’s editor of choice. For example, to set the `gedit` editor, the user could add the following to their `$HOME/.bashrc` file:

```
export EDITOR=gedit
```

The following command would then open the `wallHeatFlux.C` file in the `gedit` editor:

```
foamFind wallHeatFlux.C -edit
```

The `foamFind` script includes the `-search` option to print lines of a file matching a string, *e.g.* to search “heat”:

```
foamFind wallHeatFlux.C -search heat
```

The `-isearch` option make the search case insensitive, *e.g.* so would also match “Heat” (upper case) in the example.

```
foamFind wallHeatFlux.C -isearch heat
```

The `-numbers` option writes output with line numbers, *e.g.*

```
foamFind wallHeatFlux.C -isearch heat -numbers
```

The source code is ultimately compiled with `wmake` as described in section 3.2. The instructions for `wmake` are included in the *files* and *options* files in an associated *Make* directory. The `foamFind` script include two options, `-files` and `-options`, which return the *files* and *options* files, respectively, associated with the source code file. The following command would locate the *options* file associated with `wallHeatFlux`:

```
foamFind wallHeatFlux.C -options
```

By default, `foamFind` is searching for source code files so begins its search from the `$FOAM_SRC` directory within the OpenFOAM installation. Alternative directories for the search can be specified with the following options:

- `-applications`: the `$FOAM_APP` (*applications*) directory;

- `-modules`: the `$FOAM_MODULES` (*modules*) directory;
- `-tutorial`: the `$FOAM_TUTORIALS` (*tutorials*) directory;

The `-dir` option also allows the user to specify any other directory.

The `foamFind` script could find multiple files of the same name in a single search, *e.g.* looking for `createFields.H` in the `$FOAM_APP` directory by:

```
foamFind -applications createFields.H -print
```

This returns a selection of over 20 files, with each one numbered. The user can then enter the number of one particular file, or `ENTER` to print them all.

4.7.9 The foamMergeCase script

There is often a need to run several CFD simulations which are generally similar but include small differences. The `foamMergeCase` script is designed to help maintain cases in these circumstances. It operates on the basis that there first exists a working “source” case from which “variant” cases are created.

A variant case only contains files which include modifications from those in the source case, organised in the same directory structure. The files are given a special `.orig` extension and only need to include the modified entries. The `foamMergeCase` script can then be executed from within the variant case directory with the source case directory provided as an argument.

There is an accompanying `foamUnMergeCase` script which removes files that the `foamMergeCase` scripts copies from the source case. The variant case should **include their own** `Allrun` and `Allclean` scripts which include the respective `foamMergeCase` and `foamUnMergeCase` commands.

Let us provide an example that uses the `pitzDaily` case as the source. The variant case, `pitzDailyWater`, will be identical to `pitzDaily` except that it will run with a kinematic viscosity of $\nu = 10^{-6} \text{ m}^2 \text{ s}^{-1} = 1 \text{ cSt}$ (centistokes).

The first step of setting up a variant case is to copy the source case and then delete directories and files that remain unchanged during the merge. In our example, we can start from the `run` directory using the `run` alias, copy the `pitzDaily` case, renaming it `pitzDailyWater`, and then change into that directory.

```
run
cp -r $FOAM_TUTORIALS/incompressibleFluid/pitzDaily pitzDailyWater
cd pitzDailyWater
```

The user should then delete the directories and files that are not going to be merged, as follows.

```
rm -rf 0 system constant/momentumTransport
```

The kinematic viscosity is configured by the `nu` in the *physicalProperties* file in the *constant* directory. The modification to `nu` must be stored in a file named *physicalProperties.orig* in the corresponding directory. Therefore, the user should first rename *physicalProperties* to *physicalProperties.orig*.

```
mv constant/physicalProperties constant/physicalProperties.orig
```

The *physicalProperties.orig* should then be opened in an editor. The keyword entry for `viscosityModel` should be deleted. The entry for `nu` should be changed to the following (using `cSt` units described in section 4.7.7).

```
nu          1 [cSt];
```

The current case directory is now the complete variant case to the *pitzDaily* source case. A simulation can now be run by first merging in the *pitzDaily* source case using `foamMergeCase` as follows.

```
foamMergeCase $FOAM_TUTORIALS/incompressibleFluid/pitzDaily
```

The script copies across the relevant files from *pitzDaily*, while merging the changes in *physicalProperties.orig* into the *physicalProperties* file. The case runs with the *Allrun* script.

```
./Allrun
```

The case results can be visualised using *ParaView* to demonstrate it all works as expected.

The user can subsequently use `foamUnMergeCase`, a complementary script to `foamMergeCase`, to reset the *pitzDailyWater* case to its original “variant” form, containing only the *physicalProperties.orig* file. The case should first be cleaned to removed results, log files and post-processed data using the `foamCleanCase` script.

```
foamCleanCase
```

The case directory is then reduced to the basic *0*, *system* and *constant* directories and the *Allrun* script. The case can then be “un-merged” by running the following command.

```
foamUnMergeCase $FOAM_TUTORIALS/incompressibleFluid/pitzDaily
```

This returns the case directort to its initial state, with only the *constant* directory containing the *physicalProperties.orig* file.

As discussed earlier, it is recommended that the *Allrun* and *Allclean* scripts contain the respective `foamMergeCase` and `foamUnMergeCase` commands. Therefore, the user should open the *Allrun* in an editor and add the `foamMergeCase` command as shown below.

```
1  #!/bin/sh
2  cd ${0%/*} || exit 1    # Run from this directory
3
4  # Source tutorial run functions
5  . $WM_PROJECT_DIR/bin/tools/RunFunctions
6
7  foamMergeCase $FOAM_TUTORIALS/incompressibleFluid/pitzDaily
8
9  runApplication blockMesh -dict $FOAM_TUTORIALS/resources/blockMesh/pitzDaily
10 runApplication foamRun
```

The source case does not include an `Allclean` script, so the user should copy one using the `foamGet` script as follows.

```
foamGet Allclean
```

The `foamUnMergeCase` command should be added **after** the `cleanCase` function, otherwise the un-merging will be incomplete. An example is shown below.

```
1  #!/bin/sh
2
3  # Run from this directory
4  cd "${0%/*}" || exit 1
5
6  # Source tutorial clean functions
7  . "$WM_PROJECT_DIR/bin/tools/CleanFunctions"
8
9  # ...
10 cleanCase
11
12 foamMergeCase $FOAM_TUTORIALS/incompressibleFluid/pitzDaily
```

The case can now be run and cleaned by executing the `Allrun` and `Allclean` scripts respectively.

```
./Allrun
./Allclean
```

Chapter 5

Mesh generation and conversion

This chapter describes all topics relating to the creation of meshes in OpenFOAM: section 5.1 gives an overview of the way a mesh is described in OpenFOAM; section 5.2 lists the basic data files that describe a mesh; section 5.3 discusses mesh boundaries and introduces boundary conditions; section 5.4 covers the `blockMesh` utility for generating simple meshes of blocks of hexahedral cells; section 5.5 covers the `snappyHexMesh` utility for generating complex meshes of hexahedral and split-hexahedral cells automatically from triangulated surface geometries; section 5.7 describes the options for conversion of a mesh that has been generated by a third-party product into a format that OpenFOAM can read.

5.1 Mesh description

This section provides a specification of the way OpenFOAM describes a mesh. The mesh is an integral part of the numerical solution and must satisfy certain criteria to ensure a valid, and hence accurate, solution. OpenFOAM defines a mesh of arbitrary polyhedral cells in 3-D, bounded by arbitrary polygonal faces, *i.e.* the cells can have an unlimited number of faces where, for each face, there is no limit on the number of edges nor any restriction on its alignment. This flexible description of a mesh offers great freedom in mesh generation and manipulation when the geometry of the domain is complex.

An OpenFOAM mesh begins with **points** (or **vertices**). Each point is a location in 3D space, defined by a vector. The set of points forms a list where each point can be indexed by its position in the list, starting from zero. The list does not contain points that are unused.

The points are used to form mesh **faces**, where each face is defined as an ordered list of points, described by their where a point is referred to by its label. The ordering of point labels in a face is such that each two neighbouring points are connected by an edge, *i.e.* you follow points as you travel around the circumference of the face. The set of faces forms a list where each face is referred to by its label, representing its position in the list.

Each face can be characterised by a vector whose direction is normal to the face. The normal vector follows the right-hand rule, *i.e.* looking towards a face, if the numbering of the points follows a clockwise path, the normal vector points away from you, as shown in Figure 5.1. Note that faces can be warped, *i.e.* the points of the face may not necessarily lie on a plane. There are two types of face, described below.

- *Internal faces*, which connect two cells (and it can never be more than two). For each

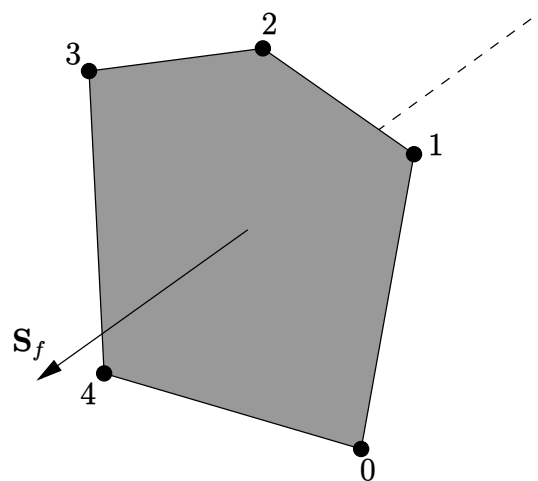


Figure 5.1: Face area vector from point numbering on the face

internal face, the ordering of the point labels is such that the face normal points into the cell with the larger label, *i.e.* for cells labelled ‘2’ and ‘5’, the normal points into ‘5’.

- *Boundary faces*, which belong to one cell since they coincide with the boundary of the domain. A boundary face is therefore addressed by one cell(only) and a boundary patch. The ordering of the point labels is such that the face normal points outside of the computational domain.

A **cell** is a list of faces in arbitrary order. Under normal circumstances, cells must have the properties listed below.

- The cells must be *contiguous*, *i.e.* completely cover the computational domain and must not overlap one another.
- Every cell must be *closed geometrically*, such that when all face area vectors are oriented to point outwards of the cell, their sum should equal the zero vector to machine accuracy;
- Every cell must be *closed topologically* such that all the edges in a cell are used by exactly two faces of the cell in question.

The **boundary** is formed by the boundary faces. It should be closed, *i.e.* the sum all boundary face area vectors equates to zero to machine tolerance. It is split into regions known as **patches** so that different boundary conditions can be applied to different parts of the boundary. A patch is defined by the labels of the faces it contains.

5.2 Mesh files

When a mesh is written out by OpenFOAM, the data files go into a *polyMesh* sub-directory. Usually the *polyMesh* directory is written to the *constant* directory, but simulations with *dynamic* meshes (*e.g.* mesh motion, refinement, *etc.*) write the modified meshes into time directories along with the field data files.

The data files are based around faces rather than cells. Each face is therefore assigned an ‘owner’ cell and ‘neighbour’ cell so that the connectivity across a given face can simply be described by the owner and neighbour cell labels. In the case of boundaries, there is no neighbour cell. With this in mind, the I/O specification consists of the following files:

points a list of vectors describing the cell vertices, where the first vector in the list represents vertex 0, the second vector represents vertex 1, *etc.*;

faces a list of faces, each face being a list of indices to vertices in the points list, where again, the first entry in the list represents face 0, *etc.*;

owner a list of owner cell labels, starting with the owner cell of face 0, then 1, 2, ...

neighbour a list of neighbour cell labels;

boundary a list of patches, containing a dictionary entry for each patch, declared using the patch name.

Critically, the **faces** list is ordered so that all internal faces are listed first, followed by the boundary faces. The boundary faces are themselves ordered so that they begin with the faces in the first patch, followed by the second, *etc.* As a consequence the patch entries in the **boundary** file are very compact, *e.g.*

```
inlet
{
    type            patch;
    nFaces          30;
    startFace       24170;
}
```

Due to the face ordering, the patch faces are simply described by: **startFace**, the index into the face list of the first face in the patch; and, **nFaces**, the number of faces in the patch.

5.3 Mesh boundary

As we saw in section 5.2, the domain boundary is defined by patches within the mesh, listed within the **boundary** mesh file. Each patch includes a **type** entry which can apply a *geometric constraint* to the patch. These geometric constraints include conditions that represent a geometric approximation, *e.g.* a symmetry plane, and conditions which form numerical connections between patches, *e.g.* a *cyclic* (or *periodic*) boundary. An example **boundary** file is shown below which includes some patches with geometric constraints.

```
16
17 5
18 (
19     top
20     {
21         type            wall;
22         inGroups        List<word> 1(wall);
23         nFaces          60;
24         startFace       3510;
25     }
```

```

26     inlet
27     {
28         type            patch;
29         nFaces          30;
30         startFace      3570;
31     }
32     outlet
33     {
34         type            patch;
35         nFaces          30;
36         startFace      3600;
37     }
38     bottom
39     {
40         type            symmetryPlane;
41         inGroups        List<word> 1(symmetryPlane);
42         nFaces          60;
43         startFace      3630;
44     }
45     frontAndBack
46     {
47         type            empty;
48         inGroups        List<word> 1(empty);
49         nFaces          3600;
50         startFace      3690;
51     }
52 )
53
54 // *****

```

A **type** entry is specified for every patch (*inlet*, *outlet*, *etc.*), with types assigned that include **patch**, **wall**, **symmetryPlane** and **empty**. Some patches also include an **inGroups** entry which is discussed in section 5.3.6.

5.3.1 Generic patch and wall

The patch types specified in the *boundary* file, which are **not** associated with a geometric constraint are the generic **patch** and **wall**. The **patch** type is assigned to open boundaries such as an *inlet* or *outlet* which does not involve any special handling of geometric approximation or numerical connections.

The **wall** type also provides no special geometric or numerical handling, but is used for patches which coincide with a solid wall. The wall ‘tag’ is required by some models, *e.g.* wall functions in turbulence modelling which require the distance to nearest wall.

5.3.2 1D/2D and axi-symmetric problems

OpenFOAM is designed as a code for 3D space and defines all meshes as such. However, 1D and 2D and axi-symmetric problems can be simulated in OpenFOAM by generating a mesh in 3 dimensions and applying special boundary conditions on any patch in the plane(s) normal to the direction(s) of interest. 1D and 2D problems apply the **empty** patch type to the relevant patches. Often the two regions of the boundary, on the ‘front’ and ‘back’ of the domain, are combined into a single patch, as in the **frontAndBack** patch in the quoted example above.

For axi-symmetric cases, the geometry, *e.g.* a cylinder, is approximated by a wedge-shaped mesh of small angle (*e.g.* 1°) and 1 cell thick, running along the centre line, straddling one of the coordinate planes, as shown in Figure 5.2. The axi-symmetric wedge planes must be specified as separate patches of **wedge** type. The generation of wedge geometries for axi-symmetric problems is discussed in section 5.4.11.

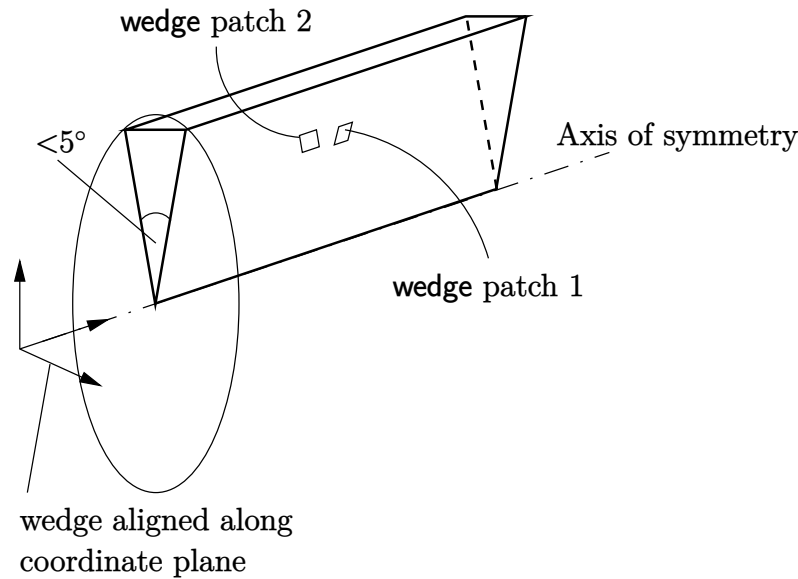


Figure 5.2: Axi-symmetric geometry using the `wedge` patch type.

5.3.3 Symmetry condition

A symmetry plane is a boundary condition that imagines the solution within the domain is ‘mirrored’ across the boundary. It can therefore be applied reliably to a domain with a plane of symmetry where the flow is believed to be symmetric across the plane. When the flow involves something like vortex shedding that breaks symmetry, the condition is less applicable.

There are two patch types relating to symmetry. Firstly, the `symmetryPlane` condition is a pure symmetry plane which can only be applied to a patch which is perfectly planar. There is then a `symmetry` condition, which can be applied to any patch, including those that are non-planar.

5.3.4 Cyclic conditions

The cyclic boundary conditions form a numerical connections between patches that are physically disconnected. The `cyclic` condition connects patches which have the same area to within a tolerance given by the `matchTolerance` keyword. Each patch specifies the name of the patch to which it connects through the `neighbourPatch` keyword. The condition can transform the field between patches, *e.g.* by a rotation, so the patches do not require the same orientation.

OpenFOAM also includes non-conformal coupling (NCC) which connects regions of a domain with independent meshes. It is and is used particularly for cases when one or more regions are moving, *e.g.* to simulate rotating geometry. Non-conformal coupling uses the `nonConformalCyclic` condition which are usually generated with the `createNonConformalCouples` utility. NCC examples can be located by searching for the `createNonConformalCouples` utility in Allrun scripts in the *tutorials* directory, *e.g.* by running

```
find $FOAM_TUTORIALS -name Allrun | \
  xargs grep -l createNonConformalCouples
```

5.3.5 Processor patches

Running applications in parallel is described in section 3.4. It involves decomposition of the mesh using `decomposePar` as described in section 3.4.1. Decomposition splits the domain which creates new patches at the exposed faces. Those patches are assigned the `processor` type which forms a numerical connection between sub-domains. Each `processor` patch entry in the `boundary` file includes a `myProcNo` entry for the processor (sub-domain) index and a `neighbProcNo` entry for the index of the matching patch on the sub-domain it connects with.

5.3.6 Patch groups

The `boundary` file example shows some patches include an `inGroups` entry, *e.g.* the `top` patch:

```
top
{
    type            wall;
    inGroups        List<word> 1(wall);
    nFaces          60;
    startFace       3510;
}
```

The `inGroups` entry is optional. It specifies one or more patch groups to which a patch can belong. A patch group is specified by a name which the user can choose. Group names can be used to specify boundary conditions in field files, simplifying the configuration. For example, if all inlet patches can be included in an `inlet` group, then one `inlet` entry can specify a boundary condition for all the patches.

Every non-generic patch, *i.e.* one which is not `patch` type, is *included in a patch group of the same name as its type*. For example, a patch of type `wall` is included in a `wall` group, one of type `symmetry` is included in a `symmetry` group, *etc.* The use of group names to specify boundary conditions is described further in section 6.3.

5.3.7 Constraint type examples

The user can scan the tutorials for mesh generation configuration files, *e.g.* `blockMeshDict` for `blockMesh` (see section 5.4) and `snappyHexMeshDict` for `snappyHexMesh` (see section 5.5, for examples of different types being used. The following example provides documentation and lists cases that use the `symmetryPlane` condition.

```
foamInfo -a symmetryPlane
```

The next example searches for `snappyHexMeshDict` files that specify the `wall` condition.

```
find $FOAM_TUTORIALS -name snappyHexMeshDict | \
xargs grep -El "type[\t ]*wall"
```

5.4 Mesh generation with the blockMesh utility

This section describes the mesh generation utility, `blockMesh`, supplied with OpenFOAM. The `blockMesh` utility creates parametric meshes with grading and curved edges. The mesh is generated from a dictionary file named *blockMeshDict* located in the *system* directory of a case. `blockMesh` reads this dictionary, generates the mesh and writes out the mesh data to *points*, *faces*, *cells* and *boundary* files in the *polyMesh* directory.

The principle behind `blockMesh` is to decompose the domain geometry into a set of 1 or more 3D hexahedral blocks. Edges of the blocks can be straight lines, arcs or splines. The mesh is ostensibly specified as a number of cells in each direction of the block, sufficient information for `blockMesh` to generate the mesh data.

5.4.1 Overview of a blockMeshDict file

The *blockMeshDict* file is a dictionary including keywords described below.

- **units** and **scale**: (optional) specify an alternative set of units, *e.g.* [mm], by which the resulting mesh is scaled to the base unit, *i.e.* metre [m], see section 5.4.2.
- **scale**: (optional) specify a factor by which the resulting mesh is scaled, see section 5.4.2.
- **vertices**: list of vertex coordinates, see section 5.4.3.
- **edges**: optional entry to describe curved geometry, see section 5.4.4.
- **blocks**: ordered list of vertex labels and mesh size, see section 5.4.5.
- **boundary**: sub-dictionary of boundary patches, see section 5.4.7.
- **defaultPatch**: optional entry describing a default patch, see section 5.4.7.
- **mergePatchPairs**: optional list of patches to be merged, see section 5.4.8.

5.4.2 Units and scaling

The *blockMeshDict* file contains parameters such as **vertices** that ultimately specify the size of the solution domain. By default, the values of the parameters correspond to the unit system of the simulation, which is Système International (SI) by default in OpenFOAM. In other words, mesh dimensions are in units of metre (m).

However, using the optional **units** keyword, a user can specify that the values correspond to an alternative set of units (in dimensions of length). For example, if the user sets

```
units      [mm];
```

all the specified values will correspond to millimetre (mm) units. The mesh is then generated by `blockMesh` in those units, *but* the generated mesh is written to file *in the base units* of the unit system, *i.e.* metre (m) by default.

Alternatively, the user can specify a factor with the optional **scale** keyword, which is applied to the final mesh. For example in the **damBreakLaminar** tutorial, the **scale** entry reads

```
scale      0.146;
```

5.4.3 The vertices

The vertices of the blocks of the mesh are given next as a standard list named **vertices**. An example set of vertices, corresponding to a block in Figure 5.3, is provided below.

```
vertices
(
    ( 0    0    0 )    // vertex number 0
    ( 1    0    0.1)   // vertex number 1
    ( 1.1  1    0.1)   // vertex number 2
    ( 0    1    0.1)   // vertex number 3
    (-0.1 -0.1  1 )    // vertex number 4
    ( 1.3  0    1.2)   // vertex number 5
    ( 1.4  1.1  1.3)   // vertex number 6
    ( 0    1    1.1)   // vertex number 7
);
```

5.4.4 The edges

Each edge joining 2 vertex points is assumed to be straight by default. However any edge may be specified to be curved by entries in a list named **edges**. The list is optional; if the geometry contains no curved edges, it may be omitted.

Each entry for a curved edge begins with a keyword specifying the type of curve from those listed below.

- **arc**: a circular arc with a single interpolation point or angle + axis (see below).
- **spline**: spline curve using a list of interpolation points
- **polyLine**: a set of lines with list of interpolation points
- **BSpline**: a B-spline curve with list of interpolation points
- **line**: a straight line, the default which requires no edge specification.

The keyword is then followed by the labels of the 2 vertices that the edge connects. Following that, interpolation points must be specified through which the edge passes. For an **arc**, either of the following is required: a single interpolation point, which the circular arc will intersect; or an angle and rotation axis for the arc. For **spline**, **polyLine** and **BSpline**, a list of interpolation points is required. For our example block in Figure 5.3 we specify an **arc** edge connecting vertices 1 and 5 as follows through the interpolation point (1.1, 0.0, 0.5):

```
edges
(
    arc 1 5 (1.1 0.0 0.5)
);
```

For the angle and axis specification of an arc, the syntax is of the form:

```
edges
(
    arc 1 5 25 (0 1 0) // 25 degrees, y-normal
);
```

5.4.5 The blocks

The block definitions are contained in a list named **blocks**. Each block of the geometry is defined by 8 vertices, one at each corner of a hexahedron. An example block is shown in Figure 5.3 with each vertex numbered according to the list in section 5.4.3. The edge connecting vertices 1 and 5 is curved as a reminder that edges can be curved in blockMesh.

Each block has a local coordinate system (x_1, x_2, x_3) that must be right-handed. A right-handed set of axes is defined such that to an observer looking down the Oz axis, with O nearest them, the arc from a point on the Ox axis to a point on the Oy axis is in a clockwise sense.

The local coordinate system is defined by the order in which the vertices are presented in the block definition according to:

- the axis origin is the first entry in the block definition, vertex 0 in our example;
- the x_1 direction is described by moving from vertex 0 to vertex 1;
- the x_2 direction is described by moving from vertex 1 to vertex 2;
- vertices 0, 1, 2, 3 define the plane $x_3 = 0$;
- vertex 4 is found by moving from vertex 0 in the x_3 direction;
- vertices 5,6 and 7 are similarly found by moving in the x_3 direction from vertices 1,2 and 3 respectively.

An example block specification is shown below.

```
blocks
(
    hex (0 1 2 3 4 5 6 7)    // vertex numbers
    (10 10 10)               // numbers of cells in each direction
    simpleGrading (1 2 3)    // cell expansion ratios
);
```

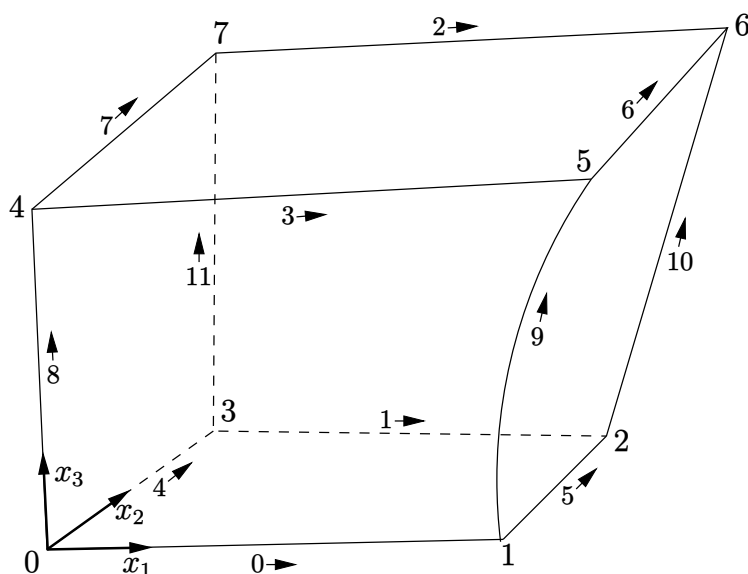


Figure 5.3: A single block

It begins with the shape identifier of the block (defined in the `$FOAM_ETC/cellModels` file). The shape is always `hex` since the blocks are always hexahedra. The list of vertex numbers follows, ordered in the manner described above.

The second entry gives the number of cells in each of the x_1 x_2 and x_3 directions for that block. The third entry gives the cell expansion ratios for each direction in the block. The expansion ratio enables the mesh to be graded, or refined, in specified directions. The ratio is that of the width of the end cell δ_e along one edge of a block to the width of the start cell δ_s along that edge, as shown in Figure 5.4.

There are two types of grading specification available in `blockMesh`. The most common one is `simpleGrading` which specifies uniform expansions in the local x_1 , x_2 and x_3 directions respectively with only 3 expansion ratios, *e.g.*

```
simpleGrading (1 2 3)
```

The more complex alternative is `edgeGrading`. This full cell expansion description gives a ratio for each edge of the block, numbered according to the scheme shown in Figure 5.3 with the arrows representing the direction from first cell... to last cell *e.g.*

```
edgeGrading (1 1 1 1 2 2 2 2 3 3 3 3)
```

This example is directly equivalent to the `simpleGrading` example given above because it uses a ratio of cell widths of 1 along edges 0-3, 2 along edges 4-7 and 3 along 8-11. Note

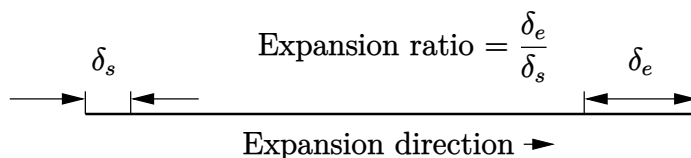


Figure 5.4: Mesh grading along a block edge

that it is possible to generate blocks with fewer than 8 vertices by collapsing one or more pairs of vertices on top of each other, as described in section 5.4.11.

5.4.6 Multi-grading of a block

Using a single expansion ratio to describe mesh grading permits only “one-way” grading within a mesh block. In some cases, it reduces complexity and effort to be able to control grading within separate divisions of a single block, rather than have to define several blocks with one grading per block. For example, to mesh a channel with two opposing walls and grade the mesh towards the walls requires three regions: two with grading to the wall with one in the middle without grading.

blockMesh includes multi-grading functionality that can divide a block in an given direction and apply different grading within each division. This multi-grading is specified by replacing any single value expansion ratio in the grading specification of the block, *e.g.* “1”, “2”, “3” in

```
blocks
(
    hex (0 1 2 3 4 5 6 7) (100 300 100)
    simpleGrading (1 2 3);
);
```

We will present multi-grading for the following example:

- split the block into 3 divisions in the y -direction, representing 20%, 60% and 20% of the block length;
- include 30% of the total cells in the y -direction (300) in *each* divisions 1 and 3 and the remaining 40% in division 2;
- apply 1:4 expansion in divisions 1 and 3, and zero expansion in division 2.

We can specify this by replacing the y -direction expansion ratio “2” in the example above with the following:

```
blocks
(
    hex (0 1 2 3 4 5 6 7) (100 300 100)
    simpleGrading
    (
        1                // x-direction expansion ratio
        (
            (0.2 0.3 4)    // 20% y-dir, 30% cells, expansion = 4
            (0.6 0.4 1)    // 60% y-dir, 40% cells, expansion = 1
            (0.2 0.3 0.25) // 20% y-dir, 30% cells, expansion = 0.25 (1/4)
        )
        3                // z-direction expansion ratio
```

```
)
);
```

Both the fraction of the block and the fraction of the cells are normalized automatically. They can be specified as percentages, fractions, absolute lengths, *etc.* and do not need to sum to 100, 1, *etc.* The example above can be specified using percentages, *e.g.*

```
blocks
(
  hex (0 1 2 3 4 5 6 7) (100 300 100)
  simpleGrading
  (
    1
    (
      (20 30 4)    // 20%, 30%...
      (60 40 1)
      (20 30 0.25)
    )
    3
  )
);
```

5.4.7 The boundary

The boundary of the mesh is given in a list named **boundary**. The boundary is broken into patches (regions), where each patch in the list has its name as the keyword, which is the choice of the user, although we recommend something that conveniently identifies the patch, *e.g. inlet*; the name is used as an identifier for setting boundary conditions in the field data files. The patch information is then contained in sub-dictionary with:

- **type**: the patch type, either a generic patch on which some boundary conditions are applied or a particular geometric condition, as listed in section 5.3;
- **faces**: a list of block faces that make up the patch and whose name is the choice of the user, although we recommend something that conveniently identifies the patch, *e.g. inlet*; the name is used as an identifier for setting boundary conditions in the field data files.

blockMesh collects *block* faces that are omitted from the patches in the **boundary** list and assigns them to a default patch. The default patch can be configured through a **defaultPatch** sub-dictionary, including **type** and **name**, *e.g.*

```
defaultPatch
{
    name    frontAndBack;
    type    empty;
}
```


In absence of any of these entries a default patch uses the name `defaultFaces` and type `empty` by default. This means that for a 2D, the user has the option to omit block faces lying in the 2D plane, knowing that they will be collected into an `empty` patch as required.

Returning to the example block in Figure 5.3, if it has an inlet on the left face, an output on the right face and the four other faces are walls then the patches could be defined as follows:

```

boundary                // keyword
(
    inlet                // patch name
    {
        type patch;     // patch type for patch 0
        faces
        (
            (0 4 7 3)    // block face in this patch
        );
    }                    // end of 0th patch definition

    outlet               // patch name
    {
        type patch;     // patch type for patch 1
        faces
        (
            (1 2 6 5)
        );
    }

    walls
    {
        type wall;
        faces
        (
            (0 1 5 4)
            (0 3 2 1)
            (3 7 6 2)
            (4 5 6 7)
        );
    }
);

```

Each block face is defined by a list of 4 vertex numbers. The list can begin with any vertex but needs to follow a sequence through connecting edges, with no restriction on the direction.

Where a patch type requires additional data in the resulting *boundary* file, the data is simply added in the patch entry in *blockMeshDict*. For example, with the cyclic patch, the user must specify the name of the related patch through the `neighbourPatch` keyword, *e.g.*

```

left

```

```

{
    type            cyclic;
    neighbourPatch  right;
    faces           ((0 4 7 3));
}
right
{
    type            cyclic;
    neighbourPatch  left;
    faces           ((1 5 6 2));
}

```

5.4.8 Multiple blocks

A mesh can be created using more than 1 block. In such circumstances, the mesh is created as described in the preceding text. The only additional issue is the connection between blocks. Firstly, if a face of one block also belongs to another block, the block face will not form an external patch but instead a set of internal faces of the cells in the resulting mesh.

Alternatively if the user wishes to combine block faces which do not exactly match one another, *i.e.* through shared vertices, they can first include the block faces within the **patches** list. Each pair of patches whose faces are to be merged can then be included in an optional list named **mergePatchPairs**. The format of **mergePatchPairs** is:

```

mergePatchPairs
(
    ( <masterPatch> <slavePatch> ) // merge patch pair 0
    ( <masterPatch> <slavePatch> ) // merge patch pair 1
    ...
)

```

See for example `$FOAM_TUTORIALS/multiphaseEuler/LBend`. The pairs of patches are interpreted such that the first patch becomes the *master* and the second becomes the *slave*. The rules for merging are as follows:

- the faces of the master patch remain as originally defined, with all vertices in their original location;
- the faces of the slave patch are projected onto the master patch where there is some separation between slave and master patch;
- the location of any vertex of a slave face might be adjusted by **blockMesh** to eliminate any face edge that is shorter than a minimum tolerance;
- if patches overlap as shown in Figure 5.5, each face that does not merge remains as an external face of the original patch, on which boundary conditions must then be applied;

- if all the faces of a patch are merged, then the patch itself will contain no faces and is removed.

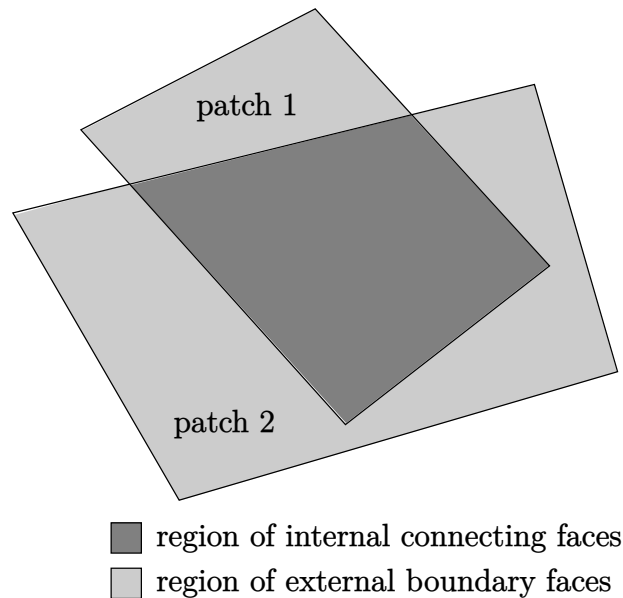


Figure 5.5: Merging overlapping patches

The consequence is that the original geometry of the slave patch will not necessarily be completely preserved during merging. Therefore in a case, say, where a cylindrical block is being connected to a larger block, it would be wise to assign the master patch to the cylinder, so that its cylindrical shape is correctly preserved. There are some additional recommendations to ensure successful merge procedures:

- in 2 dimensional geometries, the size of the cells in the third dimension, *i.e.* out of the 2D plane, should be similar to the width/height of cells in the 2D plane;
- it is inadvisable to merge a patch twice, *i.e.* include it twice in `mergePatchPairs`;
- where a patch to be merged shares a common edge with another patch to be merged, both should be declared as a master patch.

5.4.9 Projection of vertices, edges and faces

`blockMesh` can be configured to create body fitted meshes using projection of vertices, edges and/or faces onto specified geometry. The functionality can be used to mesh spherical and cylindrical geometries such as pipes and vessels conveniently. The user can specify within the `blockMeshDict` file within an optional `geometry` dictionary with the same format as used in the `snappyHexMeshDict` file. For example to specify a cylinder using the built in geometric type the user could configure with the following:

```
geometry
{
    cylinder
    {
        type searchableCylinder;
```

```

        point1 (0 -4 0);
        point2 (0 4 0);
        radius 0.7;
    }
};

```

The user can then project vertices, edges and/or faces onto the cylinder surface with the **project** keyword using example syntax shown below:

```

vertices
(
    project (-1 -0.1 -1) (cylinder)
    project ( 1 -0.1 -1) (cylinder)
    ...
);

edges
(
    project 0 1 (cylinder)
    ...
);

faces
(
    project (0 4 7 3) cylinder
    ...
);

```

The use of this functionality is demonstrated in tutorials which can be located by searching for the **project** keyword in all the *blockMeshDict* files by:

```
find $FOAM_TUTORIALS -name blockMeshDict | xargs grep -l project
```

5.4.10 Naming vertices, edges, faces and blocks

Vertices, edges, faces and blocks can be named in the configuration of a *blockMeshDict* file, which can make it easier to manage more complex examples. It is done simply using the **name** keyword. The following syntax shows naming using the example for projection in the previous subsection:

```

vertices
(
    name v0 project (-1 -0.1 -1) (cylinder)
    name v1 project ( 1 -0.1 -1) (cylinder)
    ...
);

edges
(
    project v0 v1 (cylinder)
    ...
);

```

When a name is provided for a given entity, it can be used to replace the index. In the example about, rather than specify the edge using vertex indices 0 and 1, the names **v0** and **v1** are used.

5.4.11 Blocks with fewer than 8 vertices

It is possible to collapse one or more pair(s) of vertices onto each other in order to create a block with fewer than 8 vertices. The most common example of collapsing vertices is when creating a 6-sided wedge shaped block for 2-dimensional axi-symmetric cases that use the `wedge` patch type described in section 5.3.2. The process is best illustrated by using a simplified version of our example block shown in Figure 5.6. Let us say we wished to create a wedge shaped block by collapsing vertex 7 onto 4 and 6 onto 5. This is simply done by exchanging the vertex number 7 by 4 and 6 by 5 respectively so that the block numbering would become:

```
hex (0 1 2 3 4 5 5 4)
```

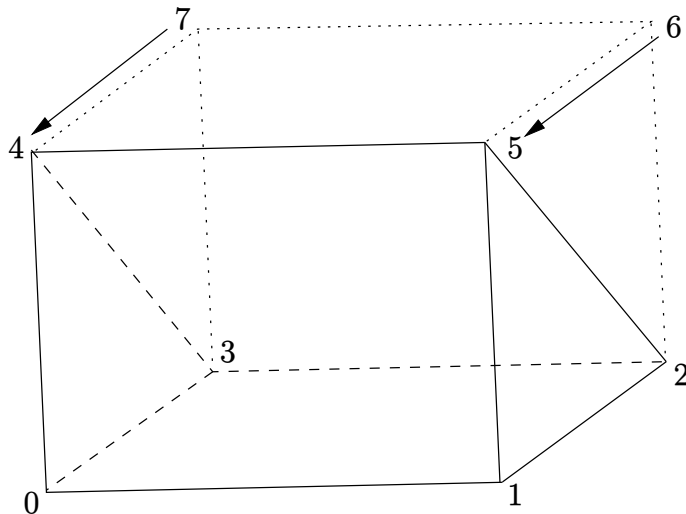


Figure 5.6: Creating a wedge shaped block with 6 vertices

The same applies to the patches with the main consideration that the block face containing the collapsed vertices, previously (4 5 6 7) now becomes (4 5 5 4). This is a block face of zero area which creates a patch with no faces in the `polyMesh`, as the user can see in a *boundary* file for such a case. The patch should be specified as `empty` in the *blockMeshDict* and the boundary condition for any fields should consequently be `empty` also.

5.4.12 Running blockMesh

As described in section 3.3, `blockMesh` can be run from within the case directory by:

```
blockMesh
```

Like many utilities, it can also be run using a configuration file named differently from *blockMeshDict*. Several examples in the *tutorials* directory for example use the *pitzDaily* geometry. They use a common `blockMesh` configuration file named *pitzDaily* in the `$FOAM_TUTORIALS/resources/blockMesh`. The meshes for these cases are generated using the `-dict` option by

```
blockMesh -dict $FOAM_TUTORIALS/resources/blockMesh/pitzDaily
```

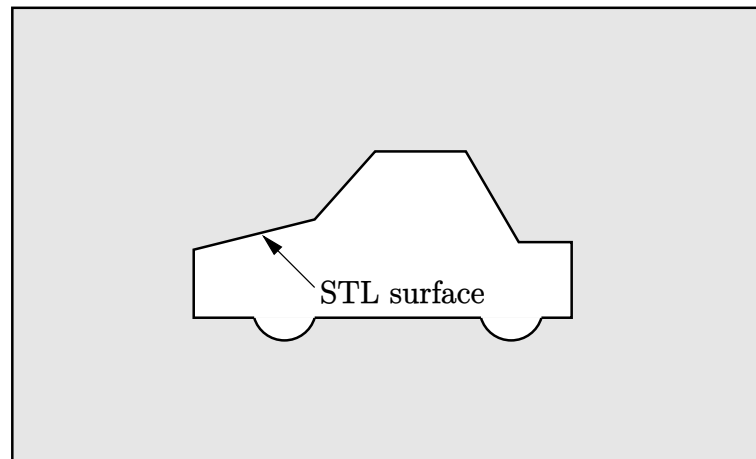


Figure 5.7: Schematic 2D meshing problem for `snappyHexMesh`

5.5 Mesh generation with `snappyHexMesh`

This section describes the mesh generation utility, `snappyHexMesh`, supplied with OpenFOAM. The `snappyHexMesh` utility generates 3-dimensional meshes containing hexahedra (hex) and split-hexahedra (split-hex) automatically from triangulated surface geometries, or tri-surfaces, in Stereolithography (STL) or Wavefront Object (OBJ) format. The mesh approximately conforms to the surface by iteratively refining a starting mesh and morphing the resulting split-hex mesh to the surface. An optional phase will shrink back the resulting mesh and insert cell layers. The specification of mesh refinement level is very flexible and the surface handling is robust with a pre-specified final mesh quality. It runs in parallel with a load balancing step every iteration.

5.5.1 The mesh generation process of `snappyHexMesh`

The process of generating a mesh using `snappyHexMesh` will be described using the schematic in Figure 5.7. The objective is to mesh a rectangular shaped region (shaded grey in the figure) surrounding an object described by a tri-surface, *e.g.* typical for an external aerodynamics simulation. Note that the schematic is 2-dimensional to make it easier to understand, even though the `snappyHexMesh` is a 3D meshing tool.

In order to run `snappyHexMesh`, the user requires the following:

- one or more tri-surface files located in a *constant/geometry* sub-directory of the case directory;
- a background hex mesh which defines the extent of the computational domain and a base level mesh density; typically generated using `blockMesh`, discussed in section 5.5.2.
- a *snappyHexMeshDict* dictionary, with appropriate entries, located in the *system* sub-directory of the case.

The *snappyHexMeshDict* dictionary includes: switches at the top level that control the various stages of the meshing process; and, individual sub-directories for each process. The entries are listed below.

- `castellatedMesh`: to switch on creation of the castellated mesh.
- `snap`: to switch on surface snapping stage.
- `addLayers`: to switch on surface layer insertion.
- `mergeTolerance`: merge tolerance as fraction of bounding box of initial mesh.
- `geometry`: sub-dictionary of all surface geometry used.
- `castellatedMeshControls`: sub-dictionary of controls for castellated mesh.
- `snapControls`: sub-dictionary of controls for surface snapping.
- `addLayersControls`: sub-dictionary of controls for layer addition.
- `meshQualityControls`: sub-dictionary of controls for mesh quality.

All the geometry used by `snappyHexMesh` is specified in a *geometry* sub-dictionary in the *snappyHexMeshDict* dictionary. The geometry can be specified through a tri-surface or bounding geometry entities in OpenFOAM. An example is given below:

```
geometry
{
    duct          // User defined region name
    {
        type      triSurfaceMesh;
        file      "duct.obj";      // surface geometry OBJ file
        regions
        {
            leftOpening      // Named region in the OBJ file
            {
                name inlet;    // User-defined patch name
            }                 // otherwise given sphere1_secondSolid
        }
    }

    box           // User defined region name
    {
        type      searchableBox;    // region defined by bounding box
        min       (1.5 1 -0.5);
        max       (3.5 2 0.5);
    }

    sphere        // User defined region name
    {
        type      searchableSphere; // region defined by bounding sphere
        centre    (1.5 1.5 1.5);
        radius     1.03;
    }
};
```

5.5.2 Creating the background hex mesh

Before `snappyHexMesh` is executed the user must create a background mesh of hexahedral cells that fills the entire region within by the external boundary as shown in Figure 5.8. This can be done simply using `blockMesh`. The following criteria must be observed when creating the background mesh:

- the mesh must consist purely of hexes;

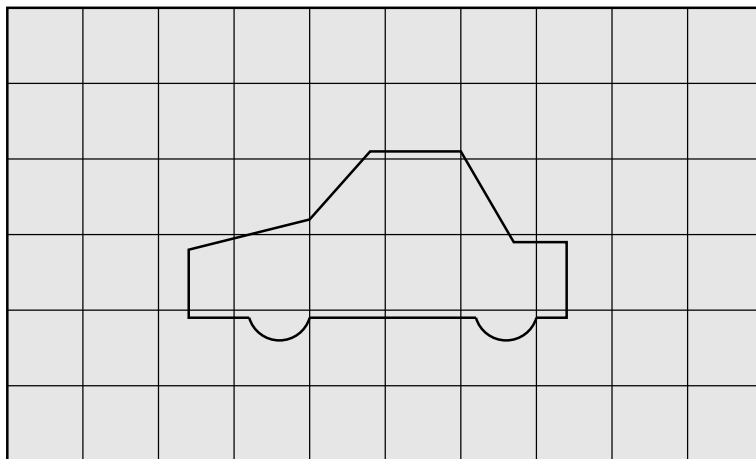


Figure 5.8: Initial mesh generation in `snappyHexMesh` meshing process

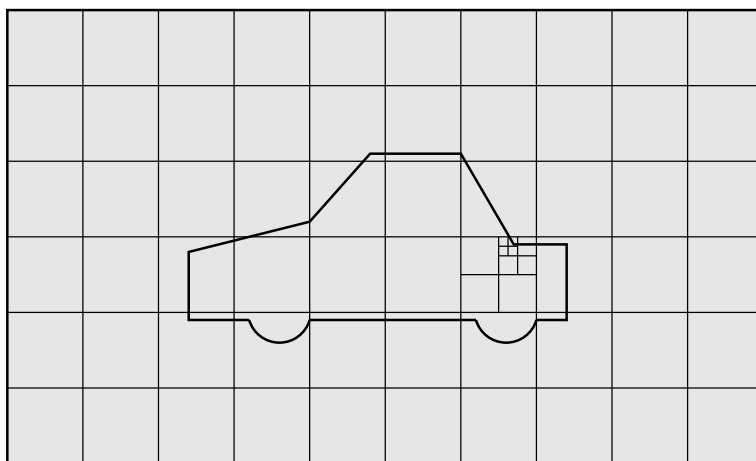


Figure 5.9: Cell splitting by feature edge in `snappyHexMesh` meshing process

- the cell aspect ratio should be approximately 1, at least near surfaces at which the subsequent snapping procedure is applied, otherwise the convergence of the snapping procedure is slow, possibly to the point of failure;
- there must be at least one intersection of a cell edge with the tri-surface, *i.e.* a mesh of one cell will not work.

5.5.3 Cell splitting at feature edges and surfaces

Cell splitting is performed according to the specification supplied by the user in the *castellatedMeshControls* sub-dictionary in the *snappyHexMeshDict*. The entries for *castellatedMeshControls* are presented below.

- **insidePoint**: location vector inside the region to be meshed; vector must not coincide with a cell face either before or during refinement.
- **maxLocalCells**: max number of cells per processor during refinement.

- **maxGlobalCells**: overall cell limit during refinement (*i.e.* before removal).
- **minRefinementCells**: if **minRefinementCells** $\geq$ number of cells to be refined, surface refinement stops.
- **nCellsBetweenLevels**: number of buffer layers of cells between successive levels of refinement (typically set to 3).
- **resolveFeatureAngle**: applies maximum level of refinement to cells that can see intersections whose angle exceeds **resolveFeatureAngle** (typically set to 30).
- **features**: list of features for refinement.
- **refinementSurfaces**: dictionary of surfaces for refinement.
- **refinementRegions**: dictionary of regions for refinement.

The splitting process begins with cells being selected according to specified edge features first within the domain as illustrated in Figure 5.9. The **features** list in the *castellatedMeshControls* sub-dictionary permits dictionary entries containing a name of an **edgeMesh** file and the level of refinement, *e.g.*:

```
features
(
    {
        file "features.eMesh"; // file containing edge mesh
        level 2;               // level of refinement
    }
);
```

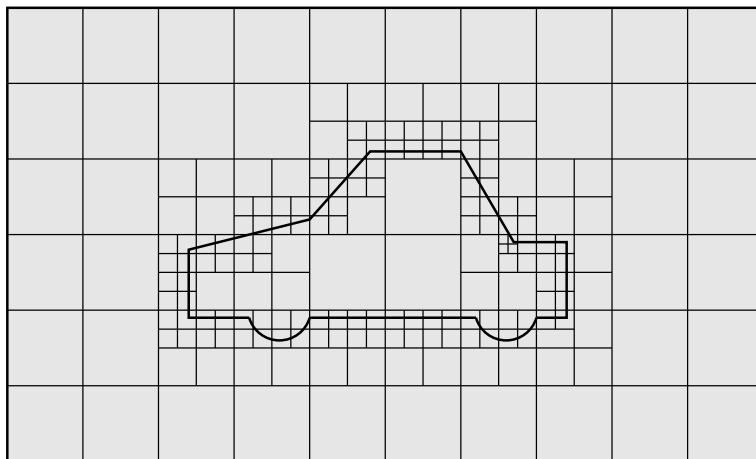
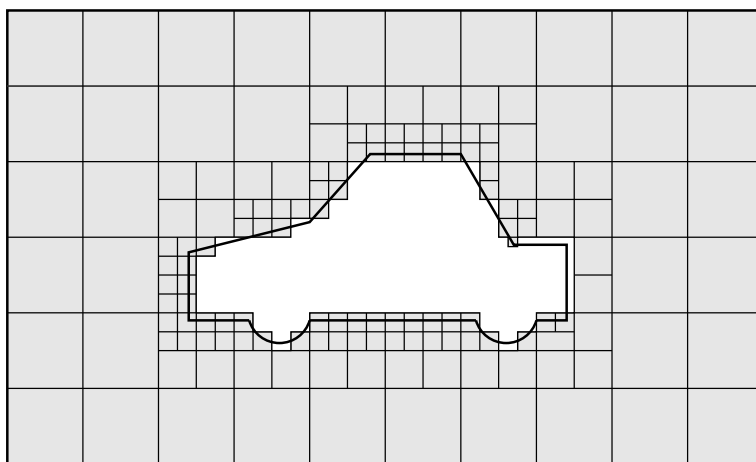
The **edgeMesh** containing the features can be extracted from the tri-surface file using the **surfaceFeatures** utility which specifies the tri-surface and controls such as included angle through a *surfaceFeaturesDict* configuration file, examples of which can be found in several tutorials and at *\$FOAM_ETC/caseDicts/surface/surfaceFeaturesDict*. The utility is simply run by executing the following in a terminal

```
surfaceFeatures
```

Following feature refinement, cells are selected for splitting in the locality of specified surfaces as illustrated in Figure 5.10. The **refinementSurfaces** dictionary in *castellatedMeshControls* requires dictionary entries for each STL surface and a default **level** specification of the minimum and maximum refinement in the form (**<min>** **<max>**). The minimum level is applied generally across the surface; the maximum level is applied to cells that can see intersections that form an angle in excess of that specified by **resolveFeatureAngle**.

The refinement can optionally be overridden on one or more specific region of an STL surface. The region entries are collected in a **regions** sub-dictionary. The keyword for each region entry is the name of the region itself and the refinement level is contained within a further sub-dictionary. An example is given below:

```
refinementSurfaces
{
    duct
    {
```

Figure 5.10: Cell splitting by surface in `snappyHexMesh` meshing processFigure 5.11: Cell removal in `snappyHexMesh` meshing process

```

level (2 2); // default (min max) refinement for whole surface
regions
{
    leftOpening
    {
        level (3 3); // optional refinement for secondSolid region
    }
}
}

```

5.5.4 Cell removal

Once the feature and surface splitting is complete a process of cell removal begins. Cell removal requires one or more regions enclosed entirely by a bounding surface within the domain. The region in which cells are retained are simply identified by a location vector within that region, specified by the `insidePoint` keyword in *castellatedMeshControls*. Cells are retained if, approximately speaking, 50% or more of their volume lies within the region. The remaining cells are removed accordingly as illustrated in Figure 5.11.

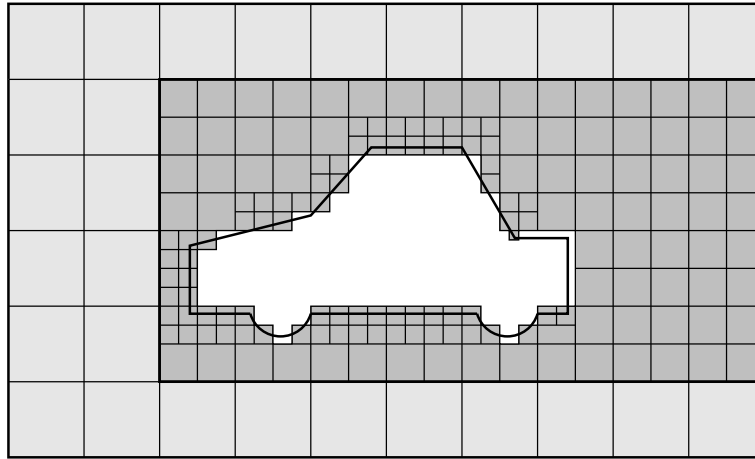


Figure 5.12: Cell splitting by region in `snappyHexMesh` meshing process

5.5.5 Cell splitting in specified regions

Those cells that lie within one or more specified volume regions can be further split as illustrated in Figure 5.12 by a rectangular region shown by dark shading. The `refinementRegions` sub-dictionary in `castellatedMeshControls` contains entries for refinement of the volume regions specified in the `geometry` sub-dictionary. A refinement `mode` is applied to each region which can be:

- `inside` refines inside the volume region;
- `outside` refines outside the volume region
- `distance` refines according to distance to the surface; and can accommodate different levels at multiple distances with the `levels` keyword.

For the `refinementRegions`, the refinement level is specified by the `level` keyword for `inside` and `outside` refinement. For `distance` refinement, the keyword is `levels` (**plural!**) which contains list of entries with the format (`<distance> <level>`). Examples are shown below:

```
refinementRegions
{
    box
    {
        mode    inside;
        level 4;           // refinement level 4
    }

    sphere
    {
        mode    distance;           // refinement level 5 within 1.0 m
        levels ((1.0 5) (2.0 3)); // refinement level 3 within 2.0 m
                                   // levels must be ordered nearest first
    }
}
```

5.5.6 Cell splitting based on local span

Refinement of cells can also be specified to guarantee a specified number of cells across the span between opposing surfaces. This refinement option can ensure that there are sufficient

cells to resolve the flow in a region of the domain, e.g. across a narrow pipe. The method requires closeness data to be provided on the surface geometry. The closeness can be calculated by the `surfaceFeatures` utility with the following entry in the *surfaceFeaturesDict* file:

```
surfaces
(
    "pipeWall.obj"
);

closeness
{
    pointCloseness    yes;
}
```

This writes closeness data to a file named *pipeWall.closeness.internalPointCloseness* into the *constant/geometry* directory. The closeness is then be used for span-based refinement by the addition of an entry in the *refinementRegions* sub-dictionary in *snappyHexMeshDict*, e.g.:

```
refinementRegions
{
    pipeWall
    {
        mode    insideSpan;
        level    (1000 2);
        cellsAcrossSpan 40;
    }
}
```

The example shows a refinement region inside the `pipeWall` surface in which a maximum 2 levels of refinement is guaranteed within a specified distance of 1000 from the wall. The span-based refinement, specified by the `insideSpan` mode, enables the user to guarantee at least 40 `cellsAcrossSpan`, *i.e.* across the pipe diameter.

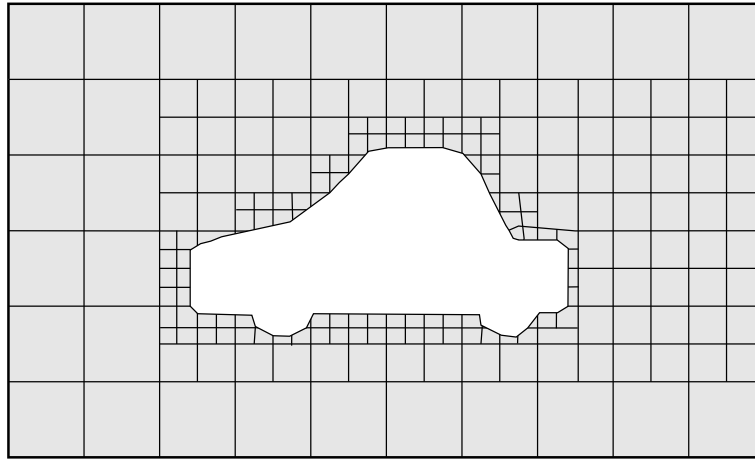
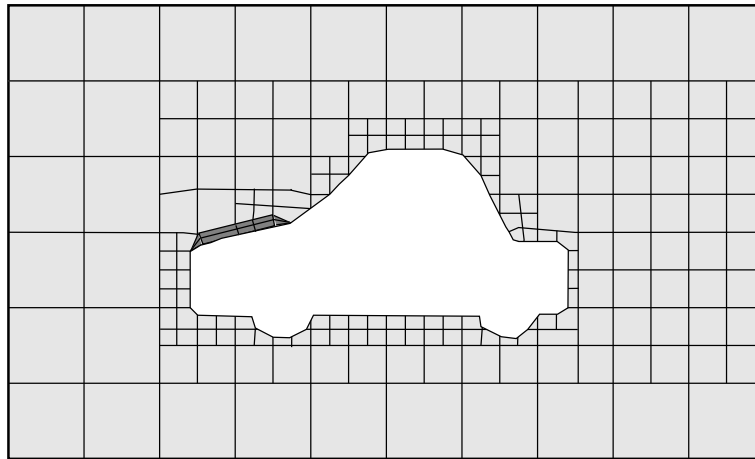
5.5.7 Snapping to surfaces

The next stage of the meshing process involves moving cell vertex points onto surface geometry to remove the jagged castellated surface from the mesh. The process is:

1. displace the vertices in the castellated boundary onto the STL surface;
2. solve for relaxation of the internal mesh with the latest displaced boundary vertices;
3. find the vertices that cause mesh quality parameters to be violated;
4. reduce the displacement of those vertices from their initial value (at 1) and repeat from 2 until mesh quality is satisfied.

The method uses the settings in the *snapControls* sub-dictionary in *snappyHexMeshDict*, listed below.

- **nSmoothPatch**: number of patch smoothing iterations before finding correspondence to surface (typically 3).
- **tolerance**: ratio of distance for points to be attracted by surface feature point or edge, to local maximum edge length (typically 2.0).

Figure 5.13: Surface snapping in `snappyHexMesh` meshing processFigure 5.14: Layer addition in `snappyHexMesh` meshing process

- `nSolveIter`: number of mesh displacement relaxation iterations (typically 30-100).
- `nRelaxIter`: maximum number of snapping relaxation iterations (typically 5).

An example is illustrated in the schematic in Figure 5.13 (albeit with mesh motion that looks slightly unrealistic).

5.5.8 Mesh layers

The mesh output from the snapping stage may be suitable for the purpose, although it can produce some irregular cells along boundary surfaces. There is an optional stage of the meshing process which introduces additional layers of hexahedral cells aligned to the boundary surface as illustrated by the dark shaded cells in Figure 5.14.

The process of mesh layer addition involves shrinking the existing mesh from the boundary and inserting layers of cells, broadly as follows:

1. the mesh is projected back from the surface by a specified thickness in the direction normal to the surface;

2. solve for relaxation of the internal mesh with the latest projected boundary vertices;
3. check if validation criteria are satisfied otherwise reduce the projected thickness and return to 2; if validation cannot be satisfied for any thickness, do not insert layers;
4. if the validation criteria can be satisfied, insert mesh layers;
5. the mesh is checked again; if the checks fail, layers are removed and we return to 2.

The layer addition procedure uses the settings in the *addLayersControls* sub-dictionary in *snappyHexMeshDict*; entries are listed below. The user has the option of 4 different layer thickness parameters — **expansionRatio**, **finalLayerThickness**, **firstLayerThickness**, **thickness** — *from which they must specify 2 only*; more than 2, and the problem is over-specified.

- **layers**: dictionary specifying layers to be inserted.
- **relativeSizes**: switch that sets whether the specified layer thicknesses are relative to undistorted cell size outside layer or absolute.
- **expansionRatio**: expansion factor for layer mesh, increase in size from one layer to the next.
- **finalLayerThickness**: thickness of layer furthest from the wall, usually in combination with relative sizes according to the **relativeSizes** entry.
- **firstLayerThickness**: thickness of layer nearest the wall, usually in combination with absolute sizes according to the **relativeSizes** entry.
- **thickness**: total thickness of all layers of cells, usually in combination with absolute sizes according to the
- **relativeSizes** entry.
- **minThickness**: minimum thickness of cell layer, either relative or absolute (as above).
- **nGrow**: number of layers of connected faces that are not grown if points do not get extruded; helps convergence of layer addition close to features.
- **featureAngle**: angle above which surface is not extruded.
- **nRelaxIter**: maximum number of snapping relaxation iterations (typically 5).
- **nSmoothSurfaceNormals**: number of smoothing iterations of surface normals (typically 1).
- **nSmoothNormals**: number of smoothing iterations of interior mesh movement direction (typically 3).
- **nSmoothThickness**: smooth layer thickness over surface patches (typically 10).
- **maxFaceThicknessRatio**: stop layer growth on highly warped cells (typically 0.5).

- **maxThicknessToMedialRatio**: reduce layer growth where ratio thickness to medial distance is large (typically 0.3)
- **minMedianAxisAngle**: angle used to pick up medial axis points (typically 90).
- **nBufferCellsNoExtrude**: create buffer region for new layer terminations (typically 0).
- **nLayerIter**: overall max number of layer addition iterations (typically 50).
- **nRelaxedIter**: max number of iterations after which the controls in the *relaxed* sub-dictionary of **meshQuality** are used (typically 20).

The **layers** sub-dictionary contains entries for each *patch* on which the layers are to be applied and the number of surface layers required. The patch name is used because the layers addition relates to the existing mesh, not the surface geometry; hence applied to a patch, not a surface region. An example **layers** entry is as follows:

```
layers
{
    sphere1_firstSolid
    {
        nSurfaceLayers 1;
    }
    maxY
    {
        nSurfaceLayers 1;
    }
}
```

5.5.9 Mesh quality controls

The mesh quality is controlled by the entries in the *meshQualityControls* sub-dictionary in *snappyHexMeshDict*; entries are listed below.

- **maxNonOrtho**: maximum non-orthogonality allowed (degrees, typically 65).
- **maxBoundarySkewness**: max boundary face skewness allowed (typically 20).
- **maxInternalSkewness**: max internal face skewness allowed (typically 4).
- **maxConcave**: max concaveness allowed (typically 80).
- **minFlatness**: ratio of minimum projected area to actual area (typically 0.5)
- **minTetQuality**: minimum quality of tetrahedral cells from cell decomposition; generally deactivated by setting large negative number since v5.0 when new barycentric tracking was introduced, which could handle negative tets.
- **minVol**: minimum cell pyramid volume (typically 1e-13, large negative number disables).
- **minArea**: minimum face area (typically -1).
- **minTwist**: minimum face twist (typically 0.05).

- **minDeterminant**: minimum normalised cell determinant; $1 = \text{hex}$; $\leq 0 = \text{illegal cell}$ (typically 0.001).
- **minFaceWeight**: $0 \rightarrow 0.5$ (typically 0.05).
- **minVolRatio**: $0 \rightarrow 1.0$ (typically 0.01).
- **minTriangleTwist**: > 0 for Fluent compatibility (typically -1).
- **nSmoothScale**: number of error distribution iterations (typically 4).
- **errorReduction**: amount to scale back displacement at error points (typically 0.75).
- **relaxed**: sub-dictionary that can include modified values for the above keyword entries to be used when **nRelaxedIter** is exceeded in the layer addition process.

5.6 Mesh Zones

This section describes mesh *zones* and some popular tools which use zones as part of the meshing process. A zone is a list (or collection) of points or faces or cells within a mesh which is identifiable by a name. The list contains exclusively the indices of points, faces or cells, meaning there are three types of zone, represented in OpenFOAM by the classes `pointZone`, `faceZone` and `cellZone`. A unique name is required for each zone of a particular type, but the same name can be used for one `pointZone`, `faceZone` and `cellZone`.

A `cellZone` has many uses. One example is that it can define a region of the mesh to which a source term, *e.g.* a heat source, can be applied using an `fvModel`. Another is to represent a region of a mesh over which some quantity is calculated, *e.g.* the average temperature. A `cellZone` is often used to identify cells for some kind of modification, *e.g.* refinement, within a mesh generation process.

A `faceZone` can optionally include information about the orientation of the faces, in addition to their indices. The information is contained in a `flipMap`, whose purpose is to provide a consistent orientation for all faces, *e.g.* to make all faces “point” in a downstream direction. The `flipMap` contains a boolean (`true/false`) for each face. The sign (+/-) of normal vectors of faces marked `true` can then be flipped so that they are oriented in a consistent direction with normal vectors of faces marked `false`.

A `pointZone` is much less commonly used than a `cellZone` and `faceZone`. It is generally used for a prescribed mesh motion, in which some or all points in a mesh are re-positioned.

5.6.1 Creating zones

A zone can be created in two ways. First, it can be created as part of the meshing process, when the zone is included within the mesh within a relevant file in the *polyMesh* directory. The relevant files are *cellZones*, *faceZones* and *pointZones*, which can each contain a set of one or more zones (in fact, a file can exist but contain zero zones).

These “static” zones can be created by a mesh generation tool, *e.g.* `blockMesh` or `snapPyHexMesh`. Other utilities can also create them, especially the dedicated `createZones` tool, where the zones are configured through a *createZonesDict* file.

A zone can also be generated “dynamically” within an application, especially during a CFD simulation itself, *i.e.* within `foamRun` or `foamMultiRun`. These zones usually remain unchanged, following their generation at the beginning of a simulation. Alternatively, if a zone represents a fixed region of space, and the mesh is moving, the list of indices describing a zone can be updated dynamically during a simulation.

Dynamic zones are configured through a `zonesGenerator` file. The file can both include the configuration of new zones and may also involve “static” zones which have already been created as part of the mesh.

5.6.2 Zone generators

Zones can be configured in a `zonesGenerator` file, `createZonesDict` file and other configuration files for utilities described in following sections. There are several types of `zoneGenerator`, which are listed in this section. For detailed information about their configuration, the user can simply run the `foamInfo` script, *e.g.*

```
foamInfo union
```

Zones are always updated when there are topological changes within the mesh, *e.g.* meshing refinement and unrefinement. When the mesh moves, users have the option to retain the same elements (cells, faces, points) within the zone, such that the zone moves with the mesh. Alternatively, they can update the elements of a zone during a simulation by setting the `moveUpdate` switch to `on/yes/true`. This allows the zones to represent regions fixed in space with the mesh moving within it.

The first set of zone generators describe volume regions within the domain. They can create a zone of cells, faces or points contained within the volume region. Cells and faces are included when their centres are within the volume; points are included which are within the volume. The volume zone generators are listed below.

- **annulus**: selects cells, faces or points inside an annulus.
- **box**: selects cells, faces or points inside a box.
- **cylinder**: selects cells, faces or points inside a cylinder.
- **hemisphere**: selects cells, faces or points inside a hemisphere.
- **insideSurface**: selects cells, faces or points inside a closed surface described by a surface geometry file, *e.g.* `surface.obj`.
- **sphere**: selects cells, faces or points inside a sphere.
- **truncatedCone**: selects cells, faces or points inside a truncated cone.

An example configuration of a `hemisphere` zone generator is presented below (taking from running “`foamInfo hemisphere`”).

```

hemisphere1
{
    type            hemisphere;
    zoneType        cell;

    centre          (-0.001 0.001 0);
    axis            (0 1 0);
    radius          0.001;
}

```

There are then zone generators which are specific to cells, faces and points. The zone generators specifically for faces are listed below.

- **face**: creates a face zone from point, cell and face zones provided by a list of **zoneGenerators**.
- **flip**: inverts the **flipMap** of a face zone from a given **zoneGenerator**.
- **normal**: selects faces from a given **zoneGenerator** that are aligned with a specified normal direction.
- **orient**: sets the face orientation **flipMap** of a face zone.
- **patch**: creates a face zone from a set of patches.
- **plane**: creates a face zone from faces, whose vector connecting its adjacent cell centres, intersects a specified plane.
- **surface**: creates a face zone from faces whose vector, , whose vector connecting its adjacent cell centres, intersects a surface geometry, *e.g.* **surface.obj**.

As discussed in the introduction to this section, a face zone can optionally include a **flipMap**. A **flipMap** is required, for example, to calculate the flow rate through an area described by a face zone by summing the fluxes ϕ_f on all the faces in the face zone. That calculation is only meaningful if the face area vectors, used to calculate ϕ_f , point in a consistent direction. To ensure consistency, the sign of ϕ_f is inverted for faces marked **true** in the **flipMap**.

Specific cell zone generators are listed below.

- **cellZone**: selects and/or generates a cell zone for a tool that requires one, *e.g.* a **functionObject**, **fvModel**, **fvConstraint** or utility application.
- **cell**: creates a cell zone from existing **zoneGenerators**, by copying cells from any exiting cell zone(s), selecting cells containing faces of any face zones, and/or selecting cells containing points of any point zones.
- **containsPoints**: creates a cell zone containing the specified points.

- **nearPatchCells:** creates a cell zone from cells within a specified distance from a set of patches.
- **patchCells:** creates a cell zone from cells adjacent to a set of patches.

Specific point zone generators are listed below, some of which are specifically used in engine combustion

- **cylinderHeadPoints:** selects the points within the engine cylinder head.
- **externalPoints:** selects static points outside the interior of the cylinder liner in an engine simulation.
- **pistonBowlPoints:** selects the points within the engine piston bowl.
- **point:** creates a point zone from face, cell and point zones provided by list of zone-Generators.

The remaining zone generators create, manipulate and combine zones. The following ones perform “combination” operations. When used to create a face zone, they have specific rules about creating a **flipMap**.

- **difference:** selects cell, face or point elements in the first specified zone and removes the elements in subsequent specified zones.
- **intersection:** selects cell, face or point elements common to all the specified zones; can contain a **flipMap** if one face zone includes one.
- **invert:** selects cell, face or point elements in the mesh and removes the elements from the specified zones.
- **periodic:** activates a specified set of zones for a given period with optional repetition. equivalent zone; no **flipMap** in a face zone.
- **union:** selects all cell, face or point elements from all the specified zones; can contain a **flipMap** if all face zones include one.

The final set of zone generators perform general operations.

- **all:** creates a zone using all cells, faces or points of the mesh; no **flipMap** for a face zone.
- **clear:** clears all zones from a mesh.
- **lookup:** general zoneGenerator which looks up and returns a set of face, point and/or cell zones
- **remove:** removes specified zones from a mesh.
- **print:** prints a summary of zones into the terminal.
- **set:** converts a legacy cell, face or point set to an equivalent zone; no **flipMap** a the face zone.
- **write:** writes existing zones to the mesh, *e.g.* for visualisation.

5.6.3 Mesh zone configuration

There are some points of syntax that require explanation when configuring mesh zones. In a *createZonesDict* file, the following configuration would be valid to create a face zone named **baffle1** using the **surface** zone generator.

```
faceZone
{
    name          baffle1;
    type          plane;
    zoneType      face;

    point         (0 0 0);
    normal        (0 1 0);
}
```

The example above includes the formal entries read when generating a zone, including:

- **name:** (optional) name of the zone;
- **zoneType:** (can be omitted where possible) type of the zone.

The **name** entry is optional; when omitted, the name of the sub-dictionary will be used for the zone name. Also, when the zone generator **type** is one which generates one specific type of zone, the **zoneType** entry can be omitted. Therefore, the entry above can instead be written.

```
baffle1          // name of zone
{
    type          plane;
    point         (0 0 0);
    normal        (0 1 0);
}
```

Below is an example of another face zone named **baffle**, created by the union of the **baffle1** face zone and another face zone named **baffle2**.

```
baffle           // name of zone
{
    type          union;

    baffle2
    {
        type      surface;
        surface    triSurface;
        file       "baffles.obj";
    }
}
```

```

    baffles1
    {
        type        lookup;
    }
}

```

The syntax above indicates that the **baffle2** face zone is generated and combined with the **baffle1** face zone using the **union** zone generator. The **baffle1** face zone is assumed already generated, and is lookup up with the **lookup** zone generator.

For the **lookup zone generator**, the user should be aware of the **alternative syntax** available, which is shorter and commonly appears in example cases. The shorthand syntax simply specifies the name of the zone, followed by a semicolon, *i.e.* as follows.

```

baffle          // name of zone
{
    type        union;

    baffles2
    {
        type        surface;
        surface     triSurface;
        file        "baffles.obj";
    }

    baffles1;
}

```

The names of each class of zone — cell, face and point — must be unique. However, the same name can be used across classes of zone, *e.g.* **baffle1** could be used from both a face zone and a point zone. In such circumstance, the shorthand syntax cannot be used because the **zoneType** must be specified as shown below.

```

baffle          // name of zone
{
    type        union;

    baffles2
    {
        type        surface;
        surface     triSurface;
        file        "baffles.obj";
    }

    baffles1
    {

```

```

        type      lookup;
        zoneType   face;
    }
}

```

5.6.4 The createZones utility

The `createZones` utility creates “static” zones that form part of a mesh. There are many example cases in OpenFOAM that use `createZones` that can be inspected to learn about its use. One example is the `coolingSphere` case (in `$FOAM_TUTORIALS/multiRegion/CHT`) which generates a cell zone that is subsequently used to form a separate mesh region for a solid sphere. The configuration from the `createPatchDict` file is shown below.

```

16
17 solid
18 {
19     type      sphere;
20     zoneType   cell;
21
22     centre     (0 0 0);
23     radius      $blockMeshDict!geometry/sphere/radius;
24 }
25
26 // *****

```

It generates a zone named `solid` using the `sphere` zone generator. The zone type is a `cell` zone. The remaining parameters describe the sphere, namely the `centre` and `radius`. The latter is read from the `blockMeshDict` file using a macro expansion, described in section 4.2.10.

The mesh and cell zone are generated by running `blockMesh` and `createZones` respectively. `blockMesh` first produces the standard mesh files in the `constant/polyMesh`. `createZones` then generates an additional `cellZones` file in that directory, containing the list of cell indices that make up the zone.

5.6.5 The refineMesh utility

The `refineMesh` utility refines cells in a mesh. The refinement can be applied to cells in the entire mesh or to specified regions of the mesh. The refinement is applied by splitting cells, which can be applied in all three directions or on one or two specified directions.

To apply refinement across the entire mesh and in all three directions, `refineMesh` can be run using the `-all` option, *i.e.*

```
refineMesh -all
```

Otherwise, the region of the mesh and/or refinement directions must be configured using a `refineMeshDict` file.

There are many example cases in OpenFOAM which include `refineMeshDict` files. One is the `pipeCyclic` case (in `$FOAM_TUTORIALS/incompressibleFluid`) whose `refineMeshDict` file is shown below.

```

16
17 hexRef8      yes;
18
19 zone

```

```

20 {
21     type    box;
22     box      (-1e6 -1e6 -1e6) (1e6 -0.15 0.3);
23 }
24
25 // *****

```

The file includes a **zone** entry to describe a cell zone configured with a **box** zone generator. The refinement is specified through the **hexRef8** switch, set to on. **hexRef8** refines cells in all directions, splitting each selected cell $2 \times 2 \times 2$. It is the method used by **snappyHexMesh** which keeps a record of the refinement history in a *refinementHistory* file in the *polyMesh* directory.

The DTCHull case (in *\$FOAM_TUTORIALS/incompressibleVoF*) is another example that uses the **refineMesh** utility. It is a case of ship hydrodynamics which generates an initial mesh which has high aspect ratio cells, stretched in the horizontal plane, in the vicinity of the water-air interface. Before meshing the ship hull with **snappyHexMesh**, the aspect ratio of cells is gradually reduced from the far field to the hull, by successive refinements of cells in the horizontal directions. The *refineMeshDict* file for this case is shown below.

```

16
17 coordinates
18 {
19     type      global;
20
21     e1        (1 0 0);
22     e2        (0 1 0);
23
24     directions (e1 e2);
25 }
26
27 zones
28 {
29     level1
30     {
31         type    box;
32         box      (-10 -6 -3) (10 0 3);
33     }
34
35     level2
36     {
37         type    box;
38         box      (-5 -3 -2.5) (9 0 2);
39     }
40
41     level3
42     {
43         type    box;
44         box      (-3 -1.5 -1) (8 0 1.5);
45     }
46
47     level4
48     {
49         type    box;
50         box      (-2 -1 -0.6) (7 0 1);
51     }
52
53     level5
54     {
55         type    box;
56         box      (-1 -0.6 -0.3) (6.5 0 0.8);
57     }
58
59     level6
60     {
61         type    box;
62         box      (-0.5 -0.55 -0.15) (6.25 0 0.65);
63     }
64 }
65
66 // *****

```

It first describes the refinement directions by the **coordinates** sub-dictionary. The coordinates must be specified by two orthogonal directions described by the **e1** and **e2** vectors. The **directions** parameter then lists the refinement directions, which could theoretically include **e3**, the vector calculated orthogonal to **e1** and **e2** (forming a right-handed set of axes).

The refinement is then specified in zones described in the a **zones** sub-dictionary. Six **box** zones are listed from largest to smallest. Refinement is applied successively to each zone so that it produces six levels of refinement in the final box, five levels in the region described by the fifth box beyond the final box, and so on.

5.6.6 The createPatch utility

The **createPatch** utility creates new patches from collections of boundary faces. It enables users to make changes to the patch configuration in a mesh, often in the final stages of a meshing workflow. The utility is configured by a *createPatchDict* file.

There are some examples in OpenFOAM which use **createPatch**. Most of them create a patch from faces defined by a face zone. The *hotRoomComfort* example (in *\$FOAM_TUTORIALS/fluid*) contains a *createPatchDict* file which creates an **inlet** and an **outlet** patch in a mesh whose boundary initially contains a single **wall** patch. The *createPatchDict* file is given below, showing each patch is generated from a **box** zone.

```

16
17 patches
18 {
19     inlet
20     {
21         // Dictionary to construct new patch from
22         patchInfo
23         {
24             type patch;
25         }
26
27         // Construct from zone
28         constructFrom zone;
29
30         // Generate zone
31         zone
32         {
33             type          box;
34             box            (-0.001 0.25 1.1)(0.001 0.75 1.3);
35         }
36     }
37
38     outlet
39     {
40         // Dictionary to construct new patch from
41         patchInfo
42         {
43             type patch;
44         }
45
46         // Construct from zone
47         constructFrom zone;
48
49         // Generate zone
50         zone
51         {
52             type          box;
53             box            (1.75 2.999 0.3)(2.25 3.001 0.5);
54         }
55     }
56 }
57
58 // ***** //
```


The `potentialFreeSurfaceMovingOscillatingBox` case (in `$FOAM_TUTORIALS/isothermalFluid`) uses the `createPatchDict` shown below.

```

16
17 patches
18 {
19     floatingObjectBottom
20     {
21         // Dictionary to construct new patch from
22         patchInfo
23         {
24             type wall;
25         }
26         // Construct patch from a faceZone
27         constructFrom zone;
28     }
29     zone
30     {
31         type normal;
32         normal (0 1 0);
33         tol 0.01;
34         floatingObject
35         {
36             type patch;
37             patch floatingObject;
38         }
39     }
40 }
41 }
42 }
43 }
44 }
45 // *****
46
```

This example uses a `normal` zone generator applied to a zone generated from `patch`. The `patch` zone generation is configured within the `normal` zone configuration, resulting in a zone containing faces, oriented with the specified normal direction, that are extracted from the patch.

5.6.7 The subsetMesh utility

The `subsetMesh` utility creates a new mesh which is a subset of an existing mesh. It is commonly used in a meshing workflow which creates a mesh, then uses `subsetMesh` to remove part of the mesh.

The `floatingObject` tutorial (in `$FOAM_TUTORIALS/incompressibleVoF`) provides a good example of its use. A mesh is initially created of a box-shaped domain. A floating object is then introduced corresponding to a zone configured in `subsetMeshDict`, shown below.

```

16
17 zone
18 {
19     type box;
20     select outside;
21     box (0.35 0.35 0.1) (0.65 0.55 0.6);
22 }
23
24 patch floatingObject;
25
26 // *****

```

The `zone` is configured by defining a `box` with the addition of the optional `selection` keyword which selects cells `outside` the box. Consequently, the new mesh becomes the original mesh without the cells contained within the box, which represents the floating object. A `patch` entry provides a name for the patch formed by the faces exposed by removal of the cells.

5.7 Mesh conversion

The user can generate meshes using other packages and convert them into the format that OpenFOAM uses. There are numerous mesh conversion utilities listed in section 3.7.3. Some of the more popular mesh converters are listed below and their use is presented in this section.

`fluentMeshToFoam` reads a `Fluent.msh` mesh file, working for both 2-D and 3-D cases;

`starToFoam` reads STAR-CD/PROSTAR mesh files.

`gambitToFoam` reads a `GAMBIT.neu` neutral file;

`ideasToFoam` reads an I-DEAS mesh written in `ANSYS.ans` format;

`cfx4ToFoam` reads a CFX mesh written in `.geo` format;

5.7.1 `fluentMeshToFoam`

Fluent writes mesh data to a single file with a `.msh` extension. The file must be written in ASCII format, which is not the default option in Fluent. It is possible to convert single-stream Fluent meshes, including the 2 dimensional geometries. In OpenFOAM, 2 dimensional geometries are currently treated by defining a mesh in 3 dimensions, where the front and back plane are defined as the `empty` boundary patch type. When reading a 2 dimensional Fluent mesh, the converter automatically extrudes the mesh in the third direction and adds the empty patch, naming it `frontAndBackPlanes`.

The following features should also be observed.

- The OpenFOAM converter will attempt to capture the Fluent boundary condition definition as much as possible; however, since there is no clear, direct correspondence between the OpenFOAM and Fluent boundary conditions, the user should check the boundary conditions before running a case.
- Creation of axi-symmetric meshes from a 2 dimensional mesh is currently not supported but can be implemented on request.
- Multiple material meshes are not permitted. If multiple fluid materials exist, they will be converted into a single OpenFOAM mesh; if a solid region is detected, the converter will attempt to filter it out.
- Fluent allows the user to define a patch which is internal to the mesh, *i.e.* consists of the faces with cells on both sides. Such patches are not allowed in OpenFOAM and the converter will attempt to filter them out.
- There is currently no support for embedded interfaces and refinement trees.

The procedure of converting a `Fluent.msh` file is first to create a new OpenFOAM case by creating the necessary directories/files: the case directory containing a `controlDict` file in a `system` subdirectory. Then at a command prompt the user should execute:

```
fluentMeshToFoam <meshFile>
```

where `<meshFile>` is the name of the `.msh` file, including the full or relative path.

5.7.2 starToFoam

This section describes how to convert a mesh generated on the STAR-CD code into a form that can be read by OpenFOAM mesh classes. The mesh can be generated by any of the packages supplied with STAR-CD, *i.e.* PROSTAR, SAMM, ProAM and their derivatives. The converter accepts any single-stream mesh including integral and arbitrary couple matching and all cell types are supported. The features that the converter does not support are:

- multi-stream mesh specification;
- baffles, *i.e.* zero-thickness walls inserted into the domain;
- partial boundaries, where an uncovered part of a couple match is considered to be a boundary face;
- sliding interfaces.

For multi-stream meshes, mesh conversion can be achieved by writing each individual stream as a separate mesh and reassemble them in OpenFOAM.

OpenFOAM adopts a policy of only accepting input meshes that conform to the fairly stringent validity criteria specified in section 5.1. It will simply not run using invalid meshes and cannot convert a mesh that is itself invalid. The following sections describe steps that must be taken when generating a mesh using a mesh generating package supplied with STAR-CD to ensure that it can be converted to OpenFOAM format. To avoid repetition in the remainder of the section, the mesh generation tools supplied with STAR-CD will be referred to by the collective name STAR-CD.

We strongly recommend that the user run the STAR-CD mesh checking tools before attempting a **starToFoam** conversion and, after conversion, the **checkMesh** utility should be run on the newly converted mesh. Alternatively, **starToFoam** may itself issue warnings containing PROSTAR commands that will enable the user to take a closer look at cells with problems. Problematic cells and matches should be checked and fixed before attempting to use the mesh with OpenFOAM. Remember that an invalid mesh will not run with OpenFOAM, but it may run in another environment that does not impose the validity criteria.

Some problems of tolerance matching can be overcome by the use of a matching tolerance in the converter. However, there is a limit to its effectiveness and an apparent need to increase the matching tolerance from its default level indicates that the original mesh suffers from inaccuracies.

When mesh generation is completed, remove any extraneous vertices and compress the cells boundary and vertex numbering, assuming that fluid cells have been created and all other cells are discarded. This is done with the following PROSTAR commands:

```
CSET NEWS FLUID
CSET INVE
```

The CSET should be empty. If this is not the case, examine the cells in CSET and adjust the model. If the cells are genuinely not desired, they can be removed using the PROSTAR command:

```
CDEL CSET
```

Similarly, vertices will need to be discarded as well:

```
CSET NEWS FLUID
VSET NEWS CSET
VSET INVE
```

Before discarding these unwanted vertices, the unwanted boundary faces have to be collected before purging:

```
CSET NEWS FLUID
VSET NEWS CSET
BSET NEWS VSET ALL
BSET INVE
```

If the BSET is not empty, the unwanted boundary faces can be deleted using:

```
BDEL BSET
```

At this time, the model should contain only the fluid cells and the supporting vertices, as well as the defined boundary faces. All boundary faces should be fully supported by the vertices of the cells, if this is not the case, carry on cleaning the geometry until everything is clean.

By default, STAR-CD assigns wall boundaries to any boundary faces not explicitly associated with a boundary region. The remaining boundary faces are collected into a **default** boundary region, with the assigned boundary type 0. OpenFOAM deliberately does not have a concept of a **default** boundary condition for undefined boundary faces since it invites human error, *e.g.* there is no means of checking that we meant to give all the unassociated faces the default condition.

Therefore **all** boundaries for each OpenFOAM mesh must be specified for a mesh to be successfully converted. The **default** boundary needs to be transformed into a real one using the procedure described below:

1. Plot the geometry with **Wire Surface** option.
2. Define an extra boundary region with the same parameters as the **default** region 0 and add all visible faces into the new region, say 10, by selecting a zone option in the boundary tool and drawing a polygon around the entire screen draw of the model. This can be done by issuing the following commands in PROSTAR:

```
RDEF 10 WALL
BZON 10 ALL
```

3. We shall remove all previously defined boundary types from the set. Go through the boundary regions:

```

BSET NEWS REGI 1
BSET NEWS REGI 2
... 3, 4, ...

```

Collect the vertices associated with the boundary set and then the boundary faces associated with the vertices (there will be twice as many of them as in the original set).

```

BSET NEWS REGI 1
VSET NEWS BSET
BSET NEWS VSET ALL
BSET DELE REGI 1
REPL

```

This should give the faces of boundary Region 10 which have been defined on top of boundary Region 1. Delete them with BDEL BSET. Repeat these for all regions.

Renumber and check the model using the commands:

```

CSET NEW FLUID
CCOM CSET

VSET NEWS CSET
VSET INVE (Should be empty!)
VSET INVE
VCOM VSET

BSET NEWS VSET ALL
BSET INVE (Should be empty also!)
BSET INVE
BCOM BSET

CHECK ALL
GEOM

```

Internal PROSTAR checking is performed by the last two commands, which may reveal some other unforeseeable error(s). Also, take note of the scaling factor because PROSTAR only applies the factor for STAR-CD and not the geometry. If the factor is not 1, use the `scalePoints` utility in OpenFOAM.

Once the mesh is completed, place all the integral matches of the model into the couple type 1. All other types will be used to indicate arbitrary matches.

```

CPSET NEWS TYPE INTEGRAL
CPMOD CPSET 1

```

The components of the computational grid must then be written to their own files. This is done using PROSTAR for boundaries by issuing the command

BWRITE

by default, this writes to a *.23* file (versions prior to 3.0) or a *.bnd* file (versions 3.0 and higher). For cells, the command

CWRITE

outputs the cells to a *.14* or *.cel* file and for vertices, the command

VWRITE

outputs to file a *.15* or *.vrt* file. The current default setting writes the files in ASCII format. If couples are present, an additional couple file with the extension *.cpl* needs to be written out by typing:

CPWRITE

After outputting to the three files, exit PROSTAR or close the files. Look through the panels and take note of all STAR-CD sub-models, material and fluid properties used – the material properties and mathematical model will need to be set up by creating and editing OpenFOAM dictionary files.

The procedure of converting the PROSTAR files is first to create a new OpenFOAM case by creating the necessary directories. The PROSTAR files must be stored within the same directory and the user must change the file extensions: from *.23*, *.14* and *.15* (below STAR-CD version 3.0), or *.pcs*, *.cls* and *.vtx* (STAR-CD version 3.0 and above); to *.bnd*, *.cel* and *.vrt* respectively.

The *.vrt* file is written in columns of data of specified width, rather than free format. A typical line of data might be as follows, giving a vertex number followed by the coordinates:

```
19422      -0.105988957      -0.413711881E-02 0.000000000E+00
```

If the ordinates are written in scientific notation and are negative, there may be no space between values, *e.g.*:

```
19423      -0.953953117E-01-0.338810333E-02 0.000000000E+00
```

The *starToFoam* converter reads the data using spaces to delimit the ordinate values and will therefore object when reading the previous example. Therefore, OpenFOAM includes a simple script, *foamCorrectVrt* to insert a space between values where necessary, *i.e.* it would convert the previous example to:

```
19423      -0.953953117E-01 -0.338810333E-02 0.000000000E+00
```

The *foamCorrectVrt* script should therefore be executed if necessary before running the *starToFoam* converter, by typing:

```
foamCorrectVrt <file>.vrt
```

The translator utility `starToFoam` can now be run to create the boundaries, cells and points files necessary for a OpenFOAM run:

```
starToFoam <meshFilePrefix>
```

where `<meshFilePrefix>` is the name of the prefix of the mesh files, including the full or relative path. After the utility has finished running, OpenFOAM boundary types should be specified by editing the *boundary* file by hand.

5.7.3 gambitToFoam

GAMBIT writes mesh data to a single file with a *.neu* extension. The procedure of converting a GAMBIT *.neu* file is first to create a new OpenFOAM case, then at a command prompt, the user should execute:

```
gambitToFoam <meshFile>
```

where `<meshFile>` is the name of the *.neu* file, including the full or relative path.

The GAMBIT file format does not provide information about type of the boundary patch, *e.g.* wall, symmetry plane, cyclic. Therefore all the patches have been created as type patch. Please reset after mesh conversion as necessary.

5.7.4 ideasToFoam

OpenFOAM can convert a mesh generated by I-DEAS but written out in ANSYS format as a *.ans* file. The procedure of converting the *.ans* file is first to create a new OpenFOAM case, then at a command prompt, the user should execute:

```
ideasToFoam <meshFile>
```

where `<meshFile>` is the name of the *.ans* file, including the full or relative path.

5.7.5 cfx4ToFoam

CFX writes mesh data to a single file with a *.geo* extension. The mesh format in CFX is block-structured, *i.e.* the mesh is specified as a set of blocks with glueing information and the vertex locations. OpenFOAM will convert the mesh and capture the CFX boundary condition as best as possible. The 3 dimensional ‘patch’ definition in CFX, containing information about the porous, solid regions *etc.* is ignored with all regions being converted into a single OpenFOAM mesh. CFX supports the concept of a ‘default’ patch, where each external face without a defined boundary condition is treated as a **wall**. These faces are collected by the converter and put into a `defaultFaces` patch in the OpenFOAM mesh and given the type **wall**; of course, the patch type can be subsequently changed.

Like, OpenFOAM 2 dimensional geometries in CFX are created as 3 dimensional meshes of 1 cell thickness. If a user wishes to run a 2 dimensional case on a mesh created by CFX, the boundary condition on the front and back planes should be set to **empty**; the user should ensure that the boundary conditions on all other faces in the plane of the calculation are set correctly. Currently there is no facility for creating an axi-symmetric geometry from a 2 dimensional CFX mesh.

The procedure of converting a CFX.*geo* file is first to create a new OpenFOAM case, then at a command prompt, the user should execute:

```
cfx4ToFoam <meshFile>
```

where <meshFile> is the name of the *.geo* file, including the full or relative path.

5.8 Mapping fields between different geometries

The **mapFields** utility maps one or more fields relating to a given geometry onto the corresponding fields for another geometry. It is completely generalised in so much as there does not need to be any similarity between the geometries to which the fields relate. However, for cases where the geometries are consistent, **mapFields** can be executed with a special option that simplifies the mapping process.

For our discussion of **mapFields** we need to define a few terms. First, we say that the data is mapped from the *source* to the *target*. The fields are deemed *consistent* if the geometry *and* boundary types, or conditions, of both source and target fields are identical. The field data that **mapFields** maps are those fields within the time directory specified by **startFrom/startTime** in the *controlDict* of the target case. The data is read from the equivalent time directory of the source case and mapped onto the equivalent time directory of the target case.

5.8.1 Mapping consistent fields

A mapping of consistent fields is simply performed by executing **mapFields** on the (target) case using the **-consistent** command line option as follows:

```
mapFields <source dir> -consistent
```

5.8.2 Mapping inconsistent fields

When the fields are not consistent, as shown in Figure 5.15, **mapFields** requires a *mapFieldsDict* dictionary in the *system* directory of the target case. The following rules apply to the mapping:

- the field data is mapped from source to target wherever possible, *i.e.* in our example all the field data within the target geometry is mapped from the source, except those in the shaded region which remain unaltered;

- the patch field data is left unaltered unless specified otherwise in the *mapFieldsDict* dictionary.

The *mapFieldsDict* dictionary contains two lists that specify mapping of patch data. The first list is *patchMap* that specifies mapping of data between pairs of target and source patches that are geometrically coincident, as shown in Figure 5.15. The list contains each pair of names of target patch (first) and source patch (second). The second list is *cuttingPatches* that contains names of target patches whose values are to be mapped from the source internal field through which the target patch cuts. In the situation where the target patch only cuts through part of the source internal field, *e.g.* bottom left target patch in our example, those values within the internal field are mapped and those outside remain unchanged. An example

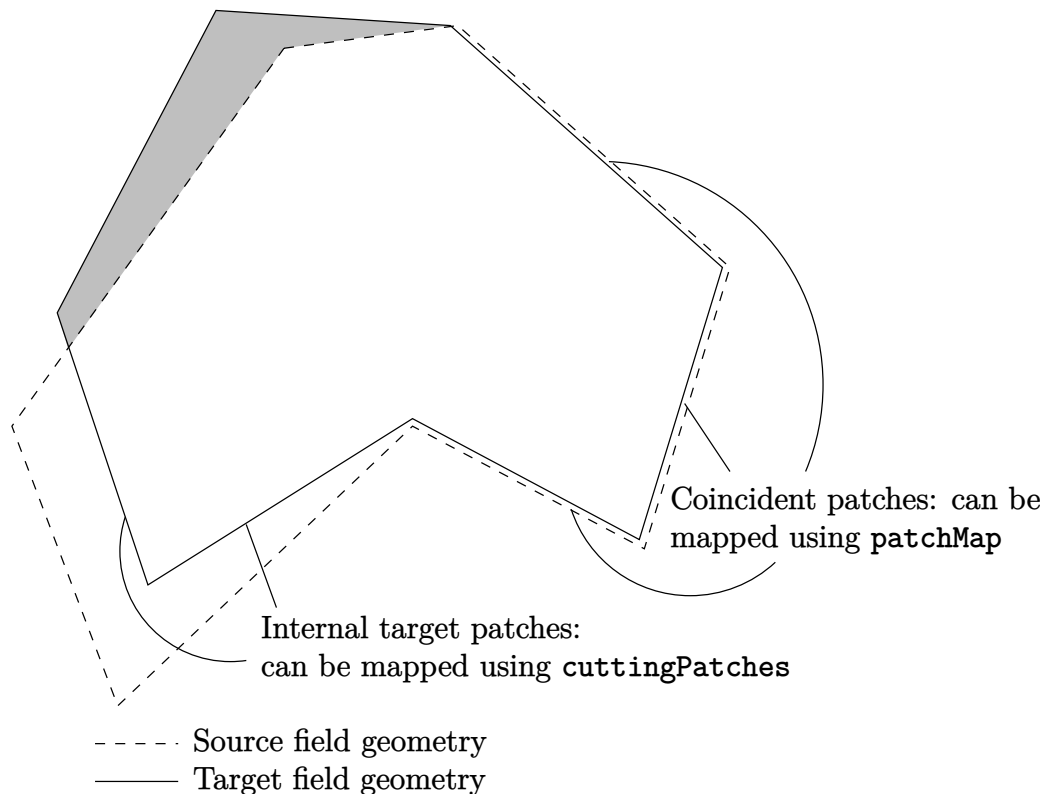


Figure 5.15: Mapping inconsistent fields

mapFieldsDict dictionary is shown below:

```

16
17 patchMap      (lid movingWall);
18
19 cuttingPatches ();
20
21
22 // *****

```

mapFields <source dir>

5.8.3 Mapping parallel cases

If either or both of the source and target cases are decomposed for running in parallel, additional options must be supplied when executing *mapFields*:

- parallelSource if the source case is decomposed for parallel running;
- parallelTarget if the target case is decomposed for parallel running.

Chapter 6

Field and boundary initialisation

This chapter describes the initialisation of fields and the configuration of the field's boundary conditions. Fields and boundary conditions are specified in field files, *e.g.* p , U . They are stored in time directories, with the initialised fields almost invariably configured in the 0 directory, corresponding to the start time $t = 0$.

The *syntax* of field files is introduced in a tutorial example in section 2.1.4 and then described more comprehensively in section 4.2.9. They include: the **dimensions** entry for the dimensional units; **internalField** for the initial internal field values; **boundaryField**, where the boundary conditions are specified; and, optionally, **sources** to specify the conditions on fluid that is introduced by a mass or volume source.

This chapter begins with initialisation of the **internalField** using *field functions* in section 6.1. The configuration of **sources** is then explained in section 6.2. The rest of the chapter then discusses popular boundary conditions for setting the **boundaryField** section of field files.

6.1 Field functions

In the configuration of a CFD simulation, it is quite common for some fields to be initialised with an **internalField** that is nonuniform. Similarly in the **boundaryField**, initial values on the faces of some patches may also need to be nonuniform. There are many examples, where the initialisation of nonuniform fields is required, but a very common example occurs in multiphase flows where the initial phase fraction field, *e.g.* ***alpha.water***, determines the position of the phase interface, and with it, the height of the liquid phase.

Prior to OpenFOAM v14, nonuniform fields were created firstly by configuring a uniform field, *e.g.* with ***alpha.water*** = 0 corresponding to “all air”. Then a pre-processing utility, typically **setFields**, would need to be configured and executed. It would read the initial uniform field, modify it and write out a new nonuniform field. Usually the initial field file would be created using the *.orig* extension, *e.g.* ***alpha.water.orig***, so that it would not be overwritten (and thus, deleted) when the generated nonuniform field was written using conventional name ***alpha.water***.

In OpenFOAM v14 however, field functions have been introduced that initialise nonuniform fields directly within the field files. This “local” initialisation is more self-contained, avoids the need to run a pre-processing utility, and offers more options to specify the initial-

isation up to even the most complex functions.

Field functions are specified for an `internalField` by replacing the `uniform/nonuniform` keyword in the usual syntax, *e.g.*

```
internalField uniform 0;
```

with a sub-dictionary entry that begins with the `type` keyword that specifies the choice of field function, *e.g.*

```
internalField
{
    type    <field function type>;
    ...
```

There are six general field function that can be selected, listed below.

- **zonal**: field values are specified in different zones of the mesh.
- **surfaces**: field values are specified in different regions of mesh, separated by surfaces, *e.g.* searchable surfaces, STL/OBJ files, *etc.*
- **coded**: field values are defined by some code which is dynamically compiled.
- **distanceFunction**: field values are calculated using a `Function1` of distance across the domain.
- **fieldFunction**: field values are calculated using a `Function1` of another field, *e.g.* a function of the temperature field `T`.
- **timeFunction**: field values are calculated using a `Function1` of time.

The following sections explain the use of each field function, with examples.

6.1.1 The zonal field function

The **zonal** field function is illustrated by the `$FOAM_TUTORIALS/shockFluid/shockTube 1D` example case. The `0/p` file is initialised with $p = 1$ bar for $x < 0$ and $p = 0.1$ bar for $x \geq 0$. The configuration of the `internalField` is as follows.

```
internalField
{
    type            zonal;
    defaultValue    1 [bar];

    zones
    {
        lowPressure
        {
```

```

        type    box;
        box      (0 -1 -1) (5 1 1);
        value    0.1 [bar];
    }
}

```

This works by first setting the `defaultValue` of the field to 1 bar. A cell zone is then created, named `lowPressure`, using a `box zoneGenerator`, which bounds the region of the mesh $x \geq 0$. Within the `lowPressure` zone, the pressure is set to 0.1 bar.

6.1.2 The surfaces field function

The `$FOAM_TUTORIALS/compressibleVoF/depthCharge3D` example case demonstrates the use of the `surfaces` field function. In the case this field function initialises the phase fraction field in the `0/alpha.water` file. It does so by initially setting the `defaultValue` to 1, *i.e.* to the water phase, before overriding the `value` to 0 to represent air both above an air-water interface and inside a bubble within the water region. The configuration of the `internalField` is as follows.

```

internalField
{
    type          surfaces;
    defaultValue  1 ;

    surfaces
    {
        air
        {
            type    plane;
            point    (0 1 0);
            normal    (0 -1 0);
            value     0;
        }

        bubble
        {
            type      sphere;
            centre     (0.5 0.5 0.5);
            radius      0.1;
            value       0;
        }
    }
}

```

The **plane** surface is an infinite plane, positioned at $y = 1$ m and with its normal in the $-y$ -direction. Since normal vectors conventionally point out of a surface, the notional “inside” is above the surface at $y > 1$ m, which is set to a value of 0, representing air.

The **sphere** surface represents a sphere, defined by the **centre** and **radius**. Inside the sphere the value is 0, representing an air bubble, as shown in Figure 6.1.

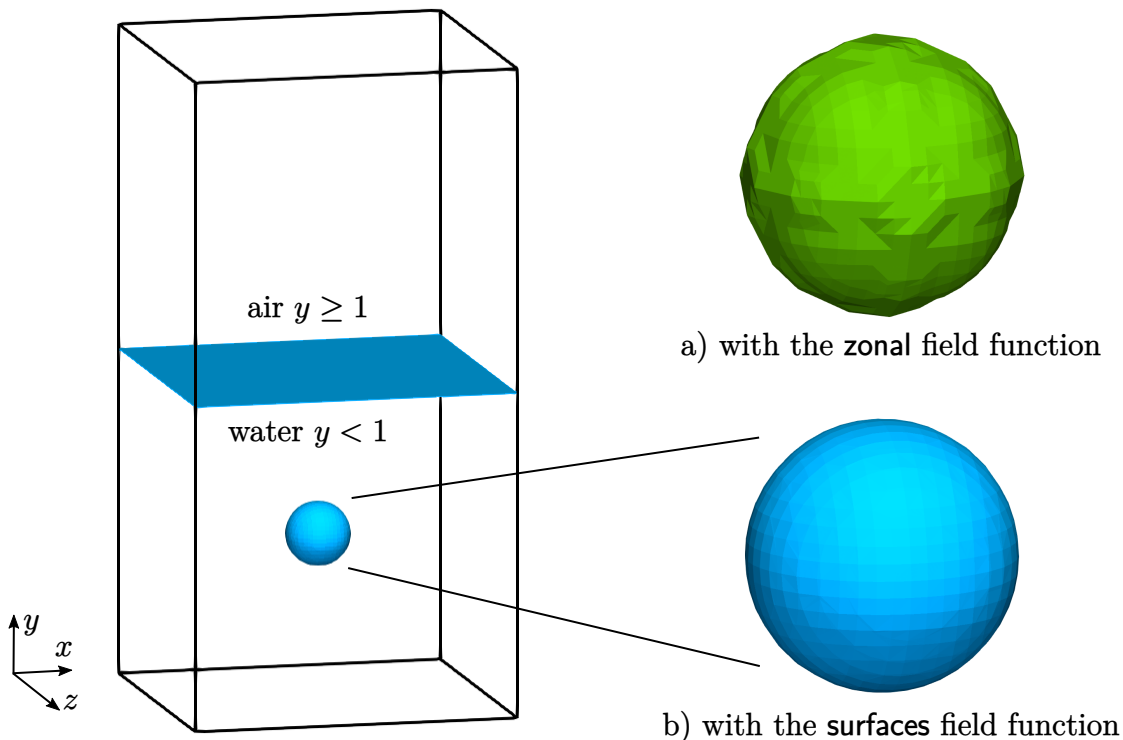


Figure 6.1: Field initialisation with the **surfaces** function object.

The **surfaces** field function performs cell cutting and calculates intermediate values in the cut cells. Cells intersected by the sphere surface therefore have intermediate values between 0 and 1. Consequently, the air-water interface, visualised by an iso-surface of `alpha.water = 0.5`, appears smooth as in Figure 6.1b).

The **zonal** field function, by contrast, does not perform surface cutting. If it were to initialise the bubble with a **sphere** zone, the values of `alpha.water` would be either 0 or 1, depending on whether a given cell is inside the zone or not. Without intermediate values in around the interface, the iso-surface appears rough in Figure 6.1a). Evidently the initialisation with the **surfaces** field function is more accurate than using **zonal**.

6.1.3 The coded field function

The **coded** field function allows the user to write code to describe the initial distribution of a field. This provides complete flexibility in initialising a field but can naturally be more complex than other options. So before attempting to use the **coded** field function, it is recommended to check whether one of the other options will work.

If the initialisation is not zonal in nature, the `zonal` and `surfaces` field functions are unlikely to be useful. Other than `coded`, all the remaining field function use a `Function1` to describe the distribution. `Function1` provides a wide range of pre-defined functions of one variable, discussed in section 6.4.4, where the “one variable” is distance across the domain, time and another field for the `distanceFunction`, `timeFunction` and `fieldFunction` field functions, respectively.

If none of those field functions are suitable, the `coded` is likely the best option. One potential starting point for configuring such a field function is the `$FOAM_TUTORIALS/incompressibleFluid/pipeCyclic` example case. The case uses a quarter section of a pipe to demonstrate the behaviour of the `cyclic` patch types. To encourage the fluid through the `cyclic` patch, a swirl flow is applied at the `inlet` patch and also the `internalField` is initialised with the same swirl flow, as shown in Figure 6.2.

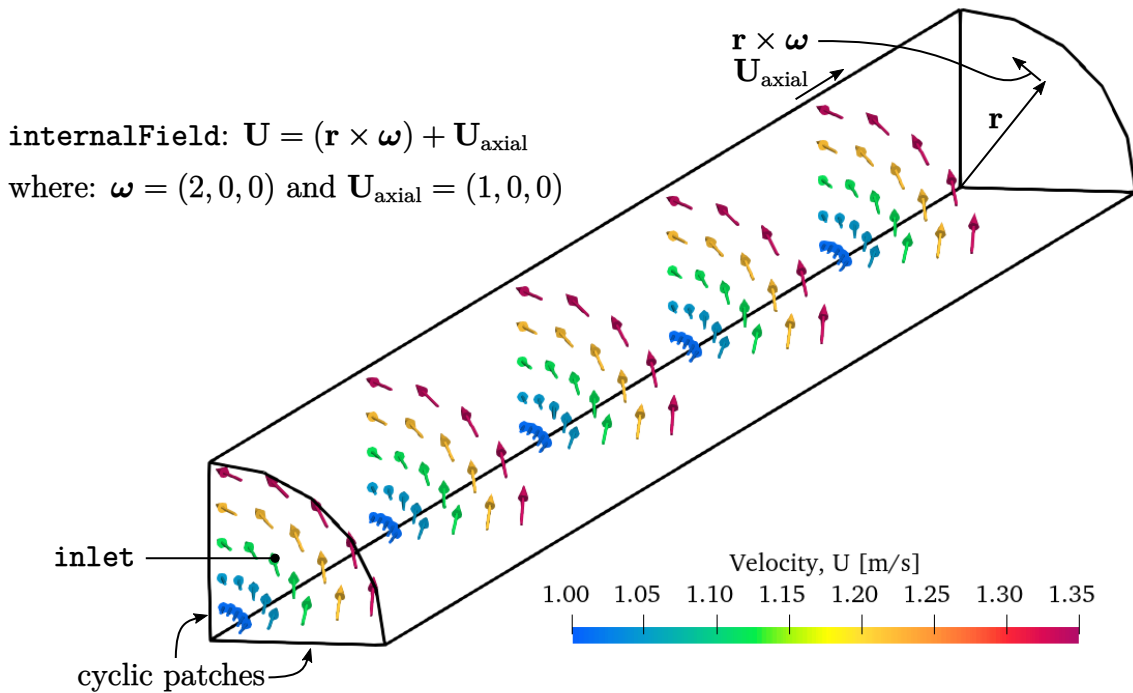


Figure 6.2: Field initialisation with the `coded` function object.

The swirl flow is defined by

$$\mathbf{U} = (\mathbf{r} \times \boldsymbol{\omega}) + \mathbf{U}_{\text{axial}}, \quad (6.1)$$

where $\mathbf{U}_{\text{axial}} = (1, 0, 0) \text{ m s}^{-1}$ is the axial velocity, $\boldsymbol{\omega} = (2, 0, 0) \text{ rad s}^{-1}$ is the rotational speed and $\mathbf{r}$ is the position vector.

The `coded` field function is implemented in for `internalField` in the `0/U` file. It includes a remarkably compact piece of code that represents Equation 6.1 using OpenFOAM’s equation syntax that includes ‘ $\wedge$ ’ to denote the cross product ($\times$).

`internalField`

```

{
    type          coded;

    evaluate
    #{
        field = velocity(1, 0, 0) + (C ^ rate(2, 0, 0));
    #};
}

```

Within the coded framework, there are two code block that can be defined.

- `evaluate #{...#};` to initialise the field at the beginning of the simulation.
- `update #{...#};` to update the field each time step during the simulation.

In the example, only the former is defined since the field is only initialised and requires no subsequent update during the simulation. To make code implementation easier the coded framework providing the following references.

- `field`: reference to the `dimensionedField`, which includes the `internalField` and `dimensions`.
- `C`: the cell centre vectors, also a `dimensionedField`, denoting ‘position’ in the domain.
- `t`: the time, as a `dimensionedScalar`.

In the code, therefore, the user can calculate any function of position and time assign the result to the field with `field = ...`. Calculations, and the assignment, include dimensions so are checked for dimensional consistency.

Consequently, the code generally requires parameters to be defined with dimensions. Often there are single valued parameters with dimensions, *i.e.* `dimensioned` values. The `dimensionSet` element can be specified using named dimensions — whose names are listed by running `foamUnits`, *e.g.* `velocity`, `kinematicViscosity`, *etc.* These names exist within a `dimensions` namespace in OpenFOAM, so would normally be specified with the qualifier, *e.g.* `dimensions::velocity`. However, the coded framework includes the directive

```
using namespace dimensions;
```

so the named dimensions do not require a qualifier, *i.e.* “`velocity`” is sufficient to specify the dimensions of velocity in the code. The `dimensioned` parameter for velocity would then normally be constructed as follows:

```
dimensioned<vector>(velocity, vector(1, 0, 0))
```

To simplify this code, OpenFOAM v14 includes a variadic template function to create a `dimensioned` parameter using the syntax

```
<named dimension>(<cmpt0>, <cmpt1>, ...)
```


where components `cmpt0`, *etc.* can include any number of components for scalar, vector, tensor, *etc.* For example, for a vector with 3 components and dimensions of velocity, the code combines the `velocity` dimensions with the 3 components of the vector, as follows.

```
velocity(1, 0, 0)
```

In addition to the specific “hooks”, *i.e.* the `field`, `C` and `t`, a coded field function can include any OpenFOAM code to perform the calculations required. For example, the mesh and its data can be accessed directly from the `field` reference, by calling the standard `mesh()` access function with `field.mesh()`. The user therefore has complete freedom to initialise a field within the limits of the knowledge and skill of programming with OpenFOAM.

6.1.4 The distanceFunction field function

The `distanceFunction` is the first of the field functions where the initialisation is performed using a `Function1`. As mentioned earlier, `Function1` provides a wide range of pre-defined functions of one variable, described in section 6.4.4. In this case the variable is *distance across the domain*. An example of a `distanceFunction` is shown below.

```
internalField
{
    type          distanceFunction;
    direction      (1 0 0);

    function
    {
        type      table;
        values     ((0 0.3) (10 0));
    }
}
```

This field function requires a `direction` entry to define the direction in which the distance function variable is defined. Specifically, the function variable is $\mathbf{C} \cdot \mathbf{n}$, where $\mathbf{C}$ is the position vector for cell centres and $\mathbf{n}$ is the unit direction vector. The distance variable is therefore the x -ordinate of the position.

The `function` entry then contains the specification of a `Function1`. The example uses a `table` function with two value pairs. The entries specify a value of 0.3 at $x = 0$ (the distance variable here) and a value of 0 at $x = 10$, with table data interpolated linearly at points between value pairs.

Applying this field function the `internalField` in *0/p* in the previous example case, gives the initialisation of pressure as shown in Figure 6.3.

6.1.5 The fieldFunction field function

An example of the `fieldFunction` field function is provided in the `$FOAM_TUTORIALS/multi-Region/CHT/hotWire` case. In that case, the field function is used to specify a field used in an `fvModel` for calculating electrical conduction.

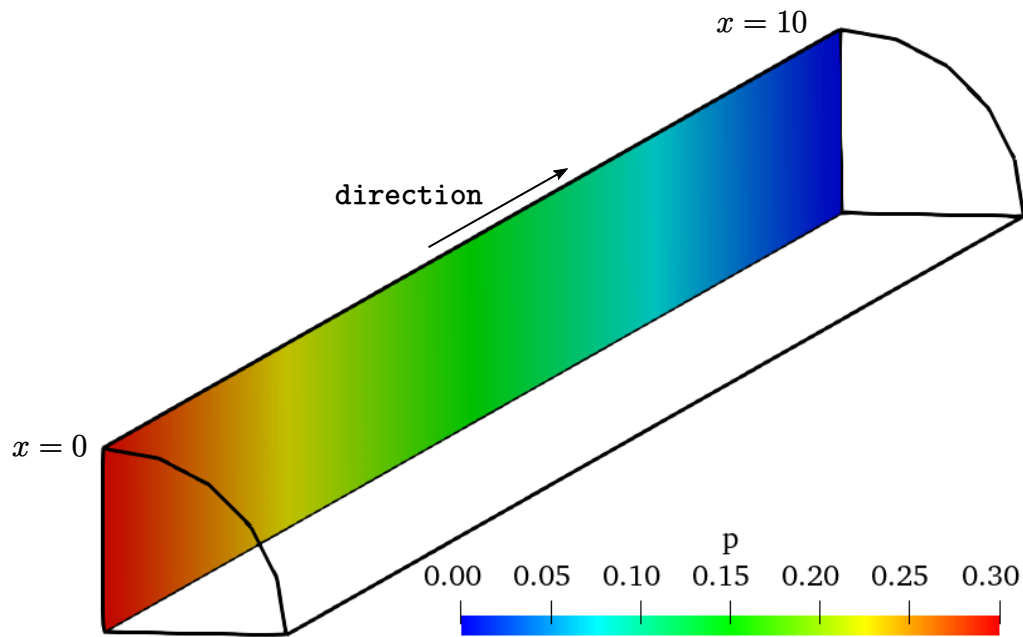


Figure 6.3: Field initialisation with the `distanceFunction` function object.

The field concerned is **sigma**, denoting electrical conductivity σ , which varies as a function of temperature T . Since T varies with time, the field also varies in time. Similarly, for the `timeFunction` field function in the following section, the field varies over time. Inevitably time-varying fields tend to appear in models, configured in the *fvModels* file, rather than as fields in the *0* directory, which are generally modified as part of the solution of governing equations.

The field function for **sigma** is in the *constant/fvModels* file within the `solidElectricalConduction` model and is reproduced below.

```
// Temperature-dependent electrical conductivity
sigma
{
    type          fieldFunction;
    field          T;
    function
    {
        type          polynomial;
        coeffs         (7.61e7 -1.38e5);
    }
}
```

It uses the `polynomial Function1`, as a function of T , meaning the applied function is $\sigma = 7.61 \times 10^7 - 1.38 \times 10^5 T$.

6.1.6 The timeFunction field function

The `timeFunction` function object is applied similarly to `fieldFunction` above. If, for example, a user wished to prescribe an electrical conductivity field which varies as a prescribed function of time $\sigma = \sigma(t)$, then `timeFunction` would be suitable. A hypothetical configuration, similar to that of the `fieldFunction` above, is given as follows.

```
// Time-dependent electrical conductivity
sigma
{
    type            timeFunction;
    function
    {
        type        polynomial;
        coeffs       (8e7 -1e5);
    }
}
```

This would represent a function $\sigma = 8 \times 10^7 - 1 \times 10^5 t$.

6.1.7 Applied field functions

As stated in the introduction to section 6.1, direct initialisation of field files with field functions largely replaces running pre-processing utilities, *e.g.* `setFields`, to initialise fields. As well as `setFields`, there are some bespoke pre-processing utilities to initialise fields for specific applications. In particular: the `setAtmBoundaryLayer` utility initialises fields corresponding to the velocity profile of an atmospheric boundary layer; and, the `setWaves` utility initialises surface waves in a liquid.

In OpenFOAM v14, both these utilities can be replaced using bespoke field functions. For an atmospheric boundary layer, there are `atmosphericBoundaryLayerVelocity`, `atmosphericBoundaryLayerTurbulentKineticEnergy` and `atmosphericBoundaryLayerEpsilon` field functions to initialise the U , k and ϵ fields, respectively. For surface waves, the `waveAlpha` and `waveVelocity` field functions to initialise the phase fraction field, *e.g.* `alpha.water`, and the U field, respectively.

These field functions read their configuration file, *atmosphericBoundaryLayerProperties* or *waveProperties*, from the *constant* directory. They use the parameters from the file to construct the distribution of their respective field; since the configuration of the parameters is in a separate file, the syntax within a field file itself is very compact, *e.g.* as follows for $0/U$.

```
internalField
{
    type        atmosphericBoundaryLayerVelocity;
    libs        ("libatmosphericModels.so");
}
```

The `libs` entry above loads the relevant library containing the field function. If in doubt, running `foamToC` with the `-search` option will provide the library name.

```
foamToC -search atmosphericBoundaryLayerVelocity
```

Example cases that include an atmospheric boundary layer and those that include surface waves can be located using `foamFind` as follows.

```
foamFind -tutorials atmosphericBoundaryLayer
foamFind -tutorials waveAlpha
```

The field functions for atmospheric boundary layers and surface waves have equivalent *boundary conditions* of the *same name*. These are inlet-outlet type conditions that specify a fixed value for inflow using the values prescribed by the field function, and zero gradient for outflow.

The syntax for specifying the field function and the boundary condition are the same, so the boundary condition can be specified using a macro expansion of the `internalField` keyword. For a patch named `inlet`, the syntax would be as follows.

```
boundaryField
{
    inlet
    {
        $internalField;
    }
    ...
}
```

6.2 Field sources

Field sources, specified by the optional `sources` entry in field files, relate to cases which include a mass or volume source that introduces fluid at some internal region of the domain. A good example to demonstrate field sources is `damBreakInjection`, which is an extension of the `damBreakLaminar` tutorial from section 2.2. The case can be run by the following commands.

```
run
cp -r $FOAM_TUTORIALS/incompressibleVoF/damBreakInjection
cd damBreakInjection
./Allrun
```

Note that the case originally only contains files with the additional entries from the `damBreakLaminar` case. The `Allrun` script executes the `foamMergeCase` script to merge the original files from the `damBreakLaminar` case, before meshing and running the simulation. The first main addition in `damBreakInjection` case is an `fvModel` defined in `constant/fvModels`, reproduced below.

```

injection
{
    type                volumeSource;
    phase                water;
    cellZone
    {
        type            containsPoints;
        points           ( (0.438 0.438 0.0073) );
    }
    volumetricFlowRate 0.0003;
}

```

This `volumeSource` model injects water at a volumetric flow rate of $0.0003 \text{ m}^3/\text{s}$ within a defined cell zone consisting of a single cell. When such a model is applied, some field values for the injected fluid must be specified, *e.g.* its velocity. The `sources` entries enable this by including them in the fields that require values. For the `damBreakInjection`, the velocity of the injected fluid is specified by a `sources` entry in the *U* file which is reproduced below.

```

sources
{
    injection
    {
        type            uniformFixedValue;
        uniformValue     (-1 0 0);
    }
}

```

Through the common name of the model, *i.e.* `injection`, the entry corresponds to the `volumeSource` model in the *fvModels* file. The velocity is specified as a `uniformFixedValue` condition with a value of `(-1 0 0)`. The resulting injection is shown in an image of phase fraction below, displays the phase fraction `alpha.water` in each cell. The water is injected in the upper right quadrant of the image, in the cell which is red, corresponding to `alpha.water = 1`. It initially follows a path in the negative *x* direction, from right to left, according to the direction specified in the `sources` entry. But soon afterwards, gravity causes the injected water to move downward, *i.e.* from top to bottom in the image.

6.3 Introduction to boundary conditions

Boundary conditions are specified in field files, *e.g.* *p*, *U*, in time directories. The structure of these files is introduced in sections 2.1.4 and 4.2.9, with the keyword entries `dimensions`, `internalField`, `boundaryField`, and, optionally `sources`. The `boundaryField` sub-dictionary is the place where boundary conditions are specified. An entry for each patch in the mesh, which are specified in the *boundary* file; below is a sample file from a 2D incompressibleFluid example in OpenFOAM.

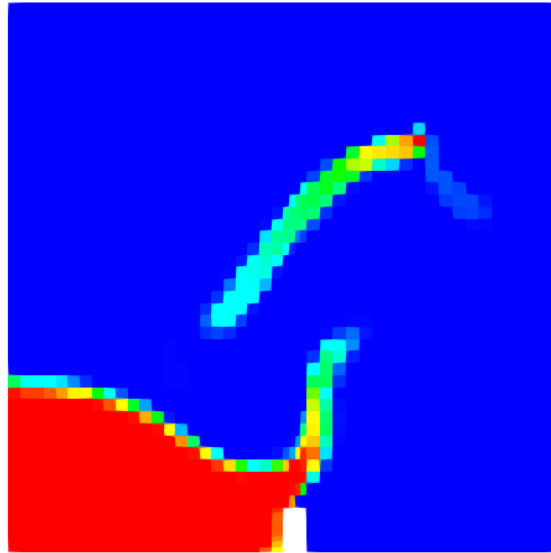


Figure 6.4: damBreak with an injection source upper right.

```

5
(
  outlet
  {
    type            patch;
    nFaces          320;
    startFace       198740;
  }
  up
  {
    type            symmetry;
    inGroups        List<word> 1(symmetry);
    nFaces          760;
    startFace       199060;
  }
  hole
  {
    type            wall;
    inGroups        List<word> 1(wall);
    nFaces          1120;
    startFace       199820;
  }
  frontAndBack
  {
    type            empty;
    inGroups        List<word> 1(empty);
    nFaces          200000;
    startFace       200940;
  }
  inlet
  {

```

```

        type          patch;
        nFaces        320;
        startFace      400940;
    }
)

```

The corresponding pressure field file, p , is shown below.

```

16
17  dimensions      [kinematicPressure];
18
19  internalField    uniform 0;
20
21  boundaryField
22  {
23      inlet
24      {
25          type          zeroGradient;
26      }
27      outlet
28      {
29          type          fixedValue;
30          value          uniform 0;
31      }
32      up
33      {
34          type          symmetry;
35      }
36      hole
37      {
38          type          zeroGradient;
39      }
40      frontAndBack
41      {
42          type          empty;
43      }
44  }
45
46  // *****

```

Each entry in the `boundaryField` sub-dictionary must include a `type` entry which specifies the type of boundary condition. The examples above include `zeroGradient` and `fixedValue` conditions corresponding to generic patches defined in the *boundary* file. They also include `symmetry` and `empty` types corresponding to equivalent constraint patches, *e.g.* the `up` patch is defined as `symmetry` in the mesh and uses a `symmetry` condition in the field file.

For details about the main boundary conditions used in OpenFOAM, refer to [Chapter 4 of *Notes on Computational Fluid Dynamics: General Principles*](#).

6.3.1 Patch selection in field files

There are three different ways an entry can be specified for a patch in the `boundaryField` of a field file: 1) by patch name; 2) by group name; 3) matching a patch name with a regular expression. They are listed here in order of precedence which is obeyed if multiple entries are valid of a particular patch. The different specifications can be illustrated by imagining a mesh with the following patches.

- `inlet`: a generic patch.
- `lowerWall` and `upperWall`: two wall patches.
- `outletSmall`, `outletMedium` and `outletLarge`: three outlet patches of generic type, all in a patch group named `outlet`.

Then imagine the following `boundaryField` for a field, *e.g.* p , corresponding to the patches above.

```
boundaryField
{
    inlet
    {
        type            zeroGradient;
    }
    ".*Wall"
    {
        type            zeroGradient;
    }
    outletSmall
    {
        type            fixedValue;
        value            uniform 1;
    }
    outlet
    {
        type            fixedValue;
        value            uniform 0;
    }
}
```

In this example, the `inlet` field entry is read for the `inlet` patch, following rule 1 above (matching patch name). Similarly, the `outletSmall` entry will be read for the patch of the same name.

The `outletMedium` and `outletLarge` patches do not have matching entries in the field file, so they instead the `outlet` entry will be applied (rule 2), since it matches the group name to which the patches belong. Note that the `outletSmall` patch does not use the `outlet` entry because a matching patch entry takes precedence over a matching group entry.

Finally, the `lowerWall` and `upperWall` match the regular expression `".*Wall"`. Regular expressions are described in section 4.2.13; they must be included in double quotations `"..."`. The `".*"` component matches any expression (including nothing), so matches the wall patch names here. The regular expression could use word grouping to provide a more precise match to the patch names, *e.g.*

```
"(lower|upper)Wall"
{
    type            zeroGradient;
}
```

Alternatively a patch entry could cover the wall patches taking advantage of the fact that every non-generic patch is automatically placed in a group of the same name as its type, as discussed in section 5.3.6. In this case, all `wall` patches are placed in a group named `wall`, so the following entry would be read for both patches.

```
wall
{
    type            zeroGradient;
}
```


6.3.2 Geometric constraints

Section 5.3 describes the mesh boundary, which is split into patches and written in the mesh *boundary* file. Each patch includes a **type** entry which can be specified as a generic patch, a **wall** or a geometric constraint, *e.g.* **empty**, **symmetry**, **cyclic** *etc.*

For each geometric constraint type for a patch in the mesh, there is an equivalent boundary condition type that must be applied to the same patch in the **boundaryField** of a field file. The type names in the mesh and **boundaryField** are the same, *e.g.* the **symmetry** boundary condition must be applied to a **symmetry** patch.

To simplify the configuration of field files, OpenFOAM includes a file named *setConstraintTypes* in the *\$FOAM_ETC/caseDicts* of the installation. The *setConstraintTypes* file contains the following entries.

```

 9  cyclic
10  {
11      type  cyclic;
12  }
13  cyclicSlip
14  {
15      type  cyclicSlip;
16  }
17  nonConformalCyclic
18  {
19      type  nonConformalCyclic;
20  }
21  nonConformalError
22  {
23      type  nonConformalError;
24  }
25  empty
26  {
27      type  empty;
28  }
29  processor
30  {
31      type  processor;
32  }
33  processorCyclic
34  {
35      type  processorCyclic;
36  }
37  nonConformalProcessorCyclic
38  {
39      type  nonConformalProcessorCyclic;
40  }
41  symmetryPlane
42  {
43      type  symmetryPlane;
44  }
45  symmetry
46  {
47      type  symmetry;
48  }
49  wedge
50  {
51      type  wedge;
52  }

```

```

63
64  internal
65  {
66      type    internal;
67  }
68
69
70  // *****

```

The file exploits the fact that a patch which is a geometric constraint is automatically included in a group of the constraint name, *e.g.* a **symmetry** patch is in a group named **symmetry**. The entries therefore set a boundary type for each constraint group (to the name of the group). All constraint conditions are covered by an entry for each condition.

The user can then include this file inside the **boundaryField** of their field files. Since the file is in the *\$FOAM_ETC* directory it can be included using the special **#includeEtc** directive, *e.g.* in the **boundaryField** entry below.

```

boundaryField
{
    inlet
    {
        type            zeroGradient;
    }

    outlet
    {
        type            fixedValue;
        value            uniform 0;
    }

    wall
    {
        type            zeroGradient;
    }

    #includeEtc "caseDicts/setConstraintTypes"
}

```

With the *setConstraintTypes* file included in the field files, the only patches that generally need to be configured are: the generic patches, corresponding to open boundaries; and, wall patches.

6.3.3 Basic boundary conditions

The main basic boundary condition types available in OpenFOAM are summarised below using a patch field named Ψ . This is not a complete list; for all types see *\$FOAM_SRC/finiteVolume/fields/fvPatchFields/basic*.

- **fixedValue**: value of Ψ is specified by **value**.
- **fixedGradient**: normal gradient of Ψ ($\partial\Psi/\partial n$) is specified by **gradient**.
- **zeroGradient**: normal gradient of Ψ is zero.

- **calculated**: patch field Ψ calculated from other patch fields.
- **mixed**: mixed **fixedValue**/ **fixedGradient** condition depending on **valueFraction** ($0 \leq \text{valueFraction} \leq 1$) where

$$\text{valueFraction} = \begin{cases} 1 & \text{corresponds to } \Psi = \text{refValue}, \\ 0 & \text{corresponds to } \partial\Psi/\partial n = \text{refGradient}. \end{cases} \quad (6.2)$$

- **directionMixed**: mixed condition with tensorial **valueFraction**, to allow different conditions in normal and tangential directions of a vector patch field, *e.g.* **fixedValue** in the tangential direction, **zeroGradient** in the normal direction.

6.4 Derived boundary conditions

There are numerous more complex boundary conditions derived from the basic conditions. For example, many complex conditions are derived from **fixedValue**, where the value is calculated by a function of other patch fields, time, geometric information, *etc.* Some other conditions derived from **mixed**/**directionMixed** switch between **fixedValue** and **fixedGradient** (usually with a zero gradient).

The available boundary conditions can be listed with **foamToC** using the **-scalarBCs** and **-vectorBCs** options, corresponding to boundary conditions for scalar fields and vector fields, respectively. For example, for scalar fields, boundary conditions are listed by

```
foamToC -scalarBCs
```

These produce long lists which the user can scan through. If the user wants more information of a particular condition, they can run the **foamInfo** script which provides a description of the boundary condition and lists example cases where it is used. For example, for the **totalPressure** boundary condition, run the following.

```
foamInfo totalPressure
```

In the following sections we will highlight some particular important, commonly used boundary conditions.

6.4.1 The inlet/outlet condition

The **inletOutlet** condition is one derived from **mixed**, which switches between **zeroGradient** when the fluid flows out of the domain at a patch face, and **fixedValue**, when the fluid is flowing into the domain. For inflow, the inlet value is specified by an **inletValue** entry. A good example of its use can be seen in the **damBreakLaminar** tutorial, where it is applied to the phase fraction on the upper **atmosphere** boundary. Where there is outflow, the condition is well posed, where there is inflow, the phase fraction is fixed with a value of 0, corresponding to 100% air.

```

16
17 dimensions      [];
18
19 internalField
20 {
21     type          zonal;
22
23     defaultValue  0;
24
25     zones
26     {
27         waterColumn
28         {
29             type          box;
30
31             box           (0 0 -1) (0.1461 0.292 1);
32
33             value         1;
34         }
35     }
36 }
37
38 boundaryField
39 {
40     #includeEtc "caseDicts/setConstraintTypes"
41
42     wall
43     {
44         type          zeroGradient;
45     }
46
47     atmosphere
48     {
49         type          inletOutlet;
50         inletValue     uniform 0;
51     }
52 }
53
54 // *****
55 // *****

```

6.4.2 Entrainment boundary conditions

The combination of the `totalPressure` condition on pressure and `pressureInletOutletVelocity` on velocity is extremely common for patches where some inflow occurs and the inlet flow velocity is not known. The conditions are used on the `atmosphere` boundary in the `damBreak` tutorial, inlet conditions where only pressure is known, outlets where flow reversal may occur, and where flow is entrained, *e.g.* on boundaries surrounding a jet through a nozzle.

The idea behind this combination is that the condition is a standard combination in the case of outflow, but for inflow the normal velocity is allowed to find its own value. Under these circumstances, a rapid rise in velocity presents a risk of instability, but the rise is moderated by the reduction of inlet pressure, and hence driving pressure gradient, as the inflow velocity increases.

The `totalPressure` condition specifies:

$$p = \begin{cases} p_0 & \text{for outflow} \\ p_0 - \frac{1}{2}\rho|\mathbf{U}^2| & \text{for inflow (dynamic pressure, subsonic)} \end{cases} \quad (6.3)$$

where the user specifies p_0 through the `p0` keyword.

The `entrainmentPressure` condition also exists which is arguably more robust than `totalPressure` by using the normal component of velocity $\mathbf{U}_n \equiv \mathbf{n} \cdot \mathbf{U}$ in the calculation for inflow,

i.e.

$$p = \begin{cases} p_0 & \text{for outflow} \\ p_0 - \frac{1}{2}\rho|\mathbf{U}_n^2| & \text{for inflow (dynamic pressure, subsonic)} \end{cases} \quad (6.4)$$

Solver applications which include buoyancy effects, though a gravitational force $\rho\mathbf{g}$ (per unit volume) source term, tend to solve for a pressure field $p_{\rho gh} = p - \rho|\mathbf{g}|\Delta h$, where the hydrostatic component is subtracted based on a height Δh above some reference. For such solvers, *e.g.* `interFoam`, an equivalent `prghTotalPressure` condition is applied which specifies:

$$p_{\rho gh} = \begin{cases} p_0 & \text{for outflow} \\ p_0 - \rho|\mathbf{g}|\Delta h - \frac{1}{2}\rho|\mathbf{U}^2| & \text{for inflow (dynamic pressure, subsonic)} \end{cases} \quad (6.5)$$

The `pressureInletOutletVelocity` condition specifies `zeroGradient` at all times, except on the tangential component which is set to `fixedValue` for inflow, specified by the `tangentialVelocity` keyword. This parameter defaults to 0 but is otherwise a field function, so can be specified with a distribution over the boundary that can also change in time, as described in section 6.5.1.

The specification of these boundary conditions in the `U` and `p_rgh` files, in the `damBreak` case, are shown below.

```

16
17 dimensions      [velocity];
18
19 internalField    uniform (0 0 0);
20
21 boundaryField
22 {
23     #includeEtc "caseDicts/setConstraintTypes"
24
25     wall
26     {
27         type      noSlip;
28     }
29
30     atmosphere
31     {
32         type      pressureInletOutletVelocity;
33     }
34 }
35
36 // *****
37

16
17 dimensions      [pressure];
18
19 internalField    uniform 0;
20
21 boundaryField
22 {
23     #includeEtc "caseDicts/setConstraintTypes"
24
25     wall
26     {
27         type      fixedFluxPressure;
28         value      $internalField;
29     }
30
31     atmosphere
32     {
33         type      prghTotalPressure;
34         p0         $internalField;

```

```

35     }
36 }
37
38
39 // *****

```

6.4.3 Fixed flux pressure

In the above example, it can be seen that all the wall boundaries use a boundary condition named `fixedFluxPressure`. This boundary condition is used for pressure in situations where `zeroGradient` is generally used, but where body forces such as gravity and surface tension are present in the solution equations. The condition adjusts the gradient accordingly.

6.4.4 Time-varying boundary conditions

There are several boundary conditions for which some input parameters are specified by a function of time (using `Function1` functionality) class. Examples of boundary conditions and field functions containing `Function1` parameters can be listed by the following command.

```

foamFind -dir $FOAM_SRC/finiteVolume/fields -H -search Function1
// select 'l' option to list files only

```

They include conditions such as `uniformFixedValue`, which is a `fixedValue` condition which applies a single value which is a function of time through a `uniformValue` keyword entry.

The `Function1` is specified by a keyword following the `uniformValue` entry, followed by parameters that relate to the particular function. The `Function1` options can be listed by `foamToC` by

```

foamToC -table scalarFunction1

```

The most relevant functions are those in the core `OpenFOAM` library which can be filtered using `grep`

```

foamToC -table scalarFunction1 | grep OpenFOAM

```

Most of those function objects are described below.

- `constant`: constant value.
- `table`: inline list of (time value) pairs; interpolates values linearly between times.
- `tableFile`: as above, but with data supplied in a separate file.
- `square`: square-wave function.
- `squarePulse`: single square pulse.
- `sine`: sine function.
- `one` and `zero`: constant one and zero values.

- **polynomial**: polynomial function using a list (**coeff** **exponent**) pairs.
- **coded**: function specified by user coding.
- **scale**: scales a given **value** function by a scalar **scale** function; both entries can be themselves **Function1**; **scale** function is often a ramp function (below), with **value** controlling the ramp value.
- **linearRamp**, **quadraticRamp**, **exponentialSqrRamp**, **halfCosineRamp**, **quarterCosineRamp** and **quarterSineRamp**: monotonic ramp functions which ramp from 0 to 1 over specified duration.
- **reverseRamp**: reverses the values of a ramp function, e.g. from 1 to 0.
- **add**: adds two **Function1**s together.
- **normalise**: scales a given **Function1** so that the integral over time equals 1 (the integral value can be changed with the **scale** **Function1**).
- **repeat**: repeats a given 'value' function with a given period or frequency, and with an optional shift along the time axis.

There are some other function objects, more commonly to describe properties as a function of something, *e.g.* temperature.

- **uniformTable**: Tabulated property function that linearly interpolates between given values.
- **nonUniformTable**: Non-uniform tabulated property function that linearly interpolates between the values.

Examples of a time-varying inlet for a scalar are shown below.

```
inlet
{
    type            uniformFixedValue;
    uniformValue    constant 2;
}

inlet
{
    type            uniformFixedValue;
    uniformValue    table ((0 0) (10 2));
}

inlet
{
    type            uniformFixedValue;
    uniformValue    polynomial ((1 0) (2 2)); // = 1*t^0 + 2*t^2
}

inlet
{
    type            uniformFixedValue;
    uniformValue
    {
        type            tableFile;
        format          csv;
        nHeaderLine     4;           // number of header lines
    }
}
```

```

        refColumn      0;                // time column index
        componentColumns (1);            // data column index
        separator      ",";              // optional (defaults to ",")
        mergeSeparators no;              // merge multiple separators
        file            "dataTable.csv";
    }
}

inlet
{
    type            uniformFixedValue;
    uniformValue
    {
        type            square;
        frequency        10;
        amplitude        1;
        scale            2; // Scale factor for wave
        level            1; // Offset
    }
}

inlet
{
    type            uniformFixedValue;
    uniformValue
    {
        type            sine;
        frequency        10;
        amplitude        1;
        scale            2; // Scale factor for wave
        level            1; // Offset
    }
}

input // ramp from 0 -> 2, from t = 0 -> 0.4
{
    type            uniformFixedValue;
    uniformValue
    {
        type            scale;
        scale            linearRamp;
        start            0;
        duration         0.4;
        value            2;
    }
}

input // ramp from 2 -> 0, from t = 0 -> 0.4
{
    type            uniformFixedValue;
    uniformValue
    {
        type            scale;
        scale            reverseRamp;
        ramp            linearRamp;
        start            0;
        duration         0.4;
        value            2;
    }
}

inlet // pulse with value 2, from t = 0 -> 0.4
{
    type            uniformFixedValue;
    uniformValue
    {
        type            scale;
        scale            squarePulse
        start            0;
    }
}

```



```

        duration      0.4;
        value         2;
    }
}

inlet
{
    type              uniformFixedValue;
    uniformValue      coded;
    name              pulse;
    codeInclude
    #{
        #include "mathematicalConstants.H"
    #};

    code
    #{
        return scalar
        (
            0.5*(1 - cos(constant::mathematical::twoPi*min(x/0.3, 1)))
        );
    #};
}

```

6.5 Field functions in boundary conditions

Field functions are described in section 6.1, where their use is demonstrated in initialising non-uniform `internalFields` within field files in a time directory, and in initialising fields within an `fvModel` configuration. Field functions can also be used to set non-uniform fields on *boundary patches*, *i.e.* to set different values on different faces of a patch. Field functions are used in boundary conditions firstly through the `functionalFixedValue` boundary condition which uses a field function to specify a non-uniform fixed value across a patch, see section 6.5.1. They are specify particular parameters in a small number of boundary conditions described in section 6.5.2 and 6.6.

6.5.1 The `functionalFixedValue` boundary condition

The `functionalFixedValue` boundary condition accompanies field functions, described in section 6.1. Field functions allow the user to initialise non-uniform fields. While they are typically applied to the `internalField`, `functionalFixedValue` is a fixed value boundary condition where the values are specified with a field function.

The `coded` field function from section 6.1.3 provides a good example to illustrate the `functionalFixedValue` condition. Recall that the example initialises swirl flow according to Equation 6.1, by the following configuration in the *0/U* file.

```

internalField
{
    type          coded;

    evaluate
    #{
        field = velocity(1, 0, 0) + (C ^ rate(2, 0, 0));
    #};
}

```

```
}
```

The case specifically requires the same velocity distribution at the `inlet` patch. The `functionalFixedValue` enables this, with a `value` entry that is a field function. Since `internalField` is the same field function, the `value` entry can be specified using a macro expansion of `internalField` as follows.

```
boundaryField
{
    inlet
    {
        type      functionalFixedValue;

        value
        {
            $internalField;
        }
    }
    ...
}
```

This will initialise the boundary condition according to the function specified above in the “`evaluate #{...#};`” block. The boundary values can also vary in time by specifying a function in an optional “`update #{...#};`” block. Recall that field functions include a reference named ‘`t`’ to the current time, which can be incorporated into a coded field function to make it a direct function of time.

6.5.2 Parameters as field functions

A small number of general boundary conditions have parameters which are implemented as field functions. The following `foamFind` command locates them.

```
foamFind -dir $FOAM_SRC/finiteVolume/fields/fvPatchFields \
        -search UFunctional
```

One of the conditions is the `pressureInletOutletVelocity` condition, discussed in section 6.4.2, where the `tangentialVelocity` parameter is a field function. Another is the `freeStream` condition where the `freeStreamVelocity` is a field function. The latter is used in the `$FOAM_TUTORIALS/fluid/stackPlume` example case to initialise the `freeStreamVelocity` in the far field with the `atmosphericBoundaryLayerVelocity` field function.

6.6 Thermal boundary conditions

For solutions involving thermal energy, boundary conditions are required for the temperature field T in the $0/T$ file. While conventional conditions can be used at inlet and outlet patches, *e.g.* `fixedValue` and `zeroGradient`, respectively, the following bespoke conditions are available for wall patches.

- **externalTemperature**: models heat transfer at an external wall, with various options to specify the heat flux.
- **coupledTemperature**: models heat transfer between regions (fluid or solid) in a conjugate heat transfer simulation.

The **externalTemperature** condition, illustrated in Figure 6.5, can include one or more of the following heat fluxes.

- Prescribed power Q (W) or power flux q (W m^{-2}).
- Thermal radiation from inside the domain, **qr** specifies the radiation field name.
- Convective heat transfer with a coefficient h and ambient temperature T_a , both a **Function1** of time.
- The coefficient h can alternatively be configured using a specialised **externalWallLayersHeatTransferCoefficient** field function, which can emulate conduction through additional layers of material on the outside of the wall patch.
- External thermal radiation with a specified **emissivity**.

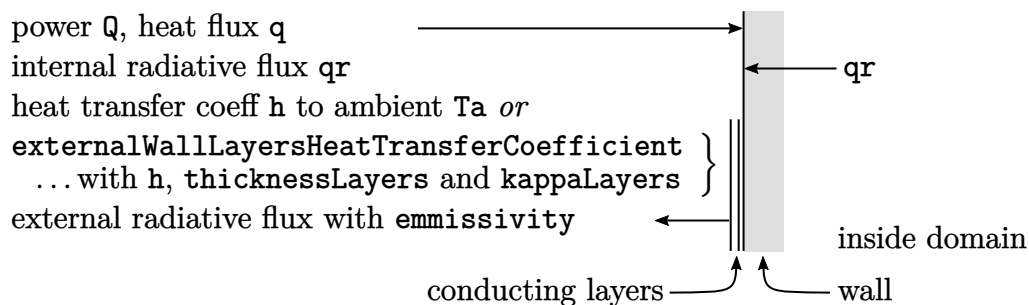


Figure 6.5: The **externalTemperature** boundary condition.

The **foamInfo** script provides further details and examples of the use of the **externalTemperature** condition.

```
foamInfo externalTemperature
```

The following example shows the use with a straightforward heat transfer coefficient h .

```
wall
{
    type      externalTemperature;
    Ta        constant 300;
    h          constant 10;
    value      uniform 400;
}
```

The example below demonstrates the `externalWallLayersHeatTransferCoefficient` field function which includes conduction through layers of material. The conducting layers are described by a list of thicknesses and corresponding conductivities via the `thicknessLayers` and `kappaLayers` lists, respectively.

```

wall
{
    type      externalTemperature;
    Ta        constant 300;
    h
    {
        type      externalWallLayersHeatTransferCoefficient;
        h          10.0;
        thicknessLayers (0.1 0.2 0.3 0.4);
        kappaLayers   (1 2 3 4);
    }
    value      uniform 400;
}

```

The `coupledTemperature` condition calculates the heat flux between regions in a conjugate heat transfer simulation. It is shown in Figure 6.6 with regions labelled 1 and 2. The condition is applied at the interface patches of both regions. If we imagine the condition applied to region 1 in the figure, `qr` specifies the name of the variable for thermal radiation from within region 1, and `qrNbr` is the name of the variable in the neighbour region, *i.e.* region 2. The condition is similar to `externalTemperature` and can include additional conducting

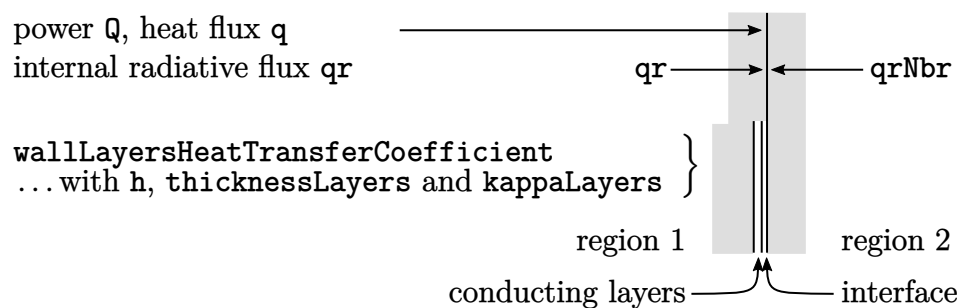


Figure 6.6: The `coupledTemperature` boundary condition.

layers at the interface between the regions, *e.g.* to simulate a contact material, contaminant, liquid film, *etc.* The conducting layers are specified by the `wallLayersHeatTransferCoefficient` field function which includes the `thicknessLayers` and `kappaLayers` parameters. An example of a `coupledTemperature` condition with conducting layers is shown below.

```

wall
{
    type      coupledTemperature;
    h

```

```
{
    type          wallLayersHeatTransferCoefficient;
    thicknessLayers (0.1 0.2 0.3 0.4);
    kappaLayers    (1 2 3 4);
}
value    uniform 400;
}
```

The conducting layers are implemented as a field function so that they can be extending to cases where the layer thickness is non-uniform. This is demonstrated in the example case in the *\$FOAM_TUTORIALS/multiRegion/CHT/thermalBending*, which uses a specialised `displacementGapHeatTransferCoefficient` field function which models a gap of non-uniform thickness between two regions, due to a displacement of one region due to thermal bending.

Chapter 7

Post-processing

This chapter describes options for post-processing with OpenFOAM. Post-processing in its most general sense involves data processing (processing results) and visualisation. The functionality for data processing is described in sections 7.2, 7.3 and 7.4. For visualisation, OpenFOAM relies on ParaView, a third-party open source application described in the example cases in chapter 2, with some additional information provided in the following section 7.1. Other methods of visualisation using third party software are described in section 7.5.

7.1 ParaView/paraFoam graphical user interface (GUI)

OpenFOAM includes a native reader module to visualise data with ParaView, an open-source, visualisation application. The module comprises of the PVFoamReader and vtkPVFoam libraries, which currently supports version 5.10.1 of ParaView. It is recommended that this version of ParaView is used, although it is possible that the latest binary release of the software will run adequately. Further details about ParaView can be found at <http://www.paraview.org>.

ParaView uses the Visualisation Toolkit (VTK) as its data processing and rendering engine and can therefore read any data in VTK format. OpenFOAM includes a variety of tools which can write data in VTK and other supported formats, which can be read directly by ParaView. Entire case data can be converted to VTK using the foamToVTK utility if the user wishes to process their results without the OpenFOAM reader.

In summary, we recommend the reader module for ParaView as the primary visualisation option for OpenFOAM. Alternatively OpenFOAM data can be converted into VTK format to be read by ParaView or any other VTK-based graphics tools.

7.1.1 Overview of ParaView/paraFoam

paraFoam is a script that launches ParaView using the reader module supplied with OpenFOAM. It is executed like any of the OpenFOAM utilities either by the single command from within the case directory or with the `-case` option with the case path as an argument, *e.g.*:

```
paraFoam -case <caseDir>
```

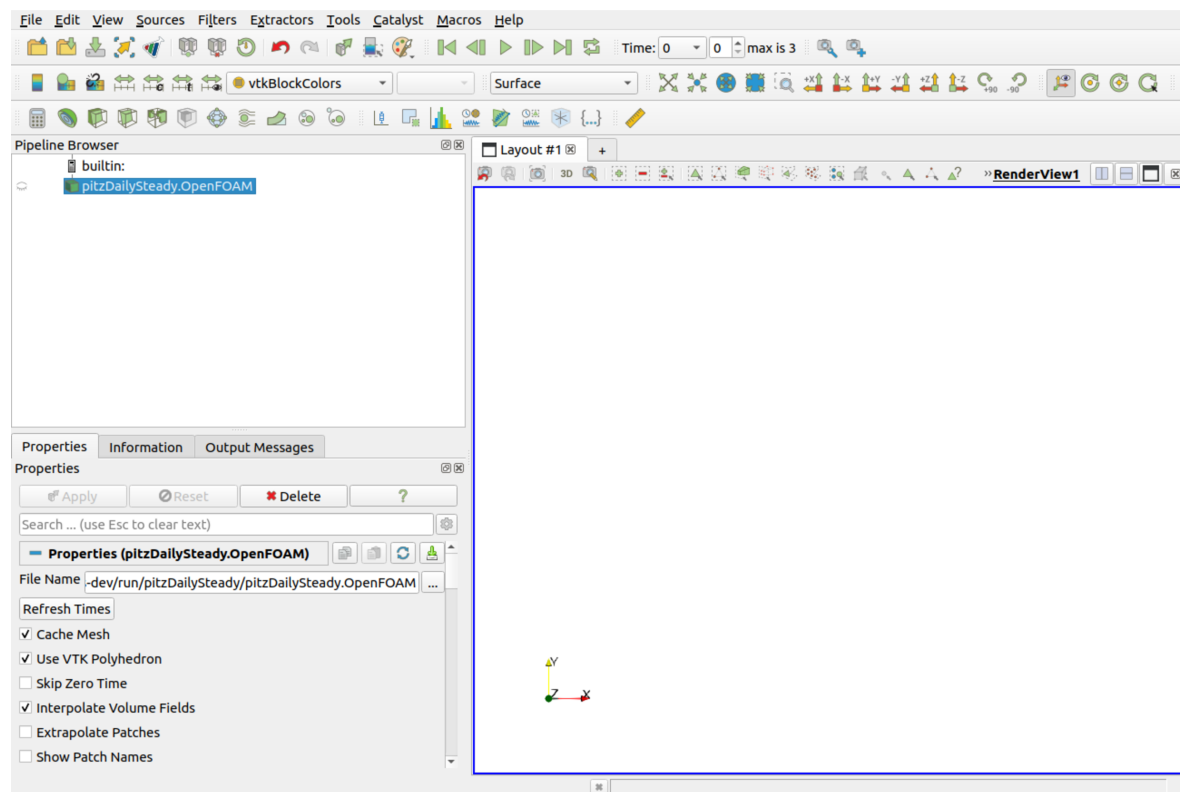


Figure 7.1: The ParaView window

ParaView is launched and opens the window shown in Figure 7.1. The case is controlled from the left panel, which contains the following:

- The Pipeline Browser lists the *modules* opened in ParaView, where the selected modules are highlighted in blue and the graphics for the given module can be enabled/disabled by clicking the eye button alongside;
- The Properties panel contains the input selections for the case, such as times, regions and fields; it includes the Display panel that controls the visual representation of the selected module, *e.g.* colours;
- Other panels can be selected from the View menu, including the Information panel which gives case statistics such as mesh geometry and size.

ParaView operates a tree-based structure in which data can be filtered from the top-level case module to create sets of sub-modules. For example, a contour plot of, say, pressure could be a sub-module of the case module which contains all the pressure data. The strength of ParaView is that the user can create a number of sub-modules and display whichever ones they need to create the desired image or animation. For example, they may add some solid geometry, mesh and velocity vectors, to a contour plot of pressure, switching any of the items on and off as necessary.

The general operation of the system is based on the user making a selection and then clicking the green Apply button in the Properties panel. The additional buttons are: the

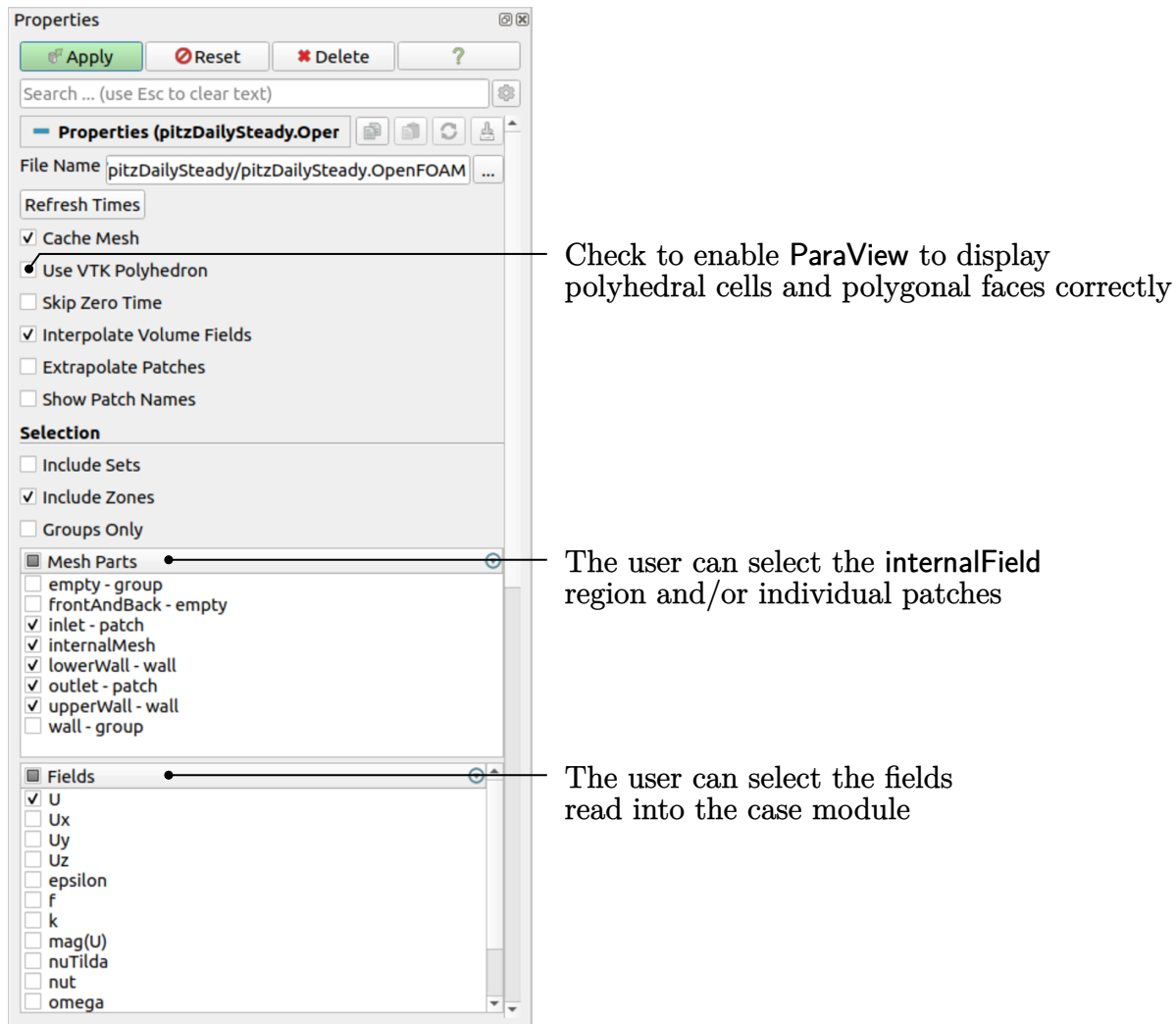


Figure 7.2: The Properties panel for the case module

Reset button which can be used to reset the settings if necessary; and, the Delete button that will delete the active module.

7.1.2 The Parameters panel

The Properties window for the case module includes the Parameters panel that contains the settings for mesh, fields and global controls. The controls are described in Figure 7.2. The user can select mesh and field data which is loaded for all time directories into ParaView. The buttons in the Current Time Controls and VCR Controls toolbars then select the time data to be displayed, as shown in section 7.1.4.

As with any operation in ParaView, the user must click Apply after making any changes to any selections. The Apply button is highlighted in green to alert the user if changes have been made but not accepted. This method of operation has the advantage of allowing the user to make a number of selections before accepting them, which is particularly useful in large cases where data processing is best kept to a minimum.

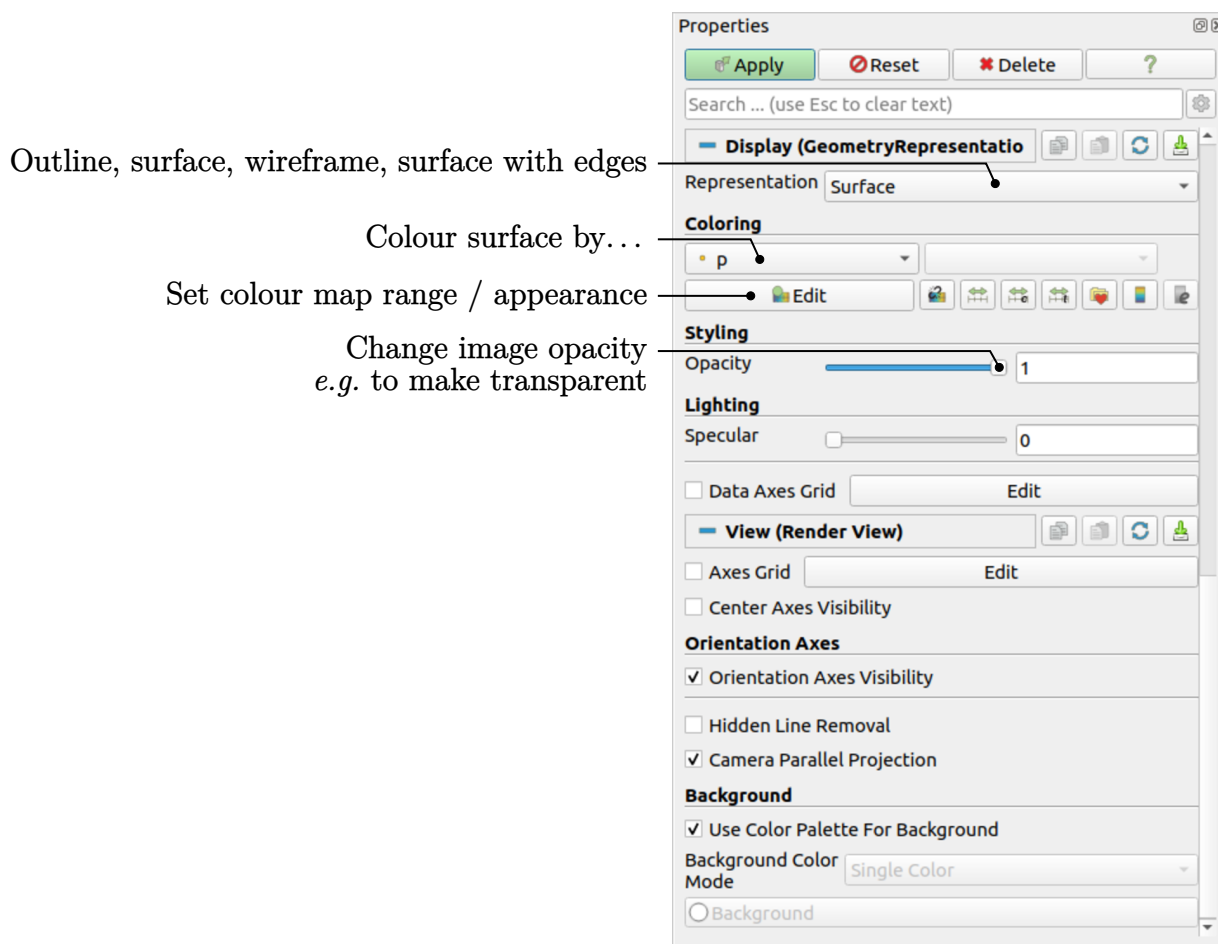


Figure 7.3: The Display panel

If new data is written to time directories while the user is running ParaView, the user must load the additional time directories by checking the **Refresh Times** button. Where there are occasions when the case data changes on file and ParaView needs to load the changes, the user can also toggle the **Cache Mesh** button in the **Parameters** panel and apply the changes.

7.1.3 The Display panel

The **Properties** window contains the **Display** panel that includes the settings for visualising the data for a given case module. The following points are particularly important:

- the data range may not be automatically updated to the max/min limits of a field, so the user should take care to select **Rescale** at appropriate intervals, in particular after loading the initial case module;
- clicking the **Edit Color Map** button, brings up a window in which there are two panels:
 1. The **Color Scale** panel in which the colours within the scale can be chosen. The standard blue to red colour scale for CFD can be selected by clicking **Choose Preset** and searching for **Blue to Red Rainbow** and selecting.

2. The **Color Legend** panel has a toggle switch for a colour bar legend and contains settings for the layout of the legend, *e.g.* font.
- the underlying mesh can be represented by selecting **Wireframe** in the **Representation** menu of the **Style** panel;
 - the geometry, *e.g.* a mesh (if **Wireframe** is selected), can be visualised as a single colour by selecting **Solid Color** from the **Color By** menu and specifying the colour in the **Set Ambient Color** window;
 - the image can be made translucent by editing the value in the **Opacity** text box (1 = solid, 0 = invisible) in the **Style** panel.

7.1.4 The button toolbars

ParaView duplicates functionality from pull-down menus at the top of the main window and the major panels, within the toolbars below the main pull-down menus. The displayed toolbars can be selected from **Toolbars** in the main **View** menu. The default layout with all toolbars is shown in Figure 7.4 with each toolbar labelled. The function of many of the buttons is clear from their icon and, with tooltips enabled in the **Help** menu, the user is given a concise description of the function of any button.

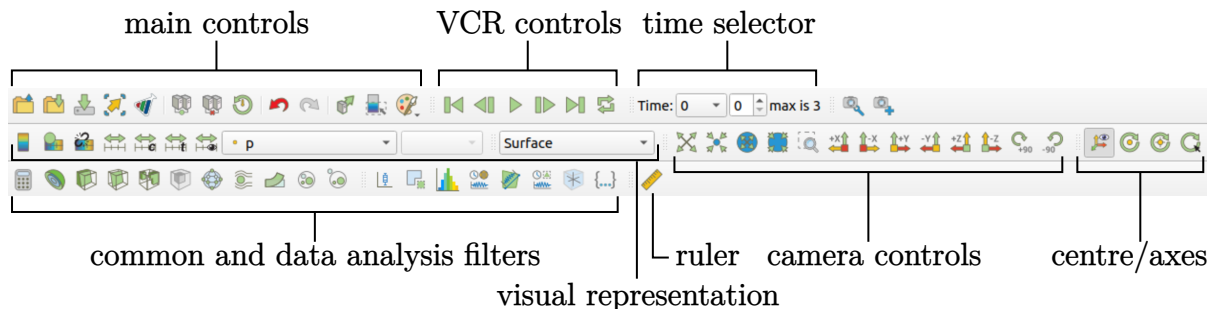


Figure 7.4: Toolbars in ParaView

7.1.5 Manipulating the view

This section describes operations for setting and manipulating the view in ParaView. Firstly, the **View Settings** are available in the **Render View** panel below the **Display** panel in the **Properties** window. Settings that are generally important only appear when the user checks the gearwheel button at the top of the **Properties** window, next to the search bar. These *advanced properties* include setting the background colour, where white is often a preferred choice for creating images for printed and website material.

The **Lights** button opens detailed lighting controls within the **Light Kit** panel. A separate **Headlight** panel controls the direct lighting of the image. Checking the **Headlight** button with white light colour of strength 1 seems to help produce images with strong bright colours, *e.g.* with an isosurface.

The **Camera Parallel Projection** is the usual choice for CFD, especially for 2D cases, and so should generally be checked. Other settings include **Cube Axes** which displays axes on the selected object to show its orientation and geometric dimensions.

The general **Settings** are selected from the **Edit** menu, which opens a general **Options** window with **General**, **Camera**, **Render View Color Arrays** and **Color Palette** menu items.

The **General** panel controls some default behaviour of **ParaView**. In particular, there is an **Auto Apply** button that enables **ParaView** to accept changes automatically without clicking the green **Apply** button in the **Properties** window. For larger cases, this option is generally not recommended: the user does not generally want the image to be re-rendered between each of a number of changes he/she selects, but be able to apply a number of changes to be re-rendered in their entirety once.

The **Render View** panel contains level of detail (LOD) which controls the rendering of the image while it is being manipulated, *e.g.* translated, resized, rotated; lowering the levels set by the sliders, allows cases with large numbers of cells to be re-rendered quickly during manipulation.

The **Camera** panel includes control settings for 3D and 2D movements. This presents the user with a map of rotation, translate and zoom controls using the mouse in combination with Shift- and Control-keys. The map can be edited to suit by the user.

7.1.6 Contour plots

A contour plot is created by selecting **Contour** from the **Filter** menu at the top menu bar. The filter acts on a given module so that, if the module is the 3D case module itself, the contours will be a set of 2D surfaces that represent a constant value, *i.e.* isosurfaces. The **Properties** panel for contours contains an **Isosurfaces** list that the user can edit, most conveniently by the **New Range** window. The chosen scalar field is selected from a pull down menu.

Very often a user will wish to create a contour plot across a plane rather than producing isosurfaces. To do so, the user must first use the **Slice** filter to create the cutting plane, on which the contours can be plotted. The **Slice** filter allows the user to specify a cutting **Plane**, **Box** or **Sphere** in the **Slice Type** menu by a **center** and **normal/radius** respectively. The user can manipulate the cutting plane like any other using the mouse.

The user can then run the **Contour** filter on the cut plane to generate contour lines.

7.1.7 Vector plots

Vector plots are created using the **Glyph** filter. The filter reads the field selected in **Vectors** and offers a range of **Glyph Types** for which the **Arrow** provides a clear vector plot images. Each glyph has a selection of graphical controls in a panel which the user can manipulate to best effect.

The remainder of the **Properties** panel contains mainly the **Scale Mode** menu for the glyphs. The most common options for **Scale Mode** are: **Vector**, where the glyph length is proportional to the vector magnitude; and, **Off** where each glyph is the same length. The **Set Scale Factor** parameter controls the base length of the glyphs.

Vectors are by default plotted on cell vertices but, very often, we wish to plot data at cell centres. This is done by first applying the **Cell Centers** filter to the case module, and

then applying the **Glyph** filter to the resulting cell centre data.

7.1.8 Streamlines

Streamlines are created by first creating tracer lines using the **Stream Tracer** filter. The tracer **Seed** panel specifies a distribution of tracer points over a **Line Source** or **Point Cloud**. The user can view the tracer source, *e.g.* the line, but it is displayed in white, so they may need to change the background colour in order to see it.

The distance the tracer travels and the length of steps the tracer takes are specified in the text boxes in the main **Stream Tracer** panel. The process of achieving desired tracer lines is largely one of trial and error in which the tracer lines obviously appear smoother as the step length is reduced but with the penalty of a longer calculation time.

Once the tracer lines have been created, the **Tubes** filter can be applied to the *Tracer* module to produce high quality images. The tubes follow each tracer line and are not strictly cylindrical but have a fixed number of sides and given radius. When the number of sides is set above, say, 10, the tubes do however appear cylindrical, but again this adds a computational cost.

7.1.9 Image output

The simplest way to output an image to file from **ParaView** is to select **Save Screenshot** from the **File** menu. On selection, a window appears in which the user can select the resolution for the image to save. There is a button that, when clicked, locks the aspect ratio, so if the user changes the resolution in one direction, the resolution is adjusted in the other direction automatically. After selecting the pixel resolution, the image can be saved. To achieve high quality output, the user might try setting the pixel resolution to 1000 or more in the *x*-direction so that when the image is scaled to a typical size of a figure in an A4 or US letter document, perhaps in a PDF document, the resolution is sharp.

7.1.10 Animation output

To create an animation, the user should first select **Save Animation** from the **File** menu. A dialogue window appears in which the user can specify a number of things including the image resolution. The user should specify the resolution as required. The other noteworthy setting is number of frames per timestep. While this would intuitively be set to 1, it can be set to a larger number in order to introduce more frames into the animation artificially. This technique can be particularly useful to produce a slower animation because some movie players have limited speed control, particularly over **mpeg** movies.

On clicking the **Save Animation** button, another window appears in which the user specifies a file name *root* and file format for a set of images. On clicking **OK**, the set of files will be saved according to the naming convention “<fileRoot>_<imageNo>.<fileExt>”, *e.g.* the third image of a series with the file root “**animation**”, saved in **jpg** format would be named “**animation_0002.jpg**” (<imageNo> starts at 0000).

Once the set of images are saved the user can convert them into a movie using their software of choice. One option is to use the built in **foamCreateVideo** script from the command line whose usage is shown with the **-help** option.

7.2 Post-processing command line interface (CLI)

Post-processing is provided directly within OpenFOAM through the command line including data processing, sampling (*e.g.* probes, graph plotting) visualisation, case control and run-time I/O. Functionality can be executed by:

- conventional *post-processing*, a data processing activity that occurs *after* a simulation has run;
- *run-time processing*, data processing that is performed *during* the running of a simulation.

Both approaches have advantages. Conventional post-processing allows the user to choose how to analyse data after the results are obtained. Run-time processing offers greater flexibility because it has access to *all* the data in the database of the run at all times, rather than just the data written during the simulation. It also allows the user to monitor processed data during a simulation and provides a greater level of convenience because the processed results can be available immediately to the user when the simulation ends.

There are 3 methods of post-processing that cover the options described above.

- The case can be configured to include run-time processing during the simulation.
- The `foamPostProcess` utility provides conventional post-processing of data after a simulation is completed.
- The `foamPostProcess` utility is run with a `-solver` which provides additional access to data available on the database for the particular solver.

All modes of post-processing access the same functionality implemented in OpenFOAM in the *function object* framework. Function objects can be listed using `foamToC` by the following command.

```
foamToC -functionObjects
```

The list represents the underlying post-processing functionality. Almost all the functionality is packaged into a set of configured tools that are conveniently integrated within the post-processing CLI. Those tools are located in `$FOAM_ETC/caseDicts/functions` and are listed by running `foamPostProcess` with the `-list` option.

```
foamPostProcess -list
```

This produces a list of tools catalogued in section 7.3.

7.2.1 Run-time data processing

When a user wishes to process data during a simulation, they need to configure the case accordingly. The configuration process is as follows, using an example of monitoring flow rate at an outlet patch named `outlet`.

Firstly, the user should include the `patchFlowRate` function in a *functions* file in the case *system* directory. The user can copy a template *functions* file using `foamGet` as follows

foamGet functions

Within the *functions* file, they include the `patchFlowRate` function with the `#includeFunc` directive (*e.g.* by un-commenting the example entry):

```
#includeFunc patchFlowRate
```

Note: prior to v12, the functions were included in a `functions` sub-dictionary in the case *controlDict* file, as shown below.

```
functions
{
    #includeFunc patchFlowRate
    ... other function objects here ...
}
```

It is still possible to do this, but using a separate *functions* file is advantageous because it avoids reading function objects in applications that do not use them. That will include the functionality in the *patchFlowRate* configuration file, located in the directory hierarchy beginning with *\$FOAM_ETC/caseDicts/postProcessing*.

The configuration of *patchFlowRate* requires the **name** of the patch to be supplied. **Option 1** for doing this is that the user copies the *patchFlowRate* file into their case *system* directory. The `foamGet` script copies the file conveniently, *e.g.*

```
foamGet patchFlowRate
```

The patch **name** can be edited in the copied file to be `outlet`. When the solver is run, it will pick up an included function in the local case *system* directory, in precedence over *\$FOAM_ETC/caseDicts/postProcessing*. The flow rate through the patch will be calculated and written out into a file within a directory named *postProcessing*.

Option 2 for specifying the patch name is to provide the name as an argument to the `patchFlowRate` in the `#includeFunc` directive, using the syntax `keyword=entry`.

```
#includeFunc patchFlowRate(patch=outlet)
```

In the case where the keyword is `field` or `fields`, only the entry is needed when specifying an argument to a function. For example, if the user wanted to calculate and write out the magnitude of velocity into time directories during a simulation they could simply add the following to the `functions` sub-dictionary in *controlDict*.

```
#includeFunc mag(U)
```

This works because the function's argument `U` is represented by the keyword `field`, see *\$FOAM_ETC/caseDicts/postProcessing/fields/mag*.

Some functions require the setting of many parameters, *e.g.* to calculate forces and generate elements for visualisation, *etc.* For those functions, it is more reliable and convenient to copy and configure the function using option 1 (above) rather than through arguments.

7.2.2 The foamPostProcess utility

The user can execute post-processing functions after the simulation is complete using the `foamPostProcess` utility. We can illustrate the use of `foamPostProcess` using the `pitzDailySteady` case from section 2.1. The tutorial does not need to be run to use the case, it can instead be copied into the user's `run` directory and run using its accompanying `Allrun` script as follows.

```
run
cp -r $FOAM_TUTORIALS/incompressibleFluid/pitzDailySteady .
cd pitzDaily
./Allrun
```

Now the user can run execute post-processing functions with `foamPostProcess`. The `-help` option provides a summary of its use.

```
foamPostProcess -help
```

Simple functions like `mag` can be executed using the `-func` option; text on the command line generally needs to be quoted ("`...`") if it contains punctuation characters.

```
foamPostProcess -func "mag(U)"
```

This operation calculates and writes the field of magnitude of velocity into a file named *mag(U)* in each time directory. Similarly, the `patchFlowRate` example can be executed using `foamPostProcess`.

```
foamPostProcess -func "patchFlowRate(name=outlet)"
```

Let us say the user now wants to calculate total pressure $= p + |\mathbf{U}|^2/2$ for incompressible flow with kinematic pressure, p . The function is available, named `totalPressureIncompressible`, which requires a `rhoInf` parameter to be specified. The user could attempt first to run as follows.

```
foamPostProcess -func "totalPressureIncompressible(rhoInf=1.2)"
```

This returns the following error message.

```
--> FOAM FATAL IO ERROR:
request for volVectorField U from objectRegistry region0 failed
```

The error message is telling the user that the velocity field U is not loaded. For the function to work, both the field needs to be loaded using the `-field` option as follows.

```
foamPostProcess -func "totalPressureIncompressible(rhoInf=1.2)" -field U
```


A more complex example is calculating wall shear stress using the `wallShearStress` function.

```
foamPostProcess -fields "(p U)" -func wallShearStress
```

Even loading relevant fields, the post-processing fails with the following message.

```
--> FOAM FATAL ERROR:
Unable to find turbulence model in the database
```

The message is telling us that the `foamPostProcess` utility has not constructed the necessary models, *i.e.* a turbulence model, that the `incompressibleFluid` solver module used when running the simulation. This is a situation where we need to post-process (as opposed to run-time process) using the `-solver` option modelling will be available that the post-processing function needs.

```
foamPostProcess -solver incompressibleFluid -func wallShearStress
```

Note that no fields need to be supplied, *e.g.* using `"-field U"`, because `incompressibleFluid` module constructs and stores the required fields. Functions can also be selected by the `#includeFunc` directive in *functions* file, instead of the `-func` option.

7.3 Post-processing functionality

The packaged function objects are catalogued in this section. Each packaged function object is a configuration file stored in `$FOAM_ETC/caseDicts/postProcessing`. As a reminder, they can be listed by the following command.

```
foamPostProcess -list
```

7.3.1 Field calculation

age Calculates and writes out the time taken for a particle to travel from an inlet to the location.

components Writes the component scalar fields (*e.g.* U_x , U_y , U_z) of a field (*e.g.* U).

CourantNo Calculates the Courant Number field from the flux field.

cylindrical Transforms a vector field into cylindrical coordinates.

ddt Calculates the Eulerian time derivative of a field.

div Calculates the divergence of a field.

enstrophy Calculates the enstrophy of the velocity field.

energyFlux Calculate the total energy-flux (*i.e.* advective plus diffusive) and write it as a **surfaceScalarField** *energyFlux*.

energyAdvectiveFlux Calculate the advective part of the energy-flux and write it as a **surfaceScalarField** *energyAdvectiveFlux*.

fieldAverage Calculates and writes the time averages of a given list of fields.

flowType Calculates and writes the **flowType** of velocity field where: -1 = rotational flow; 0 = simple shear flow; +1 = planar extensional flow.

grad Calculates the gradient of a field.

heatFlux Calculate the heat-flux (*i.e.* the diffusive part of the energy flux) and write it as a **surfaceScalarField** *heatFlux*.

Lambda2 Calculates and writes the second largest eigenvalue of the sum of the square of the symmetrical and anti-symmetrical parts of the velocity gradient tensor.

log Calculates the natural logarithm of the specified scalar field.

MachNo Calculates the Mach Number field from the velocity field.

mag Calculates the magnitude of a field.

magSqr Calculates the magnitude-squared of a field.

massFractions Calculates mass-fraction fields from mole-fraction fields, or moles fields, and a multi-component thermophysical model.

moleFractions Calculates mole-fraction fields from the mass-fraction fields of a multi-component thermophysical model.

power Calculates power fields, corresponding to pressure, shear and total stress.

PecletNo Calculates the Peclet Number field from the flux field.

Q Calculates the second invariant of the velocity gradient tensor.

randomise Adds a random component to a field, with a specified perturbation magnitude.

reconstruct Calculates the reconstruction of a field; *e.g.* to construct a cell-centred velocity **U** from the face-centred flux **phi**.

scale Multiplies a field by a scale factor

shearStress Calculates the shear stress, outputting the data as a **volSymmTensorField**.

specieAdvectiveFlux Calculate the advective flux of a specified species as a **surfaceScalarField**

specieDiffusiveFlux Calculate the diffusive flux of a specified species as a **surfaceScalarField**

specieFlux Calculate the combined flux of a specified species as a **surfaceScalarField**

streamFunction Writes the stream-function **pointScalarField**, calculated from the specified **fluxsurfaceScalarField**.

surfaceInterpolate Calculates the surface interpolation of a field.

totalEnthalpy Calculates and writes the total enthalpy $h_a + K$ as the **volScalarField** Ha .

tr Calculates the trace of a tensor field.

turbulenceFields Calculates specified turbulence fields and stores it on the database.

turbulenceIntensity Calculates and writes the turbulence intensity field I .

volField Converts a **volField::Internal** into a **volField**.

vorticity Calculates the vorticity field, i.e. the curl of the velocity field.

wallHeatFlux Calculates the heat flux at wall patches, outputting the data as a **volVectorField**.

wallHeatTransferCoeff Calculates the estimated incompressible flow heat transfer coefficient at wall patches, outputting the data as a **volScalarField**.

wallShearStress Calculates the shear stress at wall patches, outputting the data as a **volVectorField**.

writeCellCentres Writes the cell-centres **volVectorField** and the three component fields as **volScalarFields**; useful for post-processing thresholding.

writeCellVolumes Writes the cell-volumes **volScalarField**

writeVTK Writes out specified objects in VTK format, *e.g.* fields, stored on the case database.

yPlus Calculates the turbulence y^+ , outputting the data as a **yPlus** field.

7.3.2 Field operations

add Add a list of fields.

divide From the first field, divide the remaining fields in the list.

multiply Multiply a list of fields.

subtract From the first field, subtracts the remaining fields in the list.

uniform Create a uniform field.

7.3.3 Forces and force coefficients

forceCoeffsCompressible Calculates lift, drag and moment coefficients by summing forces on specified patches for a case where the solver is compressible (pressure is in units $M/(LT^2)$, *e.g.* Pa).

forceCoeffsIncompressible Calculates lift, drag and moment coefficients by summing forces on specified patches for a case where the solver is incompressible (pressure is kinematic, *e.g.* m^2/s^2).

forcesCompressible Calculates pressure and viscous forces over specified patches for a case where the solver is compressible (pressure is in units $M/(LT^2)$, *e.g.* Pa).

forcesIncompressible Calculates pressure and viscous forces over specified patches for a case where the solver is incompressible (pressure is kinematic, *e.g.* m^2/s^2).

7.3.4 Sampling for graph plotting

graphCell Writes graph data for specified fields along a line, specified by start and end points. One graph point is generated in each cell that the line intersects.

graphCellFace Writes graph data for specified fields along a line, specified by start and end points. One graph point is generated on each face and in each cell that the line intersects.

graphCutLayerAverage Writes graphs of cell values, volume-averaged in planes perpendicular to a given direction or in contours of a given distance field, adaptively grading the distribution of graph points to match the resolution of the mesh.

graphFace Writes graph data for specified fields along a line, specified by start and end points. One graph point is generated on each face that the line intersects.

graphLayerAverage Generates plots of fields averaged over the layers in the mesh.

graphPatchCutLayerAverage Writes graphs of patch face values, area-averaged in planes perpendicular to a given direction. It adaptively grades the distribution of graph points to match the resolution of the mesh

graphUniform Writes graph data for specified fields along a line, specified by start and end points. A specified number of graph points are used, distributed uniformly along the line.

7.3.5 Lagrangian data

dsmcFields Calculate intensive fields **UMean**, **translationalT**, **internalT**, **overallT** from averaged extensive fields from a DSMC calculation.

stopAtEmptyClouds Stops the run when all clouds are empty, *i.e.* have no particles.

7.3.6 Volume fields

cellMax Writes out the maximum cell value for one or more fields.

cellMaxMag Writes out the maximum cell value magnitude for one or more fields.

cellMin Writes out the minimum cell value for one or more fields.

cellMinMag Writes out the maximum cell value magnitude for one or more fields.

cellSum Writes out the sum of cell values of one or more fields.

volAverage Writes out the volume-weighted average of one or more fields.

volIntegrate Writes out the volume integral of one or more fields.

7.3.7 Numerical data

residuals For specified fields, writes out the initial residuals for the first solution of each time step; for non-scalar fields (*e.g.* vectors), writes the largest of the residuals for each component (*e.g.* x, y, z).

7.3.8 Control

adjustTimeStepToChemistry Adjusts the time step to a chemistry model's bulk chemical time scales

adjustTimeStepToReaction Adjusts the time step to a reaction model's bulk reaction time scales

removeObjects Removes the specified objects, *e.g.* fields, stored in the case database.

stopAtClockTime Stops the run when the specified clock time in second has been reached and optionally write results before stopping.

stopAtFile Stops the run when the file *stop* is created in the case directory.

stopAtTimeStep Stops the run if the time-step drops below the specified value in seconds and optionally write results before stopping.

time Writes run time, CPU time and clock time and optionally the CPU and clock times per time step.

timeStep Writes the time step to a file for monitoring.

userTimeStep Writes the user time step to a file for monitoring.

writeObjects Writes out specified objects, *e.g.* fields, stored on the case database.

7.3.9 Pressure tools

staticPressureIncompressible Calculates the pressure field in normal units, *i.e.* Pa in SI, from kinematic pressure by scaling by a specified density.

totalPressureCompressible Calculates the total pressure field in normal units, *i.e.* Pa in SI, for a case where the solver is compressible.

totalPressureIncompressible Calculates the total pressure field for a case where the solver is incompressible, in kinematic units, *i.e.* m^2/s^2 in SI.

7.3.10 Combustion and chemistry

bXiProgress Writes the combustion progress of the Weller b-Xi combustion model.

bXiQdot Writes the heat release rate of the Weller b-Xi combustion model.

Qdot Calculates and outputs the heat release rate for the current combustion model.

reactionRates Writes volume averaged reaction rates in $\text{kmol}/\text{m}^3/\text{s}$ for each reaction.

specieReactionRates Writes volume averaged reaction rates in $\text{kmol}/\text{m}^3/\text{s}$ for each species and each reaction.

ubAverage Both the corresponding unburnt and burnt gas fields must be available in the database.

ubDWEA Calculates the density weighted ensemble unburnt, burnt gas phase average of a field.

XiReactionRate Writes the turbulent flame-speed and reaction-rate **volScalarFields** for the Xi-based combustion models.

7.3.11 Multiphase

phaseForces Calculates the blended interfacial forces acting on a given phase, *i.e.* drag, virtual mass, lift, wall-lubrication and turbulent dispersion. Note that it works only in solver post-processing mode and in combination with **multiphaseEulerFoam**. For a simulation involving more than two phases, the accumulated force is calculated by looping over all **phasePairs** the phase is a part of.

phaseMap Writes the phase-fraction map field **alpha.map** with incremental value ranges for each phase e.g., with values 0 for water, 1 for air, 2 for oil, *etc.*

populationBalanceSetPhaseSizeDistribution Sets the population balance size distribution for a single phase by overwriting the values in the size-group fraction fields with values obtained by integrating a given distribution

populationBalanceSetSizeDistribution Sets the population balance size distribution by overwriting the values in the size-group fraction fields with values obtained by integrating a given distribution.

populationBalanceMoments Calculates and writes out integral (integer moments) or mean properties (mean, variance, standard deviation) of a size distribution computed with **multiphaseEulerFoam**. Requires solver post-processing.

populationBalanceSizeDistribution Writes out the size distribution computed with **multiphaseEulerFoam** for the entire domain or a volume region. Requires solver post-processing.

wallBoilingProperties Looks up wall boiling wall functions and collects and writes out out fields of bubble departure diameter, bubble departure frequency, nucleation site density, effective liquid fraction at the wall, quenching heat flux, and evaporative heat flux.

wallBoilingProperty Looks up the wall boiling model and collects and writes out a field of one of the wall boiling properties.

7.3.12 Probes

boundaryProbes Writes out values of fields at a cloud of points, interpolated to specified boundary patches.

interfaceHeight Reports the height of the interface above a set of locations. For each location, it writes the vertical distance of the interface above both the location and the lowest boundary. It also writes the point on the interface from which these heights are computed.

internalProbes Writes out values of fields interpolated to a specified cloud of points.

probes Writes out values of fields from cells nearest to specified locations.

7.3.13 Surface fields

faceZoneAverage Calculates the average value of one or more fields on a **faceZone**.

faceZoneFlowRate Calculates the flow rate through a specified face zone by summing the flux on patch faces. For solvers where the flux is volumetric, the flow rate is volumetric; where flux is mass flux, the flow rate is mass flow rate.

faceZoneFlux Calculates the total flux through a specified face zone by summing the flux on the faces.

patchAverage Calculates the average value of one or more fields on a patch.

patchDifference Calculates the difference between the average values of fields on two specified patches. Calculates the average value of one or more fields on a patch.

patchFlowRate Calculates the flow rate through a specified patch by summing the flux on patch faces. For solvers where the flux is volumetric, the flow rate is volumetric; where flux is mass flux, the flow rate is mass flow rate.

patchFlux Calculates the total flux through a specified patch by summing the flux on patch faces.

patchIntegrate Calculates the surface integral of one or more fields on a patch.

triSurfaceAverage Calculates the average on a specified triangulated surface by interpolating onto the triangles and integrating over the surface area. Triangles need to be small ($\leq$ cell size) for an accurate result.

triSurfaceDifference Calculates the difference between the average values of fields on two specified triangulated surfaces.

triSurfaceVolumetricFlowRate Calculates volumetric flow rate through a specified triangulated surface by interpolating velocity onto the triangles and integrating over the surface area. Triangles need to be small ($\leq$ cell size) for an accurate result.

7.3.14 Meshing

checkMesh Executes `primitiveMesh::checkMesh` to check the distortion of moving meshes.

multiValveEngineState Writes the multi-valve engine motion state providing details of the piston and valve position, speed etc.

7.3.15 ‘Pluggable’ solvers

particles Tracks a cloud of parcels driven by the flow of the continuous phase.

phaseScalarTransport Solves a transport equation for a scalar field within one phase of a multiphase simulation.

scalarTransport Solves a transport equation for a scalar field.

7.3.16 Sampling surfaces

cutPlaneSurface Writes out cut-plane surface files with interpolated field data in VTK format.

isoSurface Writes out iso-surface files with interpolated field data in VTK format.

patchSurface Writes out patch surface files with interpolated field data in VTK format.

7.3.17 Streamlines

streamlinesLine Writes out files of stream lines with interpolated field data in VTK format, with initial points uniformly distributed along a line.

streamlinesPatch Writes out files of stream lines with interpolated field data in VTK format, with initial points randomly selected within a patch.

streamlinesPoints Writes out files of stream lines with interpolated field data in VTK format, with specified initial points.

streamlinesSphere Writes out files of stream lines with interpolated field data in VTK format, with initial points randomly selected within a sphere.

7.4 Sampling and monitoring data

There are a set of general post-processing functions for sampling data across the domain for graphs and visualisation. Several functions also provide data in a single file, in the form of time versus values, that can be plotted onto graphs. This time-value data can be monitored during a simulation with the `foamMonitor` script.

7.4.1 Probing data

The functions for probing data are `boundaryProbes`, `internalProbes` and `probes` as listed in section 7.3.12. All functions work on the basis that the user provides some point locations and a list of fields, and the function writes out values of the fields at those locations. The differences between the functions are as follows.

- `probes` identifies the nearest cells to the probe locations and writes out the cell values; data is written into a single file in time-value format, suitable for plotting a graph.
- `boundaryProbes` and `internalProbes` interpolate field data to the probe locations, with the locations being snapped onto boundaries for `boundaryProbes`; data sets are written to separate files at scheduled write times (like fields). data.

Generally `probes` is more suitable for monitoring values at smaller numbers of locations, whereas the other functions are typically for sampling at large numbers of locations.

As an example, the user could use the `pitzDailySteady` case set up in section 2.1. The `probes` function is best configured by copying the file to the local system directory using `foamGet`.

```
foamGet probes
```

The user can modify the `probeLocations` in the `probes` file as follows.

```
12
13 #includeEtc "caseDicts/postProcessing/probes/probes.cfg"
14
15 fields (p U);
16 probeLocations
17 (
18     (0.01 0 0)
19 );
20
21 // ***** //
```

The configuration is completed by adding the `#includeFunc` directive to a `functions` file in the `system` directory.

```
#includeFunc probes
```

When the simulation runs, time-value data is written into `p` and `U` files in `postProcessing/probes/0`.

7.4.2 Sampling for graphs

The `graphUniform` function samples data for graph plotting. To use it, the *graphUniform* file can be copied into the *system* directory to be configured. We will configure it here using the *pitzDaily* case as before. The file is simply copied using `foamGet`.

```
foamGet graphUniform
```

The start and end points of the line, along which data is sampled, should be edited; the entries below provide a vertical line across the full height of the geometry 0.01 m beyond the back step.

```

14
15 start      (0.01 -0.025 0);
16 end        (0.01  0.025 0);
17 nPoints    100;
18
19 fields      (U p);
20
21 axis        distance; // The independent variable of the graph. Can be "x",
22                // "y", "z", "xyz" (all coordinates written out), or
23                // "distance" (from the start point).
24
25 #includeEtc "caseDicts/postProcessing/graphs/graphUniform.cfg"
26
27 // ***** //
```

The configuration is completed by adding the `#includeFunc` directive to a *functions* file in the *system* directory.

```
#includeFunc graphUniform
```

The simulation can be then re-run or the user could run the post-processing with the following command.

```
foamPostProcess -solver incompressibleFluid
```

Either way, distance-value data is written into files in time directories within *postProcessing/graphUniform*. The user can quickly display the data for x -component of velocity, U_x in the last time *e.g.* 285, by running `gnuplot` and plotting values.

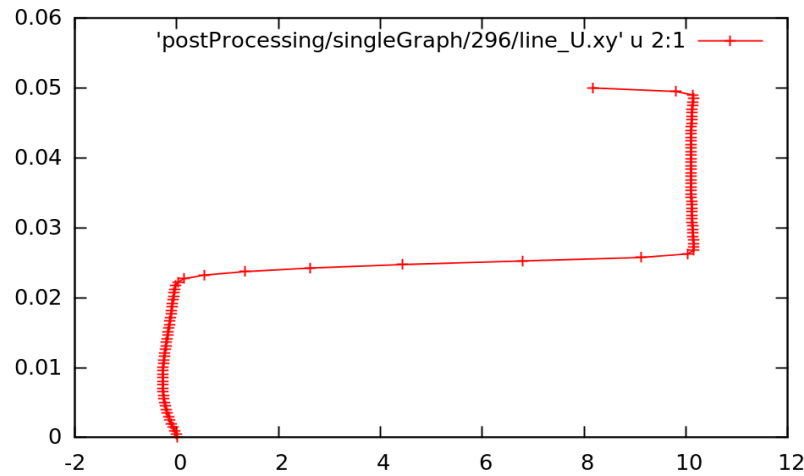
```

gnuplot
gnuplot> set style data linespoints
gnuplot> plot "postProcessing/graphUniform/285/line_U.xy" u 2:1
```

This produces the graph shown in Figure 7.5. This graph corresponds to the velocity inlet with a uniform profile, rather than a boundary layer profile. The formatting of the graph is specified in configuration files in *\$FOAM_ETC/caseDicts/postProcessing/graphs*. The *graphUniform.cfg* file in that directory includes the configuration as follows.

```

8
9 #includeEtc "caseDicts/functions/graphs/graph.cfg"
10
11 sets
12 {
13     line
```

Figure 7.5: Graph of U_x at $x = 0.01$, uniform sampling

```

14     {
15         type          lineUniform;
16         axis           $axis;
17         start          $start;
18         end            $end;
19         nPoints        $nPoints;
20     }
21 }
22
23 // *****

```

It shows that the sampling type is `lineUniform`, meaning the sampling uses a uniform distribution of points along a line. The other parameters are included by macro expansion from the main file and specify the line start and end, the number of points and the distance parameter specified on the horizontal axis of the graph.

An alternative graph function object, *graphCell*, samples the data at locations nearest to the cell centres. The user can copy that function object file and configure it as shown below.

```

13
14 start  (0.01 -0.025 0);
15 end    (0.01  0.025 0);
16 fields (U p);
17
18 axis   distance; // The independent variable of the graph. Can be "x",
19          // "y", "z", "xyz" (all coordinates written out), or
20          // "distance" (from the start point).
21
22 #includeEtc "caseDicts/postProcessing/graphs/graphCell.cfg"
23
24 // *****

```

Running the post-processing produces the graph in Figure 7.6.

7.4.3 Live monitoring of data

Functions like `probes` produce a single file of time-value data, suitable for graph plotting. When the function is executed during a simulation, the user may wish to monitor the data live on screen. The `foamMonitor` script enables this; to discover its functionality, the user run it with the `-help` option. The help option includes an example of monitoring residuals that we can demonstrate in this section.

Firstly, include the `residuals` function in the *functions* file.

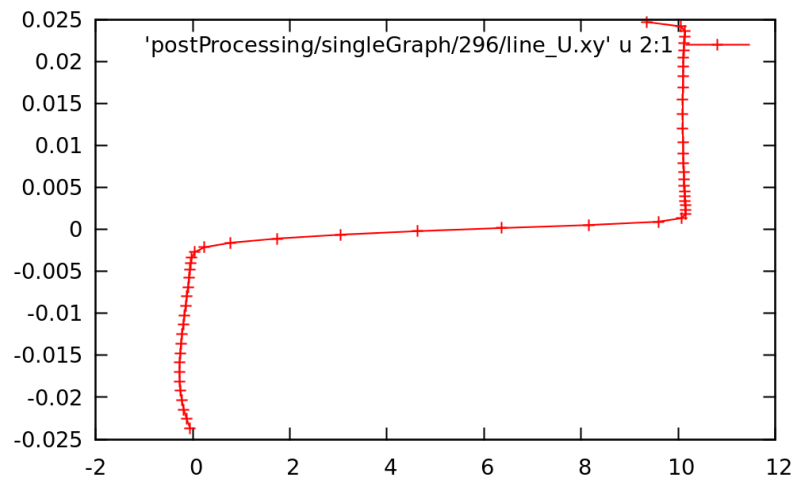


Figure 7.6: Graph of U_x at $x = 0.01$, mid-point sampling

```
#includeFunc residuals
```

The default fields whose residuals are captured are p and U . Should the user wish to configure other fields, they should make copy the *residuals* file in their *system* and edit the *fields* entry accordingly. All functions files are within the $\$FOAM_ETC/caseDicts$ directory. The *residuals* file can be located using `foamInfo`:

```
foamInfo residuals
```

It can then be copied into the *system* directory conveniently using `foamGet`:

```
foamGet residuals
```

The user can then run the case in the background.

```
foamRun > log &
```

The user should then run `foamMonitor` using the `-l` option for a log scale y -axis on the *residuals* file as follows. If the command is executed before the simulation is complete, they can see the graph being updated live.

```
foamMonitor -l postProcessing/residuals/0/residuals.dat &
```

It produces the graph of residuals for pressure and velocity in Figure 7.7.

7.4.4 Sampling for visualisation

There are several surfaces and streamlines functions, listed in sections 7.3.16 and 7.3.17, that can be used to generate files for visualisation. The use of `streamlinesLine` is already configured in the `pitzDailySteady` case.

To generate a cutting plane, the `cutPlaneSurface` function can be configured by copying the *cutPlaneSurface* file to the *system* directory using `foamGet`.

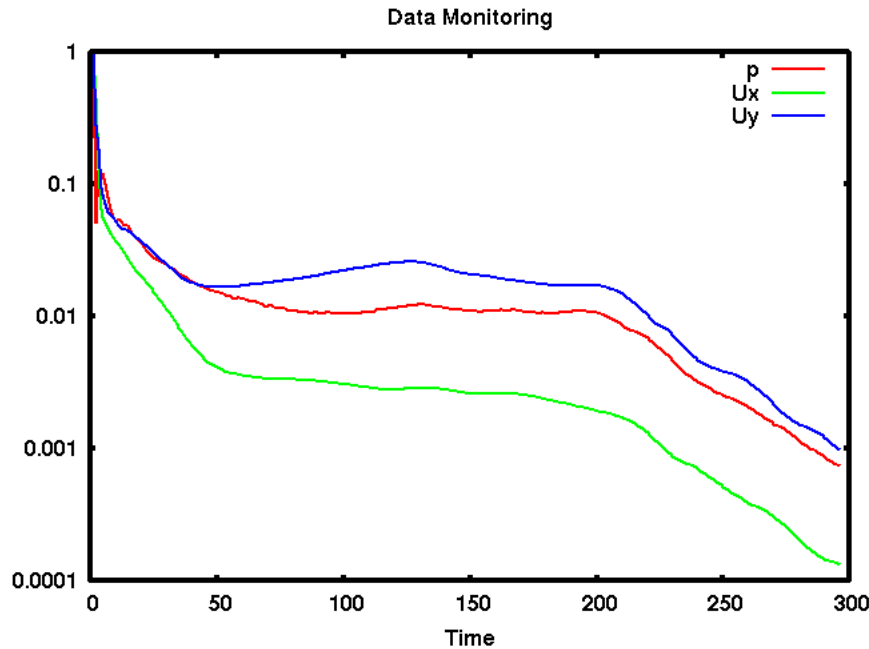


Figure 7.7: Live plot of residuals with foamMonitor

```
foamGet cutPlaneSurface
```

The file is configured by setting the origin and normal of the plane and the field data to be sampled. We can edit the file to produce a cutting plane along the `pitzDaily` geometry, normal to the z -direction.

```
16 fields      (p U);
17
18 interpolate true; // If false, write cell data to the surface triangles.
19                  // If true, write interpolated data at the surface points.
20
21 #includeEtc "caseDicts/postProcessing/surface/cutPlaneSurface.cfg"
22
23 // *****
```

The function can be included as normal by adding the `#includeFunc` to a *functions* file in the *system* directory. Alternatively, the user could test running the function using the solver post-processing by the following command.

```
foamPostProcess -solver incompressibleFluid -func cutPlaneSurface
```

This produces VTK format files of the cutting plane with pressure and velocity data in time directories in the *postProcessing/cutPlaneSurface* directory. The user can display the cutting plane by opening ParaView (type `paraview`), then doing `File->Open` and selecting one of the files, *e.g.* *postProcessing/cutPlaneSurface/285/U_zNormal.vtk* as shown in Figure 7.8.

7.4.5 The foamVTKSeries script

The previous section describes how to create surface image files in VTK format which can be read into ParaView and processed. The image files are distributed across time directo-

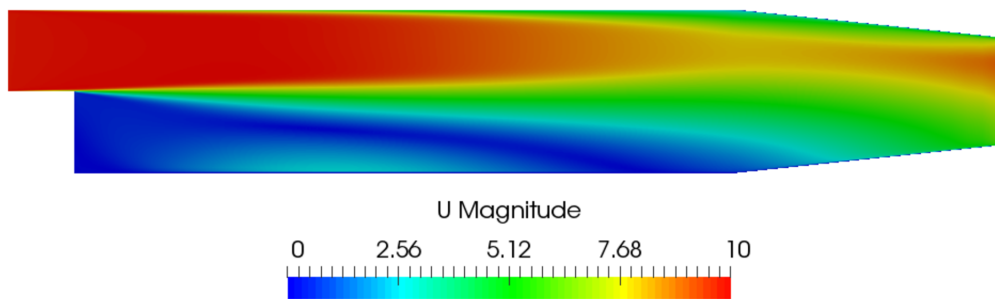


Figure 7.8: Cutting plane with velocity

ries within a sub-directory of the *postProcessing* directory, *e.g.* *postProcessing/cutPlaneSurface/0*, *postProcessing/cutPlaneSurface/100*, *postProcessing/cutPlaneSurface/200* and *postProcessing/cutPlaneSurface/285* from the case in the previous section.

If the user wishes to read a collection of files into ParaView, *e.g.* from time directories within the *cutPlaneSurface* directory, it would be a somewhat laborious job. In addition, ParaView would see no connection between the files and that they form a sequence. Without sequencing it is then not possible to create an animation from the files (without some manual effort).

The *foamVTKSeries* script overcomes this challenge quickly and easily. The user simply needs to run the script as follows.

```
foamVTKSeries
```

The script interrogates the contents of the *postProcessing* directory and creates a VTK series file, with extension *.vtk.series* for each collection of files it encounters. In the *cutPlaneSurface* example, *foamVTKSeries* would create a file named *U_zNormal.vtk.series* in the *postProcessing/cutPlaneSurface* directory. From ParaView, the user can then load the series of files by going to File->Open and selecting the *U_zNormal.vtk.series* file. The files appear in the Pipeline Browser as a single entity, with each file selectable by changing time in the Time selector.

7.5 Third-Party post-processing

OpenFOAM includes the following applications for converting data to formats for post-processing with several third-party tools. For EnSight, it additionally includes a reader module, described in the next section.

foamDataToFluent Translates OpenFOAM data to Fluent format.

foamToEnight Translates OpenFOAM data to EnSight format.

foamToEnightParts Translates OpenFOAM data to Enight format. An Enight part is created for each *cellZone* and patch.

foamToGMV Translates foam output to GMV readable files.

foamToTetDualMesh Converts **polyMesh** results to **tetDualMesh**.

foamToVTK Legacy VTK file format writer.

smapToFoam Translates a STAR-CD SMAP data file into OpenFOAM field format.

7.5.1 Post-processing with EnSight

OpenFOAM offers the capability for post-processing OpenFOAM cases with EnSight, with a choice of 2 options:

- converting the OpenFOAM data to EnSight format with the **foamToEnSight** utility;
- reading the OpenFOAM data directly into EnSight using the **ensight74FoamExec** module.

The **foamToEnSight** utility converts data from OpenFOAM to EnSight file format. For a given case, **foamToEnSight** is executed like any normal application. **foamToEnSight** creates a directory named *EnSight* in the case directory, *deleting any existing EnSight directory in the process*. The converter reads the data in all time directories and writes into a case file and a set of data files. The case file is named *EnSight_Case* and contains details of the data file names. Each data file has a name of the form *EnSight_nn.ext*, where *nn* is an incremental counter starting from 1 for the first time directory, 2 for the second and so on and *ext* is a file extension of the name of the field that the data refers to, as described in the case file, *e.g.* T for temperature, **mesh** for the mesh. Once converted, the data can be read into EnSight by the normal means:

1. from the EnSight GUI, the user should select **Data (Reader)** from the **File** menu;
2. the appropriate *EnSight_Case* file should be highlighted in the **Files** box;
3. the **Format** selector should be set to **Case**, the EnSight default setting;
4. the user should click **(Set) Case** and **Okay**.

EnSight provides the capability of using a user-defined module to read data from a format other than the standard EnSight format. OpenFOAM includes its own reader module **ensightFoamReader** that is compiled into a library named **libuserd-foam**. It is this library that EnSight needs to use which means that it must be able to locate it on the filing system as described in the following section.

In order to run the EnSight reader, it is necessary to set some environment variables correctly. The settings are made in the *bashrc* (or *cshrc*) file in the *\$WM_PROJECT_DIR/etc/apps/ensightFoam* directory. The environment variables associated with EnSight are prefixed by **\$CEI_** or **\$ENSIGHT7_** and listed in Table 7.1. With a standard user setup, only **\$CEI_HOME** may need to be set manually, to the path of the EnSight installation.

The principal difficulty in using the EnSight reader lies in the fact that EnSight expects that a case to be defined by the contents of a particular file, rather than a directory as it is in OpenFOAM. Therefore in following the instructions for the using the reader below, the user should pay particular attention to the details of case selection, since EnSight does not permit selection of a directory name.

Environment variable	Description and options
<code>\$CEI_HOME</code>	Path where EnSight is installed, eg <code>/usr/local/ensight</code> , added to the system path by default
<code>\$CEI_ARCH</code>	Machine architecture, from a choice of names corresponding to the machine directory names in <code>\$CEI_HOME/ensight74/machines</code> ; default settings include <code>linux_2.4</code> and <code>sgi_6.5_n32</code>
<code>\$ENSIGHT7_READER</code>	Path that EnSight searches for the user defined <code>libuserd-foam</code> reader library, set by default to <code>\$FOAM_LIBBIN</code>
<code>\$ENSIGHT7_INPUT</code>	Set by default to <code>dummy</code>

Table 7.1: Environment variable settings for **EnSight**.

1. from the **EnSight** GUI, the user should select **Data (Reader)** from the **File** menu;
2. The user should now be able to select the **OpenFOAM** from the **Format** menu; if not, there is a problem with the configuration described above.
3. The user should find their case directory from the **File Selection** window, highlight one of top 2 entries in the **Directories** box ending in `/.` or `/..` and click **(Set) Geometry**.
4. The path field should now contain an entry for the case. The **(Set) Geometry** text box should contain a `/`.
5. The user may now click **Okay** and **EnSight** will begin reading the data.
6. When the data is read, a new **Data Part Loader** window will appear, asking which part(s) are to be read. The user should select **Load all**.
7. When the mesh is displayed in the **EnSight** window the user should close the **Data Part Loader** window, since some features of **EnSight** will not work with this window open.

Chapter 8

Models and physical properties

OpenFOAM includes a large range of solvers, each designed for a specific class of flow, as described in section 3.6. Each solver uses a particular set of models which calculate physical properties and simulate phenomena like transport, turbulence, thermal radiation, *etc.*

From OpenFOAM v10 onwards, a distinction is made between material *properties* and models for phenomena such as those mentioned above. **Properties are specified in *physicalProperties* file in the *constant* directory.** In the case of fluids, properties in *physicalProperties* relate to a fluid at rest. They are the properties you might look up from a table in a book, so can be dependent on temperature T , based on some function.

Properties described in *physicalProperties* do not include any dependency on the flow itself. For example, turbulence, visco-elasticity and the variation of viscosity ν with strain-rate, are all specified in a *momentumTransport* file in the *constant* directory. This chapter includes a description of models for viscosity which are dependent on strain-rate in section 8.3 and turbulence models in section 8.2. Thermophysical models, which are specified in the *physicalProperties* file (since they represent temperature dependency of properties) are described in section 8.1.

8.1 Thermophysical models

Thermophysical models are concerned with: thermodynamics, *e.g.* relating internal energy e to temperature T ; transport, *e.g.* the dependence of properties such as ν on temperature; and state, *e.g.* dependence of density ρ on T and pressure p . Thermophysical models are specified in the *physicalProperties* dictionary.

A thermophysical model required an entry named **thermoType** which specifies the package of thermophysical modelling that is used in the simulation. OpenFOAM includes a large set of pre-compiled combinations of modelling, built within the code using C++ templates. It can also compile on-demand a combination which is not pre-compiled during a simulation.

Thermophysical modelling packages begin with the equation of state and then adding more layers of thermophysical modelling that derive properties from the previous layer(s). The keyword entries in **thermoType** reflects the multiple layers of modelling and the underlying framework in which they combined. Below is an example entry for **thermoType**:

```
thermoType
{
```

```

    type          hePsiThermo;
    mixture        pureMixture;
    transport      const;
    thermo         hConst;
    equationOfState perfectGas;
    specie         specie;
    energy         sensibleEnthalpy;
}

```

The keyword entries specify the choice of thermophysical models, *e.g.* **transport constant** (constant viscosity, thermal diffusion), **equationOfState perfectGas**, *etc.* In addition there is a keyword entry named **energy** that allows the user to specify the form of energy to be used in the solution and thermodynamics. The following sections explain the entries and options in the **thermoType** package.

8.1.1 Thermophysical and mixture models

Each solver that uses thermophysical modelling constructs an object of a specific thermophysical model class. The model classes are listed below.

fluidThermo Thermophysical model for a general fluid with fixed composition used by the **isoThermalFluid** and **fluid** solver modules.

rhoThermo Thermophysical model for liquids and solids, used by the **isothermalFilm** and **film** solver module.

psiThermo Thermophysical model for gases only, with fixed composition, used by the **shock-Fluid** solver module.

fluidMulticomponentThermo Thermophysical model for fluid of varying composition used by the **multicomponentFluid** solver module.

psiuMulticomponentThermo Thermophysical model for combustion that modelled by a laminar flame speed and regress variable used by the **XiFluid** solver module.

compressibleMultiphaseVoFMixtureThermo Thermophysical models for multiple phases used by the **compressibleMultiphaseVoF** solver module.

solidThermo and **solidDisplacementThermo** Thermophysical models for solids used by the **solid** and **solidDisplacement** solver modules, respectively.

The **type** keyword (in the *thermoType* sub-dictionary) specifies the underlying thermophysical model used by the solver. The user can select from the following.

- **hePsiThermo**: available for solvers that construct **fluidThermo**, **psiThermo**, **fluidMulticomponentThermo** and .
- **heRhoThermo**: available for solvers that construct **fluidThermo**, **rhoThermo**, **fluidMulticomponentThermo**, **compressibleMultiphaseVoFMixtureThermo**.

- **heheuPsiThermo**: for solvers that construct **psiuMulticomponentThermo**.
- **heSolidThermo**: for solvers that construct **solidThermo** or **solidDisplacementThermo**.

The **mixture** specifies the mixture composition. The options available are listed below.

- **pureMixture**: mixture with fixed composition, which reads properties from a a sub-dictionary called **mixture**.
- **multicomponentMixture**: mixture with variable composition, with species, *e.g.* O2, N2, listed by the **species** keyword, and properties specified for each specie within sub-dictionaries named after each specie.
- **coefficientWilkeMulticomponentMixture**: as **multicomponentMixture**, but applies Wilke's equation to calculate transport properties for the mixture.
- **valueMulticomponentMixture**: as **multicomponentMixture**, but applies mole-fraction weighting to calculate transport properties for the mixture.
- **homogeneousMixture**, **inhomogeneousMixture** and **veryInhomogeneousMixture**: for combustion based on laminar flame speed and regress variables, constituents are a set of mixtures, such as **fuel**, **oxidant** and **burntProducts**.

8.1.2 Transport model

The transport modelling concerns evaluating dynamic viscosity μ , thermal conductivity κ and thermal diffusivity α (for internal energy and enthalpy equations). The current transport models are as follows:

const assumes a constant μ and Prandtl number $Pr = c_p \mu / \kappa$ which is simply specified by a two keywords, **mu** and **Pr**, respectively.

sutherland calculates μ as a function of temperature T from a Sutherland coefficient A_s and Sutherland temperature T_s , specified by keywords **As** and **Ts**; μ is calculated according to:

$$\mu = \frac{A_s \sqrt{T}}{1 + T_s/T}. \quad (8.1)$$

polynomial calculates μ and κ as a function of temperature T from a polynomial of any order N , *e.g.*:

$$\mu = \sum_{i=0}^{N-1} a_i T^i. \quad (8.2)$$

logPolynomial calculates $\ln(\mu)$ and $\ln(\kappa)$ as a function of $\ln(T)$ from a polynomial of any order N ; from which μ , κ are calculated by taking the exponential, *e.g.*:

$$\ln(\mu) = \sum_{i=0}^{N-1} a_i [\ln(T)]^i. \quad (8.3)$$

Andrade calculates $\ln(\mu)$ and $\ln(\kappa)$ as a polynomial function of T , *e.g.* for μ :

$$\ln(\mu) = a_0 + a_1T + a_2T^2 + \frac{a_3}{a_4 + T}. \quad (8.4)$$

tabulated uses uniform tabulated data for viscosity and thermal conductivity as a function of pressure and temperature.

icoTabulated uses non-uniform tabulated data for viscosity and thermal conductivity as a function of temperature.

WLF (Williams-Landel-Ferry) calculates μ as a function of temperature from coefficients C_1 and C_2 and reference temperature T_r specified by keywords **C1**, **C2** and **Tr**; μ is calculated according to:

$$\mu = \mu_0 \exp \left(\frac{-C_1(T - T_r)}{C_2 + T - T_r} \right) \quad (8.5)$$

8.1.3 Thermodynamic models

The thermodynamic models are concerned with evaluating the specific heat c_p from which other properties are derived. The current **thermo** models are as follows:

eConst assumes a constant c_v and a heat of fusion H_f which is simply specified by a two values c_v H_f , given by keywords **Cv** and **Hf**.

elcoTabulated calculates c_v by interpolating non-uniform tabulated data of (T, c_p) value pairs, *e.g.*:
 ((200 1005) (400 1020));

ePolynomial calculates c_v as a function of temperature by a polynomial of any order N :

$$c_v = \sum_{i=0}^{N-1} a_i T^i. \quad (8.6)$$

ePower calculates c_v as a power of temperature according to:

$$c_v = c_0 \left(\frac{T}{T_{\text{ref}}} \right)^{n_0}. \quad (8.7)$$

eTabulated calculates c_v by interpolating uniform tabulated data of (T, c_p) value pairs, *e.g.*:
 ((200 1005) (400 1020));

hConst assumes a constant c_p and a heat of fusion H_f which is simply specified by a two values c_p H_f , given by keywords **Cp** and **Hf**.

hlcoTabulated calculates c_p by interpolating non-uniform tabulated data of (T, c_p) value pairs, *e.g.*:
 ((200 1005) (400 1020));

hPolynomial calculates c_p as a function of temperature by a polynomial of any order N :

$$c_p = \sum_{i=0}^{N-1} a_i T^i. \quad (8.8)$$

hPower calculates c_p as a power of temperature according to:

$$c_p = c_0 \left(\frac{T}{T_{\text{ref}}} \right)^{n_0}. \quad (8.9)$$

hTabulated calculates c_p by interpolating uniform tabulated data of (T, c_p) value pairs, *e.g.*:
((200 1005) (400 1020));

janaf calculates c_p as a function of temperature T from a set of coefficients taken from JANAF tables of thermodynamics. The ordered list of coefficients is given in Table 8.1. The function is valid between a lower and upper limit in temperature T_l and T_h respectively. Two sets of coefficients are specified, the first set for temperatures above a common temperature T_c (and below T_h), the second for temperatures below T_c (and above T_l). The function relating c_p to temperature is:

$$c_p = R(((a_4 T + a_3)T + a_2)T + a_1)T + a_0). \quad (8.10)$$

In addition, there are constants of integration, a_5 and a_6 , both at high and low temperature, used to evaluating h and s respectively.

Description	Entry	Keyword
Lower temperature limit	T_l (K)	Tlow
Upper temperature limit	T_h (K)	Thigh
Common temperature	T_c (K)	Tcommon
High temperature coefficients	$a_0 \dots a_4$	highCpCoeffs (a0 a1 a2 a3 a4...
High temperature enthalpy offset	a_5	a5...
High temperature entropy offset	a_6	a6)
Low temperature coefficients	$a_0 \dots a_4$	lowCpCoeffs (a0 a1 a2 a3 a4...
Low temperature enthalpy offset	a_5	a5...
Low temperature entropy offset	a_6	a6)

Table 8.1: JANAF thermodynamics coefficients.

8.1.4 Composition of each constituent

There is currently only one option for the **specie** model which specifies the composition of each constituent. That model is itself named **specie**, which is specified by the following entries.

- **nMoles**: number of moles of component. This entry is only used for combustion modelling based on regress variable with a homogeneous mixture of reactants; otherwise it is set to 1.
- **molWeight** in grams per mole of specie.

8.1.5 Equation of state

The following equations of state are available in the thermophysical modelling library.

`adiabaticPerfectFluid` Adiabatic perfect fluid:

$$\rho = \rho_0 \left(\frac{p + B}{p_0 + B} \right)^{1/\gamma}, \quad (8.11)$$

where ρ_0, p_0 are reference density and pressure respectively, and B is a model constant.

`Boussinesq` Boussinesq approximation

$$\rho = \rho_0 [1 - \beta (T - T_0)] \quad (8.12)$$

where β is the coefficient of volumetric expansion and ρ_0 is the reference density at reference temperature T_0 .

`icoPolynomial` Incompressible, polynomial equation of state:

$$\rho = \sum_{i=0}^{N-1} a_i T^i, \quad (8.13)$$

where a_i are polynomial coefficients of any order N .

`icoTabulated` Tabulated data for an incompressible fluid using (T, ρ) value pairs, *e.g.*

`rho ( (200 1010) (400 980) );`

`incompressiblePerfectGas` Perfect gas for an incompressible fluid:

$$\rho = \frac{1}{RT} p_{\text{ref}}, \quad (8.14)$$

where p_{ref} is a reference pressure.

`linear` Linear equation of state:

$$\rho = \psi p + \rho_0, \quad (8.15)$$

where ψ is compressibility (not necessarily $(RT)^{-1}$).

`PengRobinsonGas` Peng Robinson equation of state:

$$\rho = \frac{1}{zRT} p, \quad (8.16)$$

where the complex function $z = z(p, T)$ can be referenced in the source code in *Peng-RobinsonGasI.H*, in the `$FOAM_SRC/thermophysicalModels/specie/equationOfState/` directory.

`perfectFluid` Perfect fluid:

$$\rho = \frac{1}{RT} p + \rho_0, \quad (8.17)$$

where ρ_0 is the density at $T = 0$.

perfectGas Perfect gas:

$$\rho = \frac{1}{RT}p. \quad (8.18)$$

rhoConst Constant density:

$$\rho = \text{constant}. \quad (8.19)$$

rhoTabulated Uniform tabulated data for a compressible fluid, calculating ρ as a function of T and p .

rPolynomial Reciprocal polynomial equation of state for liquids and solids:

$$\frac{1}{\rho} = C_0 + C_1T + C_2T^2 - C_3p - C_4pT \quad (8.20)$$

where C_i are coefficients.

8.1.6 Selection of energy variable

The user must specify the form of energy to be used in the solution, either internal energy e and enthalpy h , and in forms that include the heat of formation Δh_f or not. This choice is specified through the **energy** keyword.

We refer to *absolute* energy where heat of formation is included, and *sensible* energy where it is not. For example absolute enthalpy h is related to sensible enthalpy h_s by

$$h = h_s + \sum_i c_i \Delta h_f^i \quad (8.21)$$

where c_i and h_f^i are the molar fraction and heat of formation, respectively, of specie i . In most cases, we use the sensible form of energy, for which it is easier to account for energy change due to reactions. Keyword entries for **energy** therefore include *e.g.* **sensibleEnthalpy**, **sensibleInternalEnergy** and **absoluteEnthalpy**.

8.1.7 Thermophysical property data

The basic thermophysical properties are specified for each species from input data. Data entries must contain the name of the specie as the keyword, *e.g.* **O2**, **H2O**, **mixture**, followed by sub-dictionaries of coefficients, including:

specie containing *i.e.* number of moles, **nMoles**, of the specie, and molecular weight, **molWeight** in units of g/mol;

thermodynamics containing coefficients for the chosen thermodynamic model (see below);

transport containing coefficients for the chosen transport model (see below).

The following is an example entry for a specie named **fuel** modelled using **sutherland** transport and **janaf** thermodynamics:

```

fuel
{
    specie
    {
        nMoles      1;
        molWeight    16.0428;
    }
    thermodynamics
    {
        Tlow        200;
        Thigh       6000;
        Tcommon     1000;
        highCpCoeffs (1.63543 0.0100844 -3.36924e-06 5.34973e-10
                      -3.15528e-14 -10005.6 9.9937);
        lowCpCoeffs  (5.14988 -0.013671 4.91801e-05 -4.84744e-08
                      1.66694e-11 -10246.6 -4.64132);
    }
    transport
    {
        As          1.67212e-06;
        Ts          170.672;
    }
}

```

The following is an example entry for a specie named `air` modelled using `const` transport and `hConst` thermodynamics:

```

air
{
    specie
    {
        nMoles      1;
        molWeight    28.96;
    }
    thermodynamics
    {
        Cp          1004.5;
        Hf          2.544e+06;
    }
    transport
    {
        mu          1.8e-05;
        Pr          0.7;
    }
}

```


8.2 Turbulence models

Turbulence modelling is part of general **momentum transport** which is concerned with models for the viscous stress in a fluid. Momentum transport is configured through the *momentumTransport* file in the *constant* directory of a case. The file includes the mandatory **simulationType** keyword that specifies how turbulence is modelled, which includes the following options:

laminar uses no turbulence models;

RAS uses Reynolds-averaged simulation (RAS) modelling;

LES uses large-eddy simulation (LES) modelling.

The file then includes a sub-dictionary of the same name as the chosen **simulationType** which contains the model selections. A typical example is shown below that uses the $k-\epsilon$ (k-epsilon) turbulence model.

```

16
17  simulationType RAS;
18
19  RAS
20  {
21      model          kEpsilon;
22
23      turbulence      on;
24
25      #print
26  }
27
28  // ***** //
```

The file shows the selected RAS simulation followed by the RAS sub-dictionary containing the model selections, in particular the **model** which is set to **kEpsilon**. The choice of RAS models is described in section 8.2.1 and more information can be found in [Chapter 7 of *Notes on Computational Fluid Dynamics: General Principles*](#). The LES models are listed in section 8.2.3.

Where the **laminar** option is selected, the sub-dictionary is optional and will default to a Newtonian model, using the viscosity specified in the *physicalProperties* file. Other models, including non-Newtonian and visco-elastic models, are described in section 8.3. Non-Newtonian models can also be combined with turbulence models (whereas visco-elastic models cannot).

For a general introduction to turbulence for CFD, the reader may also wish to consult Chapter 6 of [Notes on Computational Fluid Dynamics: General Principles](#).

8.2.1 Reynolds-averaged simulation (RAS) modelling

If RAS is selected, the choice of RAS modelling is specified in a RAS sub-dictionary which requires the following entries.

- **model**: name of RAS turbulence model.
- **turbulence**: switch to turn the solving of turbulence modelling on/off.

- `<modelName>`: optional sub-dictionary of coefficients for the respective `model`, *e.g.* `kEpsilon`, to override the default coefficients.

Entries for any of the coefficients of the respective `model` can be optionally included *either* directly in the `RAS` dictionary *or* within a further sub-dictionary named after the chosen name of the `model`, *e.g.* `kEpsilon`. This allows the default coefficients in the model to be overridden.

Turbulence models can be listed by running `foamToC` with a relevant table listed from the `RAS` or `LES` tables. For example, the `RAS` tables are listed by running the following command.

```
foamToC -table RAS
```

This returns several sub-tables. The user can then list the models within one of those tables, *e.g.* the incompressible models.

```
foamToC -table RASincompressibleMomentumTransportModel
```

The `RAS` models used in the tutorials can be listed using `foamSearch` with the following command. The lists of available models are given in the following sections.

```
foamSearch $FOAM_TUTORIALS momentumTransport RAS/model
```

Users can locate tutorials using a particular model, *e.g.* `buoyantKEpsilon`, using `foamInfo`.

```
foamInfo buoyantKEpsilon
```

8.2.2 RAS turbulence models

For **incompressible flows**, the `RAS model` can be chosen from the list below.

LRR Launder, Reece and Rodi Reynolds-stress turbulence model for incompressible flows.

LamBremhorstKE Lam and Bremhorst low-Reynolds number k-epsilon turbulence model for incompressible flows.

LaunderSharmaKE Launder and Sharma low-Reynolds k-epsilon turbulence model for incompressible flows.

LienCubicKE Lien cubic non-linear low-Reynolds k-epsilon turbulence models for incompressible flows.

LienLeschziner Lien and Leschziner low-Reynolds number k-epsilon turbulence model for incompressible flows.

RNGkEpsilon Renormalization group k-epsilon turbulence model for incompressible flows.

SSG Speziale, Sarkar and Gatski Reynolds-stress turbulence model for incompressible flows.

ShihQuadraticKE Shih's quadratic algebraic Reynolds stress k-epsilon turbulence model for incompressible flows

SpalartAllmaras Spalart-Allmaras one-eqn mixing-length model for incompressible external flows.

kEpsilon Standard k-epsilon turbulence model for incompressible flows.

kEpsilonLopesdaCosta Variant of the standard k-epsilon turbulence model with additional source terms to handle the changes in turbulence in porous regions for atmospheric flows over forested terrain.

kOmega Standard high Reynolds-number k-omega turbulence model for incompressible flows.

kOmega2006 Standard (2006) high Reynolds-number k-omega turbulence model for incompressible flows.

kOmegaSST Implementation of the k-omega-SST turbulence model for incompressible flows.

kOmegaSSTLM Langtry-Menter 4-equation transitional SST model based on the k-omega-SST RAS model.

kOmegaSSTsas Scale-adaptive URAS model based on the k-omega-SST RAS model.

kkLOmega Low Reynolds-number k-kl-omega turbulence model for incompressible flows.

qZeta Gibson and Dafa'Alla's q-zeta two-equation low-Re turbulence model for incompressible flows

realizableKE Realizable k-epsilon turbulence model for incompressible flows.

v2f Lien and Kalitzin's v2-f turbulence model for incompressible flows, with a limit imposed on the turbulent viscosity given by Davidson et al.

For **compressible flows**, the RAS model can be chosen from the list below.

LRR Launder, Reece and Rodi Reynolds-stress turbulence model for compressible flows.

LaunderSharmaKE Launder and Sharma low-Reynolds k-epsilon turbulence model for compressible and combusting flows including rapid distortion theory (RDT) based compression term.

RNGkEpsilon Renormalization group k-epsilon turbulence model for compressible flows.

SSG Speziale, Sarkar and Gatski Reynolds-stress turbulence model for compressible flows.

SpalartAllmaras Spalart-Allmaras one-eqn mixing-length model for compressible external flows.

buoyantKEpsilon Additional buoyancy generation/dissipation term applied to the k and epsilon equations of the standard k-epsilon model.

kEpsilon Standard k-epsilon turbulence model for compressible flows including rapid distortion theory (RDT) based compression term.

kOmega Standard high Reynolds-number k-omega turbulence model for compressible flows.

kOmega2006 Standard (2006) high Reynolds-number k-omega turbulence model for compressible flows.

kOmegaSST Implementation of the k-omega-SST turbulence model for compressible flows.

kOmegaSSTLM Langtry-Menter 4-equation transitional SST model based on the k-omega-SST RAS model.

kOmegaSSTsas Scale-adaptive URAS model based on the k-omega-SST RAS model.

realizableKE Realizable k-epsilon turbulence model for compressible flows.

v2f Lien and Kalitzin's v2-f turbulence model for compressible flows, with a limit imposed on the turbulent viscosity given by Davidson et al.

8.2.3 Large eddy simulation (LES) modelling

If LES is selected, the choice of LES modelling is specified in a LES sub-dictionary which requires the following entries.

- **model**: name of LES turbulence model.
- **turbulence**: switch to turn the solving of turbulence modelling on/off.
- **delta**: name of delta δ model.
- **<modelName>**: optional dictionary of coefficients for the respective model, *e.g.* **dynamicKEqn**, to override the default coefficients.
- **<deltaName>**: dictionary of coefficients for the **delta** model, *e.g.* **cubeRootVol**

As with RAS modelling, entries for any of the coefficients of the respective **model** can be optionally included *either* directly in the LES dictionary *or* within a further sub-dictionary named after the chosen name of the **model**, *e.g.* **dynamicKEqn**. The same rule applies to the coefficients of the **delta** model.

The LES models used in the tutorials can be listed using **foamSearch** with the following command. The lists of available models are given in the following sections.

```
foamSearch $FOAM_TUTORIALS momentumTransport LES/model
```

8.2.4 LES turbulence models

For **incompressible and compressible flows**, the LES model can be chosen from the list below.

DeardorffDiffStress Differential SGS Stress Equation Model for incompressible flows

Smagorinsky The Smagorinsky SGS model.

SpalartAllmarasDDES SpalartAllmaras DDES turbulence model for incompressible flows

SpalartAllmarasDES SpalartAllmarasDES DES turbulence model for incompressible flows

SpalartAllmarasIDDES SpalartAllmaras IDDES turbulence model for incompressible flows

WALE The Wall-adapting local eddy-viscosity (WALE) SGS model.

dynamicKEqn Dynamic one equation eddy-viscosity model

dynamicLagrangian Dynamic SGS model with Lagrangian averaging

kEqn One equation eddy-viscosity model

kOmegaSSTDES Implementation of the k-omega-SST-DES turbulence model for incompressible flows.

8.2.5 Model coefficients

As mentioned above, the coefficients for the RAS and LES turbulence models are given default values in their respective source code. If the user wishes to override these default values, then they can do so by adding an entry directly to the **RAS** sub-dictionary file or within a sub-dictionary whose name is that of the model, *e.g.* **kEpsilon** for the **kEpsilon** model. The same applies to LES modelling.

The user can print the sub-dictionary with all the default values by adding a **#print** directive into their *momentumTransport* file and running the case or expanding the file with **foamDictionary**, as discussed in section 4.2.20.

8.2.6 Wall functions

A range of wall function models is available in OpenFOAM that are applied as boundary conditions on individual patches. This enables different wall function models to be applied to different wall regions. The choice of wall function model is specified through the turbulent viscosity field ν_t in the *0/nut* file. For example, a *0/nut* file:

```

16
17  dimensions      [turbulentViscosity];
18
19  internalField    uniform 0;
20
21  boundaryField
22  {
23      inlet
24      {
25          type      calculated;
26          value      uniform 0;

```

```

27     }
28     outlet
29     {
30         type            calculated;
31         value            uniform 0;
32     }
33     upperWall
34     {
35         type            nutkWallFunction;
36         value            uniform 0;
37     }
38     lowerWall
39     {
40         type            nutkWallFunction;
41         value            uniform 0;
42     }
43     frontAndBack
44     {
45         type            empty;
46     }
47 }
48
49 // *****
50 // *****

```

There are a number of wall function models available in the release, *e.g.* `nutWallFunction`, `nutRoughWallFunction`, `nutUSpaldingWallFunction`, `nutkWallFunction` and `nutkAtmWallFunction`. The user can get the full list of wall function models using `foamInfo`:

```
foamToC -scalarBCs | grep nut
```

Within each wall function boundary condition the user can over-ride default settings for E , κ and C_μ through optional `E`, `kappa` and `Cmu` keyword entries.

Having selected the particular wall functions on various wall patches in the *nut* file, the user should select the following boundary conditions at wall patches for other turbulence fields.

- *epsilon* field: apply the `epsilonWallFunction` to corresponding patches.
- *omega* field: apply the `omegaWallFunction` to corresponding patches.
- *k*, *q* or *R* field: apply `kqRwallFunction` to corresponding patches.

8.3 Transport/rheology models

The *momentumTransport* file includes any model for the viscous stress in a fluid. That includes turbulence models, described in the previous section 8.2, but also non-Newtonian and visco-elastic models described in this section. These models are described as `laminar`, located in `$FOAM_SRC/MomentumTransportModels/momentumTransportModels/laminar`, including:

- a family of `generalisedNewtonian` models for a non-uniform viscosity which is a function of strain rate $\dot{\gamma} = \sqrt{2}|\text{symm}(\nabla\mathbf{U})|$, described in sections 8.3.1, 8.3.2, 8.3.3, 8.3.4, 8.3.5 and 8.3.6;
- a set of visco-elastic models, including `Maxwell`, `Giesekus` and `PTT` (Phan-Thien & Tanner), described in sections 8.3.7, 8.3.8 and 8.3.9, respectively;

- the `lambdaThixotropic` model, described in section 8.3.10.

When turbulence modelling is selected in the *momentumTransport* file, the *generalisedNewtonian* model is used by default to calculate the molecular viscosity. The choice of *generalisedNewtonian* model, specified by the `viscosityModel` keyword, is set to *Newtonian* by default, which simply uses the viscosity `nu` specified in the *physicalProperties* file. The following example exposes the default settings used with turbulence modelling.

```
simulationType RAS

RAS
{
    model            kEpsilon; // RAS model
    turbulence       on;

    // "laminar" model generalisedNewtonian is used by default
    viscosityModel   Newtonian; // default
}
```

While the `viscosityModel` entry is generally omitted when turbulence models are used, it can be included to set any of the non-Newtonian *generalisedNewtonian* models.

When turbulence modelling is not selected, by setting the `laminar` simulation type, the user can select any of the laminar models through the `model` keyword entry in the `laminar` sub-dictionary, including the visco-elastic models. The laminar models are listed by the following command.

```
foamToC -table laminarincompressibleMomentumTransportModel
```

If the *generalisedNewtonian* model is selected, the user must then specify the viscosity model through the `viscosityModel` keyword as mentioned above. The viscosity models are listed by the following command.

```
foamToC -table generalisedNewtonianViscosityModel
```

The example below shows how the Bird-Carreau viscosity model is selected in a configuration without turbulence modelling.

```
simulationType laminar

laminar
{
    model            generalisedNewtonian;
    viscosityModel   BirdCarreau;
    // ... followed by the BirdCarreau parameters
}
```

The laminar models still use the viscosity *property* ν (`nu`) specified in the *physicalProperties* file, *e.g.*

```
viscosityModel constant;
```

```
nu          1.5e-05;
```

This viscosity is a single value which is **constant** in time and **uniform** over the solution domain. The non-Newtonian models adopt ν as the zero strain-rate viscosity ν_0 . The visco-elastic models incorporate a linear viscous stress using ν , in addition to stress calculated by the respective models. The details of the models are provided in the following sections.

8.3.1 Bird-Carreau model

The Bird-Carreau `generalisedNewtonian` model is

$$\nu = \nu_{\infty} + (\nu_0 - \nu_{\infty}) [1 + (k\dot{\gamma})^a]^{(n-1)/a} \quad (8.22)$$

where the coefficient a has a default value of 2. An example specification of the model in *momentumTransport* is:

```
viscosityModel BirdCarreau;
```

```
nuInf      1e-05;
```

```
k          1;
```

```
n          0.5;
```

The constant, uniform viscosity at zero strain-rate, ν_0 , is specified by `nu` in the *physicalProperties* file.

8.3.2 Cross Power Law model

The Cross Power Law `generalisedNewtonian` model is:

$$\nu = \nu_{\infty} + \frac{\nu_0 - \nu_{\infty}}{1 + (m\dot{\gamma})^n} \quad (8.23)$$

An example specification of the model in *momentumTransport* is:

```
viscosityModel CrossPowerLaw;
```

```
nuInf      1e-05;
```

```
m          1;
```

```
n          0.5;
```

The constant, uniform viscosity at zero strain-rate, ν_0 , is specified by `nu` in the *physicalProperties* file.

8.3.3 Power Law model

The Power Law generalisedNewtonian model provides a function for viscosity, limited by minimum and maximum values, $\nu_{\min}$ and $\nu_{\max}$ respectively. The function is:

$$\nu = k\dot{\gamma}^{n-1} \quad \nu_{\min} \leq \nu \leq \nu_{\max} \quad (8.24)$$

An example specification of the model in *momentumTransport* is:

```
viscosityModel powerLaw;
```

```
nuMax    1e-03;
nuMin    1e-05;
k        1e-05;
n        0.5;
```

8.3.4 Herschel-Bulkley model

The Herschel-Bulkley generalisedNewtonian model combines the effects of Bingham plastic and power-law behaviour in a fluid. For low strain rates, the material is modelled as a very viscous fluid with viscosity ν_0 . Beyond a threshold in strain-rate corresponding to threshold stress τ_0 , the viscosity is described by a power law. The model is:

$$\nu = \min\left(\nu_0, \tau_0/\dot{\gamma} + k\dot{\gamma}^{n-1}\right) \quad (8.25)$$

An example specification of the model in *momentumTransport* is:

```
viscosityModel HerschelBulkley;
```

```
tau0     0.01;
k        0.001;
n        0.5;
```

The constant, uniform viscosity at zero strain-rate, ν_0 , is specified in the *physicalProperties* file.

8.3.5 Casson model

The Casson generalisedNewtonian model is a basic model used in blood rheology that specifies minimum and maximum viscosities, $\nu_{\min}$ and $\nu_{\max}$ respectively. Beyond a threshold in strain-rate corresponding to threshold stress τ_0 , the viscosity is described by a “square-root” relationship. The model is:

$$\nu = \left(\sqrt{\tau_0/\dot{\gamma}} + \sqrt{m}\right)^2 \quad \nu_{\min} \leq \nu \leq \nu_{\max} \quad (8.26)$$

An example specification of model parameters for blood is:

```
viscosityModel Casson;
```

```
m      3.934986e-6;
tau0   2.9032e-6;
nuMax  13.3333e-6;
nuMin  3.9047e-6;
```

8.3.6 General strain-rate function

A `strainRateFunction` generalisedNewtonian model exists that allows a user to specify viscosity as a function of strain rate at run-time. It uses the same `Function1` functionality to specify the function of strain-rate, used by time varying properties in boundary conditions described in section 6.4.4. An example specification of the model in *momentumTransport* is shown below using the `polynomial` function:

```
viscosityModel strainRateFunction;

function polynomial ((0 0.1) (1 1.3));
```

8.3.7 Maxwell model

The Maxwell laminar visco-elastic model solves an equation for the fluid stress tensor τ :

$$\frac{\partial \tau}{\partial t} + \nabla \cdot (\mathbf{U} \tau) = 2 \text{symm} [\tau \cdot \nabla \mathbf{U}] - 2 \frac{\nu_M}{\lambda} \text{symm}(\nabla \mathbf{U}) - \frac{1}{\lambda} \tau \quad (8.27)$$

where ν_M (`nuM`) is the “Maxwell” viscosity and λ (`lambda`) is the relaxation time. An example specification of model parameters is shown below:

```
simulationType laminar;

laminar
{
    model                Maxwell;

    Maxwell
    {
        nuM              0.002;
        lambda            0.03;
    }
}
```

If an additional constant, uniform viscosity at zero strain-rate, ν_0 , is specified in the *physicalProperties* file, the model becomes equivalent to an Oldroyd-B visco-elastic model. The Maxwell model includes a multi-mode option where τ is a sum of stresses, each with an associated relaxation time λ .

8.3.8 Giesekus model

The Giesekus laminar visco-elastic model is similar to the Maxwell model but includes an additional “mobility” term in the equation for τ :

$$\frac{\partial \tau}{\partial t} + \nabla \cdot (\mathbf{U}\tau) = 2 \text{symm} [\tau \cdot \nabla \mathbf{U}] - 2 \frac{\nu_M}{\lambda} \text{symm}(\nabla \mathbf{U}) - \frac{1}{\lambda} \tau - \frac{\alpha_G}{\nu_M} [\tau_i \cdot \tau_i] \quad (8.28)$$

where α_G (alphaG) is the mobility parameter. An example specification of model parameters is shown below:

```
simulationType laminar;

laminar
{
    model                Giesekus;

    Giesekus
    {
        nuM              0.002;
        lambda           0.03;
        alphaG           0.1;
    }
}
```

The Giesekus model includes a multi-mode option where τ is a sum of stresses, each with an associated relaxation time λ and mobility coefficient α_G .

8.3.9 Phan-Thien-Tanner (PTT) model

The Phan-Thien-Tanner (PTT) laminar visco-elastic model is also similar to the Maxwell model but includes an additional “extensibility” term in the equation for τ , suitable for polymeric liquids:

$$\frac{\partial \tau}{\partial t} + \nabla \cdot (\mathbf{U}\tau) = 2 \text{symm} [\tau \cdot \nabla \mathbf{U}] - 2 \frac{\nu_M}{\lambda} \text{symm}(\nabla \mathbf{U}) - \frac{1}{\lambda} \exp \left(-\frac{\varepsilon \lambda}{\nu_M} \text{tr}(\tau) \right) \tau \quad (8.29)$$

where ε (epsilon) is the extensibility parameter. An example specification of model parameters is shown below:

```
simulationType laminar;

laminar
{
    model                PTT;

    PTT
    {
```

```

        nuM          0.002;
        lambda       0.03;
        epsilon      0.25;
    }
}
```

The PTT model includes a multi-mode option where τ is a sum of stresses, each with an associated relaxation time λ and extensibility coefficient ε .

8.3.10 Lambda thixotropic model

The Lambda Thixotropic laminar model calculates the evolution of a structural parameter λ (`lambda`) according to:

$$\frac{\partial \lambda}{\partial t} + \nabla \cdot (\mathbf{U} \lambda) = a(1 - \lambda)^b - c \dot{\gamma}^d \lambda \quad (8.30)$$

with model coefficients a , b , c and d . The viscosity ν is then calculated according to:

$$\nu = \frac{\nu_\infty}{1 - K \lambda^2} \quad (8.31)$$

where the parameter $K = \sqrt{\nu_\infty / \nu_0}$. The viscosities ν_0 and ν_∞ are limiting values corresponding to $\lambda = 1$ and $\lambda = 0$.

An example specification of the model in *momentumTransport* is:

```

simulationType laminar;

laminar
{
    model          lambdaThixotropic;

    lambdaThixotropic
    {
        a          1;
        b          2;
        c          1e-3;
        d          3;
        nu0        0.1;
        nuInf      1e-4;
    }
}
```

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